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Online Instructor's Manual
for

Electronic Devices and Circuit Theory

Eleventh Edition

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PEARSON

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Chapter 1

1. Copper has 20 orbiting electrons with only one electron in the outermost shell. The fact that the outermost shell with its 29th electron is incomplete (subshell can contain 2 electrons) and distant from the nucleus reveals that this electron is loosely bound to its parent atom. The application of an external electric field of the correct polarity can easily draw this loosely bound electron from its atomic structure for conduction.

Both intrinsic silicon and germanium have complete outer shells due to the sharing (covalent bonding) of electrons between atoms. Electrons that are part of a complete shell structure require increased levels of applied attractive forces to be removed from their parent atom.

2. Intrinsic material: an intrinsic semiconductor is one that has been refined to be as pure as physically possible. That is, one with the fewest possible number of impurities.

Negative temperature coefficient: materials with negative temperature coefficients have decreasing resistance levels as the temperature increases.

Covalent bonding: covalent bonding is the sharing of electrons between neighboring atoms to form complete outermost shells and a more stable lattice structure.

3. –

4. a. $W = QV = (12 \mu C)(6 V) = \mathbf{72 \mu J}$

$$b. \quad 72 \times 10^{-6} J = \left[\frac{1 \text{ eV}}{1.6 \times 10^{-19} J} \right] = \mathbf{2.625 \times 10^{14} eV}$$

5. $48 \text{ eV} = 48(1.6 \times 10^{-19} J) = \mathbf{76.8 \times 10^{-19} J}$

$$Q = \frac{W}{V} = \frac{76.8 \times 10^{-19} J}{3.2 V} = \mathbf{2.40 \times 10^{-18} C}$$

$6.4 \times 10^{-19} C$ is the charge associated with 4 electrons.

6.

GaP	Gallium Phosphide	$E_g = \mathbf{2.24 eV}$
ZnS	Zinc Sulfide	$E_g = \mathbf{3.67 eV}$

7. An *n*-type semiconductor material has an excess of electrons for conduction established by doping an intrinsic material with donor atoms having more valence electrons than needed to establish the covalent bonding. The majority carrier is the electron while the minority carrier is the hole.

A *p*-type semiconductor material is formed by doping an intrinsic material with acceptor atoms having an insufficient number of electrons in the valence shell to complete the covalent bonding thereby creating a hole in the covalent structure. The majority carrier is the hole while the minority carrier is the electron.

8. A donor atom has five electrons in its outermost valence shell while an acceptor atom has only 3 electrons in the valence shell.

9. Majority carriers are those carriers of a material that far exceed the number of any other carriers in the material.
Minority carriers are those carriers of a material that are less in number than any other carrier of the material.
10. Same basic appearance as Fig. 1.7 since arsenic also has 5 valence electrons (pentavalent).
11. Same basic appearance as Fig. 1.9 since boron also has 3 valence electrons (trivalent).
12. –
13. –
14. For forward bias, the positive potential is applied to the p -type material and the negative potential to the n -type material.
15. a.
$$V_T = \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(20^\circ\text{C} + 273^\circ\text{C})}{1.6 \times 10^{-19} \text{ C}}$$
$$= \mathbf{25.27 \text{ mV}}$$

b.
$$I_D = I_s (e^{V_D/nV_T} - 1)$$
$$= 40 \text{ nA}(e^{(0.5 \text{ V})/(2)(25.27 \text{ mV})} - 1)$$
$$= 40 \text{ nA}(e^{9.89} - 1) = \mathbf{0.789 \text{ mA}}$$
16. a.
$$V_T = \frac{k(T_K)}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(100^\circ\text{C} + 273^\circ\text{C})}{1.6 \times 10^{-19}}$$
$$= \mathbf{32.17 \text{ mV}}$$

b.
$$I_D = I_s (e^{V_D/nV_T} - 1)$$
$$= 40 \text{ nA}(e^{(0.5 \text{ V})/(2)(32.17 \text{ mV})} - 1)$$
$$= 40 \text{ nA}(e^{7.77} - 1) = \mathbf{11.84 \text{ mA}}$$
17. a. $T_K = 20 + 273 = 293$
$$V_T = \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(293^\circ)}{1.6 \times 10^{-19} \text{ C}}$$
$$= \mathbf{25.27 \text{ mV}}$$

b.
$$I_D = I_s (e^{V_D/nV_T} - 1)$$
$$= 0.1 \mu\text{A}(e^{-10/(2)(25.27 \text{ mV})} - 1)$$
$$= 0.1 \mu\text{A}(e^{-197.86} - 1)$$
$$\cong \mathbf{0.1 \mu\text{A}}$$

$$\begin{aligned}
 18. \quad V_T &= \frac{kT_K}{q} = \frac{(1.38 \times 10^{-23} \text{ J/K})(25^\circ\text{C} + 273^\circ\text{C})}{1.6 \times 10^{-19} \text{ C}} \\
 &= 25.70 \text{ mV} \\
 I_D &= I_s(e^{V_D/nV_T} - 1) \\
 8 \text{ mA} &= I_s(e^{(0.5 \text{ V})/(1)(25.70 \text{ mV})} - 1) = I_s(28 \times 10^8) \\
 I_s &= \frac{8 \text{ mA}}{2.8 \times 10^8} = \mathbf{28.57 \text{ pA}}
 \end{aligned}$$

$$\begin{aligned}
 19. \quad I_D &= I_s(e^{V_D/nV_T} - 1) \\
 6 \text{ mA} &= 1 \text{ nA}(e^{V_D/(1)(26 \text{ mV})} - 1) \\
 6 \times 10^6 &= e^{V_D/26 \text{ mV}} - 1 \\
 e^{V_D/26 \text{ mV}} &= 6 \times 10^6 - 1 \cong 6 \times 10^6 \\
 \log_e e^{V_D/26 \text{ mV}} &= \log_e 6 \times 10^6 \\
 \frac{V_D}{26 \text{ mV}} &= 15.61 \\
 V_D &= 15.61(26 \text{ mV}) \cong \mathbf{0.41 \text{ V}}
 \end{aligned}$$

20. (a)

x	$y = e^x$
0	1
1	2.7182
2	7.389
3	20.086
4	54.6
5	148.4

(b) $y = e^0 = 1$

(c) For $x = 0$, $e^0 = 1$ and $I = I_s(1 - 1) = \mathbf{0 \text{ mA}}$

$$\begin{aligned}
 21. \quad T = 20^\circ\text{C}: I_s &= 0.1 \text{ } \mu\text{A} \\
 T = 30^\circ\text{C}: I_s &= 2(0.1 \text{ } \mu\text{A}) = 0.2 \text{ } \mu\text{A} \text{ (Doubles every } 10^\circ\text{C rise in temperature)} \\
 T = 40^\circ\text{C}: I_s &= 2(0.2 \text{ } \mu\text{A}) = 0.4 \text{ } \mu\text{A} \\
 T = 50^\circ\text{C}: I_s &= 2(0.4 \text{ } \mu\text{A}) = 0.8 \text{ } \mu\text{A} \\
 T = 60^\circ\text{C}: I_s &= 2(0.8 \text{ } \mu\text{A}) = \mathbf{1.6 \text{ } \mu\text{A}} \\
 1.6 \text{ } \mu\text{A}: 0.1 \text{ } \mu\text{A} &\Rightarrow 16:1 \text{ increase due to rise in temperature of } 40^\circ\text{C}.
 \end{aligned}$$

22. For most applications the silicon diode is the device of choice due to its higher temperature capability. Ge typically has a working limit of about 85 degrees centigrade while Si can be used at temperatures approaching 200 degrees centigrade. Silicon diodes also have a higher current handling capability. Germanium diodes are the better device for some RF small signal applications, where the smaller threshold voltage may prove advantageous.

23. From 1.19:

	-75°C	25°C	100°C	200°C
V_F @ 10 mA	1.1 V	0.85 V	1.0 V	0.6 V
I_s	0.01 pA	1 pA	1 μ A	1.05 μ A

V_F decreased with increase in temperature

$$1.1 \text{ V} : 0.6 \text{ V} \cong \mathbf{2.6:1}$$

I_s increased with increase in temperature

$$1 \mu\text{A} : 0.01 \mu\text{A} = \mathbf{20:1}$$

24. An “ideal” device or system is one that has the characteristics we would prefer to have when using a device or system in a practical application. Usually, however, technology only permits a close replica of the desired characteristics. The “ideal” characteristics provide an excellent basis for comparison with the actual device characteristics permitting an estimate of how well the device or system will perform. On occasion, the “ideal” device or system can be assumed to obtain a good estimate of the overall response of the design. When assuming an “ideal” device or system there is no regard for component or manufacturing tolerances or any variation from device to device of a particular lot.
25. In the forward-bias region the 0 V drop across the diode at any level of current results in a resistance level of zero ohms – the “on” state – conduction is established. In the reverse-bias region the zero current level at any reverse-bias voltage assures a very high resistance level – the open circuit or “off” state – conduction is interrupted.
26. The most important difference between the characteristics of a diode and a simple switch is that the switch, being mechanical, is capable of conducting current in either direction while the diode only allows charge to flow through the element in one direction (specifically the direction defined by the arrow of the symbol using conventional current flow).
27. $V_D \cong 0.7 \text{ V}$, $I_D = 4 \text{ mA}$

$$R_{DC} = \frac{V_D}{I_D} = \frac{0.7 \text{ V}}{4 \text{ mA}} = \mathbf{175 \Omega}$$
28. At $I_D = 15 \text{ mA}$, $V_D = 0.82 \text{ V}$

$$R_{DC} = \frac{V_D}{I_D} = \frac{0.82 \text{ V}}{15 \text{ mA}} = \mathbf{54.67 \Omega}$$

 As the forward diode current increases, the static resistance decreases.

$$\begin{aligned}
29. \quad & V_D = -10 \text{ V}, I_D = I_s = \mathbf{-0.1 \mu A} \\
& R_{DC} = \frac{V_D}{I_D} = \frac{10 \text{ V}}{0.1 \mu A} = \mathbf{100 \text{ M}\Omega} \\
& V_D = -30 \text{ V}, I_D = I_s = \mathbf{-0.1 \mu A} \\
& R_{DC} = \frac{V_D}{I_D} = \frac{30 \text{ V}}{0.1 \mu A} = \mathbf{300 \text{ M}\Omega}
\end{aligned}$$

As the reverse voltage increases, the reverse resistance increases directly (since the diode leakage current remains constant).

$$\begin{aligned}
30. \quad & I_D = 10 \text{ mA}, V_D = 0.76 \text{ V} \\
& R_{DC} = \frac{V_D}{I_D} = \frac{0.76 \text{ V}}{10 \text{ mA}} = \mathbf{76 \Omega} \\
& r_d = \frac{\Delta V_d}{\Delta I_d} \cong \frac{0.79 \text{ V} - 0.76 \text{ V}}{15 \text{ mA} - 5 \text{ mA}} = \frac{0.03 \text{ V}}{10 \text{ mA}} = \mathbf{3 \Omega} \\
& R_{DC} \gg r_d
\end{aligned}$$

$$\begin{aligned}
31. \quad (a) \quad & r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.79 \text{ V} - 0.76 \text{ V}}{15 \text{ mA} - 5 \text{ mA}} = \frac{0.03 \text{ V}}{10 \text{ mA}} = \mathbf{3 \Omega} \\
(b) \quad & r_d = \frac{26 \text{ mV}}{I_D} = \frac{26 \text{ mV}}{10 \text{ mA}} = \mathbf{2.6 \Omega}
\end{aligned}$$

(c) quite close

$$\begin{aligned}
32. \quad & I_D = 1 \text{ mA}, r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.72 \text{ V} - 0.61 \text{ V}}{2 \text{ mA} - 0 \text{ mA}} = \mathbf{55 \Omega} \\
& I_D = 15 \text{ mA}, r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0.8 \text{ V} - 0.78 \text{ V}}{20 \text{ mA} - 10 \text{ mA}} = \mathbf{2 \Omega}
\end{aligned}$$

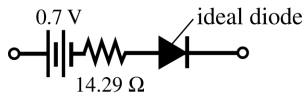
$$\begin{aligned}
33. \quad & I_D = 1 \text{ mA}, r_d = 2 \left(\frac{26 \text{ mV}}{I_D} \right) = 2(26 \Omega) = \mathbf{52 \Omega} \text{ vs } 55 \Omega \text{ (#30)} \\
& I_D = 15 \text{ mA}, r_d = \frac{26 \text{ mV}}{I_D} = \frac{26 \text{ mV}}{15 \text{ mA}} = \mathbf{1.73 \Omega} \text{ vs } 2 \Omega \text{ (#30)}
\end{aligned}$$

$$34. \quad r_{av} = \frac{\Delta V_d}{\Delta I_d} = \frac{0.9 \text{ V} - 0.6 \text{ V}}{13.5 \text{ mA} - 1.2 \text{ mA}} = \mathbf{24.4 \Omega}$$

$$35. \quad r_d = \frac{\Delta V_d}{\Delta I_d} \cong \frac{0.8 \text{ V} - 0.7 \text{ V}}{7 \text{ mA} - 3 \text{ mA}} = \frac{0.09 \text{ V}}{4 \text{ mA}} = \mathbf{22.5 \Omega}$$

(relatively close to average value of 24.4 Ω (#32))

36.
$$r_{av} = \frac{\Delta V_d}{\Delta I_d} = \frac{0.9 \text{ V} - 0.7 \text{ V}}{14 \text{ mA} - 0 \text{ mA}} = \frac{0.2 \text{ V}}{14 \text{ mA}} = \mathbf{14.29 \Omega}$$



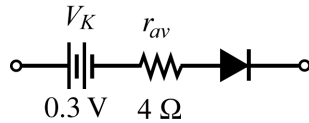
37. Using the best approximation to the curve beyond $V_D = 0.7 \text{ V}$:

$$r_{av} = \frac{\Delta V_d}{\Delta I_d} \cong \frac{0.8 \text{ V} - 0.7 \text{ V}}{25 \text{ mA} - 0 \text{ mA}} = \frac{0.1 \text{ V}}{25 \text{ mA}} = \mathbf{4 \Omega}$$



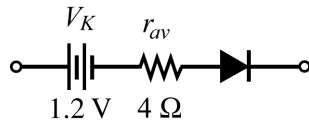
38. Germanium:

$$r_{av} = \frac{0.42 \text{ V} - 0.3 \text{ V}}{30 \text{ mA} - 0 \text{ mA}} = 4 \Omega$$



GaAs:

$$r_{av} = \frac{1.32 \text{ V} - 1.2 \text{ V}}{30 \text{ mA} - 0 \text{ mA}} = 4 \Omega$$



39. (a) $V_R = -25 \text{ V}$: $C_T \cong \mathbf{0.75 \text{ pF}}$
 $V_R = -10 \text{ V}$: $C_T \cong \mathbf{1.25 \text{ pF}}$

$$\left| \frac{\Delta C_T}{\Delta V_R} \right| = \left| \frac{1.25 \text{ pF} - 0.75 \text{ pF}}{10 \text{ V} - 25 \text{ V}} \right| = \frac{0.5 \text{ pF}}{15 \text{ V}} = \mathbf{0.033 \text{ pF/V}}$$

(b) $V_R = -10 \text{ V}$: $C_T \cong \mathbf{1.25 \text{ pF}}$
 $V_R = -1 \text{ V}$: $C_T \cong \mathbf{3 \text{ pF}}$

$$\left| \frac{\Delta C_T}{\Delta V_R} \right| = \left| \frac{1.25 \text{ pF} - 3 \text{ pF}}{10 \text{ V} - 1 \text{ V}} \right| = \frac{1.75 \text{ pF}}{9 \text{ V}} = \mathbf{0.194 \text{ pF/V}}$$

(c) 0.194 pF/V : $0.033 \text{ pF/V} = 5.88:1 \cong \mathbf{6:1}$
 Increased sensitivity near $V_D = 0 \text{ V}$

40. From Fig. 1.33
 $V_D = 0 \text{ V}$, $C_D = \mathbf{3.3 \text{ pF}}$
 $V_D = 0.25 \text{ V}$, $C_D = \mathbf{9 \text{ pF}}$

41. The transition capacitance is due to the depletion region acting like a dielectric in the reverse-bias region, while the diffusion capacitance is determined by the rate of charge injection into the region just outside the depletion boundaries of a forward-biased device. Both capacitances are present in both the reverse- and forward-bias directions, but the transition capacitance is the dominant effect for reverse-biased diodes and the diffusion capacitance is the dominant effect for forward-biased conditions.

42. $V_D = 0.2 \text{ V}$, $C_D = 7.3 \text{ pF}$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(6 \text{ MHz})(7.3 \text{ pF})} = \mathbf{3.64 \text{ k}\Omega}$$
 $V_D = -20 \text{ V}$, $C_T = 0.9 \text{ pF}$

$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(6 \text{ MHz})(0.9 \text{ pF})} = \mathbf{29.47 \text{ k}\Omega}$$

43.
$$C_T = \frac{C(0)}{(1 + |V_R/V_K|)^n} = \frac{8 \text{ pF}}{(1 + |5 \text{ V}/0.7 \text{ V}|)^{1/2}}$$

$$= \frac{8 \text{ pF}}{(1 + 7.14)^{1/2}} = \frac{8 \text{ pF}}{\sqrt{8.14}} = \frac{8 \text{ pF}}{2.85}$$

$$= \mathbf{2.81 \text{ pF}}$$

44.
$$C_T = \frac{C(0)}{(1 + |V_R/V_K|)^n}$$

$$4 \text{ pF} = \frac{10 \text{ pF}}{(1 + |V_R/0.7 \text{ V}|)^{1/3}}$$

$$= (1 + V_R/0.7 \text{ V})^{1/3} = 2.5$$

$$1 + V_R/0.7 \text{ V} = (2.5)^3 = 15.63$$

$$V_R/0.7 \text{ V} = 15.63 - 1 = 14.63$$

$$V_R = (0.7)(14.63) = \mathbf{10.24 \text{ V}}$$

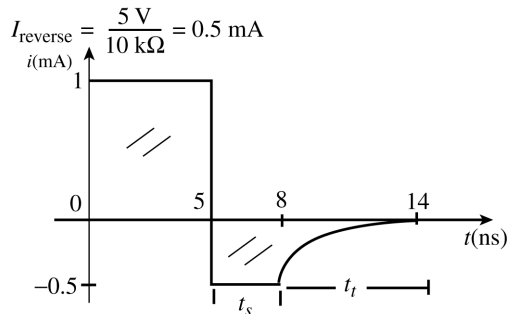
45. $I_f = \frac{10 \text{ V}}{10 \text{ k}\Omega} = 1 \text{ mA}$

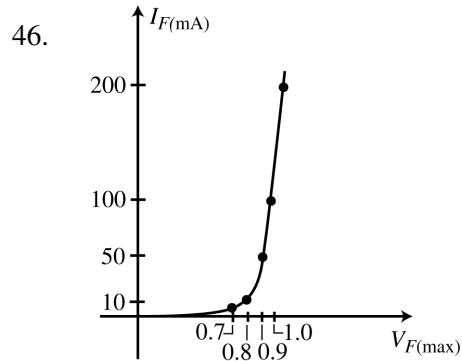
$$t_s + t_t = t_{rr} = 9 \text{ ns}$$

$$t_s + 2t_s = 9 \text{ ns}$$

$$t_s = \mathbf{3 \text{ ns}}$$

$$t_t = 2t_s = \mathbf{6 \text{ ns}}$$





47. a. As the magnitude of the reverse-bias potential increases, the capacitance drops rapidly from a level of about 5 pF with no bias. For reverse-bias potentials in excess of 10 V the capacitance levels off at about 1.5 pF.

b. 6 pF

c. At $V_R = -4$ V, $C_T = 2$ pF

$$C_T = \frac{C(0)}{(1 + |V_R/V_k|)^n}$$

$$2 \text{ pF} = \frac{6 \text{ pF}}{(1 + |4\text{V}/0.7 \text{ V}|)^n}$$

$$(1 + |4 \text{ V} + 0.7 \text{ V}|)^n = 3$$

$$(6.71)^n = 3$$

$$n \log_{10} 6.71 = \log_{10} 3$$

$$n(0.827) = 0.477$$

$$n = \frac{0.477}{0.827} \cong \mathbf{0.58}$$

48. At $V_D = -25$ V, $I_D = -0.2$ nA and at $V_D = -100$ V, $I_D \cong -0.45$ nA. Although the change in I_R is more than 100%, the level of I_R and the resulting change is relatively small for most applications.

49. Log scale: $T_A = 25^\circ\text{C}$, $I_R = \mathbf{0.5 \text{ nA}}$
 $T_A = 100^\circ\text{C}$, $I_R = \mathbf{60 \text{ nA}}$

The change is significant.

$$60 \text{ nA} : 0.5 \text{ nA} = \mathbf{120:1}$$

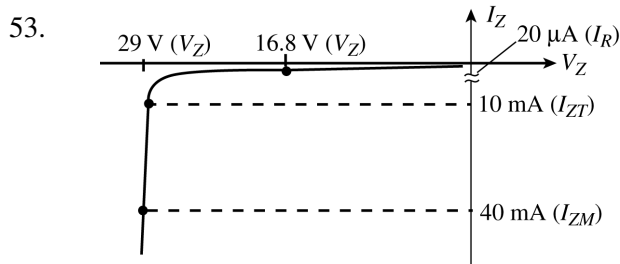
Yes, at 95°C I_R would increase to 64 nA starting with 0.5 nA (at 25°C) (and double the level every 10°C).

50. $I_F = 0.1$ mA: $r_d \cong \mathbf{700 \Omega}$
 $I_F = 1.5$ mA: $r_d \cong \mathbf{70 \Omega}$
 $I_F = 20$ mA: $r_d \cong \mathbf{6 \Omega}$

The results support the fact that the dynamic or ac resistance decreases rapidly with increasing current levels.

51. $T = 25^\circ\text{C}: P_{\max} = 500 \text{ mW}$
 $T = 100^\circ\text{C}: P_{\max} = 260 \text{ mW}$
 $P_{\max} = V_F I_F$
 $I_F = \frac{P_{\max}}{V_F} = \frac{500 \text{ mW}}{0.7 \text{ V}} = \mathbf{714.29 \text{ mA}}$
 $I_F = \frac{P_{\max}}{V_F} = \frac{260 \text{ mW}}{0.7 \text{ V}} = \mathbf{371.43 \text{ mA}}$
 $714.29 \text{ mA}: 371.43 \text{ mA} = 1.92:1 \cong \mathbf{2:1}$

52. Using the bottom right graph of Fig. 1.37:
 $I_F = 500 \text{ mA} @ T = 25^\circ\text{C}$
At $I_F = 250 \text{ mA}$, $T \cong \mathbf{104^\circ\text{C}}$



54. $T_C = +0.072\% = \frac{\Delta V_Z}{V_Z (T_1 - T_0)} \times 100\%$
 $0.072 = \frac{0.75 \text{ V}}{10 \text{ V (} T_1 - 25 \text{)}} \times 100$
 $0.072 = \frac{7.5}{T_1 - 25}$
 $T_1 - 25^\circ = \frac{7.5}{0.072} = 104.17^\circ$
 $T_1 = 104.17^\circ + 25^\circ = \mathbf{129.17^\circ}$

55. $T_C = \frac{\Delta V_Z}{V_Z (T_1 - T_0)} \times 100\%$
 $= \frac{(5 \text{ V} - 4.8 \text{ V})}{5 \text{ V (} 100^\circ - 25^\circ \text{)}} \times 100\% = \mathbf{0.053\%/^\circ\text{C}}$

56. $\frac{(20 \text{ V} - 6.8 \text{ V})}{(24 \text{ V} - 6.8 \text{ V})} \times 100\% = 77\%$

The 20 V Zener is therefore $\cong 77\%$ of the distance between 6.8 V and 24 V measured from the 6.8 V characteristic.

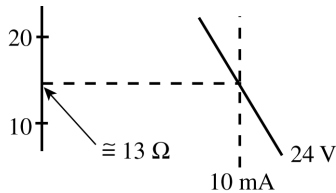
At $I_Z = 0.1 \text{ mA}$, $T_C \cong 0.06\%/^{\circ}\text{C}$

$$\frac{(5 \text{ V} - 3.6 \text{ V})}{(6.8 \text{ V} - 3.6 \text{ V})} \times 100\% = 44\%$$

The 5 V Zener is therefore $\cong 44\%$ of the distance between 3.6 V and 6.8 V measured from the 3.6 V characteristic.

At $I_Z = 0.1 \text{ mA}$, $T_C \cong -0.025\%/^{\circ}\text{C}$

57.



58.

24 V Zener:

0.2 mA: $\cong 400 \Omega$

1 mA: $\cong 95 \Omega$

10 mA: $\cong 13 \Omega$

The steeper the curve (higher dI/dV) the less the dynamic resistance.

59.

$V_K \cong 2.0 \text{ V}$, which is considerably higher than germanium ($\cong 0.3 \text{ V}$) or silicon ($\cong 0.7 \text{ V}$). For germanium it is a 6.7:1 ratio, and for silicon a 2.86:1 ratio.

60.

$$0.67 \text{ eV} \left[\frac{1.6 \times 10^{-19} \text{ J}}{1 \text{ eV}} \right] = 1.072 \times 10^{-19} \text{ J}$$

$$E_g = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E_g} = \frac{(6.626 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ m/s})}{1.072 \times 10^{-19} \text{ J}}$$

$$= 1850 \text{ nm}$$

Very low energy level.

61.

Fig. 1.53 (f) $I_F \cong 13 \text{ mA}$

Fig. 1.53 (e) $V_F \cong 2.3 \text{ V}$

62.

(a) Relative efficiency @ 5 mA $\cong 0.82$

@ 10 mA $\cong 1.02$

$$\frac{1.02 - 0.82}{0.82} \times 100\% = 24.4\% \text{ increase}$$

$$\text{ratio: } \frac{1.02}{0.82} = 1.24$$

(b) Relative efficiency @ 30 mA $\cong 1.38$

@ 35 mA $\cong 1.42$

$$\frac{1.42 - 1.38}{1.38} \times 100\% = 2.9\% \text{ increase}$$

$$\text{ratio: } \frac{1.42}{1.38} = 1.03$$

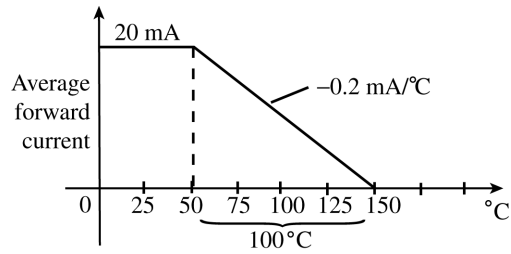
- (c) For currents greater than about 30 mA the percent increase is significantly less than for increasing currents of lesser magnitude.

63. (a) $\frac{0.75}{3.0} = 0.25$

From Fig. 1.53 (i) $\angle \cong 75^\circ$

(b) $0.5 \Rightarrow \angle = 40^\circ$

64. For the high-efficiency red unit of Fig. 1.53:



$$\frac{0.2 \text{ mA}}{^\circ\text{C}} = \frac{20 \text{ mA}}{x}$$

$$x = \frac{20 \text{ mA}}{0.2 \text{ mA/}^\circ\text{C}} = 100^\circ\text{C}$$

Chapter 2

1. The load line will intersect at $I_D = \frac{E}{R} = \frac{12 \text{ V}}{750 \Omega} = 16 \text{ mA}$ and $V_D = 12 \text{ V}$.

- (a) $V_{D_Q} \cong \mathbf{0.85 \text{ V}}$
 $I_{D_Q} \cong \mathbf{15 \text{ mA}}$
 $V_R = E - V_{D_Q} = 12 \text{ V} - 0.85 \text{ V} = \mathbf{11.15 \text{ V}}$
- (b) $V_{D_Q} \cong \mathbf{0.7 \text{ V}}$
 $I_{D_Q} \cong \mathbf{15 \text{ mA}}$
 $V_R = E - V_{D_Q} = 12 \text{ V} - 0.7 \text{ V} = \mathbf{11.3 \text{ V}}$
- (c) $V_{D_Q} \cong \mathbf{0 \text{ V}}$
 $I_{D_Q} \cong \mathbf{16 \text{ mA}}$
 $V_R = E - V_{D_Q} = 12 \text{ V} - 0 \text{ V} = \mathbf{12 \text{ V}}$

For (a) and (b), levels of V_{D_Q} and I_{D_Q} are quite close. Levels of part (c) are reasonably close but as expected due to level of applied voltage E .

2. (a) $I_D = \frac{E}{R} = \frac{6 \text{ V}}{0.2 \text{ k}\Omega} = 30 \text{ mA}$
The load line extends from $I_D = 30 \text{ mA}$ to $V_D = 6 \text{ V}$.
 $V_{D_Q} \cong \mathbf{0.95 \text{ V}}$, $I_{D_Q} \cong \mathbf{25.3 \text{ mA}}$
- (b) $I_D = \frac{E}{R} = \frac{6 \text{ V}}{0.47 \text{ k}\Omega} = 12.77 \text{ mA}$
The load line extends from $I_D = 12.77 \text{ mA}$ to $V_D = 6 \text{ V}$.
 $V_{D_Q} \cong \mathbf{0.8 \text{ V}}$, $I_{D_Q} \cong \mathbf{11 \text{ mA}}$
- (c) $I_D = \frac{E}{R} = \frac{6 \text{ V}}{0.68 \text{ k}\Omega} = 8.82 \text{ mA}$
The load line extends from $I_D = 8.82 \text{ mA}$ to $V_D = 6 \text{ V}$.
 $V_{D_Q} \cong \mathbf{0.78 \text{ V}}$, $I_{D_Q} \cong \mathbf{78 \text{ mA}}$

The resulting values of V_{D_Q} are quite close, while I_{D_Q} extends from 7.8 mA to 25.3 mA.

3. Load line through $I_{D_Q} = 10 \text{ mA}$ of characteristics and $V_D = 7 \text{ V}$ will intersect I_D axis as 11.3 mA.

$$I_D = 11.3 \text{ mA} = \frac{E}{R} = \frac{7 \text{ V}}{R}$$

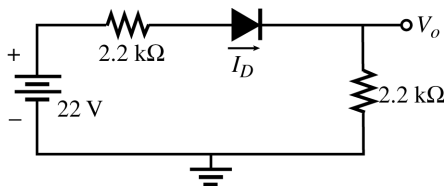
$$\text{with } R = \frac{7 \text{ V}}{11.3 \text{ mA}} = 619.47 \text{ k}\Omega \cong \mathbf{0.62 \text{ k}\Omega} \text{ standard resistor}$$

4. (a) $I_D = I_R = \frac{E - V_D}{R} = \frac{30 \text{ V} - 0.7 \text{ V}}{1.5 \text{ k}\Omega} = \mathbf{19.53 \text{ mA}}$
 $V_D = \mathbf{0.7 \text{ V}}$, $V_R = E - V_D = 30 \text{ V} - 0.7 \text{ V} = \mathbf{29.3 \text{ V}}$
- (b) $I_D = \frac{E - V_D}{R} = \frac{30 \text{ V} - 0 \text{ V}}{1.5 \text{ k}\Omega} = \mathbf{20 \text{ mA}}$
 $V_D = \mathbf{0 \text{ V}}$, $V_R = \mathbf{30 \text{ V}}$
- Yes, since $E \gg V_T$ the levels of I_D and V_R are quite close.
5. (a) $I = \mathbf{0 \text{ mA}}$; diode reverse-biased.
- (b) $V_{20\Omega} = 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V}$ (Kirchhoff's voltage law)
 $I(20 \Omega) = \frac{19.3 \text{ V}}{20 \Omega} = 0.965 \text{ A}$
 $V(10 \Omega) = 20 \text{ V} - 0.7 \text{ V} = 19.3 \text{ V}$
 $I(10 \Omega) = \frac{19.3 \text{ V}}{10 \Omega} = 1.93 \text{ A}$
 $I = I(10 \Omega) + I(20 \Omega)$
 $= \mathbf{2.895 \text{ A}}$
- (c) $I = \frac{10 \text{ V}}{10 \Omega} = \mathbf{1 \text{ A}}$; center branch open
6. (a) Diode forward-biased,
 Kirchhoff's voltage law (CW): $-5 \text{ V} + 0.7 \text{ V} - V_o = 0$
 $V_o = \mathbf{-4.3 \text{ V}}$
- $I_R = I_D = \frac{|V_o|}{R} = \frac{4.3 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{1.955 \text{ mA}}$
- (b) Diode forward-biased,
 $I_D = \frac{8 \text{ V} + 6 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega + 4.7 \text{ k}\Omega} = \mathbf{2.25 \text{ mA}}$
 $V_o = 8 \text{ V} - (2.25 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{5.3 \text{ V}}$
7. (a) $V_o = \frac{10 \text{ k}\Omega(12 \text{ V} - 0.7 \text{ V} - 0.3 \text{ V})}{2 \text{ k}\Omega + 10 \text{ k}\Omega} = \mathbf{9.17 \text{ V}}$
- (b) $V_o = \mathbf{10 \text{ V}}$

8. (a) Determine the Thevenin equivalent circuit for the 10 mA source and 2.2 k Ω resistor.

$$E_{Th} = IR = (10 \text{ mA})(2.2 \text{ k}\Omega) = 22 \text{ V}$$

$$R_{Th} = 2.2 \text{ k}\Omega$$



Diode forward-biased

$$I_D = \frac{22 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \mathbf{4.84 \text{ mA}}$$

$$V_o = I_D(1.2 \text{ k}\Omega)$$

$$= (4.84 \text{ mA})(1.2 \text{ k}\Omega)$$

$$= \mathbf{5.81 \text{ V}}$$

- (b) Diode forward-biased

$$I_D = \frac{20 \text{ V} + 20 \text{ V} - 0.7 \text{ V}}{6.8 \text{ k}\Omega} = \mathbf{5.78 \text{ mA}}$$

Kirchhoff's voltage law (CW):

$$+V_o - 0.7 \text{ V} + 20 \text{ V} = 0$$

$$V_o = \mathbf{-19.3 \text{ V}}$$

9. (a) $V_{o1} = 12 \text{ V} - 0.7 \text{ V} = \mathbf{11.3 \text{ V}}$

$$V_{o2} = \mathbf{1.2 \text{ V}}$$

- (b) $V_{o1} = \mathbf{0 \text{ V}}$

$$V_{o2} = \mathbf{0 \text{ V}}$$

10. (a) Both diodes forward-biased
Si diode turns on first and locks in 0.7 V drop.

$$I_R = \frac{12 \text{ V} - 0.7 \text{ V}}{4.7 \text{ k}\Omega} = 2.4 \text{ mA}$$

$$I_D = I_R = \mathbf{2.4 \text{ mA}}$$

$$V_o = 12 \text{ V} - 0.7 \text{ V} = \mathbf{11.3 \text{ V}}$$

- (b) Right diode forward-biased:

$$I_D = \frac{20 \text{ V} + 4 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{10.59 \text{ mA}}$$

$$V_o = 20 \text{ V} - 0.7 \text{ V} = \mathbf{19.3 \text{ V}}$$

11. (a) Si diode "on" preventing GaAs diode from turning "on":

$$I = \frac{1 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \frac{0.3 \text{ V}}{1 \text{ k}\Omega} = \mathbf{0.3 \text{ mA}}$$

$$V_o = 1 \text{ V} - 0.7 \text{ V} = \mathbf{0.3 \text{ V}}$$

- (b) $I = \frac{16 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} + 4 \text{ V}}{4.7 \text{ k}\Omega} = \frac{18.6 \text{ V}}{4.7 \text{ k}\Omega} = \mathbf{3.96 \text{ mA}}$

$$V_o = 16 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V} = \mathbf{14.6 \text{ V}}$$

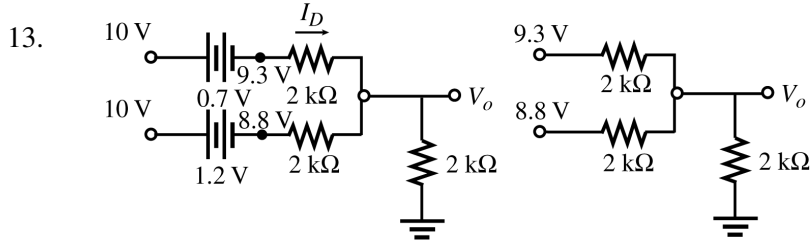
12. Both diodes forward-biased:

$$V_{o_1} = 0.7 \text{ V}, V_{o_2} = 0.7 \text{ V}$$

$$I_{1 \text{ k}\Omega} = \frac{20 \text{ V} - 0.7 \text{ V}}{1 \text{ k}\Omega} = \frac{19.3 \text{ V}}{1 \text{ k}\Omega} = 19.3 \text{ mA}$$

$$I_{0.47 \text{ k}\Omega} = 0 \text{ mA}$$

$$I = I_{1 \text{ k}\Omega} - I_{0.47 \text{ k}\Omega} = 19.3 \text{ mA} - 0 \text{ mA} \\ = \mathbf{19.3 \text{ mA}}$$



$$\text{Superposition: } V_{o_1} (9.3 \text{ V}) = \frac{1 \text{ k}\Omega (9.3 \text{ V})}{1 \text{ k}\Omega + 2 \text{ k}\Omega} = 3.1 \text{ V}$$

$$V_{o_2} (8.8 \text{ V}) = \frac{16 \text{ k}\Omega (8.8 \text{ V})}{1 \text{ k}\Omega + 2 \text{ k}\Omega} = 2.93 \text{ V}$$

$$V_o = V_{o_1} + V_{o_2} = \mathbf{6.03 \text{ V}}$$

$$I_D = \frac{9.3 \text{ V} - 6.03 \text{ V}}{2 \text{ k}\Omega} = \mathbf{1.635 \text{ mA}}$$

14. Both diodes “off”. The threshold voltage of 0.7 V is unavailable for either diode.

$$V_o = \mathbf{0 \text{ V}}$$

15. Both diodes “on”, $V_o = 10 \text{ V} - 0.7 \text{ V} = \mathbf{9.3 \text{ V}}$

16. Both diodes “on”.

$$V_o = \mathbf{0.7 \text{ V}}$$

17. Both diodes “off”, $V_o = \mathbf{10 \text{ V}}$

18. The Si diode with -5 V at the cathode is “on” while the other is “off”. The result is

$$V_o = -5 \text{ V} + 0.7 \text{ V} = \mathbf{-4.3 \text{ V}}$$

19. 0 V at one terminal is “more positive” than -5 V at the other input terminal. Therefore assume lower diode “on” and upper diode “off”.

The result:

$$V_o = 0 \text{ V} - 0.7 \text{ V} = \mathbf{-0.7 \text{ V}}$$

The result supports the above assumptions.

20. Since all the system terminals are at 10 V the required difference of 0.7 V across either diode cannot be established. Therefore, both diodes are “off” and

$$V_o = \mathbf{+10 \text{ V}}$$

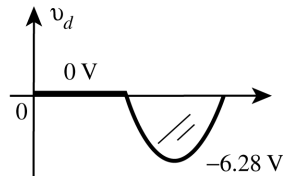
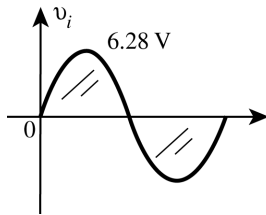
as established by 10 V supply connected to 1 kΩ resistor.

21. The Si diode requires more terminal voltage than the Ge diode to turn “on”. Therefore, with 5 V at both input terminals, assume Si diode “off” and Ge diode “on”.

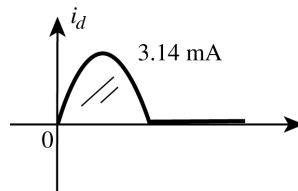
The result: $V_o = 5 \text{ V} - 0.3 \text{ V} = \mathbf{4.7 \text{ V}}$

The result supports the above assumptions.

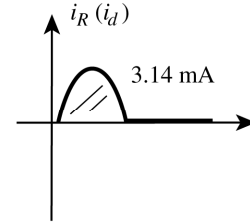
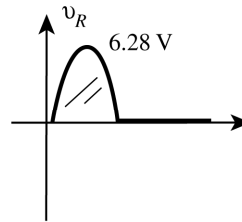
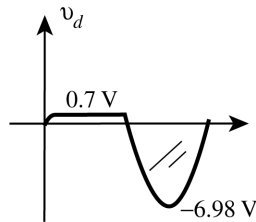
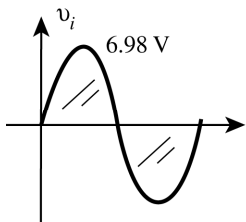
22. $V_{dc} = 0.318 V_m \Rightarrow V_m = \frac{V_{dc}}{0.318} = \frac{2 \text{ V}}{0.318} = \mathbf{6.28 \text{ V}}$



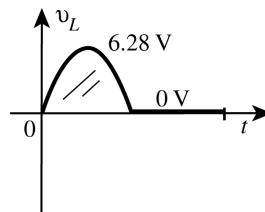
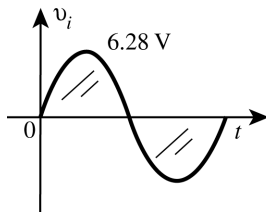
$$I_m = \frac{V_m}{R} = \frac{6.28 \text{ V}}{2 \text{ k}\Omega} = \mathbf{3.14 \text{ mA}}$$



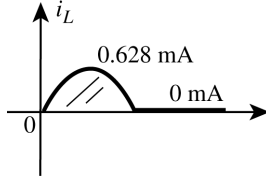
23. Using $V_{dc} \cong 0.318(V_m - V_T)$
 $2 \text{ V} = 0.318(V_m - 0.7 \text{ V})$
 Solving: $V_m = \mathbf{6.98 \text{ V}} \cong 10:1 \text{ for } V_m:V_T$



24. $V_m = \frac{V_{dc}}{0.318} = \frac{2 \text{ V}}{0.318} = \mathbf{6.28 \text{ V}}$

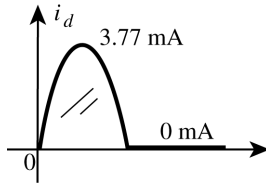


$$I_{L_{\max}} = \frac{6.28 \text{ V}}{10 \text{ k}\Omega} = \mathbf{0.628 \text{ mA}}$$

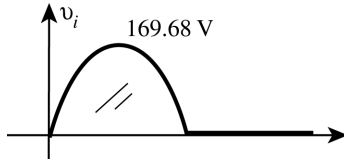


$$I_{\max}(2 \text{ k}\Omega) = \frac{6.28 \text{ V}}{2 \text{ k}\Omega} = \mathbf{3.14 \text{ mA}}$$

$$I_{D_{\max}} = I_{L_{\max}} + I_{\max}(2 \text{ k}\Omega) = 0.678 \text{ mA} + 3.14 \text{ mA} = \mathbf{3.77 \text{ mA}}$$



25. $V_m = \sqrt{2} (120 \text{ V}) = 169.68 \text{ V}$
 $V_{dc} = 0.318 V_m = 0.318(169.68 \text{ V}) = \mathbf{53.96 \text{ V}}$



26. Diode will conduct when $v_o = 0.7 \text{ V}$; that is,

$$v_o = 0.7 \text{ V} = \frac{1 \text{ k}\Omega(v_i)}{1 \text{ k}\Omega + 1 \text{ k}\Omega}$$

 Solving: $v_i = \mathbf{1.4 \text{ V}}$

For $v_i \geq 1.4 \text{ V}$ Si diode is “on” and $v_o = \mathbf{0.7 \text{ V}}$.
 For $v_i < 1.4 \text{ V}$ Si diode is open and level of v_o is determined by voltage divider rule:

$$v_o = \frac{1 \text{ k}\Omega(v_i)}{1 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.5 v_i$$

For $v_i = -10 \text{ V}$:

$$v_o = 0.5(-10 \text{ V})$$

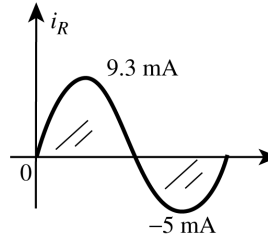
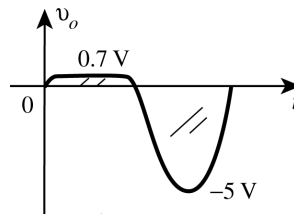
$$= \mathbf{-5 \text{ V}}$$

When $v_o = 0.7 \text{ V}$, $v_{R_{\max}} = v_{i_{\max}} - 0.7 \text{ V}$

$$= 10 \text{ V} - 0.7 \text{ V} = 9.3 \text{ V}$$

$$I_{R_{\max}} = \frac{9.3 \text{ V}}{1 \text{ k}\Omega} = 9.3 \text{ mA}$$

$$I_{\max}(\text{reverse}) = \frac{10 \text{ V}}{1 \text{ k}\Omega + 1 \text{ k}\Omega} = \mathbf{0.5 \text{ mA}}$$



27. (a) $P_{\max} = 14 \text{ mW} = (0.7 \text{ V})I_D$
 $I_D = \frac{14 \text{ mW}}{0.7 \text{ V}} = \mathbf{20 \text{ mA}}$

(b) $I_{\max} = 2 \times 20 \text{ mA} = \mathbf{40 \text{ mA}}$

(c) $4.7 \text{ k}\Omega \parallel 68 \text{ k}\Omega = 4.4 \text{ k}\Omega$
 $V_R = 160 \text{ V} - 0.7 \text{ V} = 159.3 \text{ V}$
 $I_{\max} = \frac{159.3 \text{ V}}{4.4 \text{ k}\Omega} = 36.2 \text{ mA}$
 $I_d = \frac{I_{\max}}{2} = \mathbf{18.1 \text{ mA}}$

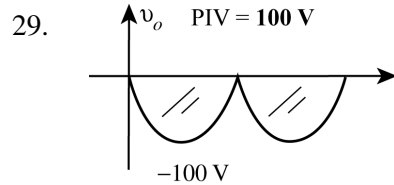
(d) Total damage, $\mathbf{36.2 \text{ mA} > 20 \text{ mA}}$

28. (a) $V_m = \sqrt{2} (120 \text{ V}) = 169.7 \text{ V}$
 $V_{L_m} = V_{i_m} - 2V_D$
 $= 169.7 \text{ V} - 2(0.7 \text{ V}) = 169.7 \text{ V} - 1.4 \text{ V}$
 $= 168.3 \text{ V}$
 $V_{dc} = 0.636(168.3 \text{ V}) = \mathbf{107.04 \text{ V}}$

(b) $\text{PIV} = V_m(\text{load}) + V_D = 168.3 \text{ V} + 0.7 \text{ V} = \mathbf{169 \text{ V}}$

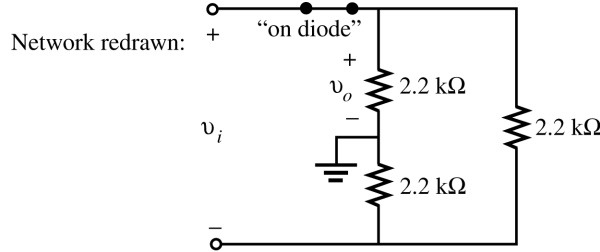
(c) $I_D(\text{max}) = \frac{V_{L_m}}{R_L} = \frac{168.3 \text{ V}}{1 \text{ k}\Omega} = \mathbf{168.3 \text{ mA}}$

(d) $P_{\max} = V_D I_D = (0.7 \text{ V})I_{\max}$
 $= (0.7 \text{ V})(168.3 \text{ mA})$
 $= \mathbf{117.81 \text{ mW}}$

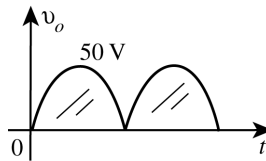
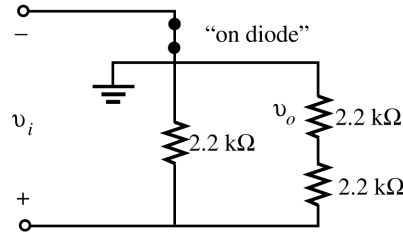


$I_{\max} = \frac{100 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{45.45 \text{ mA}}$

30. Positive half-cycle of v_i :



Negative half-cycle of v_i :



$$V_{dc} = 0.636V_m = 0.636(50 \text{ V}) = 31.8 \text{ V}$$

Voltage-divider rule:

$$\begin{aligned} V_{o_{\max}} &= \frac{2.2 \text{ k}\Omega(V_{i_{\max}})}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} \\ &= \frac{1}{2}(V_{i_{\max}}) \\ &= \frac{1}{2}(100 \text{ V}) \\ &= 50 \text{ V} \end{aligned}$$

Polarity of v_o across the $2.2 \text{ k}\Omega$ resistor acting as a load is the same.

Voltage-divider rule:

$$\begin{aligned} V_{o_{\max}} &= \frac{2.2 \text{ k}\Omega(V_{i_{\max}})}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} \\ &= \frac{1}{2}(V_{i_{\max}}) \\ &= \frac{1}{2}(100 \text{ V}) \\ &= 50 \text{ V} \end{aligned}$$

31. Positive pulse of v_i :

Top left diode "off", bottom left diode "on"

$$2.2 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = 1.1 \text{ k}\Omega$$

$$V_{o_{\text{peak}}} = \frac{1.1 \text{ k}\Omega(170 \text{ V})}{1.1 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 56.67 \text{ V}$$

Negative pulse of v_i :

Top left diode "on", bottom left diode "off"

$$V_{o_{\text{peak}}} = \frac{1.1 \text{ k}\Omega(170 \text{ V})}{1.1 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 56.67 \text{ V}$$

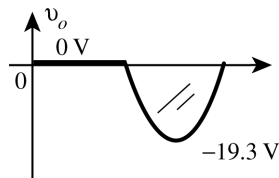
$$V_{dc} = 0.636(56.67 \text{ V}) = 36.04 \text{ V}$$

32. (a) Si diode open for positive pulse of v_i and $v_o = 0 \text{ V}$

For $-20 \text{ V} < v_i \leq -0.7 \text{ V}$ diode "on" and $v_o = v_i + 0.7 \text{ V}$.

$$\text{For } v_i = -20 \text{ V}, v_o = -20 \text{ V} + 0.7 \text{ V} = -19.3 \text{ V}$$

$$\text{For } v_i = -0.7 \text{ V}, v_o = -0.7 \text{ V} + 0.7 \text{ V} = 0 \text{ V}$$



- (b) For $v_i \leq 8 \text{ V}$ the 8 V battery will ensure the diode is forward-biased and $v_o = v_i - 8 \text{ V}$.

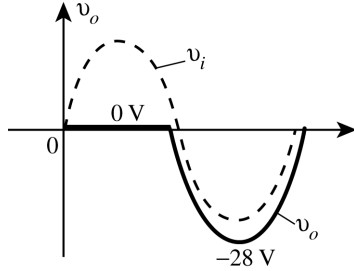
At $v_i = 8 \text{ V}$

$$v_o = 8 \text{ V} - 8 \text{ V} = \mathbf{0 \text{ V}}$$

At $v_i = -20 \text{ V}$

$$v_o = -20 \text{ V} - 8 \text{ V} = \mathbf{-28 \text{ V}}$$

For $v_i > 8 \text{ V}$ the diode is reverse-biased and $v_o = \mathbf{0 \text{ V}}$.

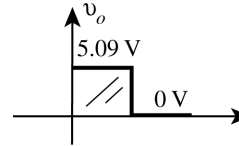


33. (a) Positive pulse of v_i :

$$V_o = \frac{1.8 \text{ k}\Omega(12 \text{ V} - 0.7 \text{ V})}{1.8 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \mathbf{5.09 \text{ V}}$$

Negative pulse of v_i :

diode "open", $v_o = \mathbf{0 \text{ V}}$

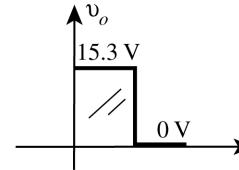


- (b) Positive pulse of v_i :

$$V_o = 12 \text{ V} - 0.7 \text{ V} + 4 \text{ V} = \mathbf{15.3 \text{ V}}$$

Negative pulse of v_i :

diode "open", $v_o = \mathbf{0 \text{ V}}$



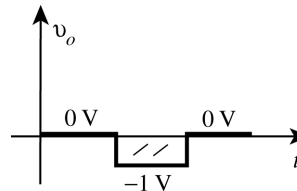
34. (a) For $v_i = 20 \text{ V}$ the diode is reverse-biased and $v_o = \mathbf{0 \text{ V}}$.

For $v_i = -5 \text{ V}$, v_i overpowers the 4 V battery and the diode is "on".

Applying Kirchhoff's voltage law in the clockwise direction:

$$-5 \text{ V} + 4 \text{ V} - v_o = 0$$

$$v_o = \mathbf{-1 \text{ V}}$$

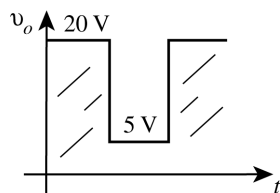


- (b) For $v_i = 20 \text{ V}$ the 20 V level overpowers the 5 V supply and the diode is "on". Using the short-circuit equivalent for the diode we find $v_o = v_i = \mathbf{20 \text{ V}}$.

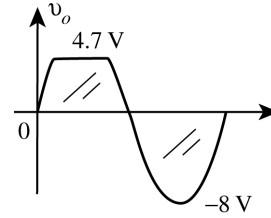
For $v_i = -5 \text{ V}$, both v_i and the 5 V supply reverse-bias the diode and separate v_i from v_o .

However, v_o is connected directly through the 2.2 k Ω resistor to the 5 V supply and

$v_o = \mathbf{5 \text{ V}}$.



35. (a) Diode “on” for $v_i \geq 4.7 \text{ V}$
 For $v_i > 4.7 \text{ V}$, $V_o = 4 \text{ V} + 0.7 \text{ V} = \mathbf{4.7 \text{ V}}$
 For $v_i < 4.7 \text{ V}$, diode “off” and $v_o = v_i$



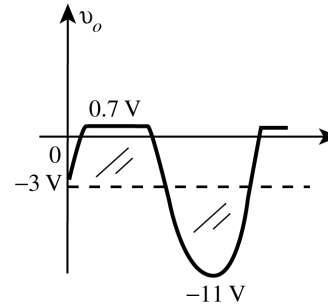
- (b) Again, diode “on” for $v_i \geq 3.7 \text{ V}$ but v_o now defined as the voltage across the diode
 For $v_i \geq 3.7 \text{ V}$, $v_o = \mathbf{0.7 \text{ V}}$

For $v_i < 3.7 \text{ V}$, diode “off”, $I_D = I_R = 0 \text{ mA}$ and $V_{2.2 \text{ k}\Omega} = IR = (0 \text{ mA})R = 0 \text{ V}$

Therefore, $v_o = v_i - 3 \text{ V}$

At $v_i = 0 \text{ V}$, $v_o = \mathbf{-3 \text{ V}}$

$v_i = -8 \text{ V}$, $v_o = -8 \text{ V} - 3 \text{ V} = \mathbf{-11 \text{ V}}$



36. For the positive region of v_i :
 The right Si diode is reverse-biased.
 The left Si diode is “on” for levels of v_i greater than $5.3 \text{ V} + 0.7 \text{ V} = 6 \text{ V}$. In fact, $v_o = \mathbf{6 \text{ V}}$ for $v_i \geq 6 \text{ V}$.

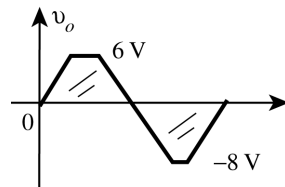
For $v_i < 6 \text{ V}$ both diodes are reverse-biased and $v_o = v_i$.

For the negative region of v_i :

The left Si diode is reverse-biased.

The right Si diode is “on” for levels of v_i more negative than $7.3 \text{ V} + 0.7 \text{ V} = 8 \text{ V}$. In fact, $v_o = \mathbf{-8 \text{ V}}$ for $v_i \leq -8 \text{ V}$.

For $v_i > -8 \text{ V}$ both diodes are reverse-biased and $v_o = v_i$.



i_R : For $-8 \text{ V} < v_i < 6 \text{ V}$ there is no conduction through the $10 \text{ k}\Omega$ resistor due to the lack of a complete circuit. Therefore, $i_R = 0 \text{ mA}$.

For $v_i \geq 6 \text{ V}$

$$v_R = v_i - v_o = v_i - 6 \text{ V}$$

For $v_i = 10 \text{ V}$, $v_R = 10 \text{ V} - 6 \text{ V} = 4 \text{ V}$

$$\text{and } i_R = \frac{4 \text{ V}}{10 \text{ k}\Omega} = \mathbf{0.4 \text{ mA}}$$

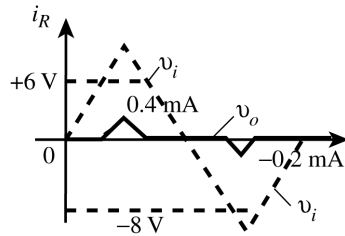
For $v_i \leq -8 \text{ V}$

$$v_R = v_i - v_o = v_i + 8 \text{ V}$$

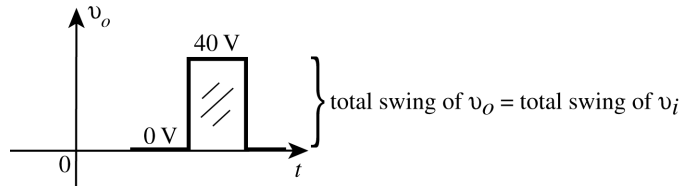
For $v_i = -10 \text{ V}$

$$v_R = -10 \text{ V} + 8 \text{ V} = -2 \text{ V}$$

$$\text{and } i_R = \frac{-2 \text{ V}}{10 \text{ k}\Omega} = -0.2 \text{ mA}$$

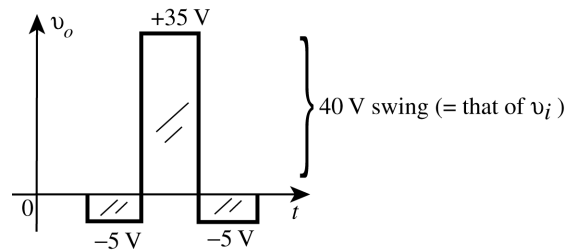


37. (a) Starting with $v_i = -20 \text{ V}$, the diode is in the “on” state and the capacitor quickly charges to $-20 \text{ V}+$. During this interval of time v_o is across the “on” diode (short-current equivalent) and $v_o = 0 \text{ V}$.
When v_i switches to the $+20 \text{ V}$ level the diode enters the “off” state (open-circuit equivalent) and $v_o = v_i + v_C = 20 \text{ V} + 20 \text{ V} = +40 \text{ V}$

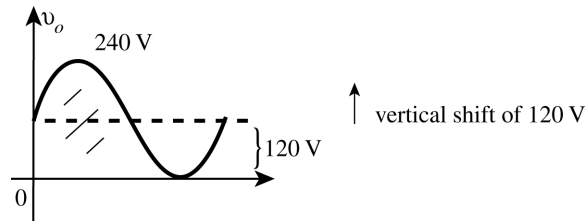


- (b) Starting with $v_i = -20 \text{ V}$, the diode is in the “on” state and the capacitor quickly charges up to $-15 \text{ V}+$. Note that $v_i = +20 \text{ V}$ and the 5 V supply are additive across the capacitor. During this time interval v_o is across “on” diode and 5 V supply and $v_o = -5 \text{ V}$.

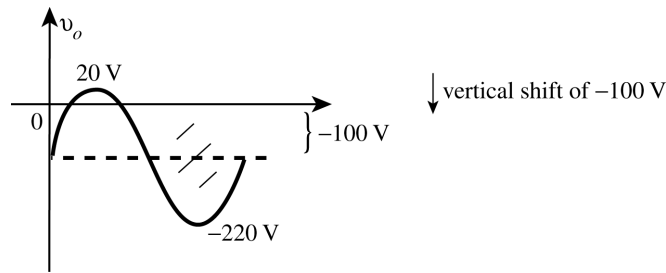
When v_i switches to the $+20 \text{ V}$ level the diode enters the “off” state and $v_o = v_i + v_C = 20 \text{ V} + 15 \text{ V} = 35 \text{ V}$.



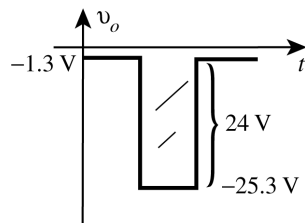
38. (a) For negative half cycle capacitor charges to peak value of $120\text{ V} = 120\text{ V}$ with polarity $(- \text{---} | \text{---} +)$. The output v_o is directly across the “on” diode resulting in $v_o = 0\text{ V}$ as a negative peak value.
For next positive half cycle $v_o = v_i + 120\text{ V}$ with peak value of $v_o = 120\text{ V} + 120\text{ V} = \mathbf{240\text{ V}}$.



- (b) For positive half cycle capacitor charges to peak value of $120\text{ V} - 20\text{ V} = 100\text{ V}$ with polarity $(+ \text{---} | \text{---} -)$. The output $v_o = 20\text{ V} = \mathbf{20\text{ V}}$
For next negative half cycle $v_o = v_i - 100\text{ V}$ with negative peak value of $v_o = -120\text{ V} - 100\text{ V} = \mathbf{-220\text{ V}}$.

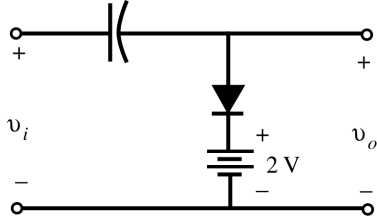


39. (a) $\tau = RC = (56\text{ k}\Omega)(0.1\text{ }\mu\text{F}) = 5.6\text{ ms}$
 $5\tau = \mathbf{28\text{ ms}}$
- (b) $5\tau = 28\text{ ms} \gg \frac{T}{2} = \frac{1\text{ ms}}{2} = \mathbf{0.5\text{ ms}}$, 56:1
- (c) Positive pulse of v_i :
Diode “on” and $v_o = -2\text{ V} + 0.7\text{ V} = -1.3\text{ V}$
Capacitor charges to $12\text{ V} + 2\text{ V} - 0.7\text{ V} = 13.3\text{ V}$
- Negative pulse of v_i :
Diode “off” and $v_o = -12\text{ V} - 13.3\text{ V} = -25.3\text{ V}$



40. Solution is network of Fig. 2.181(b) using a 10 V supply in place of the 5 V source.

41. Network of Fig. 2.178 with 2 V battery reversed.



42. (a) In the absence of the Zener diode

$$V_L = \frac{180 \Omega (20 \text{ V})}{180 \Omega + 220 \Omega} = 9 \text{ V}$$

$$V_L = 9 \text{ V} < V_Z = 10 \text{ V} \text{ and diode non-conducting}$$

$$\text{Therefore, } I_L = I_R = \frac{20 \text{ V}}{220 \Omega + 180 \Omega} = \mathbf{50 \text{ mA}}$$

$$\text{with } I_Z = \mathbf{0 \text{ mA}}$$

$$\text{and } V_L = \mathbf{9 \text{ V}}$$

- (b) In the absence of the Zener diode

$$V_L = \frac{470 \Omega (20 \text{ V})}{470 \Omega + 220 \Omega} = 13.62 \text{ V}$$

$$V_L = 13.62 \text{ V} > V_Z = 10 \text{ V} \text{ and Zener diode "on"}$$

$$\text{Therefore, } V_L = \mathbf{10 \text{ V}} \text{ and } V_{R_s} = 10 \text{ V}$$

$$I_{R_s} = V_{R_s} / R_s = 10 \text{ V} / 220 \Omega = \mathbf{45.45 \text{ mA}}$$

$$I_L = V_L / R_L = 10 \text{ V} / 470 \Omega = \mathbf{21.28 \text{ mA}}$$

$$\text{and } I_Z = I_{R_s} - I_L = 45.45 \text{ mA} - 21.28 \text{ mA} = \mathbf{24.17 \text{ mA}}$$

- (c) $P_{Z_{\max}} = 400 \text{ mW} = V_Z I_Z = (10 \text{ V})(I_Z)$

$$I_Z = \frac{400 \text{ mW}}{10 \text{ V}} = 40 \text{ mA}$$

$$I_{L_{\min}} = I_{R_s} - I_{Z_{\max}} = 45.45 \text{ mA} - 40 \text{ mA} = 5.45 \text{ mA}$$

$$R_L = \frac{V_L}{I_{L_{\min}}} = \frac{10 \text{ V}}{5.45 \text{ mA}} = \mathbf{1,834.86 \Omega}$$

Large R_L reduces I_L and forces more of I_{R_s} to pass through Zener diode.

- (d) In the absence of the Zener diode

$$V_L = 10 \text{ V} = \frac{R_L (20 \text{ V})}{R_L + 220 \Omega}$$

$$10R_L + 2200 = 20R_L$$

$$10R_L = 2200$$

$$R_L = \mathbf{220 \Omega}$$

43. (a) $V_Z = 12 \text{ V}, R_L = \frac{V_L}{I_L} = \frac{12 \text{ V}}{200 \text{ mA}} = \mathbf{60 \Omega}$

$$V_L = V_Z = 12 \text{ V} = \frac{R_L V_i}{R_L + R_s} = \frac{60 \Omega (16 \text{ V})}{60 \Omega + R_s}$$

$$720 + 12R_s = 960$$

$$12R_s = 240$$

$$R_s = \mathbf{20 \Omega}$$

(b) $P_{Z_{\max}} = V_Z I_{Z_{\max}}$
 $= (12 \text{ V})(200 \text{ mA})$
 $= \mathbf{2.4 \text{ W}}$

44. Since $I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L}$ is fixed in magnitude the maximum value of I_{R_s} will occur when I_Z is a maximum. The maximum level of I_{R_s} will in turn determine the maximum permissible level of V_i .

$$I_{Z_{\max}} = \frac{P_{Z_{\max}}}{V_Z} = \frac{400 \text{ mW}}{8 \text{ V}} = 50 \text{ mA}$$

$$I_L = \frac{V_L}{R_L} = \frac{V_Z}{R_L} = \frac{8 \text{ V}}{220 \Omega} = 36.36 \text{ mA}$$

$$I_{R_s} = I_Z + I_L = 50 \text{ mA} + 36.36 \text{ mA} = 86.36 \text{ mA}$$

$$I_{R_s} = \frac{V_i - V_Z}{R_s}$$

$$\text{or } V_i = I_{R_s} R_s + V_Z$$

$$= (86.36 \text{ mA})(91 \Omega) + 8 \text{ V} = 7.86 \text{ V} + 8 \text{ V} = \mathbf{15.86 \text{ V}}$$

Any value of v_i that exceeds 15.86 V will result in a current I_Z that will exceed the maximum value.

45. At 30 V we have to be sure Zener diode is “on”.

$$\therefore V_L = 20 \text{ V} = \frac{R_L V_i}{R_L + R_s} = \frac{1 \text{ k}\Omega (30 \text{ V})}{1 \text{ k}\Omega + R_s}$$

$$\text{Solving, } R_s = \mathbf{0.5 \text{ k}\Omega}$$

$$\text{At } 50 \text{ V, } I_{R_s} = \frac{50 \text{ V} - 20 \text{ V}}{0.5 \text{ k}\Omega} = 60 \text{ mA, } I_L = \frac{20 \text{ V}}{1 \text{ k}\Omega} = 20 \text{ mA}$$

$$I_{ZM} = I_{R_s} - I_L = 60 \text{ mA} - 20 \text{ mA} = \mathbf{40 \text{ mA}}$$

46. For $v_i = +50 \text{ V}$:

Z_1 forward-biased at 0.7 V

Z_2 reverse-biased at the Zener potential and $V_{Z_2} = 10 \text{ V}$.

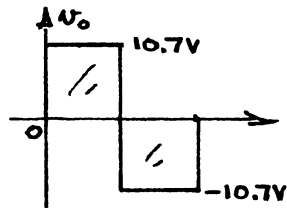
$$\text{Therefore, } V_o = V_{Z_1} + V_{Z_2} = 0.7 \text{ V} + 10 \text{ V} = \mathbf{10.7 \text{ V}}$$

For $v_i = -50$ V:

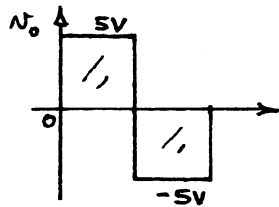
Z_1 reverse-biased at the Zener potential and $V_{Z_1} = -10$ V.

Z_2 forward-biased at -0.7 V.

Therefore, $V_o = V_{Z_1} + V_{Z_2} = -10.7$ V



For a 5 V square wave neither Zener diode will reach its Zener potential. In fact, for either polarity of v_i one Zener diode will be in an open-circuit state resulting in $v_o = v_i$.



47. $V_m = 1.414(120 \text{ V}) = 169.68 \text{ V}$
 $2V_m = 2(169.68 \text{ V}) = 339.36 \text{ V}$
48. The PIV for each diode is $2V_m$
 $\therefore \text{PIV} = 2(1.414)(V_{\text{rms}})$

Chapter 3

1. –
2. A bipolar transistor utilizes holes and electrons in the injection or charge flow process, while unipolar devices utilize either electrons or holes, but not both, in the charge flow process.
3. Forward- and reverse-biased.
4. The leakage current I_{CO} is the minority carrier current in the collector.
5. –
6. –
7. –
8. I_E the largest
 I_B the smallest
 $I_C \cong I_E$
9. $I_B = \frac{1}{100} I_C \Rightarrow I_C = 100I_B$
 $I_E = I_C + I_B = 100I_B + I_B = 101I_B$
 $I_B = \frac{I_E}{101} = \frac{8 \text{ mA}}{101} = \mathbf{79.21 \mu A}$
 $I_C = 100I_B = 100(79.21 \mu A) = \mathbf{7.921 \text{ mA}}$
10. –
11. $I_E = 5 \text{ mA}$, $V_{CB} = 1 \text{ V}$: $V_{BE} = \mathbf{800 \text{ mV}}$
 $V_{CB} = 10 \text{ V}$: $V_{BE} = \mathbf{770 \text{ mV}}$
 $V_{CB} = 20 \text{ V}$: $V_{BE} = \mathbf{750 \text{ mV}}$

The change in V_{CB} is $20 \text{ V}:1 \text{ V} = \mathbf{20:1}$
The resulting change in V_{BE} is $800 \text{ mV}:750 \text{ mV} = \mathbf{1.07:1}$ (very slight)
12. (a) $r_{av} = \frac{\Delta V}{\Delta I} = \frac{0.9 \text{ V} - 0.7 \text{ V}}{8 \text{ mA} - 0} = \mathbf{25 \Omega}$
(b) Yes, since 25Ω is often negligible compared to the other resistance levels of the network.
13. (a) $I_C \cong I_E = \mathbf{3.5 \text{ mA}}$
(b) $I_C \cong I_E = \mathbf{3.5 \text{ mA}}$
(c) negligible: change cannot be detected on this set of characteristics.
(d) $I_C \cong I_E$

14. (a) Using Fig. 3.7 first, $I_E \cong 2 \text{ mA}$
Then Fig. 3.8 results in $I_C \cong \mathbf{2 \text{ mA}}$
- (b) Using Fig. 3.8 first, $I_E \cong 5 \text{ mA}$
Then Fig. 3.7 results in $V_{BE} \cong \mathbf{0.77 \text{ V}}$
- (c) Using Fig. 3.10(b) $I_E = 5 \text{ mA}$ results in $V_{BE} \cong \mathbf{0.81 \text{ V}}$
- (d) Using Fig. 3.10(c) $I_E = 5 \text{ mA}$ results in $V_{BE} = \mathbf{0.7 \text{ V}}$
- (e) Yes, the difference in levels of V_{BE} can be ignored for most applications if voltages of several volts are present in the network.
15. (a) $I_C = \alpha I_E = (0.998)(4 \text{ mA}) = \mathbf{3.992 \text{ mA}}$
- (b) $I_E = I_C + I_B \Rightarrow I_C = I_E - I_B = 2.8 \text{ mA} - 0.02 \text{ mA} = \mathbf{2.78 \text{ mA}}$

$$\alpha_{dc} = \frac{I_C}{I_E} = \frac{2.78 \text{ mA}}{2.8 \text{ mA}} = \mathbf{0.993}$$
- (c) $I_C = \beta I_B = \left(\frac{\alpha}{1 - \alpha} \right) I_B = \left(\frac{0.98}{1 - 0.98} \right) (40 \mu\text{A}) = 1.96 \text{ mA}$

$$I_E = \frac{I_C}{\alpha} = \frac{1.96 \text{ mA}}{0.993} = \mathbf{2 \text{ mA}}$$
16. —
17. —
18. (a) Fig. 3.13(b): $I_B \cong \mathbf{35 \mu\text{A}}$
Fig. 3.13(a): $I_C \cong \mathbf{3.6 \text{ mA}}$
- (b) Fig. 3.14(a): $V_{CE} \cong \mathbf{12 \text{ V}}$
Fig. 3.14(b): $V_{BE} \cong \mathbf{0.75 \text{ V}}$
19. (a) $\beta_{dc} = \frac{I_C}{I_B} = \frac{2 \text{ mA}}{18 \mu\text{A}} = \mathbf{111.11}$
- (b) $\alpha = \frac{\beta}{\beta + 1} = \frac{111.11}{111.11 + 1} = \mathbf{0.991}$
- (c) $I_{CEO} = \mathbf{0.3 \text{ mA}}$
- (d) $I_{CBO} = (1 - \alpha)I_{CEO}$

$$= (1 - 0.991)(0.3 \text{ mA}) = \mathbf{2.7 \mu\text{A}}$$
20. (a) Fig. 3.14(a): $I_{CEO} \cong \mathbf{0.3 \text{ mA}}$
- (b) Fig. 3.14(a): $I_C \cong 1.35 \text{ mA}$

$$\beta_{dc} = \frac{I_C}{I_B} = \frac{1.35 \text{ mA}}{10 \mu\text{A}} = \mathbf{135}$$

$$(c) \quad \alpha = \frac{\beta}{\beta + 1} = \frac{135}{136} = \mathbf{0.9926}$$

$$I_{CBO} \cong (1 - \alpha)I_{CEO}$$

$$= (1 - 0.9926)(0.3 \text{ mA})$$

$$= \mathbf{2.2 \mu A}$$

$$21. \quad (a) \quad \beta_{dc} = \frac{I_C}{I_B} = \frac{5.25 \text{ mA}}{60 \mu A} = \mathbf{87.5}$$

$$(b) \quad \beta_{dc} = \frac{I_C}{I_B} = \frac{3.25 \text{ mA}}{30 \mu A} = \mathbf{108.3}$$

$$(c) \quad \beta_{dc} = \frac{I_C}{I_B} = \frac{1.35 \text{ mA}}{10 \mu A} = \mathbf{135}$$

(d) Yes, highest at lowest levels of I_C .

$$22. \quad (a) \quad \beta_{ac} = \frac{\Delta I_C}{\Delta I_B} = \frac{5.9 \text{ mA} - 4.6 \text{ mA}}{70 \mu A - 50 \mu A} = \mathbf{65}$$

$$(b) \quad \beta_{ac} = \frac{\Delta I_C}{\Delta I_B} = \frac{4.1 \text{ mA} - 2.3 \text{ mA}}{40 \mu A - 20 \mu A} = \mathbf{90}$$

$$(c) \quad \beta_{ac} = \frac{\Delta I_C}{\Delta I_B} = \frac{2.4 \text{ mA} - 0.3 \text{ mA}}{20 \mu A - 0 \mu A} = \mathbf{105}$$

(d) Yes, highest at lowest levels of I_C .

(e)	I_C	β_{dc}	β_{ac}	V_{CE}	β_{dc}/β_{ac}
a.	5.25 mA	87.5	65	4 V	1.35
b.	3.25 mA	108.3	90	7 V	1.20
c.	1.35 mA	135	105	10 V	1.29

$\beta_{dc} > \beta_{ac}$ although ratio consistent over scope of characteristics.

$$23. \quad \beta_{dc} = \frac{I_C}{I_B} = \frac{2.9 \text{ mA}}{25 \mu A} = \mathbf{116}$$

$$\alpha = \frac{\beta}{\beta + 1} = \frac{116}{116 + 1} = \mathbf{0.991}$$

$$I_E = I_C/\alpha = 2.9 \text{ mA}/0.991 = \mathbf{2.93 \text{ mA}}$$

$$24. \quad (a) \quad \beta = \frac{\alpha}{1 - \alpha} = \frac{0.980}{1 - 0.980} = \mathbf{49}$$

$$(b) \quad \alpha = \frac{\beta}{\beta + 1} = \frac{120}{120 + 1} = \mathbf{0.992}$$

$$(c) \quad I_B = \frac{I_C}{\beta} = \frac{2 \text{ mA}}{120} = 16.66 \mu\text{A}$$

$$I_E = I_C + I_B = 2 \text{ mA} + 16.66 \mu\text{A} \cong 2.017 \text{ mA}$$

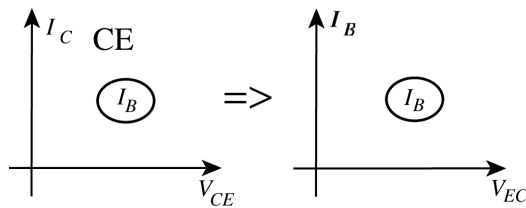
25. —

$$26. \quad V_e = V_i - V_{be} = 2 \text{ V} - 0.1 \text{ V} = 1.9 \text{ V}$$

$$A_v = \frac{V_o}{V_i} = \frac{1.9 \text{ V}}{2 \text{ V}} = 0.95 \cong 1$$

$$I_e = \frac{V_E}{R_E} = \frac{1.9 \text{ V}}{1 \text{ k}\Omega} = 1.9 \text{ mA (rms)}$$

27. Output characteristics:



Curves are essentially the same with new scales as shown.

Input characteristics:

Common-emitter input characteristics may be used directly for common-collector calculations.

$$28. \quad P_{C_{\max}} = 35 \text{ mW} = V_{CE} I_C$$

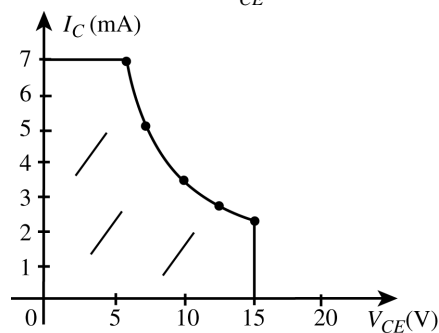
$$I_C = I_{C_{\max}}, V_{CE} = \frac{P_{C_{\max}}}{I_{C_{\max}}} = \frac{35 \text{ mW}}{6 \text{ mA}} = 5.83 \text{ V}$$

$$V_{CE} = V_{CE_{\max}}, I_C = \frac{P_{C_{\max}}}{V_{CE_{\max}}} = \frac{35 \text{ mW}}{20 \text{ V}} = 2.33 \text{ mA}$$

$$V_{CE} = 10 \text{ V}, I_C = \frac{P_{C_{\max}}}{V_{CE}} = \frac{35 \text{ mW}}{10 \text{ V}} = 3.5 \text{ mA}$$

$$I_C = 5 \text{ mA}, V_{CE} = \frac{P_{C_{\max}}}{I_C} = \frac{35 \text{ mW}}{5 \text{ mA}} = 7.00 \text{ V}$$

$$V_{CE} = 15 \text{ V}, I_C = \frac{P_{C_{\max}}}{V_{CE}} = \frac{35 \text{ mW}}{12 \text{ V}} = 2.91 \text{ mA}$$

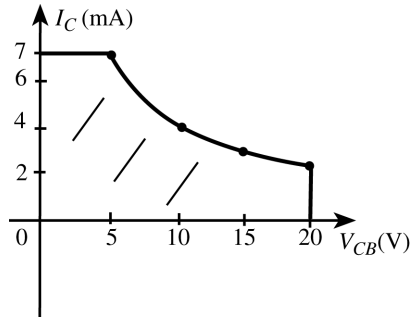


29. $I_C = I_{C_{\max}}, V_{CE} = \frac{P_{C_{\max}}}{I_{C_{\max}}} = \frac{42 \text{ mW}}{7 \text{ mA}} = \mathbf{6 \text{ V}}$

$V_{CB} = V_{CB_{\max}}, I_C = \frac{P_{C_{\max}}}{V_{CB_{\max}}} = \frac{42 \text{ mW}}{20 \text{ V}} = \mathbf{2.1 \text{ mA}}$

$I_C = 4 \text{ mA}, V_{CB} = \frac{P_{C_{\max}}}{I_C} = \frac{42 \text{ mW}}{4 \text{ mA}} = \mathbf{10.5 \text{ V}}$

$V_{CB} = 10 \text{ V}, I_C = \frac{P_{C_{\max}}}{V_{CB}} = \frac{42 \text{ mW}}{15 \text{ V}} = \mathbf{2.8 \text{ mA}}$



30. The operating temperature range is $-55^{\circ}\text{C} \leq T_J \leq 150^{\circ}\text{C}$

$$\begin{aligned} ^{\circ}\text{F} &= \frac{9}{5}^{\circ}\text{C} + 32^{\circ} \\ &= \frac{9}{5}(-55^{\circ}\text{C}) + 32^{\circ} = \mathbf{-67^{\circ}\text{F}} \\ ^{\circ}\text{F} &= \frac{9}{5}(150^{\circ}\text{C}) + 32^{\circ} = \mathbf{302^{\circ}\text{F}} \\ \therefore \mathbf{-67^{\circ}\text{F} \leq T_J \leq 302^{\circ}\text{F}} \end{aligned}$$

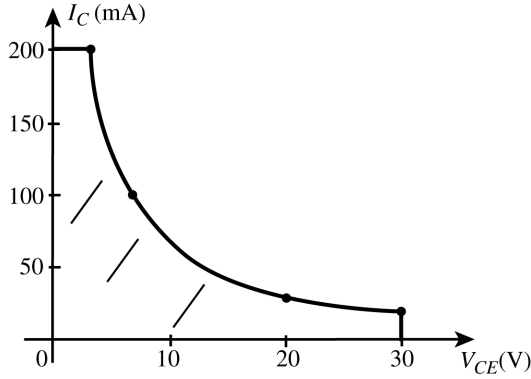
31. $I_{C_{\max}} = 200 \text{ mA}, V_{CE_{\max}} = 30 \text{ V}, P_{D_{\max}} = 625 \text{ mW}$

$$I_C = I_{C_{\max}}, V_{CE} = \frac{P_{D_{\max}}}{I_{C_{\max}}} = \frac{625 \text{ mW}}{200 \text{ mA}} = 3.125 \text{ V}$$

$$V_{CE} = V_{CE_{\max}}, I_C = \frac{P_{D_{\max}}}{V_{CE_{\max}}} = \frac{625 \text{ mW}}{30 \text{ V}} = 20.83 \text{ mA}$$

$$I_C = 100 \text{ mA}, V_{CE} = \frac{P_{D_{\max}}}{I_C} = \frac{625 \text{ mW}}{100 \text{ mA}} = 6.25 \text{ V}$$

$$V_{CE} = 20 \text{ V}, I_C = \frac{P_{D_{\max}}}{V_{CE}} = \frac{625 \text{ mW}}{20 \text{ V}} = 31.25 \text{ mA}$$



32. From Fig. 3.23 (a) $I_{CBO} = 50 \text{ nA max}$

$$\begin{aligned}\beta_{\text{avg}} &= \frac{\beta_{\min} + \beta_{\max}}{2} \\ &= \frac{50 + 150}{2} = \frac{200}{2} \\ &= 100\end{aligned}$$

$$\begin{aligned}\therefore I_{CEO} &\cong \beta I_{CBO} = (100)(50 \text{ nA}) \\ &= \mathbf{5 \mu A}\end{aligned}$$

33. $h_{FE} (\beta_{dc})$ with $V_{CE} = 1 \text{ V}, T = 25^\circ\text{C}$

$$I_C = 0.1 \text{ mA}, h_{FE} \cong 0.43(100) = 43$$

↓

$$I_C = 10 \text{ mA}, h_{FE} \cong 0.98(100) = 98$$

$$h_{fe} (\beta_{ac}) \text{ with } V_{CE} = 10 \text{ V}, T = 25^\circ\text{C}$$

$$I_C = 0.1 \text{ mA}, h_{fe} \cong 72$$

↓

$$I_C = 10 \text{ mA}, h_{fe} \cong 160$$

For both h_{FE} and h_{fe} the same increase in collector current resulted in a similar increase (relatively speaking) in the gain parameter. The levels are higher for h_{fe} but note that V_{CE} is higher also.

34. As the reverse-bias potential increases in magnitude the input capacitance C_{ibo} decreases (Fig. 3.23(b)). Increasing reverse-bias potentials causes the width of the depletion region to

increase, thereby reducing the capacitance $\left(C = \epsilon \frac{A}{d} \right)$.

35. (a) At $I_C = 1 \text{ mA}$, $h_{fe} \cong \mathbf{120}$
 At $I_C = 10 \text{ mA}$, $h_{fe} \cong \mathbf{160}$
- (b) The results confirm the conclusions of problems 21 and 22 that beta tends to increase with increasing collector current.
36. $T_j = +125^\circ\text{C}$: $1.45(\beta_{\text{norm}})$
 $T_j = +25^\circ\text{C}$: $0.95(\beta_{\text{norm}})$
 $T_j = -55^\circ\text{C}$: $0.5(\beta_{\text{norm}})$
 Yes, 34% drop between $+125^\circ\text{C}$ and 25°C . 65.5% drop between $+125^\circ\text{C}$ and -55°C .
37. (a) $\beta_{\text{ac}} = \frac{\Delta I_C}{\Delta I_B} \bigg|_{V_{CE} = 3 \text{ V}} = \frac{16 \text{ mA} - 12.2 \text{ mA}}{80 \mu\text{A} - 60 \mu\text{A}} = \frac{3.8 \text{ mA}}{20 \mu\text{A}} = \mathbf{190}$
- (b) $\beta_{\text{dc}} = \frac{I_C}{I_B} = \frac{12 \text{ mA}}{59.5 \mu\text{A}} = \mathbf{201.7}$
- (c) $\beta_{\text{ac}} = \frac{4 \text{ mA} - 2 \text{ mA}}{18 \mu\text{A} - 8 \mu\text{A}} = \frac{2 \text{ mA}}{10 \mu\text{A}} = \mathbf{200}$
- (d) $\beta_{\text{dc}} = \frac{I_C}{I_B} = \frac{3 \text{ mA}}{13 \mu\text{A}} = \mathbf{230.77}$
- (e) In both cases β_{dc} is slightly higher than β_{ac} ($\cong 10\%$)
- (f)(g) In general β_{dc} and β_{ac} increase with increasing I_C for fixed V_{CE} and both decrease for decreasing levels of V_{CE} for a fixed I_E . However, if I_C increases while V_{CE} decreases when moving between two points on the characteristics, chances are the level of β_{dc} or β_{ac} may not change significantly. In other words, the expected increase due to an increase in collector current may be offset by a decrease in V_{CE} . The above data reveals that this is a strong possibility since the levels of β are relatively close.

Chapter 4

1.
 - (a) $I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega} = \frac{15.3 \text{ V}}{510 \text{ k}\Omega} = \mathbf{30 \mu A}$
 - (b) $I_{C_Q} = \beta I_{B_Q} = (120)(30 \mu A) = \mathbf{3.6 \text{ mA}}$
 - (c) $V_{CE_Q} = V_{CC} - I_{C_Q} R_C = 16 \text{ V} - (3.6 \text{ mA})(1.8 \text{ k}\Omega) = \mathbf{9.52 \text{ V}}$
 - (d) $V_C = V_{CE_Q} = \mathbf{6.48 \text{ V}}$
 - (e) $V_B = V_{BE} = \mathbf{0.7 \text{ V}}$
 - (f) $V_E = \mathbf{0 \text{ V}}$
2.
 - (a) $I_C = \beta I_B = 80(40 \mu A) = \mathbf{3.2 \text{ mA}}$
 - (b) $R_C = \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{12 \text{ V} - 6 \text{ V}}{3.2 \text{ mA}} = \frac{6 \text{ V}}{3.2 \text{ mA}} = \mathbf{1.875 \text{ k}\Omega}$
 - (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{40 \mu A} = \frac{11.3 \text{ V}}{40 \mu A} = \mathbf{282.5 \text{ k}\Omega}$
 - (d) $V_{CE} = V_C = \mathbf{6 \text{ V}}$
3.
 - (a) $I_C = I_E - I_B = 4 \text{ mA} - 20 \mu A = \mathbf{3.98 \text{ mA}} \cong 4 \text{ mA}$
 - (b) $V_{CC} = V_{CE} + I_C R_C = 7.2 \text{ V} + (3.98 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{15.96 \text{ V}} \cong 16 \text{ V}$
 - (c) $\beta = \frac{I_C}{I_B} = \frac{3.98 \text{ mA}}{20 \mu A} = \mathbf{199} \cong 200$
 - (d) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{15.96 \text{ V} - 0.7 \text{ V}}{20 \mu A} = \mathbf{763 \text{ k}\Omega}$
4. $I_{C_{\text{sat}}} = \frac{V_{CC}}{R_C} = \frac{16 \text{ V}}{2.7 \text{ k}\Omega} = \mathbf{5.93 \text{ mA}}$
5.
 - (a) Load line intersects vertical axis at $I_C = \frac{21 \text{ V}}{3 \text{ k}\Omega} = 7 \text{ mA}$
and horizontal axis at $V_{CE} = 21 \text{ V}$.
 - (b) $I_B = 25 \mu A$: $R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{21 \text{ V} - 0.7 \text{ V}}{25 \mu A} = \mathbf{812 \text{ k}\Omega}$
 - (c) $I_{C_Q} \cong \mathbf{3.4 \text{ mA}}$, $V_{CE_Q} \cong \mathbf{10.75 \text{ V}}$

- (d) $\beta = \frac{I_C}{I_B} = \frac{3.4 \text{ mA}}{25 \mu\text{A}} = \mathbf{136}$
- (e) $\alpha = \frac{\beta}{\beta + 1} = \frac{136}{136 + 1} = \frac{136}{137} = \mathbf{0.992}$
- (f) $I_{C_{\text{sat}}} = \frac{V_{CC}}{R_C} = \frac{21 \text{ V}}{3 \text{ k}\Omega} = \mathbf{7 \text{ mA}}$
- (g) –
- (h) $P_D = V_{CE_Q} I_{C_Q} = (10.75 \text{ V})(3.4 \text{ mA}) = \mathbf{36.55 \text{ mW}}$
- (i) $P_s = V_{CC}(I_C + I_B) = 21 \text{ V}(3.4 \text{ mA} + 25 \mu\text{A}) = \mathbf{71.92 \text{ mW}}$
- (j) $P_R = P_s - P_D = 71.92 \text{ mW} - 36.55 \text{ mW} = \mathbf{35.37 \text{ mW}}$
6. (b) $I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega} = 30 \mu\text{A}$
 $V_{CE_Q} = \mathbf{8.5 \text{ V}}, I_{C_Q} = \mathbf{4.1 \text{ mA}}$
- (c) $\beta = \frac{I_C}{I_B} = \frac{4.1 \text{ mA}}{30 \mu\text{A}} = \mathbf{136.67}$
7. $I_{B_Q} = \frac{16 \text{ V} - 0.7 \text{ V}}{910 \text{ k}\Omega} = 16.81 \mu\text{A}$
 $I_{C_Q} = \beta I_{B_Q} = (120)(16.81 \mu\text{A}) = \mathbf{2.017 \text{ mA}}$
(from characteristics) $V_{E_Q} = \mathbf{11.5 \text{ V}}, I_{C_Q} = \mathbf{2.4 \text{ mA}}$
8. (a) $I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{270 \text{ k}\Omega + (126)2.2 \text{ k}\Omega} = \frac{19.3 \text{ V}}{547.2 \text{ k}\Omega}$
 $= \mathbf{35.27 \mu\text{A}}$
- (b) $I_{C_Q} = \beta I_{B_Q} = (125)(35.27 \mu\text{A}) = \mathbf{4.41 \text{ mA}}$
- (c) $V_{CE_Q} = V_{CC} - I_C(R_C + R_E) = 20 \text{ V} - (4.41 \text{ mA})(470 \text{ k}\Omega + 2.2 \text{ k}\Omega)$
 $= 20 \text{ V} - 11.77 \text{ V}$
 $= \mathbf{8.23 \text{ V}}$
- (d) $V_C = V_{CC} - I_C R_C = 20 \text{ V} - (4.41 \text{ mA})(470 \text{ k}\Omega) = 20 \text{ V} - 2.07 \text{ V}$
 $= \mathbf{17.93 \text{ V}}$
- (e) $V_B = V_{CC} - I_B R_B = 20 \text{ V} - (35.27 \mu\text{A})(270 \text{ k}\Omega)$
 $= 20 \text{ V} - 9.52 \text{ V} = \mathbf{10.48 \text{ V}}$
- (f) $V_E = V_C - V_{CE} = 17.93 \text{ V} - 8.23 \text{ V} = \mathbf{9.7 \text{ V}}$

9. (a) $I_{B_Q} = 35.27 \mu\text{A}$ using β from problem 8.
- $$I_{\text{sat}} = \frac{20 \text{ V}}{2.2 \text{ k}\Omega + 470 \Omega} = 7.49 \text{ mA}$$
- (b) $I_{C_Q} = 4.7 \text{ mA}$, $V_{CE_Q} = 7.5 \text{ V}$
- (c) $\beta = \frac{I_C}{I_B} = \frac{4.7 \text{ mA}}{35.27 \mu\text{A}} = 133.25$
- (d) reasonably close
- (e) different β and Q pt.
10. (a) $R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12 \text{ V} - 7.6 \text{ V}}{2 \text{ mA}} = \frac{4.4 \text{ V}}{2 \text{ mA}} = 2.2 \text{ k}\Omega$
- (b) $I_E \cong I_C$: $R_E = \frac{V_E}{I_E} = \frac{2.4 \text{ V}}{2 \text{ mA}} = 1.2 \text{ k}\Omega$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V} - 2.4 \text{ V}}{2 \text{ mA}/80} = \frac{8.9 \text{ V}}{25 \mu\text{A}} = 356 \text{ k}\Omega$
- (d) $V_{CE} = V_C - V_E = 7.6 \text{ V} - 2.4 \text{ V} = 5.2 \text{ V}$
- (e) $V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.4 \text{ V} = 3.1 \text{ V}$
11. (a) $I_C \cong I_E = \frac{V_E}{R_E} = \frac{2.1 \text{ V}}{0.68 \text{ k}\Omega} = 3.09 \text{ mA}$
- $$\beta = \frac{I_C}{I_B} = \frac{3.09 \text{ mA}}{20 \mu\text{A}} = 154.5$$
- (b) $V_{CC} = V_{R_C} + V_{CE} + V_E$
- $$= (3.09 \text{ mA})(2.7 \text{ k}\Omega) + 7.3 \text{ V} + 2.1 \text{ V} = 8.34 \text{ V} + 7.3 \text{ V} + 2.1 \text{ V}$$
- $$= 17.74 \text{ V}$$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{17.74 \text{ V} - 0.7 \text{ V} - 2.1 \text{ V}}{20 \mu\text{A}}$
- $$= \frac{14.94 \text{ V}}{20 \mu\text{A}} = 747 \text{ k}\Omega$$
12. $I_{C_{\text{sot}}} = \frac{V_{CC}}{R_C + R_E} = \frac{20 \text{ V}}{470 \Omega + 2.2 \text{ k}\Omega} = \frac{20 \text{ V}}{2.67 \text{ k}\Omega} = 7.49 \text{ mA}$

13. (a) $I_{C_{sat}} = 6.8 \text{ mA} = \frac{V_{CC}}{R_C + R_E} = \frac{24 \text{ V}}{R_C + 1.2 \text{ k}\Omega}$
 $R_C + 1.2 \text{ k}\Omega = \frac{24 \text{ V}}{6.8 \text{ mA}} = 3.529 \text{ k}\Omega$
 $R_C = \mathbf{2.33 \text{ k}\Omega}$
- (b) $\beta = \frac{I_C}{I_B} = \frac{4 \text{ mA}}{30 \mu\text{A}} = \mathbf{133.33}$
- (c) $R_B = \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE} - V_E}{I_B} = \frac{24 \text{ V} - 0.7 \text{ V} - (4 \text{ mA})(1.2 \text{ k}\Omega)}{30 \mu\text{A}}$
 $= \frac{18.5 \text{ V}}{30 \mu\text{A}} = \mathbf{616.67 \text{ k}\Omega}$
- (d) $P_D = V_{CE_Q} I_{C_Q}$
 $= (10 \text{ V})(4 \text{ mA}) = \mathbf{40 \text{ mW}}$
- (e) $P = I_C^2 R_C = (4 \text{ mA})^2 (2.33 \text{ k}\Omega)$
 $= \mathbf{37.28 \text{ mW}}$
14. (a) Problem 1: $I_{C_Q} = \mathbf{3.6 \text{ mA}}$, $V_{CE_Q} = \mathbf{6.48 \text{ V}}$
- (b) $I_{B_Q} = 30 \mu\text{A}$ (the same)
 $I_{C_Q} = \beta I_{B_Q} = (180)(30 \mu\text{A}) = 5.4 \text{ mA}$
 $V_{CE_Q} = V_{CC} - I_{C_Q} R_C = 16 \text{ V} - (5.4 \text{ mA})(1.8 \text{ k}\Omega) = \mathbf{6.28 \text{ V}}$
- (c) $\% \Delta I_C = \left| \frac{5.4 \text{ mA} - 3.6 \text{ mA}}{3.6 \text{ mA}} \right| \times 100\% = \mathbf{50\%}$
 $\% \Delta V_{CE} = \left| \frac{9.52 \text{ V} - 6.28 \text{ V}}{6.28 \text{ V}} \right| \times 100\% = \mathbf{51.6\%}$
About 50% change for each.
- (d) Problem 8: $I_{C_Q} = \mathbf{4.41 \text{ mA}}$, $V_{CE_Q} = \mathbf{8.23 \text{ V}}$ ($I_{B_Q} = 35.27 \mu\text{A}$)
- (e) $I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{270 \text{ k}\Omega + (187.5 + 1)(2.2 \text{ k}\Omega)} = 28.19 \mu\text{A}$
 $I_{C_Q} = \beta I_{B_Q} = (187.5)(28.19 \mu\text{A}) = \mathbf{5.29 \text{ mA}}$
 $= V_{CC} - I_C(R_C + R_E)$
 $= 20 \text{ V} - (5.29 \text{ mA})(470 \text{ k}\Omega + 2.2 \text{ k}\Omega) = \mathbf{5.88 \text{ V}}$

$$(f) \quad \% \Delta I_C = \left| \frac{5.29 \text{ mA} - 4.41 \text{ mA}}{4.41 \text{ mA}} \right| \times 100\% = \mathbf{20\%}$$

$$\% \Delta V_{CE} = \left| \frac{5.88 \text{ V} - 8.23 \text{ V}}{8.23 \text{ V}} \right| \times 100\% = \mathbf{28.6\%}$$

(g) For both I_C and V_{CE} the % change is less for the emitter-stabilized.

15. $\beta R_E \geq 10 R_2$
 $(80)(0.68 \text{ k}\Omega) \geq 10(9.1 \text{ k}\Omega)$
 $54.4 \text{ k}\Omega \not\geq 91 \text{ k}\Omega \text{ (No!)}$

(a) Use exact approach:

$$R_{Th} = R_1 \parallel R_2 = 62 \text{ k}\Omega \parallel 9.1 \text{ k}\Omega = 7.94 \text{ k}\Omega$$

$$E_{Th} = \frac{R_2 V_{CC}}{R_2 + R_1} = \frac{(9.1 \text{ k}\Omega)(16 \text{ V})}{9.1 \text{ k}\Omega + 62 \text{ k}\Omega} = 2.05 \text{ V}$$

$$I_{B_Q} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (81)(0.68 \text{ k}\Omega)} = \mathbf{21.42 \mu A}$$

(b) $I_{C_Q} = \beta I_{B_Q} = (80)(21.42 \mu A) = \mathbf{1.71 \text{ mA}}$

(c) $V_{CE_Q} = V_{CC} - I_{C_Q}(R_C + R_E)$
 $= 16 \text{ V} - (1.71 \text{ mA})(3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega)$
 $= \mathbf{8.17 \text{ V}}$

(d) $V_C = V_{CC} - I_C R_C$
 $= 16 \text{ V} - (1.71 \text{ mA})(3.9 \text{ k}\Omega)$
 $= \mathbf{9.33 \text{ V}}$

(e) $V_E = I_E R_E \cong I_C R_E = (1.71 \text{ mA})(0.68 \text{ k}\Omega)$
 $= \mathbf{1.16 \text{ V}}$

(f) $V_B = V_E + V_{BE} = 1.16 \text{ V} + 0.7 \text{ V}$
 $= \mathbf{1.86 \text{ V}}$

16. (a) $R_{Th} = 62 \text{ k}\Omega \parallel 9.1 \text{ k}\Omega = 7.94 \text{ k}\Omega$
 $E_{Th} = \frac{9.1 \text{ k}\Omega(16 \text{ V})}{9.1 \text{ k}\Omega + 62 \text{ k}\Omega} = 2.05 \text{ V}$
 $I_{B_Q} = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (140 + 1)(0.68 \text{ k}\Omega)} = \mathbf{13 \mu A \text{ (vs. } 21.42 \mu A)}$
 $I_{C_Q} = \beta I_{B_Q} = (140)(13 \mu A) = \mathbf{1.82 \text{ mA \text{ (vs. } 1.71 \text{ mA})}}$
 $V_{CE_Q} = V_{CC} - I_{C_Q}(R_C + R_E)$
 $= 16 \text{ V} - (1.82 \text{ mA})(3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega)$
 $= \mathbf{7.66 \text{ V \text{ (vs. } 8.17 \text{ V)}}$

$$\begin{aligned}
V_C &= V_{CC} - I_C R_C \\
&= 16 \text{ V} - (1.82 \text{ mA})(3.9 \text{ k}\Omega) \\
&= \mathbf{8.9 \text{ V}} \text{ (vs. } 9.33 \text{ V)} \\
V_E &= I_E R_E \cong I_C R_E = (1.82 \text{ mA})(0.68 \text{ k}\Omega) \\
&= \mathbf{1.24 \text{ V}} \text{ (vs. } 1.16 \text{ V)} \\
V_B &= V_E + V_{BE} = 1.24 \text{ V} + 0.7 \text{ V} \\
&= \mathbf{1.94 \text{ V}} \text{ (vs. } 1.86 \text{ V)}
\end{aligned}$$

(b) I_{BQ} affected the most

$$17. \quad (a) \quad I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18 \text{ V} - 12 \text{ V}}{4.7 \text{ k}\Omega} = \mathbf{1.28 \text{ mA}}$$

$$(b) \quad V_E = I_E R_E \cong I_C R_E = (1.28 \text{ mA})(1.2 \text{ k}\Omega) = \mathbf{1.54 \text{ V}}$$

$$(c) \quad V_B = V_{BE} + V_E = 0.7 \text{ V} + 1.54 \text{ V} = \mathbf{2.24 \text{ V}}$$

$$(d) \quad R_1 = \frac{V_{R_1}}{I_{R_1}} : \quad V_{R_1} = V_{CC} - V_B = 18 \text{ V} - 2.24 \text{ V} = \mathbf{15.76 \text{ V}}$$

$$I_{R_1} \cong I_{R_2} = \frac{V_B}{R_2} = \frac{2.24 \text{ V}}{5.6 \text{ k}\Omega} = 0.4 \text{ mA}$$

$$R_1 = \frac{V_{R_1}}{I_{R_1}} = \frac{15.76 \text{ V}}{0.4 \text{ mA}} = \mathbf{39.4 \text{ k}\Omega}$$

$$18. \quad (a) \quad I_C = \beta I_B = (100)(20 \mu\text{A}) = \mathbf{2 \text{ mA}}$$

$$(b) \quad I_E = I_C + I_B = 2 \text{ mA} + 20 \mu\text{A} \\ = 2.02 \text{ mA}$$

$$V_E = I_E R_E = (2.02 \text{ mA})(1.2 \text{ k}\Omega) \\ = \mathbf{2.42 \text{ V}}$$

$$(c) \quad V_{CC} = V_C + I_C R_C = 10.6 \text{ V} + (2 \text{ mA})(2.7 \text{ k}\Omega) \\ = 10.6 \text{ V} + 5.4 \text{ V} \\ = \mathbf{16 \text{ V}}$$

$$(d) \quad V_{CE} = V_C - V_E = 10.6 \text{ V} - 2.42 \text{ V} \\ = \mathbf{8.18 \text{ V}}$$

$$(e) \quad V_B = V_E + V_{BE} = 2.42 \text{ V} + 0.7 \text{ V} = \mathbf{3.12 \text{ V}}$$

$$\begin{aligned}
(f) \quad I_{R_1} &= I_{R_2} + I_B \\
&= \frac{3.12 \text{ V}}{8.2 \text{ k}\Omega} + 20 \mu\text{A} = 380.5 \mu\text{A} + 20 \mu\text{A} = 400.5 \mu\text{A} \\
R_1 &= \frac{V_{CC} - V_B}{I_{R_1}} = \frac{16 \text{ V} - 3.12 \text{ V}}{400.5 \mu\text{A}} = \mathbf{32.16 \text{ k}\Omega}
\end{aligned}$$

$$19. \quad I_{C_{\text{sat}}} = \frac{V_{CC}}{R_C + R_E} = \frac{16 \text{ V}}{3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega} = \frac{16 \text{ V}}{4.58 \text{ k}\Omega} = \mathbf{3.49 \text{ mA}}$$

$$20. \quad \begin{aligned} \text{(a) Testing } \beta R_E &\geq 10R_z \\ (140)(0.68 \text{ k}\Omega) &\geq 10(9.1 \text{ k}\Omega) \\ 95.2 \text{ k}\Omega &\geq 91 \text{ k}\Omega \text{ satisfied (barely)} \end{aligned}$$

Applying the approximation approved:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(9.1 \text{ k}\Omega)(16 \text{ V})}{62 \text{ k}\Omega + 9.1 \text{ k}\Omega} = \mathbf{2.05 \text{ V}} \text{ (vs. } 1.95 \text{ V)}$$

$$V_E = V_B - V_{BE} = 2.05 \text{ V} - 0.7 \text{ V} = \mathbf{1.35 \text{ V}} \text{ (vs. } 1.24 \text{ V)}$$

$$I_C \cong I_E = \frac{V_E}{R_E} = \frac{1.35 \text{ V}}{0.68 \text{ k}\Omega} = \mathbf{1.99 \text{ mA}} \text{ (vs. } 1.82 \text{ mA)}$$

$$V_C = V_{CC} - I_C R_C = 16 \text{ V} - (1.99 \text{ mA})(3.9 \text{ k}\Omega) = \mathbf{8.24 \text{ V}} \text{ (vs } 8.9 \text{ V)}$$

$$V_{CE} = V_C - V_E = 8.24 \text{ V} - 1.35 \text{ V} = \mathbf{6.89 \text{ V}} \text{ (vs } 7.66 \text{ V)}$$

$$21. \quad \begin{aligned} \text{(a) } \beta R_E &\geq 10R_2 \\ (120)(1 \text{ k}\Omega) &\geq 10(8.2 \text{ k}\Omega) \\ 120 \text{ k}\Omega &\geq 82 \text{ k}\Omega \text{ (checks)} \\ \therefore V_B &= \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{(8.2 \text{ k}\Omega)(18 \text{ V})}{39 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 3.13 \text{ V} \\ V_E &= V_B - V_{BE} = 3.13 \text{ V} - 0.7 \text{ V} = 2.43 \text{ V} \\ I_C \cong I_E &= \frac{V_E}{R_E} = \frac{2.43 \text{ V}}{1 \text{ k}\Omega} = \mathbf{2.43 \text{ mA}} \end{aligned}$$

$$\begin{aligned} \text{(b) } V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 18 \text{ V} - (2.43 \text{ mA})(3.3 \text{ k}\Omega + 1 \text{ k}\Omega) \\ &= \mathbf{7.55 \text{ V}} \end{aligned}$$

$$\text{(c) } I_B = \frac{I_C}{\beta} = \frac{2.43 \text{ mA}}{120} = \mathbf{20.25 \mu\text{A}}$$

$$\text{(d) } V_E = I_E R_E \cong I_C R_E = (2.43 \text{ mA})(1 \text{ k}\Omega) = \mathbf{2.43 \text{ V}}$$

$$\text{(e) } V_B = \mathbf{3.13 \text{ V}}$$

$$\begin{aligned} 22. \quad \text{(a) } R_{Th} &= R_1 \parallel R_2 = 39 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega = 6.78 \text{ k}\Omega \\ E_{Th} &= \frac{R_C V_{CC}}{R_1 + R_2} = \frac{8.2 \text{ k}\Omega(18 \text{ V})}{39 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 3.13 \text{ V} \\ I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.13 \text{ V} - 0.7 \text{ V}}{6.78 \text{ k}\Omega + (121)(1 \text{ k}\Omega)} \\ &= \frac{2.43 \text{ V}}{127.78 \text{ k}\Omega} = 19.02 \mu\text{A} \\ I_C &= \beta I_B = (120)(19.02 \mu\text{A}) = \mathbf{2.28 \text{ mA}} \text{ (vs. } 2.43 \text{ mA \#16)} \end{aligned}$$

$$(b) \quad V_{CE} = V_{CC} - I_C(R_C + R_E) = 18 \text{ V} - (2.28 \text{ mA})(3.3 \text{ k}\Omega + 1 \text{ k}\Omega) \\ = 18 \text{ V} - 9.8 \text{ V} = \mathbf{8.2 \text{ V}} \text{ (vs. } 7.55 \text{ V \#16)}$$

$$(c) \quad \mathbf{19.02 \mu A} \text{ (vs. } 20.25 \mu A \text{ \#16)}$$

$$(d) \quad V_E = I_E R_E \cong I_C R_E = (2.28 \text{ mA})(1 \text{ k}\Omega) = \mathbf{2.28 \text{ V}} \text{ (vs. } 2.43 \text{ V \#16)}$$

$$(e) \quad V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.28 \text{ V} = \mathbf{2.98 \text{ V}} \text{ (vs. } 3.13 \text{ V \#16)}$$

The results suggest that the approximate approach is valid if Eq. 4.33 is satisfied.

$$23. \quad (a) \quad V_B = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{9.1 \text{ k}\Omega(16 \text{ V})}{62 \text{ k}\Omega + 9.1 \text{ k}\Omega} = 2.05 \text{ V}$$

$$V_E = V_B - V_{BE} = 2.05 \text{ V} - 0.7 \text{ V} = 1.35 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.35 \text{ V}}{0.68 \text{ k}\Omega} = 1.99 \text{ mA}$$

$$I_{C_Q} \cong I_E = \mathbf{1.99 \text{ mA}}$$

$$V_{CE_Q} = V_{CC} - I_C(R_C + R_E) \\ = 16 \text{ V} - (1.99 \text{ mA})(3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega) \\ = 16 \text{ V} - 9.11 \text{ V} \\ = \mathbf{6.89 \text{ V}}$$

$$I_{B_Q} = \frac{I_{C_Q}}{\beta} = \frac{1.99 \text{ mA}}{80} = \mathbf{24.88 \mu A}$$

(b) From Problem 12:

$$I_{C_Q} = \mathbf{1.71 \text{ mA}}, V_{CE_Q} = \mathbf{8.17 \text{ V}}, I_{B_Q} = \mathbf{21.42 \mu A}$$

(c) The differences of about 14% suggest that the exact approach should be employed when appropriate.

$$24. \quad (a) \quad I_{C_{\text{sat}}} = 7.5 \text{ mA} = \frac{V_{CC}}{R_C + R_E} = \frac{24 \text{ V}}{3R_E + R_E} = \frac{24 \text{ V}}{4R_E}$$

$$R_E = \frac{24 \text{ V}}{4(7.5 \text{ mA})} = \frac{24 \text{ V}}{30 \text{ mA}} = \mathbf{0.8 \text{ k}\Omega}$$

$$R_C = 3R_E = 3(0.8 \text{ k}\Omega) = 2.4 \text{ k}\Omega$$

$$(b) \quad V_E = I_E R_E \cong I_C R_E = (5 \text{ mA})(0.8 \text{ k}\Omega) = \mathbf{4 \text{ V}}$$

$$(c) \quad V_B = V_E + V_{BE} = 4 \text{ V} + 0.7 \text{ V} = \mathbf{4.7 \text{ V}}$$

$$(d) \quad V_B = \frac{R_2 V_{CC}}{R_2 + R_1}, \quad 4.7 \text{ V} = \frac{R_2(24 \text{ V})}{R_2 + 24 \text{ k}\Omega} \\ R_2 = \mathbf{5.84 \text{ k}\Omega}$$

$$(e) \quad \beta_{\text{dc}} = \frac{I_C}{I_B} = \frac{5 \text{ mA}}{38.5 \mu A} = \mathbf{129.8}$$

(f) $\beta R_E \geq 10R_2$
 $(129.8)(0.8 \text{ k}\Omega) \geq 10(5.84 \text{ k}\Omega)$
 $103.84 \text{ k}\Omega \geq 58.4 \text{ k}\Omega$ (checks)

25. (a) From problem 12b, $I_C = \mathbf{1.71 \text{ mA}}$
 From problem 12c, $V_{CE} = \mathbf{8.17 \text{ V}}$

(b) β changed to 120:
 From problem 12a, $E_{Th} = 2.05 \text{ V}$, $R_{Th} = 7.94 \text{ k}\Omega$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.05 \text{ V} - 0.7 \text{ V}}{7.94 \text{ k}\Omega + (121)(0.68 \text{ k}\Omega)}$$

$$= 14.96 \mu\text{A}$$

$$I_C = \beta I_B = (120)(14.96 \mu\text{A}) = \mathbf{1.8 \text{ mA}}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

$$= 16 \text{ V} - (1.8 \text{ mA})(3.9 \text{ k}\Omega + 0.68 \text{ k}\Omega)$$

$$= \mathbf{7.76 \text{ V}}$$

(c) $\% \Delta I_C = \left| \frac{1.8 \text{ mA} - 1.71 \text{ mA}}{1.71 \text{ mA}} \right| \times 100\% = \mathbf{5.26\%}$
 $\% \Delta V_{CE} = \left| \frac{7.76 \text{ V} - 8.17 \text{ V}}{8.17 \text{ V}} \right| \times 100\% = \mathbf{5.02\%}$

	11c	11f	20c
$\% \Delta I_C$	49.83%	34.59%	5.26%
$\% \Delta V_{CE}$	48.70%	46.76%	5.02%
	$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$	$\underbrace{\hspace{1.5cm}}$
	Fixed-bias	Emitter feedback	Voltage-divider

(e) Quite obviously, the voltage-divider configuration is the least sensitive to changes in β .

26. (a) Problem 21: Approximation approach: $I_{C_Q} = \mathbf{2.43 \text{ mA}}$, $V_{CE_Q} = \mathbf{7.55 \text{ V}}$

Problem 22: Exact analysis: $I_{C_Q} = \mathbf{2.28 \text{ mA}}$, $V_{CE_Q} = \mathbf{8.2 \text{ V}}$

The exact solution will be employed to demonstrate the effect of the change of β . Using the approximate approach would result in $\% \Delta I_C = 0\%$ and $\% \Delta V_{CE} = 0\%$.

Problem 22: $E_{Th} = 3.13 \text{ V}$, $R_{Th} = 6.78 \text{ k}\Omega$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.13 \text{ V} - 0.7 \text{ V}}{6.78 \text{ k}\Omega + (180 + 1)1 \text{ k}\Omega} = \frac{2.43 \text{ V}}{187.78 \text{ k}\Omega}$$

$$= 12.94 \mu\text{A}$$

$$I_C = \beta I_B = (180)(12.94 \mu\text{A}) = \mathbf{2.33 \text{ mA}}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E) = 18 \text{ V} - (2.33 \text{ mA})(3.3 \text{ k}\Omega + 1 \text{ k}\Omega)$$

$$= \mathbf{7.98 \text{ V}}$$

$$\% \Delta I_C = \left| \frac{2.33 \text{ mA} - 2.28 \text{ mA}}{2.28 \text{ mA}} \right| \times 100\% = \mathbf{2.19\%}$$

$$\% \Delta V_{CE} = \left| \frac{7.98 \text{ V} - 8.2 \text{ V}}{8.2 \text{ V}} \right| \times 100\% = \mathbf{2.68\%}$$

For situations where $\beta R_E > 10R_2$ the change in I_C and/or V_{CE} due to significant change in β will be relatively small.

$\% \Delta I_C = 2.19\%$ vs. 49.83% for problem 14.

$\% \Delta V_{CE} = 2.68\%$ vs. 48.70% for problem 14.

Voltage-divider configuration considerably less sensitive.

(b) The resulting $\% \Delta I_C$ and $\% \Delta V_{CE}$ will be quite small.

$$27. \quad (a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{16 \text{ V} - 0.7 \text{ V}}{270 \text{ k}\Omega + (120)(3.6 \text{ k}\Omega + 1.2 \text{ k}\Omega)} = \mathbf{18.09 \mu A}$$

$$(b) \quad I_C = \beta I_B = (120)(18.09 \mu A) = \mathbf{2.17 \text{ mA}}$$

$$(c) \quad V_C = V_{CC} - I_C R_C = 16 \text{ V} - (2.17 \text{ mA})(3.6 \text{ k}\Omega) = \mathbf{8.19 \text{ V}}$$

$$28. \quad (a) \quad I_{C_Q} \cong \frac{V_{CC} - V_{BE}}{R_C + R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{3.6 \text{ k}\Omega + 1.2 \text{ k}\Omega} = \mathbf{3.19 \text{ mA}}$$

$$(b) \quad I_{C_Q} = \mathbf{3.19 \text{ mA}} \text{ vs } 2.17 \text{ mA (not too close)}$$

$$(c) \quad R' = R_C + R_E = 4.8 \text{ k}\Omega, \quad R_F / \beta = 270 \text{ k}\Omega / 120 = 2.25 \text{ k}\Omega \\ \therefore R' \cong \mathbf{2(R_F / \beta)}$$

$$(d) \quad \text{Yes, } I_{C_Q} = \beta I_{B_Q} = \frac{\beta(V_{CC} - V_{BE})}{R_F + \beta(R_C + R_E)} = \frac{V_{CC} - V_{BE}}{R_F / \beta + R_C + R_E}$$

$$\text{For } R_C + R_E \gg R_F / \beta, \quad I_{C_Q} \cong \frac{V_{CC} - V_{BE}}{R_C + R_E}$$

$$(e) \quad I_{C_Q} = \frac{16 \text{ V} - 0.7 \text{ V}}{\frac{270 \text{ k}\Omega}{240} + 3.6 \text{ k}\Omega + 1.2 \text{ k}\Omega} = \mathbf{2.58 \text{ mA}} \text{ vs. } 3.19 \text{ mA (much closer)}$$

$$29. \quad (a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{30 \text{ V} - 0.7 \text{ V}}{550 \text{ k}\Omega + 180(8.2 \text{ k}\Omega + 1.8 \text{ k}\Omega)} = 12.47 \mu\text{A}$$

$$I_C = \beta I_B = (180)(12.47 \mu\text{A}) = \mathbf{2.24 \text{ mA}}$$

$$(b) \quad V_C = V_{CC} - I_C R_C \\ = 30 \text{ V} - (2.24 \text{ mA})(8.2 \text{ k}\Omega) = 30 \text{ V} - 18.37 \text{ V} = \mathbf{11.63 \text{ V}}$$

$$(c) \quad V_E = I_E R_E \cong I_C R_E = (2.24 \text{ mA})(1.8 \text{ k}\Omega) = \mathbf{4.03 \text{ V}}$$

$$(d) \quad V_{CE} = V_{CC} - I_C(R_C + R_E) = 30 \text{ V} - (2.24 \text{ mA})(8.2 \text{ k}\Omega + 1.8 \text{ k}\Omega) \\ = \mathbf{7.6 \text{ V}}$$

$$30. \quad (a) \quad R' = R_C + R_E = 8.2 \text{ k}\Omega + 1.8 \text{ k}\Omega = 10 \text{ k}\Omega \\ R_F/\beta = 550 \text{ k}\Omega/180 = 3.06 \text{ k}\Omega \\ R' \cong \mathbf{3.27(R_F/\beta)} \text{ should be close}$$

$$(b) \quad I_{C_Q} = \frac{V_{CC} - V_{BE}}{R_C + R_E} = \frac{30 \text{ V} - 0.7 \text{ V}}{8.2 \text{ k}\Omega + 1.8 \text{ k}\Omega} = \mathbf{2.93 \text{ mA}} \\ I_{C_Q} = \mathbf{2.93 \text{ mA}} \text{ relatively close to } 2.24 \text{ mA}$$

$$31. \quad (a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (90)(9.1 \text{ k}\Omega + 9.1 \text{ k}\Omega)} \\ = 10.09 \mu\text{A} \\ I_C = \beta I_B = (90)(10.09 \mu\text{A}) = \mathbf{0.91 \text{ mA}} \\ V_{CE} = V_{CC} - I_C(R_C + R_E) = 22 \text{ V} - (0.91 \text{ mA})(9.1 \text{ k}\Omega + 9.1 \text{ k}\Omega) \\ = \mathbf{5.44 \text{ V}}$$

$$(b) \quad \beta = 135, \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{22 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (135)(9.1 \text{ k}\Omega + 9.1 \text{ k}\Omega)} \\ = 7.28 \mu\text{A}$$

$$I_C = \beta I_B = (135)(7.28 \mu\text{A}) = \mathbf{0.983 \text{ mA}} \\ V_{CE} = V_{CC} - I_C(R_C + R_E) = 22 \text{ V} - (0.983 \text{ mA})(9.1 \text{ k}\Omega + 9.1 \text{ k}\Omega) \\ = \mathbf{4.11 \text{ V}}$$

$$(c) \quad \% \Delta I_C = \left| \frac{0.983 \text{ mA} - 0.91 \text{ mA}}{0.91 \text{ mA}} \right| \times 100\% = \mathbf{8.02\%} \\ \% \Delta V_{CE} = \left| \frac{4.11 \text{ V} - 5.44 \text{ V}}{5.44 \text{ V}} \right| \times 100\% = \mathbf{24.45\%}$$

- (d) The results for the collector feedback configuration are closer to the voltage-divider configuration than to the other two. However, the voltage-divider configuration continues to have the least sensitivities to change in β .

32. $1 \text{ M}\Omega = 0 \text{ }\Omega, R_B = 150 \text{ k}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12 \text{ V} - 0.7 \text{ V}}{150 \text{ k}\Omega + (180)(4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega)}$$

$$= 7.11 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (180)(7.11 \text{ }\mu\text{A}) = 1.28 \text{ mA}$$

$$V_C = V_{CC} - I_C R_C = 12 \text{ V} - (1.28 \text{ mA})(4.7 \text{ k}\Omega)$$

$$= \mathbf{5.98 \text{ V}}$$

Full $1 \text{ M}\Omega$: $R_B = 1,000 \text{ k}\Omega + 150 \text{ k}\Omega = 1,150 \text{ k}\Omega = 1.15 \text{ M}\Omega$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + \beta(R_C + R_E)} = \frac{12 \text{ V} - 0.7 \text{ V}}{1.15 \text{ M}\Omega + (180)(4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega)}$$

$$= 4.36 \text{ }\mu\text{A}$$

$$I_C = \beta I_B = (180)(4.36 \text{ }\mu\text{A}) = 0.785 \text{ mA}$$

$$V_C = V_{CC} - I_C R_C = 12 \text{ V} - (0.785 \text{ mA})(4.7 \text{ k}\Omega)$$

$$= \mathbf{8.31 \text{ V}}$$

V_C ranges from **5.98 V to 8.31 V**

33. (a) $V_E = V_B - V_{BE} = 4 \text{ V} - 0.7 \text{ V} = \mathbf{3.3 \text{ V}}$

(b) $I_C \cong I_E = \frac{V_E}{R_E} = \frac{3.3 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{2.75 \text{ mA}}$

(c) $V_C = V_{CC} - I_C R_C = 18 \text{ V} - (2.75 \text{ mA})(2.2 \text{ k}\Omega)$

$$= \mathbf{11.95 \text{ V}}$$

(d) $V_{CE} = V_C - V_E = 11.95 \text{ V} - 3.3 \text{ V} = \mathbf{8.65 \text{ V}}$

(e) $I_B = \frac{V_{R_B}}{R_B} = \frac{V_C - V_B}{R_B} = \frac{11.95 \text{ V} - 4 \text{ V}}{330 \text{ k}\Omega} = \mathbf{24.09 \text{ }\mu\text{A}}$

(f) $\beta = \frac{I_C}{I_B} = \frac{2.75 \text{ mA}}{24.09 \text{ }\mu\text{A}} = \mathbf{114.16}$

34. (a) $I_B = \frac{V_{CC} + V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{6 \text{ V} + 6 \text{ V} - 0.7 \text{ V}}{330 \text{ k}\Omega + (121)(1.2 \text{ k}\Omega)}$

$$= 23.78 \text{ }\mu\text{A}$$

$$I_E = (\beta + 1)I_B = (121)(23.78 \text{ }\mu\text{A})$$

$$= \mathbf{2.88 \text{ mA}}$$

$$-V_{EE} + I_E R_E - V_E = 0$$

$$V_E = -V_{EE} + I_E R_E = -6 \text{ V} + (2.88 \text{ mA})(1.2 \text{ k}\Omega)$$

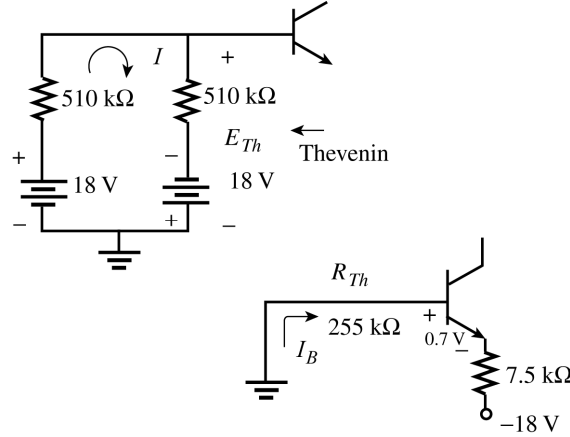
$$= \mathbf{-2.54 \text{ V}}$$

35. (a) $V_B = \frac{82 \text{ k}\Omega(12 \text{ V})}{82 \text{ k}\Omega + 22 \text{ k}\Omega} = 9.46 \text{ V}$
 $V_E = V_B - V_{BE} = 9.46 \text{ V} - 0.7 \text{ V} = 8.76 \text{ V}$
 $I_E = \frac{V_E}{R_E} = \frac{8.76 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{7.3 \text{ mA}}$
 $I_B = \frac{I_E}{\beta + 1} = \frac{7.3 \text{ mA}}{111} = \mathbf{65.77 \mu\text{A}}$
 $I_C = \beta I_B = (110)(65.77 \mu\text{A}) = \mathbf{7.23 \text{ mA}}$
- (b) $V_B = \mathbf{9.46 \text{ V}}$, $V_C = \mathbf{12 \text{ V}}$, $V_E = \mathbf{8.76 \text{ V}}$
- (c) $V_{BC} = V_B - V_C = 9.46 \text{ V} - 12 \text{ V} = \mathbf{-2.54 \text{ V}}$
 $V_{CE} = V_C - V_E = 12 \text{ V} - 8.76 \text{ V} = \mathbf{3.24 \text{ V}}$
36. (a) $I_B = \frac{V_{EE} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{12 \text{ V} - 0.7 \text{ V}}{9.1 \text{ k}\Omega + (80 + 1)15 \text{ k}\Omega} = \mathbf{9.26 \mu\text{A}}$
- (b) $I_C = \beta I_B = (80)(9.26 \mu\text{A}) = \mathbf{0.741 \text{ mA}}$
- (c) $V_{CE} = V_{CC} + V_{EE} - I_C(R_C + R_E)$
 $= 16 \text{ V} + 12 \text{ V} - (0.741 \text{ mA})(27 \text{ k}\Omega)$
 $= \mathbf{8 \text{ V}}$
- (d) $V_C = V_{CC} - I_C R_C = 16 \text{ V} - (0.741 \text{ mA})(12 \text{ k}\Omega) = \mathbf{7.11 \text{ V}}$
37. (a) $I_E = \frac{8 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}\Omega} = \frac{7.3 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{3.32 \text{ mA}}$
- (b) $V_C = 10 \text{ V} - (3.32 \text{ mA})(1.8 \text{ k}\Omega) = 10 \text{ V} - 5.976 = \mathbf{4.02 \text{ V}}$
- (c) $V_{CE} = 10 \text{ V} + 8 \text{ V} - (3.32 \text{ mA})(2.2 \text{ k}\Omega + 1.8 \text{ k}\Omega)$
 $= 18 \text{ V} - 13.28 \text{ V}$
 $= \mathbf{4.72 \text{ V}}$
38. (a) $V_E = 4 \text{ V} - 0.7 \text{ V} = 3.3 \text{ V}$
 $I_E = \frac{V_E}{R_E} = \frac{V_E}{R_E} = \frac{3.3 \text{ V}}{1.1 \text{ k}\Omega} = 3 \text{ mA}$
 $I_C \cong I_E = 3 \text{ mA}$
 $V_C = 8 \text{ V} = V_{CC} - I_C R_C = 14 \text{ V} - (3 \text{ mA})R_C$
 $R_C = \mathbf{2 \text{ k}\Omega}$
- (b) $I_E = 3 \text{ mA}$, $I_B = \frac{I_E}{\beta + 1} = \frac{3 \text{ mA}}{91} = \mathbf{32.97 \mu\text{A}}$

$$(c) \quad V_{BC} = V_B - V_C = 4 \text{ V} - 8 \text{ V} = -4 \text{ V}$$

$$V_{CE} = V_C - V_E = 8 \text{ V} - 3.3 \text{ V} = 4.7 \text{ V}$$

39. (a) $\beta R_E > 10R_2$ not satisfied $\therefore$ Use exact approach:
Network redrawn to determine the Thevenin equivalent:



$$R_{Th} = \frac{510 \text{ k}\Omega}{2} = 255 \text{ k}\Omega$$

$$I = \frac{18 \text{ V} + 18 \text{ V}}{510 \text{ k}\Omega + 510 \text{ k}\Omega} = 35.29 \mu\text{A}$$

$$E_{Th} = -18 \text{ V} + (35.29 \mu\text{A})(510 \text{ k}\Omega) = 0 \text{ V}$$

$$I_B = \frac{18 \text{ V} - 0.7 \text{ V}}{255 \text{ k}\Omega + (130 + 1)(7.5 \text{ k}\Omega)} = 13.95 \mu\text{A}$$

$$(b) \quad I_C = \beta I_B = (130)(13.95 \mu\text{A}) = 1.81 \text{ mA}$$

$$(c) \quad V_E = -18 \text{ V} + (1.81 \text{ mA})(7.5 \text{ k}\Omega)$$

$$= -18 \text{ V} + 13.58 \text{ V}$$

$$= -4.42 \text{ V}$$

$$(d) \quad V_{CE} = 18 \text{ V} + 18 \text{ V} - (1.81 \text{ mA})(9.1 \text{ k}\Omega + 7.5 \text{ k}\Omega)$$

$$= 36 \text{ V} - 30.05 \text{ V} = 5.95 \text{ V}$$

40. (a) $I_B = \frac{V_{R_B}}{R_B} = \frac{V_C - V_{BE}}{R_B} = \frac{8 \text{ V} - 0.7 \text{ V}}{560 \text{ k}\Omega} = 13.04 \mu\text{A}$

$$(b) \quad I_C = \frac{V_{CC} - V_C}{R_C} = \frac{18 \text{ V} - 8 \text{ V}}{3.9 \text{ k}\Omega} = \frac{10 \text{ V}}{3.9 \text{ k}\Omega} = 2.56 \text{ mA}$$

$$(c) \quad \beta = \frac{I_C}{I_B} = \frac{2.56 \text{ mA}}{13.04 \mu\text{A}} = 196.32$$

$$(d) \quad V_{CE} = V_C = 8 \text{ V}$$

$$\begin{aligned}
41. \quad I_B &= \frac{I_C}{\beta} = \frac{2.5 \text{ mA}}{80} = 31.25 \mu\text{A} \\
R_B &= \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{31.25 \mu\text{A}} = \mathbf{361.6 \text{ k}\Omega} \\
R_C &= \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - V_{CE_Q}}{I_{C_Q}} = \frac{12 \text{ V} - 6 \text{ V}}{2.5 \text{ mA}} = \frac{6 \text{ V}}{2.5 \text{ mA}} \\
&= \mathbf{2.4 \text{ k}\Omega}
\end{aligned}$$

Standard values:

$$R_B = \mathbf{360 \text{ k}\Omega}$$

$$R_C = \mathbf{2.4 \text{ k}\Omega}$$

$$\begin{aligned}
42. \quad I_{C_{\text{sat}}} &= \frac{V_{CC}}{R_C + R_E} = 10 \text{ mA} \\
\frac{20 \text{ V}}{4R_E + R_E} &= 10 \text{ mA} \Rightarrow \frac{20 \text{ V}}{5R_E} = 10 \text{ mA} \Rightarrow 5R_E = \frac{20 \text{ V}}{10 \text{ mA}} = 2 \text{ k}\Omega \\
R_E &= \frac{2 \text{ k}\Omega}{5} = \mathbf{400 \Omega} \\
R_C &= 4R_E = \mathbf{1.6 \text{ k}\Omega} \\
I_B &= \frac{I_C}{\beta} = \frac{5 \text{ mA}}{120} = 41.67 \mu\text{A} \\
R_B &= V_{R_B}/I_B = \frac{20 \text{ V} - 0.7 \text{ V} - 5 \text{ mA}(0.4 \text{ k}\Omega)}{41.67 \mu\text{A}} = \frac{19.3 - 2 \text{ V}}{41.67 \mu\text{A}} \\
&= \mathbf{415.17 \text{ k}\Omega}
\end{aligned}$$

Standard values: $R_E = \mathbf{390 \Omega}$, $R_C = \mathbf{1.6 \text{ k}\Omega}$, $R_B = \mathbf{430 \text{ k}\Omega}$

$$\begin{aligned}
43. \quad R_E &= \frac{V_E}{I_E} \cong \frac{V_E}{I_C} = \frac{3 \text{ V}}{4 \text{ mA}} = \mathbf{0.75 \text{ k}\Omega} \\
R_C &= \frac{V_{R_C}}{I_C} = \frac{V_{CC} - V_C}{I_C} = \frac{V_{CC} - (V_{CE_Q} + V_E)}{I_C} \\
&= \frac{24 \text{ V} - (8 \text{ V} + 3 \text{ V})}{4 \text{ mA}} = \frac{24 \text{ V} - 11 \text{ V}}{4 \text{ mA}} = \frac{13 \text{ V}}{4 \text{ mA}} = \mathbf{3.25 \text{ k}\Omega}
\end{aligned}$$

$$V_B = V_E + V_{BE} = 3 \text{ V} + 0.7 \text{ V} = 3.7 \text{ V}$$

$$V_B = \frac{R_2 V_{CC}}{R_2 + R_1} \Rightarrow 3.7 \text{ V} = \frac{R_2 (24 \text{ V})}{R_2 + R_1} \left. \vphantom{\frac{R_2 V_{CC}}{R_2 + R_1}} \right\} 2 \text{ unknowns!}$$

$\therefore$ use $\beta R_E \geq 10R_2$ for increased stability

$$(110)(0.75 \text{ k}\Omega) = 10R_2$$

$$R_2 = 8.25 \text{ k}\Omega$$

$$\text{Choose } R_2 = \mathbf{7.5 \text{ k}\Omega}$$

Substituting in the above equation:

$$3.7 \text{ V} = \frac{7.5 \text{ k}\Omega(24 \text{ V})}{7.5 \text{ k}\Omega + R_1}$$

$$R_1 = \mathbf{41.15 \text{ k}\Omega}$$

Standard values:

$$R_E = \mathbf{0.75 \text{ k}\Omega}, R_C = \mathbf{3.3 \text{ k}\Omega}, R_2 = \mathbf{7.5 \text{ k}\Omega}, R_1 = \mathbf{43 \text{ k}\Omega}$$

$$44. \quad V_E = \frac{1}{5} V_{CC} = \frac{1}{5} (28 \text{ V}) = 5.6 \text{ V}$$

$$R_E = \frac{V_E}{I_E} = \frac{5.6 \text{ V}}{5 \text{ mA}} = \mathbf{1.12 \text{ k}\Omega} \text{ (use } \mathbf{1.1 \text{ k}\Omega})$$

$$V_C = \frac{V_{CC}}{2} + V_E = \frac{28 \text{ V}}{2} + 5.6 \text{ V} = 14 \text{ V} + 5.6 \text{ V} = 19.6 \text{ V}$$

$$V_{R_C} = V_{CC} - V_C = 28 \text{ V} - 19.6 \text{ V} = 8.4 \text{ V}$$

$$R_C = \frac{V_{R_C}}{I_C} = \frac{8.4 \text{ V}}{5 \text{ mA}} = \mathbf{1.68 \text{ k}\Omega} \text{ (use } \mathbf{1.6 \text{ k}\Omega})$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 5.6 \text{ V} = 6.3 \text{ V}$$

$$V_B = \frac{R_2 V_{CC}}{R_2 + R_1} \Rightarrow 6.3 \text{ V} = \frac{R_2 (28 \text{ V})}{R_2 + R_1} \text{ (2 unknowns)}$$

$$\beta = \frac{I_C}{I_B} = \frac{5 \text{ mA}}{37 \mu\text{A}} = 135.14$$

$$\beta R_E = 10 R_2$$

$$(135.14)(1.12 \text{ k}\Omega) = 10(R_2)$$

$$R_2 = 15.14 \text{ k}\Omega \text{ (use } \mathbf{15 \text{ k}\Omega})$$

$$\text{Substituting: } 6.3 \text{ V} = \frac{(15.14 \text{ k}\Omega)(28 \text{ V})}{15.14 \text{ k}\Omega + R_1}$$

$$\text{Solving, } R_1 = 52.15 \text{ k}\Omega \text{ (use } \mathbf{51 \text{ k}\Omega})$$

Standard values:

$$R_E = \mathbf{1.1 \text{ k}\Omega}$$

$$R_C = \mathbf{1.6 \text{ k}\Omega}$$

$$R_1 = \mathbf{51 \text{ k}\Omega}$$

$$R_2 = \mathbf{15 \text{ k}\Omega}$$

$$45. \quad (a) \quad V_{B_1} = \frac{4.7 \text{ k}\Omega(20 \text{ V})}{4.7 \text{ k}\Omega + 18 \text{ k}\Omega} = \mathbf{4.14 \text{ V}}$$

$$V_{E_1} = 4.14 \text{ V} - 0.7 \text{ V} = \mathbf{3.44 \text{ V}}$$

$$I_{C_1} \cong I_{E_1} = \frac{3.44 \text{ V}}{1 \text{ k}\Omega} = 3.44 \text{ mA}$$

$$V_{C_1} = 20 \text{ V} - (3.44 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{12.43 \text{ V}}$$

$$V_{B_2} = \frac{3.3 \text{ k}\Omega(20 \text{ V})}{3.3 \text{ k}\Omega + 22 \text{ k}\Omega} = \mathbf{2.61 \text{ V}}$$

$$V_{E_2} = 2.61 \text{ V} - 0.7 \text{ V} = \mathbf{1.91 \text{ V}}$$

$$I_{E_2} \cong I_{C_2} = \frac{1.91 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{1.59 \text{ mA}}$$

$$V_{C_2} = 20 \text{ V} - (1.59 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{16.5 \text{ V}}$$

$$(b) \quad I_{B_1} = \frac{I_{C_1}}{\beta} = \frac{3.44 \text{ mA}}{160} = \mathbf{21.5 \mu\text{A}}, \quad I_{C_1} \cong I_{E_1} = \mathbf{3.44 \text{ mA}}$$

$$I_{B_2} = \frac{I_{C_2}}{\beta} = \frac{1.59 \text{ mA}}{90} = \mathbf{17.67 \mu\text{A}}, \quad I_{C_2} \cong I_{E_2} = \mathbf{1.59 \text{ mA}}$$

$$46. \quad (a) \quad \beta_D = \beta_1 \beta_2 = (50)(75) = 3750$$

$$(b) \quad I_{B_1} = \frac{V_{CC} - V_{BE_1} - V_{BE_2}}{R_B + (\beta_D + 1)R_E} = \frac{18 \text{ V} - 0.7 \text{ V} - 0.7 \text{ V}}{2.2 \text{ M}\Omega + (3750+1)470 \Omega} \\ = \mathbf{4.19 \mu\text{A}}$$

$$I_{B_2} = (\beta_1 + 1)I_{B_1} = (50 + 1)(4.19 \mu\text{A}) = \mathbf{213.69 \mu\text{A}}$$

$$(c) \quad I_{C_1} = \beta_1 I_{B_1} = (50)(4.19 \mu\text{A}) = \mathbf{0.21 \text{ mA}}$$

$$I_{C_2} = \beta_2 I_{B_2} = (75)(213.69 \mu\text{A}) = \mathbf{16.03 \text{ mA}}$$

$$(d) \quad V_{C_1} = \mathbf{18 \text{ V}}, \quad V_{C_2} = \mathbf{18 \text{ V}}$$

$$V_{E_2} = I_E R_E \cong I_{C_2} R_E = (16.03 \text{ mA})(470 \Omega) \\ = \mathbf{7.53 \text{ V}}$$

$$V_{E_1} = V_{E_2} + 0.7 \text{ V} = 7.53 \text{ V} + 0.7 \text{ V} = \mathbf{8.23 \text{ V}}$$

$$47. \quad (a) \quad V_{B_1} = \frac{3.3 \text{ k}\Omega(22 \text{ V})}{3.3 \text{ k}\Omega + 4.7 \text{ k}\Omega + 8.2 \text{ k}\Omega} = 4.48 \text{ V}$$

$$V_{E_1} = V_{B_1} - 0.7 \text{ V} = 3.78 \text{ V}$$

$$I_{E_1} \cong I_{E_2} \cong I_{C_2} \cong I_{C_1} = \frac{V_{E_1}}{R_E} = \frac{3.78 \text{ V}}{1.1 \text{ k}\Omega} = 3.44 \text{ mA}$$

$$I_{B_1} = \frac{I_{C_1}}{\beta} = \mathbf{57.33 \mu\text{A}}, \quad I_{C_1} = \mathbf{3.44 \text{ mA}}$$

$$I_{B_2} = \frac{I_{C_3}}{120} = \mathbf{28.67 \mu\text{A}}, \quad I_{C_2} = \mathbf{3.44 \text{ mA}}$$

$$(b) \quad V_{B_1} = \mathbf{4.48 \text{ V}}$$

$$V_{B_2} = \frac{(4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega)(22 \text{ V})}{3.3 \text{ k}\Omega + 4.7 \text{ k}\Omega + 8.2 \text{ k}\Omega} = \mathbf{10.86 \text{ V}}$$

$$V_{E_1} = \mathbf{3.78 \text{ V}}, \quad V_{C_1} = V_{B_2} - 0.7 \text{ V} = \mathbf{10.16 \text{ V}}$$

$$V_{E_{@}} = \mathbf{10.16 \text{ V}}, \quad V_{C_2} = 22 \text{ V} - (3.44 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{14.43 \text{ V}}$$

48. (a) $I_{B_1} = \frac{V_{CC} - V_{EB_1}}{R_B + \beta_1 \beta_2 R_C} = \frac{12 \text{ V} - 0.7 \text{ V}}{1.8 \text{ M}\Omega + (80)(160)(220 \text{ }\Omega)} = \mathbf{2.45 \text{ }\mu A}$
 $I_{C_1} = \beta_1 I_{B_1} = (80)(2.45 \text{ }\mu A) = \mathbf{196 \text{ }\mu A}$
 $I_{B_2} = I_{C_1} = \mathbf{196 \text{ }\mu A}$
 $I_{C_2} = \beta_2 I_{B_2} = (160)(196 \text{ }\mu A) = \mathbf{31.36 \text{ mA}}$
- (b) $V_{B_1} = I_{B_1} R_B = (2.45 \text{ }\mu A)(1.8 \text{ M}\Omega) = \mathbf{4.41 \text{ V}}$
 $V_{E_1} = 12 \text{ V} - I_{C_1} (220 \text{ }\Omega) = 12 \text{ V} - (31.36 \text{ mA})(220 \text{ }\Omega) = \mathbf{5.1 \text{ V}}$
 $V_{B_2} = V_{BE_2} = \mathbf{0.7 \text{ V}}, V_{C_2} = V_{E_1} = \mathbf{5.1 \text{ V}}, V_{C_1} = V_{B_2} = \mathbf{0.7 \text{ V}}, V_{E_2} = \mathbf{0 \text{ V}}$
49. $I_{2 \text{ k}\Omega} = \frac{18 \text{ V} - 0.7 \text{ V}}{2 \text{ k}\Omega} = \mathbf{8.65 \text{ mA}} \cong I$
50. For current mirror:
 $I(3 \text{ k}\Omega) = I(2.4 \text{ k}\Omega) = I = \mathbf{2 \text{ mA}}$
51. $6 \text{ V} - I_B R_B - V_{BE} - I_E R_E = 0$
 $6 \text{ V} - I_B 100 \text{ k}\Omega - 0.7 \text{ V} - (\beta + 1)I_B 1.2 \text{ k}\Omega = 0$
 $6 \text{ V} - 0.7 \text{ V} - I_B (100 \text{ k}\Omega + (120 + 1)(1.2 \text{ k}\Omega)) = 0$
 $I_B = \frac{6 \text{ V} - 0.7 \text{ V}}{100 \text{ k}\Omega + 145 \text{ k}\Omega} = 21.61 \text{ }\mu A$
 $I = I_C = \beta I_B = (120)(21.61 \text{ }\mu A) = \mathbf{2.59 \text{ mA}}$
52. $V_B \cong \frac{4.3 \text{ k}\Omega}{4.3 \text{ k}\Omega + 4.3 \text{ k}\Omega}(-18 \text{ V}) = -9 \text{ V}$
 $V_E = -9 \text{ V} - 0.7 \text{ V} = -9.7 \text{ V}$
 $I_E = \frac{-18 \text{ V} - (-9.7 \text{ V})}{1.8 \text{ k}\Omega} = \mathbf{4.6 \text{ mA}} = I$
53. $I_E = \frac{V_Z - V_{BE}}{R_E} = \frac{5.1 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{3.67 \text{ mA}}$
54. $I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{510 \text{ k}\Omega} = \frac{11.3 \text{ V}}{510 \text{ k}\Omega} = 22.16 \text{ }\mu A$
 $I_C = \beta I_B = (100)(22.16 \text{ }\mu A) = \mathbf{2.216 \text{ mA}}$
 $V_C = -V_{CC} + I_C R_C = -12 \text{ V} + (2.216 \text{ mA})(3.3 \text{ k}\Omega)$
 $= \mathbf{-4.69 \text{ V}}$
 $V_{CE} = V_C = \mathbf{-4.69 \text{ V}}$

55. $\beta R_E \geq 10 R_2$
 $(220)(0.75 \text{ k}\Omega) \geq 10(16 \text{ k}\Omega)$
 $165 \text{ k}\Omega \geq 160 \text{ k}\Omega$ (checks)
 Use approximate approach:

$$V_B \cong \frac{16 \text{ k}\Omega(-22 \text{ V})}{16 \text{ k}\Omega + 82 \text{ k}\Omega} = -3.59 \text{ V}$$

$$V_E = V_B + 0.7 \text{ V} = -3.59 \text{ V} + 0.7 \text{ V} = -2.89 \text{ V}$$

$$I_C \cong I_E = V_E / R_E = 2.89 / 0.75 \text{ k}\Omega = 3.85 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{3.85 \text{ mA}}{220} = \mathbf{17.5 \mu A}$$

$$\begin{aligned} V_C &= -V_{CC} + I_C R_C \\ &= -22 \text{ V} + (3.85 \text{ mA})(2.2 \text{ k}\Omega) \\ &= \mathbf{-13.53 \text{ V}} \end{aligned}$$

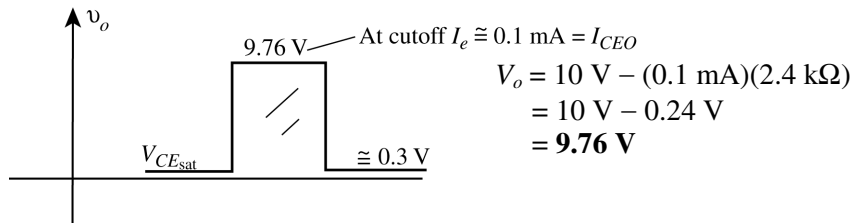
56. $I_E = \frac{V - V_{BE}}{R_E} = \frac{8 \text{ V} - 0.7 \text{ V}}{3.3 \text{ k}\Omega} = \frac{7.3 \text{ V}}{3.3 \text{ k}\Omega} = \mathbf{2.212 \text{ mA}}$
 $V_C = -V_{CC} + I_C R_C = -12 \text{ V} + (2.212 \text{ mA})(3.9 \text{ k}\Omega)$
 $= \mathbf{-3.37 \text{ V}}$

57. $I_{C_{\text{sat}}} = \frac{V_{CC}}{R_C} = \frac{10 \text{ V}}{2.4 \text{ k}\Omega} = \mathbf{4.167 \text{ mA}}$

From characteristics $I_{B_{\text{max}}} \cong 31 \mu\text{A}$

$$I_B = \frac{V_i - V_{BE}}{R_B} = \frac{10 \text{ V} - 0.7 \text{ V}}{180 \text{ k}\Omega} = 51.67 \mu\text{A}$$

$51.67 \mu\text{A} \gg 31 \mu\text{A}$, well saturated



$$58. \quad I_{C_{\text{sat}}} = 8 \text{ mA} = \frac{5 \text{ V}}{R_C}$$

$$R_C = \frac{5 \text{ V}}{8 \text{ mA}} = \mathbf{0.625 \text{ k}\Omega}$$

$$I_{B_{\text{max}}} = \frac{I_{C_{\text{sat}}}}{\beta} = \frac{8 \text{ mA}}{100} = 80 \mu\text{A}$$

Use $1.2 (80 \mu\text{A}) = 96 \mu\text{A}$

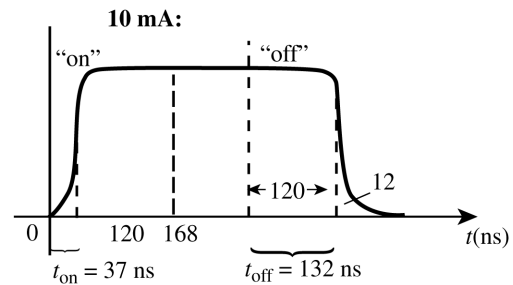
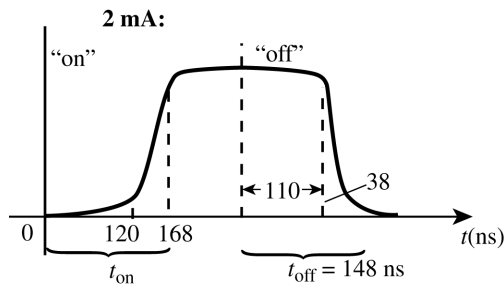
$$R_B = \frac{5 \text{ V} - 0.7 \text{ V}}{96 \mu\text{A}} = \mathbf{44.79 \text{ k}\Omega}$$

Standard values:

$$R_B = \mathbf{43 \text{ k}\Omega}$$

$$R_C = \mathbf{0.62 \text{ k}\Omega}$$

59. (a) From Fig. 3.23c:
 $I_C = 2 \text{ mA}$: $t_f = 38 \text{ ns}$, $t_r = 48 \text{ ns}$, $t_d = 120 \text{ ns}$, $t_s = 110 \text{ ns}$
 $t_{\text{on}} = t_r + t_d = 48 \text{ ns} + 120 \text{ ns} = \mathbf{168 \text{ ns}}$
 $t_{\text{off}} = t_s + t_f = 110 \text{ ns} + 38 \text{ ns} = \mathbf{148 \text{ ns}}$
- (b) $I_C = 10 \text{ mA}$: $t_f = 12 \text{ ns}$, $t_r = 15 \text{ ns}$, $t_d = 22 \text{ ns}$, $t_s = 120 \text{ ns}$
 $t_{\text{on}} = t_r + t_d = 15 \text{ ns} + 22 \text{ ns} = \mathbf{37 \text{ ns}}$
 $t_{\text{off}} = t_s + t_f = 120 \text{ ns} + 12 \text{ ns} = \mathbf{132 \text{ ns}}$
 The turn-on time has dropped dramatically
 $168 \text{ ns} : 37 \text{ ns} = \mathbf{4.54:1}$
 while the turn-off time is only slightly smaller
 $148 \text{ ns} : 132 \text{ ns} = \mathbf{1.12:1}$



60. (a) Open-circuit in the base circuit
 Bad connection of emitter terminal
 Damaged transistor
- (b) Shorted base-emitter junction
 Open at collector terminal
- (c) Open-circuit in base circuit
 Open transistor

61. (a) The base voltage of 9.4 V reveals that the 18 k Ω resistor is not making contact with the base terminal of the transistor.

If operating properly:

$$V_B \cong \frac{18 \text{ k}\Omega(16 \text{ V})}{18 \text{ k}\Omega + 91 \text{ k}\Omega} = \mathbf{2.64 \text{ V}} \text{ vs. } 9.4 \text{ V}$$

As an emitter feedback bias circuit:

$$\begin{aligned} I_B &= \frac{V_{CC} - V_{BE}}{R_1 + (\beta + 1)R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{91 \text{ k}\Omega + (100 + 1)1.2 \text{ k}\Omega} \\ &= 72.1 \mu\text{A} \\ V_B &= V_{CC} - I_B(R_1) = 16 \text{ V} - (72.1 \mu\text{A})(91 \text{ k}\Omega) \\ &= \mathbf{9.4 \text{ V}} \end{aligned}$$

- (b) Since $V_E > V_B$ the transistor should be “off”

$$\text{With } I_B = 0 \mu\text{A}, V_B = \frac{18 \text{ k}\Omega(16 \text{ V})}{18 \text{ k}\Omega + 91 \text{ k}\Omega} = 2.64 \text{ V}$$

$\therefore$ Assume base circuit “open”

The 4 V at the emitter is the voltage that would exist if the transistor were shorted collector to emitter.

$$V_E = \frac{1.2 \text{ k}\Omega(16 \text{ V})}{1.2 \text{ k}\Omega + 3.6 \text{ k}\Omega} = \mathbf{4 \text{ V}}$$

62. (a) $R_B \uparrow, I_B \downarrow, I_C \downarrow, V_C \uparrow$
 (b) $\beta \downarrow, I_C \downarrow$
 (c) Unchanged, $I_{C_{\text{sat}}}$ not a function of β
 (d) $V_{CC} \downarrow, I_B \downarrow, I_C \downarrow$
 (e) $\beta \downarrow, I_C \downarrow, V_{R_C} \downarrow, V_{R_E} \downarrow, V_{CE} \uparrow$

63. (a)
$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \cong \frac{E_{Th} - V_{BE}}{R_{Th} + \beta R_E}$$

$$I_C = \beta I_B = \beta \left[\frac{E_{Th} - V_{BE}}{R_{Th} + \beta R_E} \right] = \frac{E_{Th} - V_{BE}}{\frac{R_{Th}}{\beta} + R_E}$$

$$\text{As } \beta \uparrow, \frac{R_{Th}}{\beta} \downarrow, I_C \uparrow, V_{R_C} \uparrow$$

$$V_C = V_{CC} - V_{R_C}$$

and $V_C \downarrow$

- (b) $R_2 = \text{open}, I_B \uparrow, I_C \uparrow$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

 and $V_{CE} \downarrow$

- (c) $V_{CC} \downarrow, V_B \downarrow, V_E \downarrow, I_E \downarrow, I_C \downarrow$
- (d) $I_B = 0 \mu\text{A}, I_C = I_{CEO}$ and $I_C(R_C + R_E)$ negligible
with $V_{CE} \cong V_{CC} = 20 \text{ V}$
- (e) Base-emitter junction = short $I_B \uparrow$ but transistor action lost and $I_C = 0 \text{ mA}$ with
 $V_{CE} = V_{CC} = 20 \text{ V}$
64. (a) R_B open, $I_B = 0 \mu\text{A}, I_C = I_{CEO} \cong 0 \text{ mA}$
and $V_C \cong V_{CC} = 18 \text{ V}$
- (b) $\beta \uparrow, I_C \uparrow, V_{R_C} \uparrow, V_{R_E} \uparrow, V_{CE} \downarrow$
- (c) $R_C \downarrow, I_B \uparrow, I_C \uparrow, V_E \uparrow$
- (d) Drop to a relatively low voltage $\cong 0.06 \text{ V}$
- (e) Open in the base circuit
65. (a) $S(I_{CO}) = \beta = 120$
- (b) $S(V_{BE}) \cong -\beta/R_B - 120/510 \text{ k}\Omega = -235 \times 10^{-6} \text{ S}$
- (c) $S(\beta) = I_{C_1}/\beta_1 = 3.6 \text{ mA}/120 = 30 \times 10^{-6} \text{ A}$
- (d) $\Delta I_C = S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE} + S(\beta)\Delta\beta$
 $= (120)(10 \mu\text{A} - 0.2 \mu\text{A}) + (-2.35 \times 10^{-4} \text{ S})(0.5 \text{ V} - 0.7 \text{ V})$
 $+ (30 \times 10^{-6} \text{ A})(150 - 120)$
 $\cong 2.12 \text{ mA}$
66. (a) $S(I_{CO}) \cong \frac{\beta(1 + R_B/R_E)}{\beta + R_B/R_E} = \frac{125(1 + 270 \text{ k}\Omega/2.2 \text{ k}\Omega)}{125 + 270 \text{ k}\Omega/2.2 \text{ k}\Omega}$
 $= 62.44$
- (b) $S(V_{BE}) \cong \frac{-\beta/R_E}{\beta + R_B/R_E} = \frac{-125/2.2 \text{ k}\Omega}{125 + 270 \text{ k}\Omega/2.2 \text{ k}\Omega}$
 $= -229.3 \times 10^{-6} \text{ S}$
- (c) $S(\beta) = \frac{I_{C_1}(1 + R_B/R_E)}{\beta_1(\beta_2 + R_B/R_E)} = \frac{4.41 \text{ mA}(1 + 122.73)}{125(156.25 + 122.73)}$
 $= 15.65 \times 10^{-6} \text{ A}$

$$\begin{aligned}
(d) \quad \Delta I_C &= S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE} + S(\beta)\Delta\beta \\
&= (62.44)(10 \mu\text{A} - 0.2 \mu\text{A}) + (-229.36 \times 10^{-6} \text{ S})(0.5 \text{ V} - 0.7 \text{ V}) \\
&\quad + 15.65 \times 10^{-6} \text{ A}(156.25 - 125) \\
&= \mathbf{1.03 \text{ mA}}
\end{aligned}$$

$$\begin{aligned}
67. \quad (a) \quad S(I_{CO}) &= \frac{\beta(1 + R_{Th} / R_E)}{\beta + R_{Th} / R_E} = (80) \frac{(1 + 7.94 \text{ k}\Omega / 0.68 \text{ k}\Omega)}{80 + 7.94 \text{ k}\Omega / 0.68 \text{ k}\Omega} \\
&= \mathbf{11.06}
\end{aligned}$$

$$\begin{aligned}
(b) \quad S(V_{BE}) &= \frac{-\beta / R_E}{\beta + R_{Th} / R_E} = \frac{-80 / 0.68 \text{ k}\Omega}{80 + 7.94 \text{ k}\Omega / 0.68 \text{ k}\Omega} \\
&= \mathbf{-1280 \times 10^{-6} \text{ S}}
\end{aligned}$$

$$\begin{aligned}
(c) \quad S(\beta) &= \frac{I_{C_1}(1 + R_{Th} / R_E)}{\beta_1(\beta_2 + R_{Th} / R_E)} = \frac{1.71 \text{ mA}(1 + 7.94 \text{ k}\Omega / 0.68 \text{ k}\Omega)}{80(100 + 7.94 \text{ k}\Omega / 0.68 \text{ k}\Omega)} \\
&= \mathbf{2.43 \times 10^{-6} \text{ A}}
\end{aligned}$$

$$\begin{aligned}
(d) \quad \Delta I_C &= S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE} + S(\beta)\Delta\beta \\
&= (11.06)(10 \mu\text{A} - 0.2 \mu\text{A}) + (-1.28 \times 10^{-3} \text{ S})(0.5 \text{ V} - 0.7 \text{ V}) + (2.43 \times 10^{-6})(100 - 80) \\
&= \mathbf{0.313 \text{ mA}}
\end{aligned}$$

$$\begin{aligned}
68. \quad (a) \quad S(I_{CO}) &= (\beta + 1) \frac{\beta(1 + R_B / R_C)}{\beta + R_B / R_C} = \frac{196.32(1 + 560 \text{ k}\Omega / 3.9 \text{ k}\Omega)}{196.32 + 560 \text{ k}\Omega / 3.9 \text{ k}\Omega} \\
&= \mathbf{83.69}
\end{aligned}$$

$$\begin{aligned}
(b) \quad S(V_{BE}) &= \frac{-\beta / R_C}{\beta + R_B / R_C} = \frac{-196.32 / 3.9 \text{ k}\Omega}{196.32 + 560 \text{ k}\Omega / 3.9 \text{ k}\Omega} \\
&= \mathbf{-1.48.04 \times 10^{-6} \text{ S}}
\end{aligned}$$

$$\begin{aligned}
(c) \quad S(\beta) &= \frac{I_{C_1}(R_B + R_C)}{\beta_1(R_B + \beta_2 R_C)} = \frac{2.56 \text{ mA}(560 \text{ k}\Omega + 3.9 \text{ k}\Omega)}{196.32(560 \text{ k}\Omega + 245.4(3.9 \text{ k}\Omega))} \\
&= \mathbf{4.83 \times 10^{-6} \text{ A}}
\end{aligned}$$

$$\begin{aligned}
(d) \quad \Delta I_C &= S(I_{CO})\Delta I_{CO} + S(V_{BE})\Delta V_{BE} + S(\beta)\Delta\beta \\
&= (83.69)(10 \mu\text{A} - 0.2 \mu\text{A}) + (-1.48.04 \times 10^{-6} \text{ S})(0.5 - 0.7 \text{ V}) \\
&\quad + (4.83 \times 10^{-6} \text{ A})(245.4 - 196.32) \\
&= \mathbf{1.087 \text{ mA}}
\end{aligned}$$

69.	Type	$S(I_{CO})$	$S(V_{BE})$	$S(\beta)$
	Fixed-bias	120	$-235 \times 10^{-6} \text{S}$	$30 \times 10^{-6} \text{A}$
	Emitter-bias	62.44	$-229.36 \times 10^{-6} \text{S}$	$15.65 \times 10^{-6} \text{A}$
	Voltage-divider	11.06	$-1280 \times 10^{-6} \text{S}$	$2.43 \times 10^{-6} \text{A}$
	Collector feedback	83.69	$-148.04 \times 10^{-6} \text{S}$	$4.83 \times 10^{-6} \text{A}$

$S(I_{CO})$: Considerably less for the voltage-divider configuration compared to the other three.

$S(V_{BE})$: The voltage-divider configuration is the most sensitive .

$S(\beta)$: The voltage-divider configuration is the least sensitive with the fixed-bias configuration very sensitive.

In general, the voltage-divider configuration is the least sensitive with the fixed-bias the most sensitive.

70. (a) Fixed-bias:

$$S(I_{CO}) = 120, \Delta I_C = 1.176 \text{ mA}$$

$$S(V_{BE}) = -235 \times 10^{-6} \text{S}, \Delta I_C = 0.047 \text{ mA}$$

$$S(\beta) = 30 \times 10^{-6} \text{A}, \Delta I_C = 0.90 \text{ mA}$$

(b) Voltage-divider bias:

$$S(I_{CO}) = 11.06, \Delta I_C = 0.108 \text{ mA}$$

$$S(V_{BE}) = -1280 \times 10^{-6} \text{S}, \Delta I_C = 0.0256 \text{ mA}$$

$$S(\beta) = 2.43 \times 10^{-6} \text{A}, \Delta I_C = 0.0048 \times 10^{-6} \text{ A}$$

(c) For the fixed-bias configuration there is a strong sensitivity to changes in I_{CO} and β and much less to changes in V_{BE} .

For the voltage-divider configuration the opposite occurs with a high sensitivity to changes in V_{BE} and less to changes in I_{CO} and β .

In total the voltage-divider configuration is considerably more stable than the fixed-bias configuration.

Chapter 5

1. (a) If the dc power supply is set to zero volts, the amplification will be zero.

(b) Too low a dc level will result in a clipped output waveform.

$$\begin{aligned} \text{(c)} \quad P_o &= I^2 R = (5 \text{ mA})^2 2.2 \text{ k}\Omega = 55 \text{ mW} \\ P_i &= V_{CC} I = (18 \text{ V})(3.8 \text{ mA}) = 68.4 \text{ mW} \\ \eta &= \frac{P_o(\text{ac})}{P_i(\text{dc})} = \frac{55 \text{ mW}}{68.4 \text{ mW}} = 0.804 \Rightarrow \mathbf{80.4\%} \end{aligned}$$

2. —

3. No, because they intersect to define the Q point.

$$\begin{aligned} 4. \quad X_C &= \frac{1}{2\pi fC} = \frac{1}{2\pi(1 \text{ kHz})(10 \mu\text{F})} = \mathbf{15.92 \Omega} \\ f &= 100 \text{ kHz: } X_C = \mathbf{0.159 \Omega} \\ &\text{Yes, better at 100 kHz} \end{aligned}$$

5. —

$$6. \quad \text{(a)} \quad r_o = \frac{V_A + V_{CEQ}}{I_{CQ}} = \frac{100 \text{ V} + 8 \text{ V}}{4 \text{ mA}} = \mathbf{27 \text{ k}\Omega}$$

$$\text{(b)} \quad r_o = \frac{\Delta V_{CE}}{\Delta I_C} \Rightarrow \Delta I_C = \frac{\Delta V_{CE}}{r_o} = \frac{6 \text{ V}}{27 \text{ k}\Omega} = \mathbf{0.222 \text{ mA}}$$

$$\begin{aligned} 7. \quad \text{(a)} \quad Z_i &= \frac{V_i}{I_i} = \frac{10 \text{ mV}}{0.5 \text{ mA}} \\ &= \mathbf{20 \Omega (=r_e)} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad V_o &= I_c R_L \\ &= \alpha I_e R_L \\ &= (0.98)(0.5 \text{ mA})(1.2 \text{ k}\Omega) \\ &= \mathbf{0.588 \text{ V}} \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad A_v &= \frac{V_o}{V_i} = \frac{0.588 \text{ V}}{10 \text{ mV}} \\ &= \mathbf{58.8} \end{aligned}$$

$$\text{(d)} \quad Z_o = \mathbf{\infty \Omega}$$

- (e) $A_i = \frac{I_o}{I_i} = \frac{\alpha I_e}{I_e} = \alpha = \mathbf{0.98}$
- (f) $I_b = I_e - I_c$
 $= 0.5 \text{ mA} - 0.49 \text{ mA}$
 $= \mathbf{10 \mu A}$
8. (a) $r_e = \frac{26 \text{ mV}}{I_E(\text{dc})} = \frac{26 \text{ mV}}{2 \text{ mA}} = 13 \Omega$
 $Z_i = \beta r_e = (80)(13 \Omega)$
 $= \mathbf{1.04 \text{ k}\Omega}$
- (b) $I_b = \frac{I_c}{\beta} = \frac{\alpha I_e}{\beta} = \frac{\beta}{\beta+1} \cdot \frac{I_e}{\beta} = \frac{I_e}{\beta+1}$
 $= \frac{2 \text{ mA}}{81} = \mathbf{24.69 \mu A}$
- (c) $A_i = \frac{I_o}{I_i} = \frac{I_L}{I_b}$
 $I_L = \frac{r_o(\beta I_b)}{r_o + R_L}$
 $A_i = \frac{\frac{r_o}{r_o + R_L} \cdot \beta I_b}{I_b} = \frac{r_o}{r_o + R_L} \cdot \beta$
 $= \frac{40 \text{ k}\Omega}{40 \text{ k}\Omega + 1.2 \text{ k}\Omega} (80)$
 $= \mathbf{77.67}$
- (d) $A_v = -\frac{R_L \parallel r_o}{r_e} = -\frac{1.2 \text{ k}\Omega \parallel 40 \text{ k}\Omega}{13 \Omega}$
 $= -\frac{1.165 \text{ k}\Omega}{13 \Omega}$
 $= \mathbf{-89.6}$
9. (a) $Z_i = \beta r_e = (140)r_e = 1200$
 $r_e = \frac{1200}{140} = \mathbf{8.57 \Omega}$
- (b) $I_b = \frac{V_i}{Z_i} = \frac{30 \text{ mV}}{1.2 \text{ k}\Omega} = \mathbf{25 \mu A}$
- (c) $I_c = \beta I_b = (140)(25 \mu A) = \mathbf{3.5 \text{ mA}}$

$$(d) \quad I_L = \frac{r_o I_c}{r_o + R_L} = \frac{(50 \text{ k}\Omega)(3.5 \text{ mA})}{50 \text{ k}\Omega + 2.7 \text{ k}\Omega} = 3.321 \text{ mA}$$

$$A_i = \frac{I_L}{I_i} = \frac{3.321 \text{ mA}}{25 \text{ }\mu\text{A}} = \mathbf{132.84}$$

$$(e) \quad A_v = \frac{V_o}{V_i} = \frac{-A_i R_L}{Z_i} = -(132.84) \frac{(2.7 \text{ k}\Omega)}{1.2 \text{ k}\Omega} \\ = \mathbf{-298.89}$$

$$10. \quad (a) \quad r_e = \frac{V_i}{I_i} = \frac{48 \text{ mV}}{3.2 \text{ mA}} = \mathbf{15 \text{ }\Omega}$$

$$(b) \quad Z_i = r_e = \mathbf{15 \text{ }\Omega}$$

$$(c) \quad I_C = \alpha I_e = (0.99)(3.2 \text{ mA}) = \mathbf{3.168 \text{ mA}}$$

$$(d) \quad V_o = I_C R_L = (3.168 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{6.97 \text{ V}}$$

$$(e) \quad A_v = \frac{V_o}{V_i} = \frac{6.97 \text{ V}}{48 \text{ mV}} = \mathbf{145.21}$$

$$(f) \quad I_b = (1 - \alpha)I_e = (1 - 0.99)I_e = (0.01)(3.2 \text{ mA}) \\ = \mathbf{32 \text{ }\mu\text{A}}$$

$$11. \quad (a) \quad r_e: \quad I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega} = 51.36 \text{ }\mu\text{A}$$

$$I_E = (\beta + 1)I_B = (60 + 1)(51.36 \text{ }\mu\text{A}) \\ = 3.13 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.13 \text{ mA}} = 8.31 \text{ }\Omega$$

$$Z_i = R_B \parallel \beta r_e = 220 \text{ k}\Omega \parallel (60)(8.31 \text{ }\Omega) = 220 \text{ k}\Omega \parallel 498.6 \text{ }\Omega \\ = \mathbf{497.47 \text{ }\Omega}$$

$$r_o \geq 10R_C \therefore Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$$

$$(b) \quad A_v = -\frac{R_C}{r_e} = \frac{-2.2 \text{ k}\Omega}{8.31 \text{ }\Omega} = \mathbf{-264.74}$$

$$(c) \quad Z_i = \mathbf{497.47 \text{ }\Omega} \text{ (the same)} \\ Z_o = r_o \parallel R_C = 20 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \\ = \mathbf{1.98 \text{ k}\Omega}$$

$$A_v = \frac{-R_C \parallel r_o}{r_e} = \frac{-1.98 \text{ k}\Omega}{8.31 \text{ }\Omega} = \mathbf{-238.27}$$

$$12. \quad A_v = -\frac{R_C}{r_e} \Rightarrow r_e = -\frac{R_C}{A_v} = -\frac{4.7 \text{ k}\Omega}{(-200)} = 23.5 \text{ }\Omega$$

$$r_e = \frac{26 \text{ mV}}{I_E} \Rightarrow I_E = \frac{26 \text{ mV}}{r_e} = \frac{26 \text{ mV}}{23.5 \Omega} = 1.106 \text{ mA}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{1.106 \text{ mA}}{91} = 12.15 \mu\text{A}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} \Rightarrow V_{CC} = I_B R_B + V_{BE}$$

$$= (12.15 \mu\text{A})(1 \text{ M}\Omega) + 0.7 \text{ V}$$

$$= 12.15 \text{ V} + 0.7 \text{ V}$$

$$= \mathbf{12.85 \text{ V}}$$

13. (a) $I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{8 \text{ V} - 0.7 \text{ V}}{390 \text{ k}\Omega} = \mathbf{18.72 \mu\text{A}}$

$$I_E = (\beta + 1)I_B = (101)(18.72 \mu\text{A}) = 1.89 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.89 \text{ mA}} = \mathbf{13.76 \Omega}$$

$$I_C = \beta I_B = (100)(18.72 \mu\text{A}) = \mathbf{1.87 \text{ mA}}$$

(b) $Z_i = R_B \parallel \beta r_e = 390 \text{ k}\Omega \parallel (100)(13.76 \Omega) = 390 \text{ k}\Omega \parallel 1.38 \text{ k}\Omega$
 $= \mathbf{1.38 \text{ k}\Omega}$

$$r_o \geq 10R_C \therefore Z_o = R_C = \mathbf{5.6 \text{ k}\Omega}$$

(c) $A_v = -\frac{R_C}{r_e} = \frac{-5.6 \text{ k}\Omega}{13.76 \Omega} = \mathbf{-406.98}$

(d) $A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{(5.6 \text{ k}\Omega) \parallel (30 \text{ k}\Omega)}{13.76 \Omega} = -\frac{4.72 \text{ k}\Omega}{13.76 \Omega} = \mathbf{-343.03}$

14. $A_v = \frac{-R_C}{r_e} = \frac{-406.98}{2} = -203.49$
 $R_C = (-203.49)(13.76 \Omega) = \mathbf{2.8 \text{ k}\Omega}$

15. (a) Test $\beta R_E \geq 10R_2$
 $\quad \quad \quad ?$
 $(100)(1.2 \text{ k}\Omega) \geq 10(4.7 \text{ k}\Omega)$
 $120 \text{ k}\Omega > 47 \text{ k}\Omega \text{ (satisfied)}$

Use approximate approach:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2} = \frac{4.7 \text{ k}\Omega(16 \text{ V})}{39 \text{ k}\Omega + 4.7 \text{ k}\Omega} = 1.721 \text{ V}$$

$$V_E = V_B - V_{BE} = 1.721 \text{ V} - 0.7 \text{ V} = 1.021 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.021 \text{ V}}{1.2 \text{ k}\Omega} = 0.8507 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.8507 \text{ mA}} = \mathbf{30.56 \Omega}$$

$$\begin{aligned}
 \text{(b)} \quad Z_i &= R_1 \parallel R_2 \parallel \beta r_e \\
 &= 4.7 \text{ k}\Omega \parallel 39 \text{ k}\Omega \parallel (100)(30.56 \text{ }\Omega) \\
 &= \mathbf{1.77 \text{ k}\Omega} \\
 r_o &\geq 10R_C \therefore Z_o \cong R_C = \mathbf{3.9 \text{ k}\Omega}
 \end{aligned}$$

$$\text{(c)} \quad A_v = -\frac{R_C}{r_e} = -\frac{3.9 \text{ k}\Omega}{30.56 \text{ }\Omega} = \mathbf{-127.6}$$

$$\text{(d)} \quad r_o = 25 \text{ k}\Omega$$

$$\begin{aligned}
 \text{(b)} \quad Z_i(\text{unchanged}) &= \mathbf{1.77 \text{ k}\Omega} \\
 Z_o &= R_C \parallel r_o = 3.9 \text{ k}\Omega \parallel 25 \text{ k}\Omega = \mathbf{3.37 \text{ k}\Omega}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad A_v &= -\frac{(R_C \parallel r_o)}{r_e} = -\frac{(3.9 \text{ k}\Omega) \parallel (25 \text{ k}\Omega)}{30.56 \text{ }\Omega} = -\frac{3.37 \text{ k}\Omega}{30.56 \text{ }\Omega} \\
 &= \mathbf{-110.28} \text{ (vs. } -127.6)
 \end{aligned}$$

$$\begin{aligned}
 16. \quad & \beta R_E \geq 10R_2 \\
 & (100)(1 \text{ k}\Omega) \geq 10(5.6 \text{ k}\Omega) \\
 & 100 \text{ k}\Omega > 56 \text{ k}\Omega \text{ (checks!)} \text{ \& } r_o \geq 10R_C \Rightarrow 50 \text{ k}\Omega \geq 10(3.3 \text{ k}\Omega) = 33 \text{ k}\Omega \text{ (checks)}
 \end{aligned}$$

Use approximate approach:

$$A_v = -\frac{R_C}{r_e} \Rightarrow r_e = -\frac{R_C}{A_v} = -\frac{3.3 \text{ k}\Omega}{-160} = \mathbf{20.625 \text{ }\Omega}$$

$$r_e = \frac{26 \text{ mV}}{I_E} \Rightarrow I_E = \frac{26 \text{ mV}}{r_e} = \frac{26 \text{ mV}}{20.625 \text{ }\Omega} = 1.261 \text{ mA}$$

$$I_E = \frac{V_E}{R_E} \Rightarrow V_E = I_E R_E = (1.261 \text{ mA})(1 \text{ k}\Omega) = 1.261 \text{ V}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 1.261 \text{ V} = 1.961 \text{ V}$$

$$V_B = \frac{5.6 \text{ k}\Omega V_{CC}}{5.6 \text{ k}\Omega + 82 \text{ k}\Omega} = 1.961 \text{ V}$$

$$\begin{aligned}
 5.6 \text{ k}\Omega V_{CC} &= (1.961 \text{ V})(87.6 \text{ k}\Omega) \\
 V_{CC} &= \mathbf{30.68 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 17. \quad & \text{Test } \beta R_E \geq 10R_2 \\
 & \quad ? \\
 & (180)(2.2 \text{ k}\Omega) \geq 10(56 \text{ k}\Omega) \\
 & 396 \text{ k}\Omega < 560 \text{ k}\Omega \text{ (not satisfied)}
 \end{aligned}$$

Use exact analysis:

$$\text{(a)} \quad R_{Th} = 56 \text{ k}\Omega \parallel 220 \text{ k}\Omega = 44.64 \text{ k}\Omega$$

$$E_{Th} = \frac{56 \text{ k}\Omega(20 \text{ V})}{220 \text{ k}\Omega + 56 \text{ k}\Omega} = 4.058 \text{ V}$$

$$\begin{aligned}
 I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{4.058 \text{ V} - 0.7 \text{ V}}{44.64 \text{ k}\Omega + (181)(2.2 \text{ k}\Omega)} \\
 &= \mathbf{7.58 \text{ }\mu\text{A}}
 \end{aligned}$$

$$\begin{aligned}
I_E &= (\beta + 1)I_B = (181)(7.58 \mu\text{A}) \\
&= 1.372 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.372 \text{ mA}} = \mathbf{18.95 \Omega}
\end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad V_E &= I_E R_E = (1.372 \text{ mA})(2.2 \text{ k}\Omega) = 3.02 \text{ V} \\
V_B &= V_E + V_{BE} = 3.02 \text{ V} + 0.7 \text{ V} \\
&= \mathbf{3.72 \text{ V}} \\
V_C &= V_{CC} - I_C R_C \\
&= 20 \text{ V} - \beta I_B R_C = 20 \text{ V} - (180)(7.58 \mu\text{A})(4.7 \text{ k}\Omega) \\
&= \mathbf{13.59 \text{ V}}
\end{aligned}$$

$$\begin{aligned}
\text{(c)} \quad Z_i &= R_1 \parallel R_2 \parallel \beta r_e \\
&= 56 \text{ k}\Omega \parallel 220 \text{ k}\Omega \parallel (180)(18.95 \text{ k}\Omega) \\
&= 44.64 \text{ k}\Omega \parallel 3.41 \text{ k}\Omega \\
&= \mathbf{3.17 \text{ k}\Omega} \\
r_o &< 10R_C \\
33.33 \text{ k}\Omega &< 10(4.7 \text{ k}\Omega) = 47 \text{ k}\Omega \quad (\text{checks})
\end{aligned}$$

$$\begin{aligned}
\therefore A_v &= -\frac{R_C \parallel r_o}{r_e} \\
&= -\frac{(6.8 \text{ k}\Omega) \parallel (50 \text{ k}\Omega)}{18.95 \Omega} \\
&= \mathbf{-298.15}
\end{aligned}$$

$$\begin{aligned}
18. \quad \text{(a)} \quad \text{Test: } \beta R_E &\geq 10R_2 \\
(70)(2.2 \text{ k}\Omega) &\geq 10(68 \text{ k}\Omega) \\
154 \text{ k}\Omega &\geq 680 \text{ k}\Omega \quad \text{No!}
\end{aligned}$$

$\therefore$ Exact method:

$$\begin{aligned}
R_{Th} &= 27 \text{ k}\Omega \parallel 68 \text{ k}\Omega = 19.32 \text{ k}\Omega \\
E_{Th} &= \frac{68 \text{ k}\Omega(12 \text{ V})}{68 \text{ k}\Omega + 27 \text{ k}\Omega} = 8.59 \text{ V} \\
I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{9.59 \text{ V} - 0.7 \text{ V}}{19.32 \text{ k}\Omega + (70 + 1)(2.2 \text{ k}\Omega)} \\
&= 44.95 \mu\text{A} \\
I_E &= (\beta + 1)I_B = (70 + 1)(44.95 \mu\text{A}) = 3.19 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.19 \text{ mA}} = \mathbf{8.15 \Omega}
\end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad V_B &= V_{BE} + I_E R_E = 0.7 \text{ V} + (3.19 \text{ mA})(2.2 \text{ k}\Omega) \\
&= \mathbf{7.72 \text{ V}}
\end{aligned}$$

$$\begin{aligned}
V_C &= 24 \text{ V} - I_C R_C = 24 \text{ V} - (\beta I_B)(3.3 \text{ k}\Omega) \\
&= 24 \text{ V} - (70)(44.95 \mu\text{A})(3.3 \text{ k}\Omega) \\
&= \mathbf{13.61 \text{ V}}
\end{aligned}$$

$$V_{CB} = V_C - V_B = 13.61 \text{ V} - 7.72 \text{ V} = \mathbf{5.89 \text{ V}}$$

$$V_E = V_B - 0.7 = 7.72 \text{ V} - 0.7 \text{ V} = \mathbf{7.02 \text{ V}}$$

$$V_{CE} = V_C - V_E = 13.61 \text{ V} - 7.02 \text{ V} = \mathbf{6.59 \text{ V}}$$

$$\begin{aligned}
\text{(c)} \quad Z_i &= 68 \text{ k}\Omega \parallel 27 \text{ k}\Omega \parallel \beta R_E = 19.3 \text{ k}\Omega \parallel (70)(2.2 \text{ k}\Omega) \\
&= 19.32 \text{ k}\Omega \parallel 154 \text{ k}\Omega = \mathbf{17.17 \text{ k}\Omega}
\end{aligned}$$

$$\text{(d)} \quad A_v = \frac{-R_C}{R_E} = -\frac{3.3 \text{ k}\Omega}{2.2 \text{ k}\Omega} = \mathbf{-1.5}$$

$$\begin{aligned}
19. \quad \text{(a)} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{390 \text{ k}\Omega + (141)(1.2 \text{ k}\Omega)} \\
&= \frac{19.3 \text{ V}}{559.2 \text{ k}\Omega} = 34.51 \mu\text{A} \\
I_E &= (\beta + 1)I_B = (140 + 1)(34.51 \mu\text{A}) = 4.866 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{4.866 \text{ mA}} = \mathbf{5.34 \Omega}
\end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad Z_b &= \beta r_e + (\beta + 1)R_E \\
&= (140)(5.34 \text{ k}\Omega) + (140 + 1)(1.2 \text{ k}\Omega) = 747.6 \Omega + 169.9 \text{ k}\Omega \\
&= \mathbf{169.95 \text{ k}\Omega} \\
Z_i &= R_B \parallel Z_b = 390 \text{ k}\Omega \parallel 169.95 \text{ k}\Omega = \mathbf{118.37 \text{ k}\Omega} \\
Z_o &= R_C = \mathbf{2.2 \text{ k}\Omega}
\end{aligned}$$

$$\text{(c)} \quad A_v = -\frac{\beta R_C}{Z_b} = -\frac{(140)(2.2 \text{ k}\Omega)}{169.95 \text{ k}\Omega} = \mathbf{-1.81}$$

$$\begin{aligned}
\text{(d)} \quad Z_b &= \beta r_e + \left[\frac{(\beta + 1) + R_C / r_o}{1 + (R_C + R_E) / r_o} \right] R_E \\
&= 747.6 \Omega \left[\frac{(141) + 2.2 \text{ k}\Omega / 20 \text{ k}\Omega}{1 + (3.4 \text{ k}\Omega) / 20 \text{ k}\Omega} \right] 1.2 \text{ k}\Omega \\
&= 747.6 \Omega + 144.72 \text{ k}\Omega \\
&= \mathbf{145.47 \text{ k}\Omega}
\end{aligned}$$

$$\begin{aligned}
Z_i &= R_B \parallel Z_b = 390 \text{ k}\Omega \parallel 145.47 \text{ k}\Omega = \mathbf{105.95 \text{ k}\Omega} \\
Z_o &= R_C = \mathbf{2.2 \text{ k}\Omega} \text{ (any level of } r_o)
\end{aligned}$$

$$\begin{aligned}
A_v = \frac{V_o}{V_i} &= \frac{-\frac{\beta R_C}{Z_b} \left[1 + \frac{r_e}{r_o} \right] + \frac{R_C}{r_o}}{1 + \frac{R_C}{r_o}} \\
&= \frac{\frac{-(140)(2.2 \text{ k}\Omega)}{145.47 \text{ k}\Omega} \left[1 + \frac{5.34 \cancel{\Omega}}{20 \text{ k}\Omega} \right] + \frac{2.2 \text{ k}\Omega}{20 \text{ k}\Omega}}{1 + \frac{2.2 \text{ k}\Omega}{20 \text{ k}\Omega}} \\
&= \frac{-2.117 + 0.11}{1.11} = \mathbf{-1.81}
\end{aligned}$$

20. (a) dc analysis the same
 $\therefore r_e = \mathbf{5.34 \Omega}$ (as in #19)
- (b) $Z_i = R_B \parallel Z_b = R_B \parallel \beta r_e = 390 \text{ k}\Omega \parallel (140)(5.34 \Omega) = \mathbf{746.17 \Omega}$ vs. $118.37 \text{ k}\Omega$ in #32
 $Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$ (as in #32)
- (c) $A_v = \frac{-R_C}{r_e} = \frac{-2.2 \text{ k}\Omega}{5.34 \Omega} = \mathbf{-411.99}$ vs -1.81 in #19
- (d) $Z_i = \mathbf{746.17 \Omega}$ vs. $105.95 \text{ k}\Omega$ for #19
 $Z_o = R_C \parallel r_o = 2.2 \text{ k}\Omega \parallel 20 \text{ k}\Omega = \mathbf{1.98 \text{ k}\Omega}$ vs. $2.2 \text{ k}\Omega$ in #19
 $A_v = -\frac{R_C \parallel r_o}{r_e} = -\frac{1.98 \text{ k}\Omega}{5.34 \Omega} = \mathbf{-370.79}$ vs. -1.81 in #19
Significant difference in the results for A_v .

21. $r_o \geq 10R_C$
 $100 \text{ k}\Omega \geq 10(8.2 \text{ k}\Omega) = 82 \text{ k}\Omega$
Therefore:

$$\begin{aligned}
A_v &= -\frac{\beta R_C}{Z_b} = -\frac{\beta R_C}{\beta R_E} = -\frac{R_C}{R_E} = -10 \\
\therefore R_E &= \frac{R_C}{10} = \frac{8.2 \text{ k}\Omega}{10} = \mathbf{0.82 \text{ k}\Omega} \\
I_E &= \frac{26 \text{ mV}}{r_e} = \frac{26 \text{ mV}}{3.8 \Omega} = 6.842 \text{ mA} \\
V_E &= I_E R_E = (6.842 \text{ mA})(0.82 \text{ k}\Omega) = 5.61 \text{ V} \\
V_B &= V_E + V_{BE} = 5.61 \text{ V} + 0.7 \text{ V} = 6.31 \text{ V} \\
I_B &= \frac{I_E}{(\beta + 1)} = \frac{6.842 \text{ mA}}{121} = 56.55 \mu\text{A} \\
\text{and } R_B &= \frac{V_{R_B}}{I_B} = \frac{V_{CC} - V_B}{I_B} = \frac{20 \text{ V} - 6.31 \text{ V}}{56.55 \mu\text{A}} = \mathbf{242.09 \text{ k}\Omega}
\end{aligned}$$

$$\begin{aligned}
22. \quad (a) \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} \\
&= \frac{22 \text{ V} - 0.7 \text{ V}}{330 \text{ k}\Omega + (81)(1.2 \text{ k}\Omega + 0.47 \text{ k}\Omega)} = \frac{21.3 \text{ V}}{465.27 \text{ k}\Omega} \\
&= 45.78 \mu\text{A} \\
I_E &= (\beta + 1)I_B = (81)(45.78 \mu\text{A}) = 3.71 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.71 \text{ mA}} = \mathbf{7 \Omega}
\end{aligned}$$

$$\begin{aligned}
(b) \quad r_o &< 10(R_C + R_E) \\
\therefore Z_b &= \beta r_e + \left[\frac{(\beta + 1) + R_C / r_o}{1 + (R_C + R_E) / r_o} \right] R_E \\
&= (80)(7 \Omega) + \left[\frac{(81) + 5.6 \text{ k}\Omega / 40 \text{ k}\Omega}{1 + 6.8 \text{ k}\Omega / 40 \text{ k}\Omega} \right] 1.2 \text{ k}\Omega \\
&= 560 \Omega + \left[\frac{81 + 0.14}{1 + 0.17} \right] 1.2 \text{ k}\Omega
\end{aligned}$$

(note that $(\beta + 1) = 81 \gg R_C / r_o = 0.14$)

$$\begin{aligned}
&= 560 \Omega + [81.14 / 1.17] 1.2 \text{ k}\Omega = 560 \Omega + 83.22 \text{ k}\Omega \\
&= \mathbf{83.78 \text{ k}\Omega}
\end{aligned}$$

$$Z_i = R_B \parallel Z_b = 330 \text{ k}\Omega \parallel 83.78 \text{ k}\Omega = \mathbf{66.82 \text{ k}\Omega}$$

$$\begin{aligned}
A_v &= \frac{\frac{-\beta R_C}{Z_b} \left(1 + \frac{r_e}{r_o} \right) + \frac{R_C}{r_o}}{1 + \frac{R_C}{r_o}} \\
&= \frac{\frac{-(80)(5.6 \text{ k}\Omega)}{83.78 \text{ k}\Omega} \left(1 + \frac{7 \Omega}{40 \text{ k}\Omega} \right) + \frac{5.6 \text{ k}\Omega}{40 \text{ k}\Omega}}{1 + 5.6 \text{ k}\Omega / 40 \text{ k}\Omega} \\
&= \frac{-(5.35) + 0.14}{1 + 0.14} \\
&= \mathbf{-4.57}
\end{aligned}$$

$$\begin{aligned}
23. \quad (a) \quad \text{Test: } \beta R_E &\geq 10R_2 \\
(200)(1.2 \text{ k}\Omega) &\geq 10(120 \text{ k}\Omega) \\
240 \text{ k}\Omega &\geq 1.2 \text{ M}\Omega \quad \text{No!}
\end{aligned}$$

Exact method:

$$R_{Th} = 430 \text{ k}\Omega \parallel 120 \text{ k}\Omega = 93.82 \text{ k}\Omega$$

$$E_{Th} = \frac{120 \text{ k}\Omega(16 \text{ V})}{120 \text{ k}\Omega + 430 \text{ k}\Omega} = 3.49 \text{ V}$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.49 \text{ V} - 0.7 \text{ V}}{93.82 \text{ k}\Omega + (200 + 1)(1.2 \text{ k}\Omega)} = 8.33 \text{ }\mu\text{A}$$

$$I_E = (\beta + 1)I_B = (200 + 1)(8.33 \text{ }\mu\text{A}) = 1.674 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.674 \text{ mA}} = \mathbf{15.53 \text{ }\Omega}$$

$$(b) \quad V_B = V_E + 0.7 \text{ V} = (I_E)(R_E) + 0.7 \text{ V} + (1.674 \text{ mA})(1.2 \text{ k}\Omega) + 0.7 \text{ V} = 2.71 \text{ V}$$

$$I_C = \beta I_B = (200)(8.33 \text{ }\mu\text{A}) = 1.67 \text{ mA}$$

$$V_C = V_{CC} - I_C R_C = 16 \text{ V} - (1.67 \text{ mA})(4.7 \text{ k}\Omega) = 16 \text{ V} - 7.85 \text{ V} = \mathbf{8.15 \text{ V}}$$

$$V_E = I_E R_E = (1.674 \text{ mA})(1.2 \text{ k}\Omega) = 2.01 \text{ V}$$

$$V_{CE} = V_C - V_E = 8.15 \text{ V} - 2.01 \text{ V} = \mathbf{6.14 \text{ V}}$$

$$V_{CB} = V_C - V_B = 8.15 \text{ V} - 2.71 \text{ V} = \mathbf{5.44 \text{ V}}$$

$$(c) \quad Z_i = 120 \text{ k}\Omega \parallel 430 \text{ k}\Omega \parallel \beta R_E = 93.82 \text{ k}\Omega \parallel (200)(1.2 \text{ k}\Omega) = 93.82 \text{ k}\Omega \parallel 240 \text{ k}\Omega = \mathbf{67.45 \text{ k}\Omega}$$

$$Z_o = \mathbf{4.7 \text{ k}\Omega}$$

$$(d) \quad A_v = \frac{-R_C}{R_E} = -\frac{4.7 \text{ k}\Omega}{1.2 \text{ k}\Omega} = \mathbf{-3.92}$$

$$(e) \quad A_i = A_v \frac{Z_i}{R_C} = \frac{-(3.92)(67.45 \text{ k}\Omega)}{(4.7 \text{ k}\Omega)} = \mathbf{56.26}$$

$$24. \quad (a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{270 \text{ k}\Omega + (111)(2.7 \text{ k}\Omega)} = \frac{15.3 \text{ V}}{569.7 \text{ k}\Omega} = 26.86 \text{ }\mu\text{A}$$

$$I_E = (\beta + 1)I_B = (110 + 1)(26.86 \text{ }\mu\text{A}) = 2.98 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.98 \text{ mA}} = \mathbf{8.72 \text{ }\Omega}$$

$$\beta r_e = (110)(8.72 \text{ }\Omega) = \mathbf{959.2 \text{ }\Omega}$$

$$\begin{aligned}
\text{(b)} \quad Z_b &= \beta r_e + (\beta + 1)R_E \\
&= 959.2 \, \Omega + (111)(2.7 \, \text{k}\Omega) \\
&= 300.66 \, \text{k}\Omega \\
Z_i &= R_B \parallel Z_b = 270 \, \text{k}\Omega \parallel 300.66 \, \text{k}\Omega \\
&= 142.25 \, \text{k}\Omega \\
Z_o &= R_E \parallel r_e = 2.7 \, \text{k}\Omega \parallel 8.72 \, \Omega = \mathbf{8.69 \, \Omega}
\end{aligned}$$

$$\text{(c)} \quad A_v = \frac{R_E}{R_E + r_e} = \frac{2.7 \, \text{k}\Omega}{2.7 \, \text{k}\Omega + 8.69 \, \Omega} \cong \mathbf{0.997}$$

$$\begin{aligned}
25. \quad \text{(a)} \quad I_B &= \frac{V_{CE} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{8 \, \text{V} - 0.7 \, \text{V}}{390 \, \text{k}\Omega + (121)5.6 \, \text{k}\Omega} = 6.84 \, \mu\text{A} \\
I_E &= (\beta + 1)I_B = (121)(6.84 \, \mu\text{A}) = 0.828 \, \text{mA} \\
r_e &= \frac{26 \, \text{mV}}{I_E} = \frac{26 \, \text{mV}}{0.828 \, \text{mA}} = 31.4 \, \Omega \\
r_o &< 10R_E: \\
Z_b &= \beta r_e + \frac{(\beta + 1)R_E}{1 + R_E / r_o} \\
&= (120)(31.4 \, \Omega) + \frac{(121)(5.6 \, \text{k}\Omega)}{1 + 5.6 \, \text{k}\Omega / 40 \, \text{k}\Omega} \\
&= 3.77 \, \text{k}\Omega + 594.39 \, \text{k}\Omega \\
&= 598.16 \, \text{k}\Omega \\
Z_i &= R_B \parallel Z_b = 390 \, \text{k}\Omega \parallel 598.16 \, \text{k}\Omega \\
&= \mathbf{236.1 \, \text{k}\Omega} \\
Z_o &\cong R_E \parallel r_e \\
&= 5.6 \, \text{k}\Omega \parallel 31.4 \, \Omega \\
&= \mathbf{31.2 \, \Omega}
\end{aligned}$$

$$\begin{aligned}
\text{(b)} \quad A_v &= \frac{(\beta + 1)R_E / Z_b}{1 + R_E / r_o} \\
&= \frac{(121)(5.6 \, \text{k}\Omega) / 598.16 \, \text{k}\Omega}{1 + 5.6 \, \text{k}\Omega / 40 \, \text{k}\Omega} \\
&= \mathbf{0.994}
\end{aligned}$$

$$\begin{aligned}
\text{(c)} \quad A_v &= \frac{V_o}{V_i} = 0.994 \\
V_o &= A_v V_i = (0.994)(1 \, \text{mV}) = \mathbf{0.994 \, \text{mV}}
\end{aligned}$$

26. (a) $\beta R_E \geq 10R_2$
 $(200)(2 \text{ k}\Omega) \geq 10(8.2 \text{ k}\Omega)$
 $400 \text{ k}\Omega \geq 82 \text{ k}\Omega$ (checks)!

Use approximate approach:

$$V_B = \frac{8.2 \text{ k}\Omega(20 \text{ V})}{8.2 \text{ k}\Omega + 56 \text{ k}\Omega} = 2.5545 \text{ V}$$

$$V_E = V_B - V_{BE} = 2.5545 \text{ V} - 0.7 \text{ V} \cong 1.855 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.855 \text{ V}}{2 \text{ k}\Omega} = \mathbf{0.927 \text{ mA}}$$

$$I_B = \frac{I_E}{(\beta + 1)} = \frac{0.927 \text{ mA}}{(200 + 1)} = \mathbf{4.61 \mu\text{A}}$$

$$I_C = \beta I_B = (200)(4.61 \mu\text{A}) = \mathbf{0.922 \text{ mA}}$$

$$(b) \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.927 \text{ mA}} = \mathbf{28.05 \Omega}$$

$$(c) \quad Z_b = \beta r_e + (\beta + 1)R_E$$

$$= (200)(28.05 \Omega) + (200 + 1)2 \text{ k}\Omega$$

$$= 5.61 \text{ k}\Omega + 402 \text{ k}\Omega = 407.61 \text{ k}\Omega$$

$$Z_i = 56 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega \parallel 407.61 \text{ k}\Omega$$

$$= 7.15 \text{ k}\Omega \parallel 407.61 \text{ k}\Omega$$

$$= \mathbf{7.03 \text{ k}\Omega}$$

$$Z_o = R_E \parallel r_e = 2 \text{ k}\Omega \parallel 28.05 \Omega = \mathbf{27.66 \Omega}$$

$$(d) \quad A_v = \frac{R_E}{R_E + r_e} = \frac{2 \text{ k}\Omega}{2 \text{ k}\Omega + 28.05 \Omega} = \mathbf{0.986}$$

$$27. (a) \quad I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6 \text{ V} - 0.7 \text{ V}}{6.8 \text{ k}\Omega} = 0.779 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.779 \text{ mA}} = \mathbf{33.38 \Omega}$$

$$(b) \quad Z_i = R_E \parallel r_e = 6.8 \text{ k}\Omega \parallel 33.38 \Omega$$

$$= \mathbf{33.22 \Omega}$$

$$Z_o = R_C = \mathbf{4.7 \text{ k}\Omega}$$

$$(c) \quad A_v = \frac{\alpha R_C}{r_e} = \frac{(0.998)(4.7 \text{ k}\Omega)}{33.38 \Omega}$$

$$= \mathbf{140.52}$$

$$\begin{aligned}
28. \quad \alpha &= \frac{\beta}{\beta+1} = \frac{75}{76} = 0.9868 \\
I_E &= \frac{V_{EE} - V_{BE}}{R_E} = \frac{5 \text{ V} - 0.7 \text{ V}}{3.9 \text{ k}\Omega} = \frac{4.3 \text{ V}}{3.9 \text{ k}\Omega} = 1.1 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.1 \text{ mA}} = 23.58 \text{ }\Omega \\
A_v &= \alpha \frac{R_C}{r_e} = \frac{(0.9868)(3.9 \text{ k}\Omega)}{23.58 \text{ }\Omega} = \mathbf{163.2}
\end{aligned}$$

$$\begin{aligned}
29. \quad (a) \quad I_B &= \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{12 \text{ V} - 0.7 \text{ V}}{220 \text{ k}\Omega + 120(3.9 \text{ k}\Omega)} \\
&= 16.42 \text{ }\mu\text{A} \\
I_E &= (\beta + 1)I_B = (120 + 1)(16.42 \text{ }\mu\text{A}) \\
&= 1.987 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.987 \text{ mA}} = \mathbf{13.08 \text{ }\Omega}
\end{aligned}$$

$$\begin{aligned}
(b) \quad Z_i &= \beta r_e \parallel \frac{R_F}{|A_v|} \\
&\text{Need } A_v! \\
A_v &= \frac{-R_C}{r_e} = \frac{-3.9 \text{ k}\Omega}{13.08 \text{ }\Omega} = -298 \\
Z_i &= (120)(13.08 \text{ }\Omega) \parallel \frac{220 \text{ k}\Omega}{298} \\
&= 1.5696 \text{ k}\Omega \parallel 738 \text{ }\Omega \\
&= \mathbf{501.98 \text{ }\Omega} \\
Z_o &= R_C \parallel R_F = 3.9 \text{ k}\Omega \parallel 220 \text{ k}\Omega \\
&= \mathbf{3.83 \text{ k}\Omega}
\end{aligned}$$

$$(c) \quad \text{From above, } A_v = \mathbf{-298}$$

$$\begin{aligned}
30. \quad A_v &= \frac{-R_C}{r_e} = -160 \\
R_C &= 160(r_e) = 160(10 \text{ }\Omega) = \mathbf{1.6 \text{ k}\Omega}
\end{aligned}$$

$$\begin{aligned}
A_i &= \frac{\beta R_F}{R_F + \beta R_C} = 19 \Rightarrow 19 = \frac{200R_F}{R_F + 200(1.6 \text{ k}\Omega)} \\
19R_F + 3800R_C &= 200R_F \\
R_F &= \frac{3800R_C}{181} = \frac{3800(1.6 \text{ k}\Omega)}{181} \\
&= \mathbf{33.59 \text{ k}\Omega}
\end{aligned}$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C}$$

$$I_B(R_F + \beta R_C) = V_{CC} - V_{BE}$$

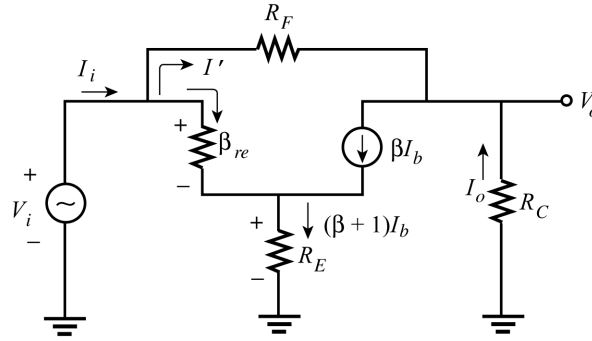
$$\text{and } V_{CC} = V_{BE} + I_B(R_F + \beta R_C)$$

$$\text{with } I_E = \frac{26 \text{ mV}}{r_e} = \frac{26 \text{ mV}}{10 \Omega} = 2.6 \text{ mA}$$

$$I_B = \frac{I_E}{\beta + 1} = \frac{2.6 \text{ mA}}{200 + 1} = 12.94 \mu\text{A}$$

$$\begin{aligned} \therefore V_{CC} &= V_{BE} + I_B(R_F + \beta R_C) \\ &= 0.7 \text{ V} + (12.94 \mu\text{A})(33.59 \text{ k}\Omega + (200)(1.6 \text{ k}\Omega)) \\ &= \mathbf{5.28 \text{ V}} \end{aligned}$$

31. (c)



$$\begin{aligned} \text{(a) } A_v: V_i &= I_b \beta r_e + (\beta + 1) I_b R_E \\ I_o + I' &= I_C = \beta I_b \\ \text{but } I_i &= I' + I_b \\ \text{and } I' &= I_i - I_b \\ \text{Substituting, } I_o + (I_i - I_b) &= \beta I_b \\ \text{and } I_o &= (\beta + 1) I_b - I_i \end{aligned}$$

$$\text{Assuming } (\beta + 1) I_b \gg I_i$$

$$I_o \cong (\beta + 1) I_b$$

$$\text{and } V_o = -I_o R_C = -(\beta + 1) I_b R_C$$

$$\text{Therefore, } \frac{V_o}{V_i} = \frac{-(\beta + 1) I_b R_C}{I_b \beta r_e + (\beta + 1) I_b R_E}$$

$$\cong \frac{\beta I_b R_C}{\beta I_b r_e + \beta I_b R_E}$$

$$\text{and } A_v = \frac{V_o}{V_i} \cong -\frac{R_C}{r_e + R_E} \cong -\frac{R_C}{R_E}$$

(b) $V_i \cong \beta I_b(r_e + R_E)$
 For $r_e \ll R_E$
 $V_i \cong \beta I_b R_E$

Now $I_i = I' + I_b$

$$= \frac{V_i - V_o}{R_F} + I_b$$

Since $V_o \gg V_i$

$$I_i = -\frac{V_o}{R_F} + I_b$$

$$\text{or } I_b = I_i + \frac{V_o}{R_F}$$

and $V_i = \beta I_b R_E$

$$V_i = \beta R_E I_i + \beta \frac{V_o}{R_F} R_E$$

but $V_o = A_v V_i$

and $V_i = \beta R_E I_i + \frac{\beta A_v V_i R_E}{R_F}$

or $V_i - \frac{A_v \beta R_E V_i}{R_F} = \beta R_E I_i$

$$V_i \left[1 - \frac{A_v \beta R_E}{R_F} \right] = [\beta R_E] I_i$$

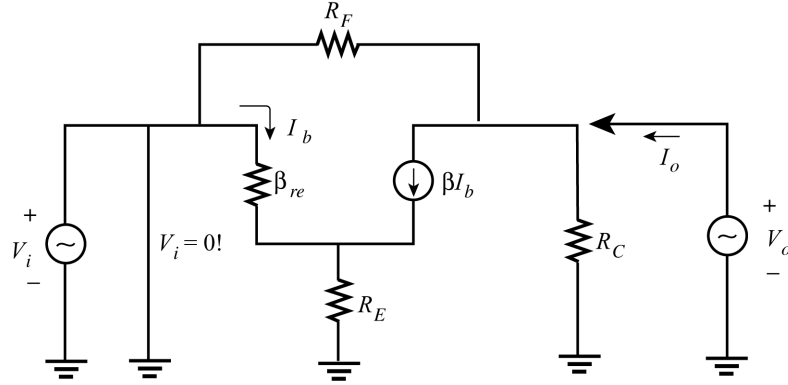
so $Z_i = \frac{V_i}{I_i} = \frac{\beta R_E}{1 - \frac{A_v \beta R_E}{R_F}} = \frac{\beta R_E R_F}{R_F + \beta(-A_v)R_E}$

$$Z_i = \frac{V_i}{I_i} = x \parallel y \quad \text{where } x = \beta R_E \text{ and } y = R_F / |A_v|$$

with $Z_i = \frac{x \cdot y}{x + y} = \frac{(\beta R_E)(R_F / |A_v|)}{\beta R_E + R_F / |A_v|}$

$$Z_i \cong \frac{\beta R_E R_F}{\beta R_E |A_v| + R_F}$$

Z_o : Set $V_i = 0$



$$V_i = I_b \beta r_e + (\beta + 1) I_b R_E$$

$$V_i \cong \beta I_b (r_e + R_E) = 0$$

$$\text{since } \beta, r_e + R_E \neq 0 \quad I_b = 0 \text{ and } \beta I_b = 0$$

$$\therefore I_o = \frac{V_o}{R_C} + \frac{V_o}{R_F} = V_o \left[\frac{1}{R_C} + \frac{1}{R_F} \right]$$

$$\text{and } Z_o = \frac{V_o}{I_o} = \frac{1}{\frac{1}{R_C} + \frac{1}{R_F}} = \frac{R_C R_F}{R_C + R_F} = R_C \parallel R_F$$

$$(c) \quad A_v \cong -\frac{R_C}{R_E} = -\frac{2.2 \text{ k}\Omega}{1.2 \text{ k}\Omega} = \mathbf{-1.83}$$

$$Z_i \cong \frac{\beta R_E R_F}{\beta R_E |A_v| + R_F} = \frac{(90)(1.2 \text{ k}\Omega)(120 \text{ k}\Omega)}{(90)(1.2 \text{ k}\Omega)(1.83) + 120 \text{ k}\Omega}$$

$$= \mathbf{40.8 \text{ k}\Omega}$$

$$Z_o \cong R_C \parallel R_F$$

$$= 2.2 \text{ k}\Omega \parallel 120 \text{ k}\Omega$$

$$= \mathbf{2.16 \text{ k}\Omega}$$

$$32. \quad (a) \quad I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C} = \frac{9 \text{ V} - 0.7 \text{ V}}{(39 \text{ k}\Omega + 22 \text{ k}\Omega) + (80)(1.8 \text{ k}\Omega)}$$

$$= \frac{8.3 \text{ V}}{61 \text{ k}\Omega + 144 \text{ k}\Omega} = \frac{8.3 \text{ V}}{205 \text{ k}\Omega} = 40.49 \mu\text{A}$$

$$I_E = (\beta + 1) I_B = (80 + 1)(40.49 \mu\text{A}) = 3.28 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.28 \text{ mA}} = 7.93 \Omega$$

$$Z_i = R_{F_1} \parallel \beta r_e$$

$$= 39 \text{ k}\Omega \parallel (80)(7.93 \Omega) = 39 \text{ k}\Omega \parallel 634.4 \Omega = \mathbf{0.62 \text{ k}\Omega}$$

$$Z_o = R_C \parallel R_{F_2} = 1.8 \text{ k}\Omega \parallel 22 \text{ k}\Omega = \mathbf{1.66 \text{ k}\Omega}$$

$$\begin{aligned}
 \text{(b)} \quad A_v &= \frac{-R'}{r_e} = \frac{-R_C \parallel R_{F_2}}{r_e} = -\frac{1.8 \text{ k}\Omega \parallel 22 \text{ k}\Omega}{7.93 \text{ }\Omega} \\
 &= \frac{-1.664 \text{ k}\Omega}{7.93 \text{ }\Omega} = \mathbf{-209.82}
 \end{aligned}$$

$$\begin{aligned}
 33. \quad \text{(a)} \quad Z_i &\cong \frac{R_E}{\left[\frac{1}{\beta} + \frac{(R_E + R_C)}{R_F} \right]} = \frac{0.68 \text{ k}\Omega}{\left[\frac{1}{80} + \frac{(0.68 \text{ k}\Omega + 1.8 \text{ k}\Omega)}{61 \text{ k}\Omega} \right]} \\
 &= \frac{0.68 \text{ k}\Omega}{\left[12.5 \times 10^{-3} + 4.066 \times 10^{-3} \right]} = \frac{0.68 \text{ k}\Omega}{53.16 \times 10^{-3}} \\
 &= \mathbf{12.79 \text{ k}\Omega}
 \end{aligned}$$

$$Z_o = R_C \parallel R_F = 1.8 \text{ k}\Omega \parallel 61 \text{ k}\Omega = \mathbf{1.75 \text{ k}\Omega}$$

$$\text{(b)} \quad A_v = -\frac{R_C}{R_E} = -\frac{1.8 \text{ k}\Omega}{0.68 \text{ k}\Omega} = \mathbf{-2.65}$$

$$\begin{aligned}
 34. \quad \text{(a)} \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B} = \frac{18 \text{ V} - 0.7 \text{ V}}{680 \text{ k}\Omega} = 25.44 \text{ }\mu\text{A} \\
 I_E &= (\beta + 1)I_B = (100 + 1)(25.44 \text{ }\mu\text{A}) \\
 &= 2.57 \text{ mA} \\
 r_e &= \frac{26 \text{ mV}}{2.57 \text{ mA}} = 10.116 \text{ }\Omega \\
 A_{v_{NL}} &= -\frac{R_C}{r_e} = -\frac{3.3 \text{ k}\Omega}{10.116 \text{ }\Omega} = \mathbf{-326.22} \\
 Z_i &= R_B \parallel \beta r_e = 680 \text{ k}\Omega \parallel (100)(10.116 \text{ }\Omega) \\
 &= 680 \text{ k}\Omega \parallel 1,011.6 \text{ }\Omega \\
 &= \mathbf{1.01 \text{ k}\Omega} \\
 Z_o &= R_C = \mathbf{3.3 \text{ k}\Omega}
 \end{aligned}$$

(b) —

$$\begin{aligned}
 \text{(c)} \quad A_{v_L} &= \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} (-326.22) \\
 &= \mathbf{-191.65}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad A_{i_L} &= -A_v \frac{Z_i}{R_L} = -(-191.65) \frac{(1.01 \text{ k}\Omega)}{4.7 \text{ k}\Omega} \\
 &= \mathbf{41.18}
 \end{aligned}$$

35. (a) $A_{v_{NL}} = -326.22$

$$A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}}$$

$$R_L = 4.7 \text{ k}\Omega: A_{v_L} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} (-326.22) = \mathbf{-191.65}$$

$$R_L = 2.2 \text{ k}\Omega: A_{v_L} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} (-326.22) = \mathbf{-130.49}$$

$$R_L = 0.5 \text{ k}\Omega: A_{v_L} = \frac{0.5 \text{ k}\Omega}{0.5 \text{ k}\Omega + 2.3 \text{ k}\Omega} (-326.22) = \mathbf{-42.92}$$

As $R_L \downarrow$, $A_{v_L} \downarrow$

(b) No change for Z_i , Z_o , and $A_{v_{NL}}$!

36. (a) $I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{1 \text{ M}\Omega} = 11.3 \mu\text{A}$

$$I_E = (\beta + 1)I_B = (181)(11.3 \mu\text{A}) = 2.045 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.045 \text{ mA}} = 12.71 \Omega$$

$$A_{v_{NL}} = -\frac{R_C}{r_e} = -\frac{3 \text{ k}\Omega}{12.71 \Omega} = \mathbf{-236}$$

$$Z_i = R_B \parallel \beta r_e = 1 \text{ M}\Omega \parallel (180)(12.71 \Omega) = 1 \text{ M}\Omega \parallel 2.288 \text{ k}\Omega$$

$$= \mathbf{2.283 \text{ k}\Omega}$$

$$Z_o = R_C = \mathbf{3 \text{ k}\Omega}$$

(b) –

(c) No-load: $A_v = A_{v_{NL}} = \mathbf{-236}$

(d) $A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_{NL}} = \frac{2.283 \text{ k}\Omega(-236)}{2.283 \text{ k}\Omega + 0.6 \text{ k}\Omega}$

$$= \mathbf{-186.9}$$

(e) No change!

(f) $A_{v_s} = \frac{Z_i}{Z_i + R_s} (A_{v_{NL}}) = \frac{2.283 \text{ k}\Omega(-236)}{2.283 \text{ k}\Omega + 1 \text{ k}\Omega} = \mathbf{-164.1}$

$R_s \uparrow, A_{v_s} \downarrow$

(g) No change!

(h) $A_i = A_v \frac{Z_i}{R_C} = -(-236) \frac{(2.283 \text{ k}\Omega)}{3 \text{ k}\Omega}$

$$= \mathbf{179.6}$$

37. (a) $I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{24 \text{ V} - 0.7 \text{ V}}{500 \text{ k}\Omega} = 41.61 \mu\text{A}$
 $I_E = (\beta + 1)I_B = (80 + 1)(41.61 \mu\text{A}) = 3.37 \text{ mA}$
 $r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.37 \text{ mA}} = 7.715 \Omega$
 $A_{v_{NL}} = -\frac{R_L}{r_e} = -\frac{4.3 \text{ k}\Omega}{7.715 \Omega} = \mathbf{-557.36}$
 $Z_i = R_B \parallel \beta r_e = 560 \text{ k}\Omega \parallel (80)(7.715 \Omega)$
 $= 560 \text{ k}\Omega \parallel 617.2 \Omega$
 $= \mathbf{616.52 \Omega}$
 $Z_o = R_C = \mathbf{4.3 \text{ k}\Omega}$
- (b) –
- (c) $A_{v_L} = \frac{V_o}{V_i} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{2.7 \text{ k}\Omega(-557.36)}{2.7 \text{ k}\Omega + 4.3 \text{ k}\Omega}$
 $= \mathbf{-214.98}$
 $A_{v_s} = \frac{V_o}{V_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s}$
 $V_i = \frac{Z_i V_s}{Z_i + R_s} = \frac{616.52 \Omega V_s}{616.52 \Omega + 1 \text{ k}\Omega} = 0.381 V_s$
 $A_{v_s} = (-214.98)(0.381)$
 $= \mathbf{-81.91}$
- (d) $A_{i_s} = -A_{v_s} \left(\frac{R_s + Z_i}{R_L} \right) = -(-81.91) \left(\frac{1 \text{ k}\Omega + 616.52 \Omega}{2.7 \text{ k}\Omega} \right)$
 $= \mathbf{49.04}$
- (e) $A_{v_L} = \frac{V_o}{V_i} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{5.6 \text{ k}\Omega(-557.36)}{5.6 \text{ k}\Omega + 4.3 \text{ k}\Omega} = -315.27$
 $\frac{V_i}{V_s}$ the same = 0.381
 $A_{v_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s} = (-315.27)(0.381) = \mathbf{-120.12}$
As $R_L \uparrow$, $A_{v_s} \uparrow$
- (f) A_{v_L} the same = -214.98
 $\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s} = \frac{616.52 \Omega}{616.52 \Omega + 0.5 \text{ k}\Omega} = 0.552$
 $A_{v_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s} = (-214.98)(0.552) = \mathbf{-118.67}$
As $R_s \downarrow$, $A_{v_s} \uparrow$
- (g) No change!

38. (a) Exact analysis:

$$E_{Th} = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{16 \text{ k}\Omega (16 \text{ V})}{68 \text{ k}\Omega + 16 \text{ k}\Omega} = 3.048 \text{ V}$$

$$R_{Th} = R_1 \parallel R_2 = 68 \text{ k}\Omega \parallel 16 \text{ k}\Omega = 12.95 \text{ k}\Omega$$

$$I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{3.048 \text{ V} - 0.7 \text{ V}}{12.95 \text{ k}\Omega + (101)(0.75 \text{ k}\Omega)} = 26.47 \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (101)(26.47 \mu\text{A}) = 2.673 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.673 \text{ mA}} = 9.726 \Omega$$

$$A_{v_{NL}} = \frac{-R_C}{r_e} = -\frac{2.2 \text{ k}\Omega}{9.726 \Omega} = \mathbf{-226.2}$$

$$\begin{aligned} Z_i &= 68 \text{ k}\Omega \parallel 16 \text{ k}\Omega \parallel \beta r_e \\ &= 12.95 \text{ k}\Omega \parallel (100)(9.726 \Omega) \\ &= 12.95 \text{ k}\Omega \parallel 972.6 \Omega \\ &= \mathbf{904.66 \Omega} \end{aligned}$$

$$Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$$

(b) –

$$(c) \quad A_{v_L} = \frac{R_L}{R_L + Z_o} (A_{v_{NL}}) = \frac{5.6 \text{ k}\Omega (-226.2)}{5.6 \text{ k}\Omega + 2.2 \text{ k}\Omega} = \mathbf{-162.4}$$

$$\begin{aligned} (d) \quad A_{i_L} &= -A_{v_L} \frac{Z_i}{R_L} \\ &= -(-162.4) \frac{(904.66 \Omega)}{5.6 \text{ k}\Omega} \\ &= \mathbf{26.24} \end{aligned}$$

$$\begin{aligned} (e) \quad A_{v_L} &= \frac{-R_C \parallel R_e}{r_e} = \frac{-2.2 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega}{9.726 \Omega} \\ &= \mathbf{-162.4} \end{aligned}$$

$$Z_i = 68 \text{ k}\Omega \parallel 16 \text{ k}\Omega \parallel \underbrace{972.6 \Omega}_{\beta r_e}$$

$$= \mathbf{904.66 \Omega}$$

$$\begin{aligned} A_{i_L} &= -A_{v_L} \frac{Z_i}{R_L} \\ &= \frac{(-162.4)(904.66 \Omega)}{5.6 \text{ k}\Omega} \\ &= \mathbf{26.24} \end{aligned}$$

$$Z_o = R_C = \mathbf{2.2 \text{ k}\Omega}$$

Same results!

39. (a) $A_{v_L} = \frac{R_L}{R_L + Z_o} A_{v_{NL}}$

$R_L = 4.7 \text{ k}\Omega$: $A_{v_L} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 2.2 \text{ k}\Omega} (-226.4) = \mathbf{-154.2}$

$R_L = 2.2 \text{ k}\Omega$: $A_{v_L} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 2.2 \text{ k}\Omega} (-226.4) = \mathbf{-113.2}$

$R_L = 0.5 \text{ k}\Omega$: $A_{v_L} = \frac{0.5 \text{ k}\Omega}{0.5 \text{ k}\Omega + 2.2 \text{ k}\Omega} (-226.4) = \mathbf{-41.93}$

$R_L \downarrow, A_{v_L} \downarrow$

(b) Unaffected!

40. (a) $I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{18 \text{ V} - 0.7 \text{ V}}{680 \text{ k}\Omega + (111)(0.82 \text{ k}\Omega)}$

$= 22.44 \text{ }\mu\text{A}$

$I_E = (\beta + 1)I_B = (110 + 1)(22.44 \text{ }\mu\text{A})$

$= 2.49 \text{ mA}$

$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.49 \text{ mA}} = 10.44 \text{ }\Omega$

$A_{v_{NL}} = -\frac{R_C}{r_e + R_E} = -\frac{3 \text{ k}\Omega}{10.44 \text{ }\Omega + 0.82 \text{ k}\Omega}$

$= \mathbf{-3.61}$

$Z_i \cong R_B \parallel Z_b = 680 \text{ k}\Omega \parallel (\beta r_e + (\beta + 1)R_E)$

$= 680 \text{ k}\Omega \parallel (610)(10.44 \text{ }\Omega) + (110 + 1)(0.82 \text{ k}\Omega)$

$= 680 \text{ k}\Omega \parallel 92.17 \text{ k}\Omega$

$= \mathbf{81.17 \text{ k}\Omega}$

$Z_o \cong R_C = \mathbf{3 \text{ k}\Omega}$

(b) –

(c) $A_{v_L} = \frac{V_o}{V_i} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega(-3.61)}{4.7 \text{ k}\Omega + 3 \text{ k}\Omega}$

$= \mathbf{-2.2}$

$A_{v_s} = \frac{V_o}{V_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s}$

$V_i = \frac{Z_i V_s}{Z_i + R_s} = \frac{81.17 \text{ k}\Omega (V_s)}{81.17 \text{ k}\Omega + 0.6 \text{ k}\Omega} = 0.992 V_s$

$A_{v_s} = (-2.2)(0.992)$

$= \mathbf{-2.18}$

(d) None!

(e) A_{v_L} – none!

$$\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s} = \frac{81.17 \text{ k}\Omega}{81.17 \text{ k}\Omega + 1 \text{ k}\Omega} = 0.988$$

$$A_{v_s} = (-2.2)(0.988)$$

$$= \mathbf{-2.17}$$

$R_s \uparrow$, $A_{v_s} \downarrow$, (but only slightly for moderate changes in R_s since Z_i is typically much larger than R_s)

$$(f) \quad A_i = A_{v_L} \frac{Z_i}{R_L} = (-2.2) \frac{(81.17 \text{ k}\Omega)}{4.7 \text{ k}\Omega}$$

$$= \mathbf{37.99}$$

41. Using the exact approach:

$$\begin{aligned} I_B &= \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} \\ &= \frac{2.33 \text{ V} - 0.7 \text{ V}}{10.6 \text{ k}\Omega + (121)(1.2 \text{ k}\Omega)} \\ &= 10.46 \mu\text{A} \end{aligned}$$

$$\begin{aligned} E_{Th} &= \frac{R_2}{R_1 + R_2} V_{CC} \\ &= \frac{12 \text{ k}\Omega}{91 \text{ k}\Omega + 12 \text{ k}\Omega} (20 \text{ V}) = 2.33 \text{ V} \\ R_{Th} &= R_1 \parallel R_2 = 91 \text{ k}\Omega \parallel 12 \text{ k}\Omega = 10.6 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} I_E &= (\beta + 1)I_B = (121)(10.46 \mu\text{A}) \\ &= 1.266 \text{ mA} \end{aligned}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.266 \text{ mA}} = 20.54 \Omega$$

$$(a) \quad A_{v_{NL}} \cong \frac{R_E}{r_e + R_E} = \frac{1.2 \text{ k}\Omega}{20.54 \Omega + 1.2 \text{ k}\Omega} = \mathbf{0.983}$$

$$\begin{aligned} Z_i &= R_1 \parallel R_2 \parallel (\beta r_e + (\beta + 1)R_E) \\ &= 91 \text{ k}\Omega \parallel 12 \text{ k}\Omega \parallel ((120)(20.54 \Omega) + (120 + 1)(1.2 \text{ k}\Omega)) \\ &= 10.6 \text{ k}\Omega \parallel (2.46 \text{ k}\Omega + 145.2 \text{ k}\Omega) \\ &= 10.6 \text{ k}\Omega \parallel 147.66 \text{ k}\Omega \\ &= \mathbf{9.89 \text{ k}\Omega} \end{aligned}$$

$$\begin{aligned} Z_o &= R_E \parallel r_e = 1.2 \text{ k}\Omega \parallel 20.54 \Omega \\ &= \mathbf{20.19 \Omega} \end{aligned}$$

(b) –

$$\begin{aligned} (c) \quad A_{v_L} &= \frac{R_L}{R_L + Z_o} A_{v_{NL}} = \frac{2.7 \text{ k}\Omega(0.983)}{2.7 \text{ k}\Omega + 20.19 \Omega} \\ &= \mathbf{0.976} \end{aligned}$$

$$\begin{aligned} A_{v_s} &= \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{9.89 \text{ k}\Omega(0.976)}{9.89 \text{ k}\Omega + 0.6 \text{ k}\Omega} \\ &= \mathbf{0.92} \end{aligned}$$

(d) $A_{v_L} = 0.976$ (unaffected by change in R_s)

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} = \frac{9.89 \text{ k}\Omega(0.976)}{9.89 \text{ k}\Omega + 1 \text{ k}\Omega}$$

$$= \mathbf{0.886} \text{ (vs. } 0.92 \text{ with } R_s = 0.6 \text{ k}\Omega)$$

As $R_s \uparrow$, $A_{v_s} \downarrow$

(e) Changing R_s will have no effect on $A_{v_{NL}}$, Z_i , or Z_o .

(f) $A_{v_L} = \frac{R_L}{R_L + Z_o} (A_{v_{NL}}) = \frac{5.6 \text{ k}\Omega(0.983)}{5.6 \text{ k}\Omega + 20.19 \text{ }\Omega}$

$$= \mathbf{0.979} \text{ (vs. } 0.976 \text{ with } R_L = 2.7 \text{ k}\Omega)$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} (A_{v_L}) = \frac{9.89 \text{ k}\Omega(0.979)}{9.89 \text{ k}\Omega + 0.6 \text{ k}\Omega}$$

$$= \mathbf{0.923} \text{ (vs. } 0.92 \text{ with } R_L = 2.7 \text{ k}\Omega)$$

As $R_L \uparrow$, $A_v \uparrow$, $A_{v_s} \uparrow$

(g) $A_i = A_{v_L} \frac{Z_i}{R_L} = (0.979) \frac{(9.89 \text{ k}\Omega)}{2.7 \text{ k}\Omega}$

$$= \mathbf{3.59}$$

42. (a) $I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{6 \text{ V} - 0.7 \text{ V}}{2.2 \text{ k}\Omega}$

$$= 2.41 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.41 \text{ mA}} = 10.79 \text{ }\Omega$$

$$A_{v_{NL}} = \frac{R_C}{r_e} = \frac{4.7 \text{ k}\Omega}{10.79 \text{ }\Omega} = \mathbf{435.59}$$

$$Z_i = R_E \parallel r_e = 2.2 \text{ k}\Omega \parallel 10.79 \text{ }\Omega = \mathbf{10.74 \text{ }\Omega}$$

$$Z_o = R_C = \mathbf{4.7 \text{ k}\Omega}$$

(b) –

(c) $A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{5.6 \text{ k}\Omega(435.59)}{5.6 \text{ k}\Omega + 4.7 \text{ k}\Omega} = \mathbf{236.83}$

$$V_i = \frac{Z_i}{Z_i + R_s} V_s = \frac{10.74 \text{ }\Omega (V_s)}{10.74 \text{ }\Omega + 100 \text{ }\Omega} = 0.097 V_s$$

$$A_{v_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s} = (236.83)(0.097)$$

$$= \mathbf{22.97}$$

(d) $V_i = I_e \cdot r_e$

$$V_o = -I_o R_L$$

$$I_o = \frac{-4.7 \text{ k}\Omega(I_e)}{4.7 \text{ k}\Omega + 5.6 \text{ k}\Omega} = -0.4563 I_e$$

$$A_{v_L} = \frac{V_o}{V_i} = \frac{+(0.4563 \cancel{r_e})R_L}{\cancel{r_e} \cdot r_e} = \frac{0.4563(5.6 \text{ k}\Omega)}{10.79 \Omega} \\ = \mathbf{236.82} \text{ (vs. 236.83 for part c)}$$

$$A_{v_s} : 2.2 \text{ k}\Omega \parallel 10.79 \Omega = 10.74 \Omega$$

$$V_i = \frac{Z_i}{Z_i + R_s} \cdot V_s = \frac{10.74 \Omega (V_s)}{10.74 \Omega + 100 \Omega} = 0.097 V_s$$

$$A_{v_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s} = (236.82)(0.097) \\ = \mathbf{22.97} \text{ (same results)}$$

$$(e) \quad A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 4.7 \text{ k}\Omega} (435.59) \\ = \mathbf{138.88}$$

$$A_{v_s} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s}, \quad \frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s} = \frac{10.74 \Omega}{10.74 \Omega + 500 \Omega} = 0.021$$

$$A_{v_s} = (138.88)(0.021) = \mathbf{2.92}$$

A_{v_s} very sensitive to increase in R_s due to relatively small Z_i ; $R_s \uparrow$, $A_{v_s} \downarrow$

A_{v_L} sensitive to R_L ; $R_L \downarrow$, $A_{v_L} \downarrow$

(f) $Z_o = R_C = \mathbf{4.7 \text{ k}\Omega}$ unaffected by value of R_s !

(g) $Z_i = R_E \parallel r_e = 10.74 \Omega$ unaffected by value of R_L !

$$(h) \quad A_i = A_{v_L} \frac{Z_i}{R_L} = \frac{(236.82)(10.74 \text{ k}\Omega)}{5.6 \text{ k}\Omega} \\ = \mathbf{0.454}$$

$$43. \quad (a) \quad A_{v_1} = \frac{R_L A_{v_{NL}}}{R_L + R_o} = \frac{1 \text{ k}\Omega(-420)}{1 \text{ k}\Omega + 3.3 \text{ k}\Omega} = \mathbf{-97.67}$$

$$A_{v_2} = \frac{R_L A_{v_{NL}}}{R_L + R_o} = \frac{2.7 \text{ k}\Omega(-420)}{2.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} = \mathbf{-189}$$

$$(b) \quad A_{v_L} = A_{v_1} \cdot A_{v_2} = (-97.67)(-189) = \mathbf{18.46 \times 10^3}$$

$$A_{v_s} = \frac{V_o}{V_s} = \frac{V_o}{V_{i_2}} \cdot \frac{V_{o_1}}{V_{i_1}} \cdot \frac{V_{i_1}}{V_s}$$

$$= A_{v_2} \cdot A_{v_1} \cdot \frac{V_i}{V_s}$$

$$V_i = \frac{Z_i V_s}{Z_i + R_s} = \frac{1 \text{ k}\Omega(V_s)}{1 \text{ k}\Omega + 0.6 \text{ k}\Omega} = 0.625$$

$$A_{v_s} = (-189)(-97.67)(0.625) \\ = \mathbf{11.54 \times 10^3}$$

$$(c) \quad A_{i_1} = -\frac{A_v Z_i}{R_L} = \frac{-(-97.67)(1 \text{ k}\Omega)}{1 \text{ k}\Omega} = \mathbf{97.67}$$

$$A_{i_2} = \frac{-A_v Z_i}{R_L} = \frac{-(-189)(1 \text{ k}\Omega)}{2.7 \text{ k}\Omega} = \mathbf{70}$$

$$(d) \quad A_{i_L} = A_{i_1} \cdot A_{i_2} = (97.67)(70) = \mathbf{6.84 \times 10^3}$$

(e) No effect!

(f) No effect!

(g) In phase

$$44. \quad (a) \quad A_{v_1} = \frac{Z_{i_2}}{Z_{i_2} + Z_{o_1}} A_{v_{1NL}} = \frac{1.2 \text{ k}\Omega}{1.2 \text{ k}\Omega + 20 \text{ }\Omega} (1) \\ = 0.984$$

$$A_{v_2} = \frac{R_L}{R_L + Z_{o_2}} A_{v_{2NL}} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 4.6 \text{ k}\Omega} (-640) \\ = \mathbf{-207.06}$$

$$(b) \quad A_{v_L} = A_{v_1} \cdot A_{v_2} = (0.984)(-207.06) \\ = \mathbf{-203.74}$$

$$A_{v_s} = \frac{Z_i}{Z_i + R_s} A_{v_L} \\ = \frac{50 \text{ k}\Omega}{50 \text{ k}\Omega + 1 \text{ k}\Omega} (-203.74) \\ = \mathbf{-199.75}$$

$$(c) \quad A_{i_1} = -A_{v_1} \frac{Z_{i_1}}{Z_{i_2}} \\ = -(0.984) \frac{(50 \text{ k}\Omega)}{1.2 \text{ k}\Omega} \\ = \mathbf{-41}$$

$$A_{i_2} = -A_{v_2} \frac{Z_{i_2}}{R_L} \\ = -(-207.06) \frac{(1.2 \text{ k}\Omega)}{2.2 \text{ k}\Omega} \\ = \mathbf{112.94}$$

$$(d) \quad A_{i_L} = -A_{v_L} \frac{Z_{i_1}}{R_L} \\ = -(-203.74) \frac{(50 \text{ k}\Omega)}{2.2 \text{ k}\Omega} \\ = \mathbf{4.63 \times 10^3}$$

- (e) A load on an emitter-follower configuration will contribute to the emitter resistance (in fact, lower the value) and therefore affect Z_i (reduce its magnitude).
- (f) The fact that the second stage is a CE amplifier will isolate Z_o from the first stage and R_s .
- (g) The emitter-follower has zero phase shift while the common-emitter amplifier has a 180° phase shift. The system, therefore, has a total phase shift of 180° as noted by the negative sign in front of the gain for A_{v_T} in part b.

45. For each stage:

$$V_B = \frac{6.2 \text{ k}\Omega}{24 \text{ k}\Omega + 6.2 \text{ k}\Omega} (15 \text{ V}) = \mathbf{3.08 \text{ V}}$$

$$V_E = V_B - 0.7 \text{ V} = 3.08 \text{ V} - 0.7 \text{ V} = \mathbf{2.38 \text{ V}}$$

$$I_E \cong I_C = \frac{V_E}{R_E} = \frac{2.38 \text{ V}}{1.5 \text{ k}\Omega} = \mathbf{1.59 \text{ mA}}$$

$$V_C = V_{CC} - I_C R_C = 15 \text{ V} - (1.59 \text{ mA})(5.1 \text{ k}\Omega) = \mathbf{6.89 \text{ V}}$$

46. (a) $r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{1.59 \text{ mA}} = 16.35 \text{ }\Omega$

$$R_{i_1} = R_{i_2} = R_1 \parallel R_2 \parallel \beta r_e = 6.2 \text{ k}\Omega \parallel 24 \text{ k}\Omega \parallel (150)(16.35 \text{ }\Omega) = 1.64 \text{ k}\Omega$$

$$A_{v_1} = -\frac{R_C \parallel R_{i_2}}{r_e} = \frac{5.1 \text{ k}\Omega \parallel 1.64 \text{ k}\Omega}{16.35 \text{ }\Omega} = \mathbf{-75.8}$$

$$A_{v_2} = -\frac{R_C}{r_e} = \frac{-5.1 \text{ k}\Omega}{16.35 \text{ }\Omega} = \mathbf{-311.9}$$

$$A_v = A_{v_1} A_{v_2} = (-75.8)(-311.9) = \mathbf{23,642}$$

(b) $A_{i_T} = \frac{I_o}{I_i} = A_{v_T} \frac{Z_i}{R_L}$

$$= \frac{(23,642)(1.64 \text{ k}\Omega)}{5.1 \text{ k}\Omega} = \mathbf{7602.5}$$

47. $V_{B_1} = \frac{3.9 \text{ k}\Omega}{3.9 \text{ k}\Omega + 6.2 \text{ k}\Omega + 7.5 \text{ k}\Omega} (20 \text{ V}) = \mathbf{4.4 \text{ V}}$

$$V_{B_2} = \frac{6.2 \text{ k}\Omega + 3.9 \text{ k}\Omega}{3.9 \text{ k}\Omega + 6.2 \text{ k}\Omega + 7.5 \text{ k}\Omega} (20 \text{ V}) = \mathbf{11.48 \text{ V}}$$

$$V_{E_1} = V_{B_1} - 0.7 \text{ V} = 4.4 \text{ V} - 0.7 \text{ V} = \mathbf{3.7 \text{ V}}$$

$$I_{C_1} \cong I_{E_1} = \frac{V_{E_1}}{R_E} = \frac{3.7 \text{ V}}{1 \text{ k}\Omega} = 3.7 \text{ mA} \cong I_{E_2} \cong I_{C_2}$$

$$V_{C_2} = V_{CC} - I_C R_C = 20 \text{ V} - (3.7 \text{ mA})(1.5 \text{ k}\Omega) = \mathbf{14.45 \text{ V}}$$

$$V_{C_1} = V_{B_2} - V_{BE_2} = 11.48 \text{ V} - 0.7 \text{ V} = \mathbf{10.78 \text{ V}}$$

$$48. \quad r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.7 \text{ mA}} = 7 \Omega$$

$$A_{v_1} = -\frac{r_e}{r_e} = -1$$

$$A_{v_2} = \frac{R_C}{r_e} = \frac{1.5 \text{ k}\Omega}{7 \Omega} \cong 214$$

$$A_{v_T} = A_{v_1} A_{v_2} = (-1)(214) = -214$$

$$V_o = A_{v_T} V_i = (-214)(10 \text{ mV}) = -2.14 \text{ V}$$

$$49. \quad R_o = R_D = 1.5 \text{ k}\Omega \quad (V_o \text{ (from problem \#48)} = -2.14 \text{ V})$$

$$V_o(\text{load}) = \frac{R_L}{R_o + R_L} (V_o) = \frac{10 \text{ k}\Omega}{10 \text{ k}\Omega + 1.5 \text{ k}\Omega} (-2.14 \text{ V})$$

$$= -1.86 \text{ V}$$

$$50. \quad (a) \quad \beta_D = \beta_1 \beta_2 = (50)(120) = 6000$$

$$I_B = \frac{V_{CC} - V_{BE_1} - V_{BE_2}}{R_B + \beta_D R_E} = \frac{16 \text{ V} - 1.4 \text{ V}}{2.4 \text{ M}\Omega + (6000)(510 \Omega)}$$

$$= 2.67 \mu\text{A}$$

$$V_{B_1} = 16 \text{ V} - (I_B)(2.4 \text{ M}\Omega) = 16 \text{ V} - (2.67 \mu\text{A})(2.4 \text{ M}\Omega)$$

$$= 9.59 \text{ V}$$

$$V_{C_1} = 16 \text{ V}$$

$$V_{E_2} = I_{E_2} R_E = \beta_D I_B R_E$$

$$= (6000)(2.67 \mu\text{A})(510 \Omega)$$

$$= 8.17 \text{ V}$$

$$V_{CB_1} = V_{C_1} - V_{B_1} = 16 \text{ V} - 9.59 \text{ V} = 6.41 \text{ V}$$

$$V_{CE_2} = 16 \text{ V} - V_{E_2} = 16 \text{ V} - 8.17 \text{ V} = 7.83 \text{ V}$$

$$(b) \quad I_{B_1} = 2.67 \mu\text{A}, \quad I_{B_2} = I_{C_1} = \beta_1 I_{B_1} = (50)(2.67 \mu\text{A}) = 133.5 \mu\text{A}$$

$$I_{E_2} = V_{E_2} / R_E = 8.17 \text{ V} / 510 \Omega = 16.02 \text{ mA}$$

$$(c) \quad Z_i = R_B \parallel \beta_D R_E = 2.4 \text{ M}\Omega \parallel (6000)(510 \Omega) = 2.4 \text{ M}\Omega \parallel 3.06 \text{ M}\Omega$$

$$= 1.35 \text{ M}\Omega$$

$$Z_o = \frac{r_{e_1}}{\beta_2} + r_{e_2} \quad \text{with} \quad I_{E_1} = (\beta + 1)I_{B_1} = (51)(2.67 \mu\text{A}) = 136.17 \mu\text{A}$$

$$\text{and} \quad r_{e_1} = \frac{26 \text{ mV}}{I_{E_1}} = \frac{26 \text{ mV}}{136.17 \mu\text{A}} = 190.94 \Omega$$

$$r_{e_2} = \frac{26 \text{ mV}}{I_{E_2}} = \frac{26 \text{ mV}}{16.02 \text{ mA}} = 1.62 \Omega$$

$$Z_o = \frac{190.94 \Omega}{120} + 1.62 \Omega = 1.59 \Omega + 1.62 \Omega$$

$$= 3.21 \Omega$$

(d) $A_v \cong 1$

$$A_i = \frac{\beta_D R_B}{R_B + \beta_D R_E} = \frac{(6000)(2.4 \text{ M}\Omega)}{2.4 \text{ M}\Omega + (6000)(510 \text{ }\Omega)} = \frac{14.4 \times 10^3 \text{ M}\Omega}{5.46 \text{ M}\Omega} = 2.64 \times 10^3$$

51. (a) and (b) the same as problem 50

(a) $V_{B_1} = 9.59 \text{ V}$, $V_{C_1} = 16 \text{ V}$, $V_{E_2} = 8.17 \text{ V}$, $V_{CB_1} = 6.41 \text{ V}$, $V_{CE_2} = 7.83 \text{ V}$

(b) $I_{B_1} = 2.67 \text{ }\mu\text{A}$, $I_{B_2} = 133.5 \text{ }\mu\text{A}$, $I_{E_2} = 16.02 \text{ mA}$

(c) $Z_i = R_B \parallel \beta(R_E \parallel R_L)$
 $= 2.4 \text{ M}\Omega \parallel (6000)(510 \text{ }\Omega \parallel 1.2 \text{ k}\Omega) = 2.4 \text{ M}\Omega \parallel 2.15 \text{ M}\Omega$
 $= 1.13 \text{ M}\Omega$

$$Z_o = \frac{r_{e_1}}{\beta_2} = r_{e_2} \text{ not affected by } R_L, Z_o = 3.21 \text{ }\Omega \text{ from problem 50}$$

(d) $A_v \cong 1$

$$A_i = \frac{\beta_D R_B}{R_B + \beta_D (R_E \parallel R_L)} = \frac{(6000)(2.4 \text{ M}\Omega)}{2.4 \text{ M}\Omega + (6000)(510 \text{ }\Omega \parallel 0.2 \text{ k}\Omega)} = 3.16 \times 10^3$$

52. From Problem 51, $Z_i = 1.35 \text{ M}\Omega$, and $A_{v_{NL}} \cong 1$

$$V_i = \frac{1.35 \text{ M}\Omega (V_s)}{1.35 \text{ M}\Omega + 1.2 \text{ k}\Omega} = 0.999 V_s$$

$$\therefore A_v = V_o / V_s \cong 1$$

53. (a) $I_B = \frac{V_{CC} - V_{BE_T}}{R_B + \beta_D R_E} = \frac{16 \text{ V} - 1.6 \text{ V}}{2.4 \text{ M}\Omega + (4000)(510 \text{ }\Omega)} = 3.24 \text{ }\mu\text{A}$

$$V_{B_1} = V_{CC} - I_B R_B = 16 \text{ V} - (3.24 \text{ }\mu\text{A})(2.4 \text{ M}\Omega) = 8.22 \text{ V}$$

$$V_{E_2} = I_{E_2} R_E = \beta_D I_B R_E = (4000)(3.24 \text{ }\mu\text{A})(510 \text{ }\Omega) = 6.61 \text{ V}$$

$$V_{C_2} = V_{CC} - I_{C_2} R_C \cong 16 \text{ V} - I_{E_2} R_C$$

$$= 16 \text{ V} - \beta_D I_B R_C = 16 \text{ V} - (4000)(3.24 \text{ }\mu\text{A})(470 \text{ }\Omega)$$

$$= 9.91 \text{ V}$$

$$V_{CE_2} = V_{C_2} - V_{E_2} = 9.91 \text{ V} - 6.61 \text{ V} = 3.3 \text{ V}$$

$$V_{CB_1} = V_{C_1} - V_{B_1} = V_{C_2} - V_{B_1} = 9.91 \text{ V} - 8.22 \text{ V} = 1.69 \text{ V}$$

$$(c) \quad I_{E_2} \cong \beta_D I_B = (4000)(3.24 \mu A) = 12.96 \text{ mA}$$

$$r_{e_2} = \frac{26 \text{ mV}}{I_{E_2}} = \frac{26 \text{ mV}}{12.96 \text{ mA}} = 2 \Omega$$

$$Z_i \cong 2.4 \text{ M}\Omega \parallel \beta_D r_{e_2} = 2.4 \text{ M}\Omega \parallel (4000)(2 \Omega) = 2.4 \text{ M}\Omega \parallel 8 \text{ k}\Omega$$

$$\cong \mathbf{8 \text{ k}\Omega}$$

$$Z_o = R_C = \mathbf{470 \Omega}$$

$$(d) \quad A_v = \frac{-\beta_D R_C}{Z_i} = \frac{-(4000)(470 \Omega)}{8 \text{ k}\Omega} = \mathbf{-235}$$

$$(e) \quad A_i = \frac{-A_v Z_i}{R_C} = \frac{-(235)(8 \text{ k}\Omega)}{470 \Omega} = \mathbf{4 \times 10^3}$$

$$54. \quad (a) \quad I_{B_1} = \frac{V_{CC} - V_{BE_1}}{R_B + \beta_1 \beta_2 R_C} = \frac{16 \text{ V} - 0.7 \text{ V}}{1.5 \text{ M}\Omega + (160)(200)(68 \Omega)} = 4.16 \mu A$$

$$V_{B_1} = I_{B_1} R_B = (4.16 \mu A)(1.5 \text{ M}\Omega) = \mathbf{6.24 \text{ V}}$$

$$V_{E_2} = \mathbf{0 \text{ V}}$$

$$V_{C_2} = V_{E_1} = 16 \text{ V} - I_C 68 \Omega = 16 \text{ V} - (\beta_1 \beta_2 I_{B_1}) 68 \Omega$$

$$= 16 \text{ V} - (160)(200)(4.16 \mu A)(68 \Omega) = \mathbf{6.95 \text{ V}}$$

$$V_{B_2} = V_{BE_2} = \mathbf{0.7 \text{ V}}$$

$$V_{C_1} = V_{B_2} = \mathbf{0.7 \text{ V}}$$

$$V_{E_1} = V_{C_2} = \mathbf{6.95 \text{ V}}$$

$$(b) \quad I_{B_1} = \mathbf{4.16 \mu A}$$

$$I_{C_1} = \beta_1 I_{B_1} = (160)(4.16 \mu A) = \mathbf{0.666 \text{ mA}}$$

$$I_{B_2} = I_{C_1} = \mathbf{0.666 \text{ mA}}, \quad I_{C_2} = I_C = \beta_1 \beta_2 I_{B_1} = \mathbf{133.12 \text{ mA}}$$

$$I_{E_2} = I_{C_2} = \mathbf{133.12 \text{ mA}}$$

$$(c) \quad Z_i = R_B \parallel \beta_1 \beta_2 R_C$$

$$= 1.5 \text{ M}\Omega \parallel (160)(200)(68 \Omega)$$

$$= \mathbf{2.17 \text{ M}\Omega}$$

$$Z_o = R_C \parallel r_{o_1} \cong \mathbf{68 \Omega}$$

$$(d) \quad A_v = \mathbf{1}$$

$$(e) \quad A_i = \frac{-\beta_1 \beta_2 R_B}{R_B + \beta_1 \beta_2 R_C} = \frac{-(160)(200)(1.5 \text{ M}\Omega)}{1.5 \text{ M}\Omega + (160)(200)(68 \Omega)} = \mathbf{-13.06 \times 10^3}$$

55. (a)
$$I_{B_1} = \frac{V_{CC} - V_{BE_1}}{R_B + \beta_1 \beta_2 R_C} = \frac{16 \text{ V} - 0.7 \text{ V}}{1.5 \text{ M}\Omega + (160)(200)(68 \text{ }\Omega)} = 4.16 \text{ }\mu\text{A}$$

$$V_{B_1} = I_{B_1} R_B = (4.16 \text{ }\mu\text{A})(1.5 \text{ M}\Omega) = \mathbf{6.24 \text{ V}}$$

$$V_{E_2} = I_{E_2} R_E \cong I_{C_2} R_E = (\beta_1 \beta_2 I_{B_1}) R_E = (160)(200)(4.16 \text{ }\mu\text{A})(22 \text{ }\Omega) = \mathbf{2.93 \text{ V}}$$

$$V_{C_2} = 16 \text{ V} - (I_{C_2})(68 \text{ }\Omega) = 16 \text{ V} - (\beta_1 \beta_2 I_{B_1})(68 \text{ }\Omega)$$

$$= 16 \text{ V} - (160)(200)(4.16 \text{ }\mu\text{A})(68 \text{ }\Omega) = \mathbf{6.95 \text{ V}}$$

$$V_{B_2} = V_{BE_2} + V_{E_2} = 0.7 \text{ V} + 2.93 \text{ V} = \mathbf{3.63 \text{ V}}$$

$$V_{C_1} = V_{B_2} = \mathbf{3.63 \text{ V}}$$

$$V_{E_1} = V_{C_2} = \mathbf{6.95 \text{ V}}$$
- (b)
$$I_{B_1} = \mathbf{4.16 \text{ }\mu\text{A}}$$

$$I_{C_1} = \beta_1 I_{B_1} = (160)(4.16 \text{ }\mu\text{A}) = \mathbf{0.666 \text{ mA}}$$

$$I_{B_2} = I_{C_1} = \mathbf{0.666 \text{ mA}}, I_{C_2} = I_C = \beta_1 \beta_2 I_{B_1} = \mathbf{133.12 \text{ mA}}$$

$$I_{E_2} = I_{C_2} = \mathbf{133.12 \text{ mA}}$$
- (c)
$$Z_i = R_B \parallel \beta_1 \beta_2 R_C$$

$$= 1.5 \text{ M}\Omega \parallel (160)(200)(68 \text{ }\Omega)$$

$$= \mathbf{0.887 \text{ M}\Omega}$$

$$Z_o = R_C \parallel r_{o_1} \cong \mathbf{68 \text{ }\Omega}$$
- (d)
$$A_v \cong \mathbf{1}$$
- (e)
$$A_i = \frac{-\beta_1 \beta_2 R_B}{R_B + \beta_1 \beta_2 R_C} = \frac{-(160)(200)(1.5 \text{ M}\Omega)}{1.5 \text{ M}\Omega + (160)(200)(68 \text{ }\Omega)} = \mathbf{13.06 \times 10^3}$$
56. (a) dc same as 54
- (b) dc same as 54
- (c)
$$Z_{i=} = R_B \parallel \beta_1 \beta_2 (R_C \parallel R_L)$$

$$= 1.5 \text{ M}\Omega \parallel (160)(200)(68 \text{ }\Omega \parallel 1.2 \text{ k}\Omega)$$

$$= 1.5 \text{ M}\Omega \parallel (160)(200)(64.35 \text{ }\Omega) = 1.5 \text{ M}\Omega \parallel 2.06 \text{ M}\Omega$$

$$= \mathbf{0.867 \text{ M}\Omega}$$

$$Z_o \cong \mathbf{68 \text{ }\Omega}$$

(d) $A_v \cong 1$

$$A_i = \frac{\beta_1 \beta_2 R_B}{R_B + \beta_1 \beta_2 (R_C \parallel R_L)} = \frac{-(160)(200)(1.5 \text{ M}\Omega)}{1.5 \text{ M}\Omega + (160)(200)(64.55 \text{ }\Omega)} \\ = -13.49 \times 10^3$$

57. $r_e = \frac{26 \text{ mV}}{I_{E(\text{dc})}} = \frac{26 \text{ mV}}{1.2 \text{ mA}} = 21.67 \text{ }\Omega$

$$\beta r_e = (120)(21.67 \text{ }\Omega) = 2.6 \text{ k}\Omega$$

58. —

59. —

60. —

61. —

62. (a) $A_v = \frac{V_o}{V_i} = -160$

$$V_o = -160 V_i$$

(b) $I_b = \frac{V_i - h_{re} V_o}{h_{ie}} = \frac{V_i - h_{re} A_v V_i}{h_{ie}} = \frac{V_i (1 - h_{re} A_v)}{h_{ie}} \\ = \frac{V_i (1 - (2 \times 10^{-4})(160))}{1 \text{ k}\Omega} \\ I_b = 9.68 \times 10^{-4} V_i$

(c) $I_b = \frac{V_i}{1 \text{ k}\Omega} = 1 \times 10^{-3} V_i$

(d) $\% \text{ Difference} = \frac{1 \times 10^{-3} V_i - 9.68 \times 10^{-4} V_i}{1 \times 10^{-3} V_i} \times 100\% \\ = 3.2 \%$

(e) Valid first approximation

63. $\% \text{ difference in total load} = \frac{R_L - R_L \parallel 1/h_{oe}}{R_L} \times 100\% \\ = \frac{2.2 \text{ k}\Omega - (2.2 \text{ k}\Omega \parallel 50 \text{ k}\Omega)}{2.2 \text{ k}\Omega} \times 100\% \\ = \frac{2.2 \text{ k}\Omega - 2.1073 \text{ k}\Omega}{2.2 \text{ k}\Omega} \times 100\% \\ = 4.2 \%$

In this case the effect of $1/h_{oe}$ can be ignored.

64. (a) $V_o = -180V_i$ ($h_{ie} = 4 \text{ k}\Omega$, $h_{re} = 4.05 \times 10^{-4}$)
- (b)
$$I_b = \frac{V_i - (4.05 \times 10^{-4})(180V_i)}{4 \text{ k}\Omega} = 2.32 \times 10^{-4}V_i$$
- (c)
$$I_b = \frac{V_i}{h_{ie}} = \frac{V_i}{4 \text{ k}\Omega} = 2.5 \times 10^{-4}V_i$$
- (d)
$$\% \text{ Difference} = \frac{2.5 \times 10^{-4}V_i - 2.32 \times 10^{-4}V_i}{2.5 \times 10^{-4}V_i} \times 100\% = 7.2\%$$
- (e) Yes, less than 10%
65. From Fig. 5.18
- min max
- $h_{oe}: 1 \text{ }\mu\text{S} \quad 30 \text{ }\mu\text{S} \quad \text{Avg} = \frac{(1 + 30)\mu\text{S}}{2} = 15.5 \text{ }\mu\text{S}$
- $1/h_{oe} = 64.53 \text{ k}\Omega$
- $$\frac{3.3 \text{ k}\Omega - (3.3 \text{ k}\Omega \parallel 64.52 \text{ k}\Omega)}{3.3 \text{ k}\Omega} \times 100\% = 4.88\%$$
- $\therefore$ Ignore effects
66. (a) $h_{fe} = \beta = 120$
 $h_{ie} \cong \beta r_e = (120)(4.5 \text{ }\Omega) = 540 \text{ }\Omega$
 $h_{oe} = \frac{1}{r_o} = \frac{1}{40 \text{ k}\Omega} = 25 \text{ }\mu\text{S}$
- (b) $r_e \cong \frac{h_{ie}}{\beta} = \frac{1 \text{ k}\Omega}{90} = 11.11 \text{ }\Omega$
 $\beta = h_{fe} = 90$
 $r_o = \frac{1}{h_{oe}} = \frac{1}{20 \text{ }\mu\text{S}} = 50 \text{ k}\Omega$
67. (a) $r_e = 8.31 \text{ }\Omega$ (from problem 26)
- (b) $h_{fe} = \beta = 60$
 $h_{ie} = \beta r_e = (60)(8.31 \text{ }\Omega) = 498.6 \text{ }\Omega$
- (c) $Z_i = R_B \parallel h_{ie} = 220 \text{ k}\Omega \parallel 498.6 \text{ }\Omega = 497.47 \text{ }\Omega$
 $Z_o = R_C = 2.2 \text{ k}\Omega$
- (d) $A_v = \frac{-h_{fe} R_C}{h_{ie}} = \frac{-(60)(2.2 \text{ k}\Omega)}{498.6 \text{ }\Omega} = -264.74$
 $A_i \cong h_{fe} = 60$

(e) $Z_i = \mathbf{497.47\ \Omega}$ (the same)

$$\begin{aligned} Z_o &= r_o \parallel R_C, \quad r_o = \frac{1}{25\ \mu\text{S}} = 40\ \text{k}\Omega \\ &= 40\ \text{k}\Omega \parallel 2.2\ \text{k}\Omega \\ &= \mathbf{2.09\ \text{k}\Omega} \end{aligned}$$

(f) $A_v = \frac{-h_{fe}(r_o \parallel R_C)}{h_{ie}} = \frac{-(60)(2.085\ \text{k}\Omega)}{498.6\ \Omega} = \mathbf{-250.90}$

$$A_i = -A_v Z_i / R_C = -(-250.90)(497.47\ \Omega) / 2.2\ \text{k}\Omega = \mathbf{56.73}$$

68. (a) $68\ \text{k}\Omega \parallel 12\ \text{k}\Omega = 10.2\ \text{k}\Omega$

$$\begin{aligned} Z_i &= 10.2\ \text{k}\Omega \parallel h_{ie} = 10.2\ \text{k}\Omega \parallel 2.75\ \text{k}\Omega \\ &= \mathbf{2.166\ \text{k}\Omega} \end{aligned}$$

$$\begin{aligned} Z_o &= R_C \parallel r_o \\ &= 2.2\ \text{k}\Omega \parallel 40\ \text{k}\Omega \\ &= \mathbf{2.085\ \text{k}\Omega} \end{aligned}$$

(b) $A_v = \frac{-h_{fe} R'_C}{h_{ie}} \quad R'_C = R_C \parallel r_o = 2.085\ \text{k}\Omega$

$$= \frac{-(180)(2.085\ \text{k}\Omega)}{2.75\ \text{k}\Omega} = \mathbf{-136.5}$$

$$\begin{aligned} A_i &= \frac{I_o}{I_i} = \frac{I_o}{I'_i} \cdot \frac{I'_i}{I_i} \\ &= \left(\frac{h_{fe}}{1 + h_{oe} R_L} \right) \left(\frac{10.2\ \text{k}\Omega}{10.2\ \text{k}\Omega + 2.68\ \text{k}\Omega} \right) \\ &= \left(\frac{180}{1 + (25\ \mu\text{S})(2.2\ \text{k}\Omega)} \right) (0.792) \\ &= \mathbf{135.13} \end{aligned}$$

69. (a) $Z_i = R_E \parallel h_{ib}$

$$\begin{aligned} &= 1.2\ \text{k}\Omega \parallel 9.45\ \Omega \\ &= \mathbf{9.38\ \Omega} \end{aligned}$$

$$Z_o = R_C \parallel \frac{1}{h_{ob}} = 2.7\ \text{k}\Omega \parallel \frac{1}{1 \times 10^{-6}\ \text{A/V}} = 2.7\ \text{k}\Omega \parallel 1\ \text{M}\Omega \cong \mathbf{2.7\ \text{k}\Omega}$$

(b) $A_v = \frac{-h_{fb}(R_C \parallel 1/h_{ob})}{h_{ib}} = \frac{-(-0.992)(\cong 2.7\ \text{k}\Omega)}{9.45\ \Omega}$

$$= \mathbf{284.43}$$

$$A_i \cong \mathbf{-1}$$

$$(c) \quad \alpha = -h_{fb} = -(-0.992) = \mathbf{0.992}$$

$$\beta = \frac{\alpha}{1 - \alpha} = \frac{0.992}{1 - 0.992} = \mathbf{124}$$

$$r_e = h_{ib} = \mathbf{9.45 \, \Omega}$$

$$r_o = \frac{1}{h_{ob}} = \frac{1}{1 \, \mu A/V} = \mathbf{1 \, M\Omega}$$

$$70. \quad (a) \quad Z_i' = h_{ie} - \frac{h_{fe} h_{re} R_L}{1 + h_{oe} R_L} = 2.75 \, k\Omega - \frac{(180)(2 \times 10^{-4})(2.2 \, k\Omega)}{(1 + 25 \, \mu S)(2.2 \, k\Omega)}$$

$$= 2.68 \, k\Omega$$

$$Z_i = 10.2 \, k\Omega \parallel Z_i' = \mathbf{2.12 \, k\Omega}$$

$$Z_o' = \frac{1}{h_{oe} - (h_{fe} h_{re} / h_{ie})} = \frac{1}{25 \, \mu S - (180)(2 \times 10^{-4}) / 2.75 \, k\Omega}$$

$$= 83.75 \, k\Omega$$

$$Z_o = 2.2 \, k\Omega \parallel 83.75 \, k\Omega = 2.14 \, k\Omega$$

$$(b) \quad A_v = \frac{-h_{fe} R_L}{h_{ie} + (h_{ie} h_{oe}) - h_{fe} h_{re}) R_L} = \frac{-(180)(2.2 \, k\Omega)}{2.75 \, k\Omega + ((2.75 \, k\Omega)(25 \, \mu S) - (180) * 2 \times 10^{-4})) 2.2 \, k\Omega}$$

$$= \mathbf{-140.3}$$

$$(c) \quad A_i' = \frac{h_{fe}}{1 + h_{oe} R_L} = \frac{(180)}{1 + (25 \, \mu S)(2.2 \, k\Omega)} = 170.62$$

$$A_i = \frac{I_o}{I_i} = \frac{I_o}{I_i'} \cdot \frac{I_i'}{I_i} = (170.62) \left(\frac{10.2 \, k\Omega}{10.2 \, k\Omega + 2.68 \, k\Omega} \right)$$

$$= \mathbf{135.13}$$

$$71. \quad (a) \quad Z_i' = h_{ie} - \frac{h_{fe} h_{re} R_L}{1 + h_{oe} R_L}$$

$$= 0.86 \, k\Omega - \frac{(140)(1.5 \times 10^{-4})(2.2 \, k\Omega)}{1 + (25 \, \mu S)(2.2 \, k\Omega)}$$

$$= 0.86 \, k\Omega - 43.79 \, \Omega$$

$$= 816.21 \, \Omega$$

$$Z_i = R_B \parallel Z_i'$$

$$= 470 \, k\Omega \parallel 816.21 \, \Omega$$

$$= \mathbf{814.8 \, \Omega}$$

$$(b) \quad A_v = \frac{-h_{fe} R_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) R_L}$$

$$= \frac{-(140)(2.2 \, k\Omega)}{0.86 \, k\Omega + ((0.86 \, k\Omega)(25 \, \mu S) - (140)(1.5 \times 10^{-4})) 2.2 \, k\Omega}$$

$$= \mathbf{-357.68}$$

$$(c) \quad A_i = \frac{I_o}{I_i} = \frac{h_{fe}}{1 + h_{oe} R_L} = \frac{140}{1 + (25 \mu S)(2.2 \text{ k}\Omega)}$$

$$= 132.70$$

$$A_i' = \frac{I_o}{I_i'} = \left(\frac{I_o}{I_i} \right) \left(\frac{I_i}{I_i'} \right) \quad I_i = \frac{470 \text{ k}\Omega I_i'}{470 \text{ k}\Omega + 0.816 \text{ k}\Omega}$$

$$= (132.70)(0.998) \quad \frac{I_i}{I_i'} = 0.998$$

$$= \mathbf{132.43}$$

$$(d) \quad Z_o = \frac{1}{h_{oe} - (h_{fe} h_{re} / (h_{ie} + R_s))} = \frac{1}{25 \mu S - ((140)(1.5 \times 10^{-4}) / (0.86 \text{ k}\Omega + 1 \text{ k}\Omega))}$$

$$= \frac{1}{13.71 \mu S} \cong \mathbf{72.9 \text{ k}\Omega}$$

$$72. \quad (a) \quad Z_i' = h_{ib} - \frac{h_{fb} h_{rb} R_L}{1 + h_{ob} R_L} = 9.45 \text{ k}\Omega - \frac{(-0.997)(1 \times 10^{-4})(2.2 \text{ k}\Omega)}{1 + (0.5 \mu A/V)(2.2 \text{ k}\Omega)}$$

$$= 9.67 \text{ }\Omega$$

$$Z_i = 1.2 \text{ k}\Omega \parallel Z_i' = 1.2 \text{ k}\Omega \parallel 9.67 \text{ }\Omega = \mathbf{9.59 \text{ }\Omega}$$

$$(b) \quad A_i' = \frac{h_{fb}}{1 + h_{ob} R_L} = \frac{0.997}{1 + (0.5 \mu A/V)(2.2 \text{ k}\Omega)} = -0.996$$

$$A_i = \frac{I_o}{I_i'} \cdot \frac{I_i'}{I_i} = (-0.996) \left(\frac{1.2 \text{ k}\Omega}{1.2 \text{ k}\Omega + 9.67 \text{ }\Omega} \right)$$

$$\cong -0.988$$

$$(c) \quad A_v = \frac{-h_{fL} R_L}{h_{ib} + (h_{ib} h_{ob} - h_{fb} h_{fb}) R_L}$$

$$= \frac{-(-0.997)(2.2 \text{ k}\Omega)}{9.45 \text{ }\Omega + ((9.45 \text{ }\Omega)(0.5 \mu A/V) - (0.997)(1 \times 10^{-4}))(2.2 \text{ k}\Omega)}$$

$$= \mathbf{226.21}$$

$$(d) \quad Z_o' = \frac{1}{h_{ob} - [h_{fb} h_{rb} / h_{ib}]}$$

$$= \frac{1}{0.5 \mu A/V - [(-0.997)(1 \times 10^{-4}) / 9.45 \text{ }\Omega]}$$

$$= 90.5 \text{ k}\Omega$$

$$Z_o = 2.2 \text{ k}\Omega \parallel Z_o' = \mathbf{2.15 \text{ k}\Omega}$$

73. —

74. (a) $h_{fe}(0.2 \text{ mA}) \cong 0.6$ (normalized)
 $h_{fe}(1 \text{ mA}) = 1.0$

$$\begin{aligned}\% \text{ change} &= \left| \frac{h_{fe}(0.2 \text{ mA}) - h_{fe}(1 \text{ mA})}{h_{fe}(0.2 \text{ mA})} \right| \times 100\% \\ &= \left| \frac{0.6 - 1}{0.6} \right| \times 100\% \\ &= \mathbf{66.7\%}\end{aligned}$$

- (b) $h_{fe}(1 \text{ mA}) = 1.0$
 $h_{fe}(5 \text{ mA}) \cong 1.5$

$$\begin{aligned}\% \text{ change} &= \left| \frac{h_{fe}(1 \text{ mA}) - h_{fe}(5 \text{ mA})}{h_{fe}(1 \text{ mA})} \right| \times 100\% \\ &= \left| \frac{1 - 1.5}{1} \right| \times 100\% \\ &= \mathbf{50\%}\end{aligned}$$

75. Log-log scale!

- (a) $I_c = 0.2 \text{ mA}$, $h_{ie} = 4$ (normalized)
 $I_c = 1 \text{ mA}$, $h_{ie} = 1$ (normalized)

$$\% \text{ change} = \left| \frac{4 - 1}{4} \right| \times 100\% = \mathbf{75\%}$$

- (b) $I_e = 5 \text{ mA}$, $h_{ie} = 0.3$ (normalized)

$$\% \text{ change} = \left| \frac{1 - 0.3}{1} \right| \times 100\% = \mathbf{70\%}$$

76. (a) $h_{oe} = 20 \mu\text{S}$ @ 1 mA
 $I_c = 0.2 \text{ mA}$, $h_{oe} = 0.2(h_{oe} \text{ @ } 1 \text{ mA})$
 $= 0.2(20 \mu\text{S})$
 $= \mathbf{4 \mu\text{S}}$

- (b) $r_o = \frac{1}{h_{oe}} = \frac{1}{4 \mu\text{S}} = 250 \text{ k}\Omega \gg 6.8 \text{ k}\Omega$
 Ignore $1/h_{oe}$

77. (a) $I_c = 10 \text{ mA}$, $h_{oe} = 10(20 \mu\text{S}) = \mathbf{200 \mu\text{S}}$

- (b) $r_o = \frac{1}{h_{oe}} = \frac{1}{200 \mu\text{S}} = 5 \text{ k}\Omega$ vs. $\mathbf{8.6 \text{ k}\Omega}$
 Not a good approximation

78. (a) $h_{re}(0.1 \text{ mA}) = 4(h_{re}(1 \text{ mA}))$
 $= 4(2 \times 10^{-4})$
 $= \mathbf{8 \times 10^{-4}}$
- (b) $h_{re}V_{ce} = h_{re}A_v \cdot V_i$
 $= (8 \times 10^{-4})(210)V_i$
 $= \mathbf{0.168 V_i}$
 In this case $h_{re}V_{ce}$ is too large a factor to be ignored.
79. (a) h_{fe}
- (b) h_{oe}
- (c) $h_{oe} \cong 30 \mu\text{S}$ (normalized) to $h_{oe} \cong 0.1 \mu\text{S}$ (normalized) at low levels of I_c
- (d) mid-region
80. (a) h_{ie} is the most temperature-sensitive parameter of Fig. 5.33.
- (b) h_{oe} exhibited the smallest change.
- (c) Normalized: $h_{fe(\text{max})} = \mathbf{1.5}$, $h_{fe(\text{min})} = \mathbf{0.5}$
 For $h_{fe} = 100$ the range would extend from 50 to 150—certainly significant.
- (d) On a normalized basis r_e increased from 0.3 at -65°C to 3 at 200°C —a significant change.
- (e) The parameters show the least change in the region $0^\circ \rightarrow 100^\circ\text{C}$.
81. (a) Test:
 $\beta R_E \geq 10R_2$
 $70(1.5 \text{ k}\Omega) \geq 10(39 \text{ k}\Omega)$
 $\quad \quad \quad ?$
 $105 \text{ k}\Omega \geq 390 \text{ k}\Omega$
 $\quad \quad \quad \text{No!}$
 $R_{Th} = 39 \text{ k}\Omega \parallel 150 \text{ k}\Omega = 30.95 \text{ k}\Omega$
 $E_{Th} = \frac{39 \text{ k}\Omega(14 \text{ V})}{39 \text{ k}\Omega + 150 \text{ k}\Omega} = 2.89 \text{ V}$
 $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.89 \text{ V} - 0.7 \text{ V}}{30.95 \text{ k}\Omega + (71)(1.5 \text{ k}\Omega)}$
 $= 15.93 \mu\text{A}$
 $V_B = E_{Th} - I_B R_{Th}$
 $= 2.89 \text{ V} - (15.93 \mu\text{A})(30.95 \text{ k}\Omega)$
 $= 2.397 \text{ V}$

$$V_E = 2.397 \text{ V} - 0.7 \text{ V} = 1.697 \text{ V}$$

$$\text{and } I_E = \frac{V_E}{R_E} = \frac{1.697 \text{ V}}{1.5 \text{ k}\Omega} = 1.13 \text{ mA}$$

$$\begin{aligned} V_{CE} &= V_{CC} - I_C(R_C + R_E) \\ &= 14 \text{ V} - 1.13 \text{ mA}(2.2 \text{ k}\Omega + 1.5 \text{ k}\Omega) \\ &= 9.819 \text{ V} \end{aligned}$$

Biasing OK

(b) R_2 not connected at base:

$$I_B = \frac{V_{CC} - 0}{R_B + (\beta + 1)R_E} = \frac{14 \text{ V} - 0.7 \text{ V}}{150 \text{ k}\Omega + (71)(1.5 \text{ k}\Omega)} = 51.85 \mu\text{A}$$

$$\begin{aligned} V_B &= V_{CC} - I_B R_B = 14 \text{ V} - (51.85 \mu\text{A})(150 \text{ k}\Omega) \\ &= \mathbf{6.22 \text{ V}} \text{ as noted in Fig. 5.194.} \end{aligned}$$

Chapter 6

1. –

2. From Fig. 6.11:

$$V_{GS} = 0 \text{ V}, I_D = 8 \text{ mA}$$

$$V_{GS} = -1 \text{ V}, I_D = 4.5 \text{ mA}$$

$$V_{GS} = -1.5 \text{ V}, I_D = 3.25 \text{ mA}$$

$$V_{GS} = -1.8 \text{ V}, I_D = 2.5 \text{ mA}$$

$$V_{GS} = -4 \text{ V}, I_D = 0 \text{ mA}$$

$$V_{GS} = -6 \text{ V}, I_D = 0 \text{ mA}$$

3. –

4. (a) $V_{DS} \cong 1.4 \text{ V}$

$$(b) \quad r_d = \frac{V}{I} = \frac{1.4 \text{ V}}{6 \text{ mA}} = 233.33 \, \Omega$$

(c) $V_{DS} \cong 1.6 \text{ V}$

$$(d) \quad r_d = \frac{V}{I} = \frac{1.6 \text{ V}}{3 \text{ mA}} = 533.33 \, \Omega$$

(e) $V_{DS} \cong 1.4 \text{ V}$

$$(f) \quad r_d = \frac{V}{I} = \frac{1.4 \text{ V}}{1.5 \text{ mA}} = 933.33 \, \Omega$$

(g) $r_o = 233.33 \, \Omega$

$$r_d = \frac{r_o}{[1 - V_{GS}/V_P]^2} = \frac{233.33 \, \Omega}{[1 - (-1 \text{ V})/(-4 \text{ V})]^2} = \frac{233.33 \, \Omega}{0.5625} = 414.81 \, \Omega$$

$$(h) \quad r_d = \frac{233.33 \, \Omega}{[1 - (-2 \text{ V})/(-4 \text{ V})]^2} = \frac{233.33 \, \Omega}{0.25} = 933.2 \, \Omega$$

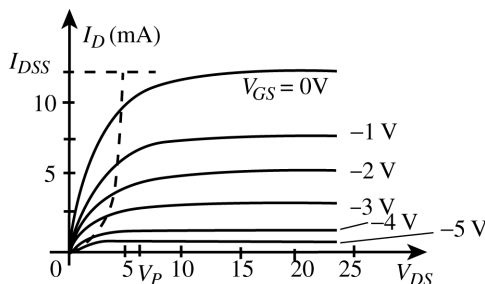
$$(i) \quad \left. \begin{array}{l} 533.33 \, \Omega \text{ vs. } 414.81 \, \Omega \\ 933.33 \, \Omega \text{ vs. } 933.2 \, \Omega \end{array} \right\} \text{Eq. (6.1) is valid!}$$

5. (a) $V_{GS} = 0 \text{ V}, I_D = 8 \text{ mA}$ (for $V_{DS} > V_P$)
 $V_{GS} = -1 \text{ V}, I_D = 4.5 \text{ mA}$
 $\Delta I_D = 3.5 \text{ mA}$

(b) $V_{GS} = -1 \text{ V}, I_D = 4.5 \text{ mA}$
 $V_{GS} = -2 \text{ V}, I_D = 2 \text{ mA}$
 $\Delta I_D = 2.5 \text{ mA}$

- (c) $V_{GS} = -2 \text{ V}$, $I_D = 2 \text{ mA}$
 $V_{GS} = -3 \text{ V}$, $I_D = 0.5 \text{ mA}$
 $\Delta I_D = 1.5 \text{ mA}$
- (d) $V_{GS} = -3 \text{ V}$, $I_D = 0.5 \text{ mA}$
 $V_{GS} = -4 \text{ V}$, $I_D = 0 \text{ mA}$
 $\Delta I_D = 0.5 \text{ mA}$
- (e) As V_{GS} becomes more negative, the change in I_D gets progressively smaller for the same change in V_{GS} .
- (f) Non-linear. Even though the change in V_{GS} is fixed at 1 V, the change in I_D drops from a maximum of 3.5 mA to a minimum of 0.5 mA—a 7:1 change in ΔI_D .

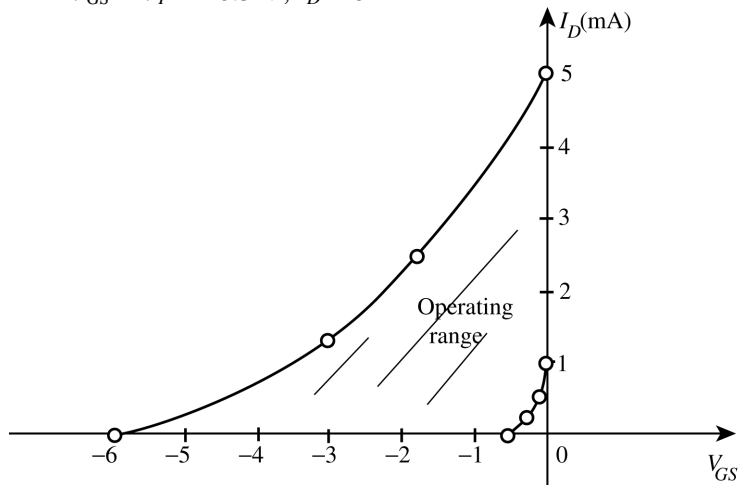
6. The collector characteristics of a BJT transistor are a plot of output current versus the output voltage for different levels of *input current*. The drain characteristics of a JFET transistor are a plot of the output current versus input voltage. For the BJT transistor increasing levels of input current result in increasing levels of output current. For JFETs, increasing magnitudes of input voltage result in lower levels of output current. The spacing between curves for a BJT are sufficiently similar to permit the use of a single beta (on an approximate basis) to represent the device for the dc and ac analysis. For JFETs, however, the spacing between the curves changes quite dramatically with increasing levels of input voltage requiring the use of Shockley's equation to define the relationship between I_D and V_{GS} . $V_{C_{sat}}$ and V_P define the region of nonlinearity for each device.
7.
 - (a) The input current I_G for a JFET is effectively zero since the JFET gate-source junction is reverse-biased for linear operation, and a reverse-biased junction has a very high resistance.
 - (b) The input impedance of the JFET is high due to the reverse-biased junction between the gate and source.
 - (c) The terminology is appropriate since it is the electric field established by the applied gate to source voltage that controls the level of drain current. The term “field” is appropriate due to the absence of a conductive path between gate and source (or drain).
8. $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS} = 12 \text{ mA}$
 $V_{GS} = V_P = -6 \text{ V}$, $I_D = 0 \text{ mA}$
 Shockley's equation: $V_{GS} = -1 \text{ V}$, $I_D = 8.33 \text{ mA}$; $V_{GS} = -2 \text{ V}$, $I_D = 5.33 \text{ mA}$; $V_{GS} = -3 \text{ V}$, $I_D = 3 \text{ mA}$; $V_{GS} = -4 \text{ V}$, $I_D = 1.33 \text{ mA}$; $V_{GS} = -5 \text{ V}$, $I_D = 0.333 \text{ mA}$.



9. For a *p*-channel JFET, all the voltage polarities in the network are reversed as compared to an *n*-channel device. In addition, the drain current has reversed direction.
10. (b) $I_{DSS} = 10 \text{ mA}$, $V_P = -6 \text{ V}$

11. $V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 12 \text{ mA}$
 $V_{GS} = V_P = -4 \text{ V}, I_D = 0 \text{ mA}$
 $V_{GS} = \frac{V_P}{2} = -2 \text{ V}, I_D = \frac{I_{DSS}}{4} = 3 \text{ mA}$
 $V_{GS} = 0.3V_P = -1.2 \text{ V}, I_D = 6 \text{ mA}$
 $V_{GS} = -3 \text{ V}, I_D = 0.75 \text{ mA}$ (Shockley's equation)
12. (a) $I_D = I_{DSS} = \mathbf{9 \text{ mA}}$
(b) $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$
 $= 9 \text{ mA}(1 - (-2)/(-4))^2$
 $= \mathbf{2.25 \text{ mA}}$
(c) $V_{GS} = V_P = -4 \text{ V}, I_D = \mathbf{0 \text{ mA}}$
(d) $|V_{GS}| \geq |V_P|, I_D = \mathbf{0 \text{ mA}}$
13. $V_{GS} = \mathbf{0 \text{ V}}, I_D = \mathbf{16 \text{ mA}}$
 $V_{GS} = 0.3V_P = 0.3(-5 \text{ V}) = \mathbf{-1.5 \text{ V}}, I_D = I_{DSS}/2 = \mathbf{8 \text{ mA}}$
 $V_{GS} = 0.5V_P = 0.5(-5 \text{ V}) = \mathbf{-2.5 \text{ V}}, I_D = I_{DSS}/4 = \mathbf{4 \text{ mA}}$
 $V_{GS} = V_P = \mathbf{-5 \text{ V}}, I_D = \mathbf{0 \text{ mA}}$
14. $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$
 $4 \text{ mA} = 12 \text{ mA}(1 - (-3)/V_P)^2$
 $0.333 = 1 - \frac{(-3)}{V_P}$
 $\sqrt{0.333} = 1 - \frac{(-3)}{V_P}$
 $0.577 = 1 - \frac{(-3)}{V_P}$
 $0.423 = \frac{-3}{V_P}$
 $V_P = \frac{-3}{0.423} \cong \mathbf{-7.1 \text{ V}}$
15. (a) $I_D = I_{DSS}(1 - V_{GS}/V_P)^2 = 6 \text{ mA}(1 - (-2 \text{ V})/(-4.5 \text{ V}))^2$
 $= \mathbf{1.852 \text{ mA}}$
 $I_D = I_{DSS}(1 - V_{GS}/V_P)^2 = 6 \text{ mA}(1 - (-3.6 \text{ V})/(-4.5 \text{ V}))^2$
 $= \mathbf{0.24 \text{ mA}}$
(b) $V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = (-4.5 \text{ V}) \left(1 - \sqrt{\frac{3 \text{ mA}}{6 \text{ mA}}} \right)$
 $= \mathbf{-1.318 \text{ V}}$
 $V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = (-4.5 \text{ V}) \left(1 - \sqrt{\frac{5.5 \text{ mA}}{6 \text{ mA}}} \right)$
 $= \mathbf{-0.192 \text{ V}}$

16. $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$
 $3 \text{ mA} = I_{DSS}(1 - (-3 \text{ V})/(-6 \text{ V}))^2$
 $3 \text{ mA} = I_{DSS}(0.25)$
 $I_{DSS} = \mathbf{12 \text{ mA}}$
17. $V_{GS} = 0 \text{ V}, I_D = I_{DSS} = \mathbf{7.5 \text{ mA}}$
 $V_{GS} = 0.3V_P = (0.3)(4 \text{ V}) = 1.2 \text{ V}, I_D = I_{DSS}/2 = 7.5 \text{ mA}/2 = \mathbf{3.75 \text{ mA}}$
 $V_{GS} = 0.5V_P = (0.5)(4 \text{ V}) = 2 \text{ V}, I_D = I_{DSS}/4 = 7.5 \text{ mA}/4 = \mathbf{1.875 \text{ mA}}$
 $V_{GS} = V_P = 4 \text{ V}, I_D = \mathbf{0 \text{ mA}}$
18. From Fig. 6.22:
 $-0.5 \text{ V} < V_P < -6 \text{ V}$
 $1 \text{ mA} < I_{DSS} < 5 \text{ mA}$
 For $I_{DSS} = 5 \text{ mA}$ and $V_P = -6 \text{ V}$:
 $V_{GS} = 0 \text{ V}, I_D = 5 \text{ mA}$
 $V_{GS} = 0.3V_P = -1.8 \text{ V}, I_D = 2.5 \text{ mA}$
 $V_{GS} = V_P/2 = -3 \text{ V}, I_D = 1.25 \text{ mA}$
 $V_{GS} = V_P = -6 \text{ V}, I_D = 0 \text{ mA}$
 For $I_{DSS} = 1 \text{ mA}$ and $V_P = -0.5 \text{ V}$:
 $V_{GS} = 0 \text{ V}, I_D = 1 \text{ mA}$
 $V_{GS} = 0.3V_P = -0.15 \text{ V}, I_D = 0.5 \text{ mA}$
 $V_{GS} = V_P/2 = -0.25 \text{ V}, I_D = 0.25 \text{ mA}$
 $V_{GS} = V_P = -0.5 \text{ V}, I_D = 0 \text{ mA}$



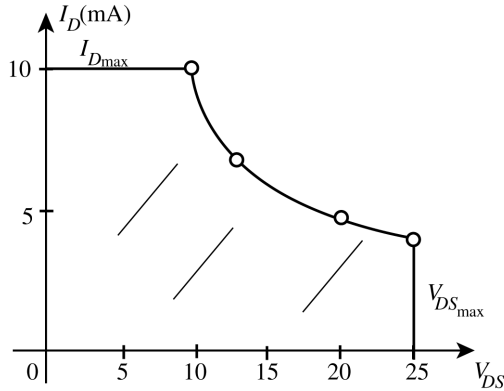
19. At 25°C , $P_D = 625 \text{ mW}$
 $45^\circ\text{C} - 25^\circ\text{C} = 20^\circ\text{C}$
 $20^\circ\text{C} \propto \frac{5 \text{ mW}}{^\circ\text{C}} = 100 \text{ mW}$
 $P'_D = 625 \text{ mW} - 100 \text{ mW}$
 $= \mathbf{525 \text{ mW}}$

$$20. \quad V_{DS} = V_{DS_{\max}} = \mathbf{25 \text{ V}}, I_D = \frac{P_{D_{\max}}}{V_{DS_{\max}}} = \frac{100 \text{ mW}}{25 \text{ V}} = \mathbf{4 \text{ mA}}$$

$$I_D = I_{DSS} = \mathbf{10 \text{ mA}}, V_{DS} = \frac{P_{D_{\max}}}{I_{DSS}} = \frac{100 \text{ mW}}{10 \text{ mA}} = \mathbf{10 \text{ V}}$$

$$V_D = 20 \text{ V}, I_D = 6.67 \text{ mA}$$

$$V_D = 15 \text{ V}, I_D = 5 \text{ mA}$$



$$21. \quad \left. \begin{array}{l} V_{GS} = -0.5 \text{ V}, I_D = 6.5 \text{ mA} \\ V_{GS} = -1 \text{ V}, I_D = 4 \text{ mA} \end{array} \right\} 2.5 \text{ mA}$$

Determine ΔI_D above 4 mA line:

$$\frac{2.5 \text{ mA}}{0.5 \text{ V}} = \frac{x}{0.3 \text{ V}} \Rightarrow x = 1.5 \text{ mA}$$

$I_D = 4 \text{ mA} + 1.5 \text{ mA} = \mathbf{5.5 \text{ mA}}$ corresponding with values determined from a purely graphical approach.

22. Yes, all knees of V_{GS} curves at or below $|V_P| = 3 \text{ V}$.

23. From Fig 6.25, $I_{DSS} \cong 9 \text{ mA}$

At $V_{GS} = -1 \text{ V}$, $I_D = 4 \text{ mA}$

$$I_D = I_{DSS}(1 - V_{GS}/V_P)^2$$

$$\sqrt{\frac{I_D}{I_{DSS}}} = 1 - V_{GS}/V_P$$

$$\frac{V_{GS}}{V_P} = 1 - \sqrt{\frac{I_D}{I_{DSS}}}$$

$$\begin{aligned} V_P &= \frac{V_{GS}}{1 - \sqrt{\frac{I_D}{I_{DSS}}}} = \frac{-1 \text{ V}}{1 - \sqrt{\frac{4 \text{ mA}}{9 \text{ mA}}}} \\ &= \mathbf{-3 \text{ V}} \text{ (an exact match)} \end{aligned}$$

$$\begin{aligned} 24. \quad I_D &= I_{DSS}(1 - V_{GS}/V_P)^2 \\ &= 9 \text{ mA}(1 - (-1 \text{ V})/(-3 \text{ V}))^2 \\ &= \mathbf{4 \text{ mA}}, \text{ which compares very well with the level obtained using Fig. 6.25.} \end{aligned}$$

25. (a) $V_{DS} \cong 0.7 \text{ V}$ @ $I_D = 4 \text{ mA}$ (for $V_{GS} = 0 \text{ V}$)

$$r = \frac{\Delta V_{DS}}{\Delta I_D} = \frac{0.7 \text{ V} - 0 \text{ V}}{4 \text{ mA} - 0 \text{ mA}} = \mathbf{175 \Omega}$$

- (b) For $V_{GS} = -0.5 \text{ V}$, @ $I_D = 3 \text{ mA}$, $V_{DS} = 0.7 \text{ V}$

$$r = \frac{0.7 \text{ V}}{3 \text{ mA}} = \mathbf{233 \Omega}$$

- (c) $r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2} = \frac{175 \Omega}{(1 - (-0.5 \text{ V})/(-3 \text{ V}))^2}$
 $= \mathbf{252 \Omega}$ vs. 233Ω from part (b)

26. —

27. The construction of a depletion-type MOSFET and an enhancement-type MOSFET are identical except for the doping in the channel region. In the depletion MOSFET the channel is established by the doping process and exists with no gate-to-source voltage applied. As the gate-to-source voltage increases in magnitude the channel decreases in size until pinch-off occurs. The enhancement MOSFET does not have a channel established by the doping sequence but relies on the gate-to-source voltage to create a channel. The larger the magnitude of the applied gate-to-source voltage, the larger the available channel.

28. —

29. At $V_{GS} = 0 \text{ V}$, $I_D = \mathbf{6 \text{ mA}}$

$$\text{At } V_{GS} = -1 \text{ V}, I_D = 6 \text{ mA}(1 - (-1 \text{ V})/(-3 \text{ V}))^2 = \mathbf{2.66 \text{ mA}}$$

$$\text{At } V_{GS} = +1 \text{ V}, I_D = 6 \text{ mA}(1 - (+1 \text{ V})/(-3 \text{ V}))^2 = 6 \text{ mA}(1.333)^2 = \mathbf{10.667 \text{ mA}}$$

$$\text{At } V_{GS} = +2 \text{ V}, I_D = 6 \text{ mA}(1 - (+2 \text{ V})/(-3 \text{ V}))^2 = 6 \text{ mA}(1.667)^2 = \mathbf{16.67 \text{ mA}}$$

$$\Delta I_D = 4.67 \text{ mA}$$

V_{GS}	I_D	
-1 V	2.66 mA	} $\Delta I_D = \mathbf{3.34 \text{ mA}}$
0	6.0 mA	
+1 V	10.67 mA	} $\Delta I_D = \mathbf{4.67 \text{ mA}}$
+2 V	16.67 mA	
		} $\Delta I_D = \mathbf{6 \text{ mA}}$

From -1 V to 0 V , $\Delta I_D = 3.34 \text{ mA}$

while from $+1 \text{ V}$ to $+2 \text{ V}$, $\Delta I_D = 6 \text{ mA}$ — almost a 2:1 margin.

In fact, as V_{GS} becomes more and more positive, I_D will increase at a faster and faster rate due to the squared term in Shockley's equation.

30. $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS} = 12 \text{ mA}$; $V_{GS} = -8 \text{ V}$, $I_D = 0 \text{ mA}$; $V_{GS} = \frac{V_P}{2} = -4 \text{ V}$, $I_D = 3 \text{ mA}$;

$$V_{GS} = 0.3V_P = -2.4 \text{ V}, I_D = 6 \text{ mA}; V_{GS} = -6 \text{ V}, I_D = 0.75 \text{ mA}$$

31. From problem 20:

$$V_P = \frac{V_{GS}}{1 - \sqrt{\frac{I_D}{I_{DSS}}}} = \frac{+1 \text{ V}}{1 - \sqrt{\frac{14 \text{ mA}}{9.5 \text{ mA}}}} = \frac{+1 \text{ V}}{1 - \sqrt{1.473}} = \frac{+1 \text{ V}}{1 - 1.21395}$$

$$= \frac{1}{-0.21395} \cong -4.67 \text{ V}$$

32. $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$

$$I_{DSS} = \frac{I_D}{(1 - V_{GS}/V_P)^2} = \frac{4 \text{ mA}}{(1 - (-2 \text{ V})/(-5 \text{ V}))^2} = 11.11 \text{ mA}$$

33. From problem 14(b):

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right) = (-5 \text{ V}) \left(1 - \sqrt{\frac{20 \text{ mA}}{2.9 \text{ mA}}} \right)$$

$$= (-5 \text{ V})(1 - 2.626) = (-5 \text{ V})(-1.626)$$

$$= 8.13 \text{ V}$$

34. From Fig. 6.34, $P_{D_{\max}} = 200 \text{ mW}$, $I_D = 8 \text{ mA}$

$$P = V_{DS}I_D$$

$$\text{and } V_{DS} = \frac{P_{\max}}{I_D} = \frac{200 \text{ mW}}{8 \text{ mA}} = 25 \text{ V}$$

35. (a) In a depletion-type MOSFET the channel exists in the device and the applied voltage V_{GS} controls the size of the channel. In an enhancement-type MOSFET the channel is not established by the construction pattern but induced by the applied control voltage V_{GS} .

(b) –

(c) Briefly, an applied gate-to-source voltage greater than V_T will establish a channel between drain and source for the flow of charge in the output circuit.

36. (a) $I_D = k(V_{GS} - V_T)^2 = 0.4 \times 10^{-3}(V_{GS} - 3.5)^2$

V_{GS}	I_D
3.5 V	0
4 V	0.1 mA
5 V	0.9 mA
6 V	2.5 mA
7 V	4.9 mA
8 V	8.1 mA

(b) $I_D = 0.8 \times 10^{-3}(V_{GS} - 3.5)^2$

V_{GS}	I_D	
3.5 V	0	For same levels of V_{GS} , I_D attains twice the current level as part (a). Transfer curve has steeper slope. For both curves, $I_D = 0$ mA for $V_{GS} < 3.5$ V.
4 V	0.2 mA	
5 V	1.8 mA	
6 V	5.0 mA	
7 V	9.8 mA	
8 V	16.2 mA	

37. (a) $k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2} = \frac{4 \text{ mA}}{(6 \text{ V} - 4 \text{ V})^2} = \mathbf{1 \text{ mA/V}^2}$

$I_D = k(V_{GS} - V_T)^2 = \mathbf{1 \times 10^{-3}(V_{GS} - 4 \text{ V})^2}$

(b)

V_{GS}	I_D	For $V_{GS} < V_T = 4 \text{ V}$, $I_D = 0 \text{ mA}$
4 V	0 mA	
5 V	1 mA	
6 V	4 mA	
7 V	9 mA	
8 V	16 mA	

(c)

V_{GS}	I_D	$(V_{GS} < V_T)$
2 V	0 mA	
5 V	1 mA	
10 V	36 mA	

38. From Fig. 6.58, $V_T = 2.0 \text{ V}$
At $I_D = 6.5 \text{ mA}$, $V_{GS} = 5.5 \text{ V}$:

$$I_D = k(V_{GS} - V_T)^2$$

$$6.5 \text{ mA} = k(5.5 \text{ V} - 2 \text{ V})^2$$

$$k = \mathbf{5.31 \times 10^{-4}}$$

$I_D = \mathbf{5.31 \times 10^{-4}(V_{GS} - 2)^2}$

39. $I_D = k(V_{GS(\text{on})} - V_T)^2$

and $(V_{GS(\text{on})} - V_T)^2 = \frac{I_D}{k}$

$$V_{GS(\text{on})} - V_T = \sqrt{\frac{I_D}{k}}$$

$$V_T = V_{GS(\text{on})} - \sqrt{\frac{I_D}{k}}$$

$$= 4 \text{ V} - \sqrt{\frac{3 \text{ mA}}{0.4 \times 10^{-3}}} = 4 \text{ V} - \sqrt{7.5} \text{ V}$$

$$= 4 \text{ V} - 2.739 \text{ V}$$

$$= \mathbf{1.261 \text{ V}}$$

$$\begin{aligned}
40. \quad I_D &= k(V_{GS} - V_T)^2 \\
\frac{I_D}{k} &= (V_{GS} - V_T)^2 \\
\sqrt{\frac{I_D}{k}} &= V_{GS} - V_T \\
V_{GS} &= V_T + \sqrt{\frac{I_D}{k}} = 5 \text{ V} + \sqrt{\frac{30 \text{ mA}}{0.06 \times 10^{-3}}} \\
&= \mathbf{27.36 \text{ V}}
\end{aligned}$$

41. Enhancement-type MOSFET:

$$\begin{aligned}
I_D &= k(V_{GS} - V_T)^2 \\
\frac{dI_D}{dV_{GS}} &= 2k(V_{GS} - V_T) \frac{d}{dV_{GS}} (V_{GS} - V_T) \\
&= 2k(V_{GS} - V_T)
\end{aligned}$$

Depletion-type MOSFET:

$$\begin{aligned}
I_D &= I_{DSS}(1 - V_{GS}/V_P)^2 \\
\frac{dI_D}{dV_{GS}} &= I_{DSS} \frac{d}{dV_{GS}} \left(1 - \frac{V_{GS}}{V_P}\right)^2 \\
&= I_{DSS} 2 \left(1 - \frac{V_{GS}}{V_P}\right) \frac{d}{dV_{GS}} \left(1 - \frac{V_{GS}}{V_P}\right) \\
&\quad \underbrace{\left(-\frac{1}{V_P}\right)}_{-\frac{1}{V_P}} \\
&= 2I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right) \left(-\frac{1}{V_P}\right) \\
&= -\frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right) \\
&= -\frac{2I_{DSS}}{V_P} \frac{V_P}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right) \\
\frac{dI_D}{dV_{GS}} &= \frac{2I_{DSS}}{V_P^2} (V_{GS} - V_P)
\end{aligned}$$

For both devices $\frac{dI_D}{dV_{GS}} = k_1(V_{GS} - K_2)$

revealing that the drain current of each will increase at about the same rate.

$$\begin{aligned}
42. \quad I_D &= k(V_{GS} - V_T)^2 = 0.45 \times 10^{-3} (V_{GS} - (-5 \text{ V}))^2 \\
&= 0.45 \times 10^{-3} (V_{GS} + 5 \text{ V})^2 \\
V_{GS} &= -5 \text{ V}, I_D = 0 \text{ mA}; V_{GS} = -6 \text{ V}, I_D = 0.45 \text{ mA}; V_{GS} = -7 \text{ V}, I_D = 1.8 \text{ mA}; \\
V_{GS} &= -8 \text{ V}, I_D = 4.05 \text{ mA}; V_{GS} = -9 \text{ V}, I_D = 7.2 \text{ mA}; V_{GS} = -10 \text{ V}, I_D = 11.25 \text{ mA}
\end{aligned}$$

43. —

44. —

45. —

46. —

(b) For the “on” transistor: $R = \frac{V}{I} = \frac{0.1 \text{ V}}{4 \text{ mA}} = \mathbf{25 \text{ ohms}}$

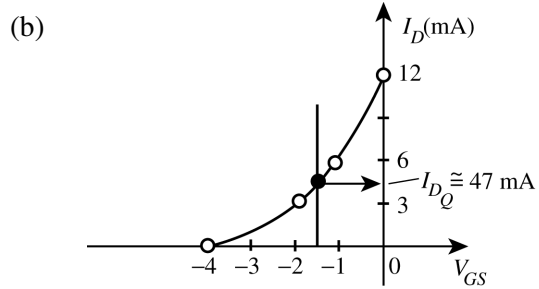
For the “off” transistor: $R = \frac{V}{I} = \frac{4.9 \text{ V}}{0.5 \text{ } \mu\text{A}} = \mathbf{9.8 \text{ M}\Omega}$

Absolutely, the high resistance of the “off” resistance will ensure V_o is very close to 5 V.

47. —

Chapter 7

1. (a) $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS} = 12 \text{ mA}$
 $V_{GS} = V_P = -4 \text{ V}$, $I_D = 0 \text{ mA}$
 $V_{GS} = V_P/2 = -2 \text{ V}$, $I_D = I_{DSS}/4 = 3 \text{ mA}$
 $V_{GS} = 0.3V_P = -1.2 \text{ V}$, $I_D = I_{DSS}/2 = 6 \text{ mA}$



- (c) $I_{DQ} \cong 4.7 \text{ mA}$
 $V_{DSQ} = V_{DD} - I_{DQ} R_D = 14 \text{ V} - (4.7 \text{ mA})(1.8 \text{ k}\Omega)$
 $= 5.54 \text{ V}$
- (d) $I_{DQ} = I_{DSS}(1 - V_{GS}/V_P)^2 = 12 \text{ mA}(1 - (-1.5 \text{ V})/(-4 \text{ V}))^2$
 $= 4.69 \text{ mA}$

$$V_{DSQ} = V_{DD} - I_{DQ} R_D = 14 \text{ V} - (4.69 \text{ mA})(1.8 \text{ k}\Omega)$$

$$= 5.56 \text{ V}$$

excellent comparison

2. (a) $I_{DQ} = I_{DSS}(1 - V_{GS}/V_P)^2$
 $= 10 \text{ mA}(1 - (-3 \text{ V})/(-4.5 \text{ V}))^2$
 $= 10 \text{ mA}(0.333)^2$
 $I_{DQ} = 1.11 \text{ mA}$

- (b) $V_{GSQ} = -3 \text{ V}$

- (c) $V_{DS} = V_{DD} - I_D(R_D + R_S)$
 $= 16 \text{ V} - (1.11 \text{ mA})(2.2 \text{ k}\Omega)$
 $= 16 \text{ V} - 2.444 \text{ V}$
 $= 13.56 \text{ V}$
 $V_D = V_{DS} = 13.56 \text{ V}$
 $V_G = V_{GSQ} = -3 \text{ V}$
 $V_S = 0 \text{ V}$

3. (a) $I_{D_Q} = \frac{V_{DD} - V_D}{R_D} = \frac{12 \text{ V} - 6 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{2.727 \text{ mA}}$

(b) $V_{DS} = V_D - V_S = 6 \text{ V} - 0 \text{ V} = \mathbf{6 \text{ V}}$

(c) $I_D = I_{DSS}(1 - V_{GS}/V_P)^2 \Rightarrow V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$

$$V_{GS} = (-4 \text{ V}) \left(1 - \sqrt{\frac{2.727 \text{ mA}}{8 \text{ mA}}}\right)$$

$$= -1.66 \text{ V}$$

$$\therefore V_{GG} = \mathbf{1.66 \text{ V}}$$

4. $V_{GS_Q} = \mathbf{0 \text{ V}}, I_D = I_{DSS} = 5 \text{ mA}$

$$\begin{aligned} V_D &= V_{DD} - I_D R_D \\ &= 20 \text{ V} - (5 \text{ mA})(2.2 \text{ k}\Omega) \\ &= 20 \text{ V} - 11 \text{ V} \\ &= \mathbf{9 \text{ V}} \end{aligned}$$

5. $V_{GS} = V_P = \mathbf{-4 \text{ V}}$

$$\therefore I_{D_Q} = 0 \text{ mA}$$

$$\begin{aligned} \text{and } V_D &= V_{DD} - I_{D_Q} R_D = 18 \text{ V} - (0)(2 \text{ k}\Omega) \\ &= \mathbf{18 \text{ V}} \end{aligned}$$

6. (a)(b) $V_{GS} = 0 \text{ V}, I_D = 10 \text{ mA}$

$$V_{GS} = V_P = -4 \text{ V}, I_D = 0 \text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = -2 \text{ V}, I_D = 2.5 \text{ mA}$$

$$V_{GS} = 0.3 V_P = -1.2 \text{ V}, I_D = 5 \text{ mA}$$

$$V_{GS} = -I_D R_S$$

$$I_D = 5 \text{ mA:}$$

$$\begin{aligned} V_{GS} &= -(5 \text{ mA})(0.75 \text{ k}\Omega) \\ &= \mathbf{-3.75 \text{ V}} \end{aligned}$$

(c) $I_{D_Q} \cong \mathbf{2.7 \text{ mA}}$

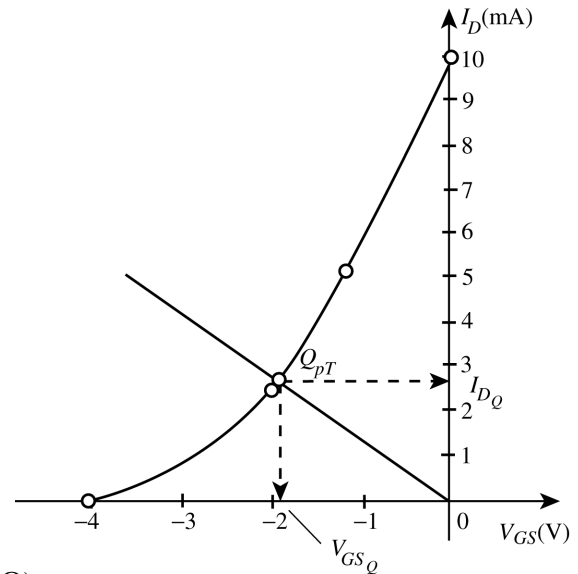
$$V_{GS_Q} \cong \mathbf{-1.9 \text{ V}}$$

(d) $\begin{aligned} V_{DS} &= V_{DD} - I_D(R_D + R_S) \\ &= 18 \text{ V} - (2.7 \text{ mA})(1.5 \text{ k}\Omega + 0.75 \text{ k}\Omega) \\ &= \mathbf{11.93 \text{ V}} \end{aligned}$

$$\begin{aligned} V_D &= V_{DD} - I_D R_D \\ &= 18 \text{ V} - (2.7 \text{ mA})(1.5 \text{ k}\Omega) \\ &= \mathbf{13.95 \text{ V}} \end{aligned}$$

$$V_G = \mathbf{0 \text{ V}}$$

$$\begin{aligned} V_S &= I_S R_S = I_D R_S \\ &= (2.7 \text{ mA})(0.75 \text{ k}\Omega) \\ &= \mathbf{2.03 \text{ V}} \end{aligned}$$



$$7. \quad I_D = I_{DSS}(1 - V_{GS}/V_P)^2 = I_{DSS} \left[1 + \frac{2I_D R_S}{V_P} + \frac{I_D^2 R_S^2}{V_P^2} \right]$$

$$\frac{I_{DSS} R_S^2}{V_P^2} I_D^2 + \frac{2I_{DSS} R_S}{V_P} I_D - I_{DSS} = 0$$

$$\text{Substituting: } 351.56 I_D^2 - 4.75 I_D + 10 \text{ mA} = 0$$

$$I_D = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = 10.91 \text{ mA}, 2.60 \text{ mA}$$

$$I_{DQ} = \mathbf{2.6 \text{ mA}} \text{ (exact match \#6)}$$

$$V_{GS} = -I_D R_S = -(2.60 \text{ mA})(0.75 \text{ k}\Omega) \\ = -1.95 \text{ V vs. } -2 \text{ V (\#6)}$$

$$8. \quad V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 6 \text{ mA}$$

$$V_{GS} = V_P = -6 \text{ V}, I_D = 0 \text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = -3 \text{ V}, I_D = 1.5 \text{ mA}$$

$$V_{GS} = 0.3 V_P = -1.8 \text{ V}, I_D = 3 \text{ mA}$$

$$V_{GS} = -I_D R_S$$

$$I_D = 2 \text{ mA:}$$

$$V_{GS} = -(2 \text{ mA})(1.6 \text{ k}\Omega) \\ = -3.2 \text{ V}$$

$$(a) \quad I_{DQ} = \mathbf{1.7 \text{ mA}}$$

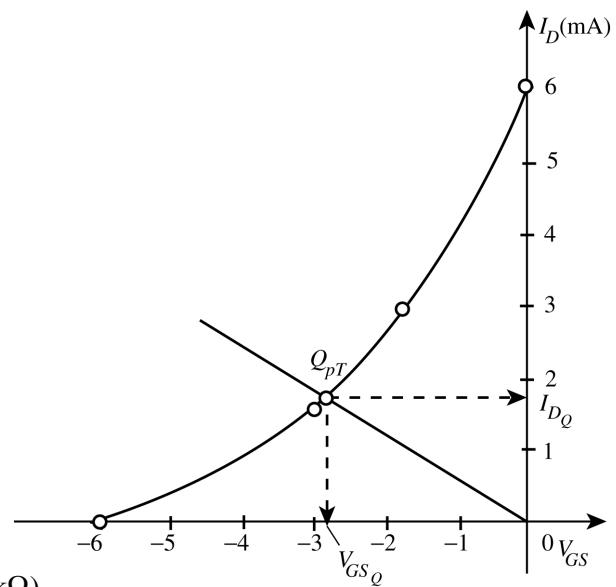
$$V_{GSQ} = \mathbf{-2.8 \text{ V}}$$

$$(b) \quad V_{DS} = V_{DD} - I_D(R_D + R_S) \\ = 12 \text{ V} - (1.7 \text{ mA})(2.2 \text{ k}\Omega + 1.6 \text{ k}\Omega) \\ = \mathbf{5.54 \text{ V}}$$

$$V_D = V_{DD} - I_D R_D \\ = 12 \text{ V} - (1.7 \text{ mA})(2.2 \text{ k}\Omega) \\ = \mathbf{8.26 \text{ V}}$$

$$V_G = \mathbf{0 \text{ V}}$$

$$V_S = I_S R_S = I_D R_S \\ = (1.7 \text{ mA})(1.6 \text{ k}\Omega) \\ = \mathbf{2.72 \text{ V}} \text{ (vs. } 2.8 \text{ V from } V_S = (V_{GSQ}))$$



$$9. \quad (a) \quad I_{DQ} = I_S = \frac{V_S}{R_S} = \frac{1.7 \text{ V}}{0.51 \text{ k}\Omega} = \mathbf{3.33 \text{ mA}}$$

$$(b) \quad V_{GSQ} = -I_{DQ} R_S = -(3.33 \text{ mA})(0.51 \text{ k}\Omega) \\ \cong \mathbf{-1.7 \text{ V}}$$

$$\begin{aligned}
 (c) \quad I_D &= I_{DSS}(1 - V_{GS}/V_P)^2 \\
 3.33 \text{ mA} &= I_{DSS}(1 - (-1.7 \text{ V})/(-4 \text{ V}))^2 \\
 3.33 \text{ mA} &= I_{DSS}(0.331) \\
 I_{DSS} &= \mathbf{10.06 \text{ mA}}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad V_D &= V_{DD} - I_{D_Q} R_D \\
 &= 18 \text{ V} - (3.33 \text{ mA})(2 \text{ k}\Omega) = 18 \text{ V} - 6.66 \text{ V} \\
 &= \mathbf{11.34 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad V_{DS} &= V_D - V_S = 11.34 \text{ V} - 1.7 \text{ V} \\
 &= \mathbf{9.64 \text{ V}}
 \end{aligned}$$

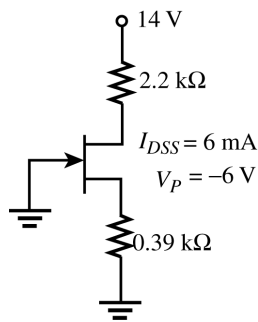
$$\begin{aligned}
 10. \quad (a) \quad V_{GS} &= 0 \text{ V} \\
 \therefore I_D &= I_{DSS} = \mathbf{4.5 \text{ mA}}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad V_{DS} &= V_{DD} - I_D(R_D + R_S) \\
 &= 20 \text{ V} + 4 \text{ V} - (4.5 \text{ mA})(2.2 \text{ k}\Omega + 0.68 \text{ k}\Omega) \\
 &= 24 \text{ V} - 12.96 \\
 &= \mathbf{11.04 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad V_D &= V_{DD} - I_D R_D \\
 &= 20 \text{ V} - (4.5 \text{ mA})(2.2 \text{ k}\Omega) \\
 &= \mathbf{10.1 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad -4 \text{ V} + I_S R_S - V_S &= 0 \\
 V_S = I_S R_S - 4 \text{ V} &= I_D R_S - 4 \text{ V} = (4.5 \text{ mA})(0.68 \text{ k}\Omega) - 4 \text{ V} \\
 &= 3.06 \text{ V} - 4 \text{ V} \\
 &= \mathbf{-0.94 \text{ V}}
 \end{aligned}$$

11. Network redrawn:

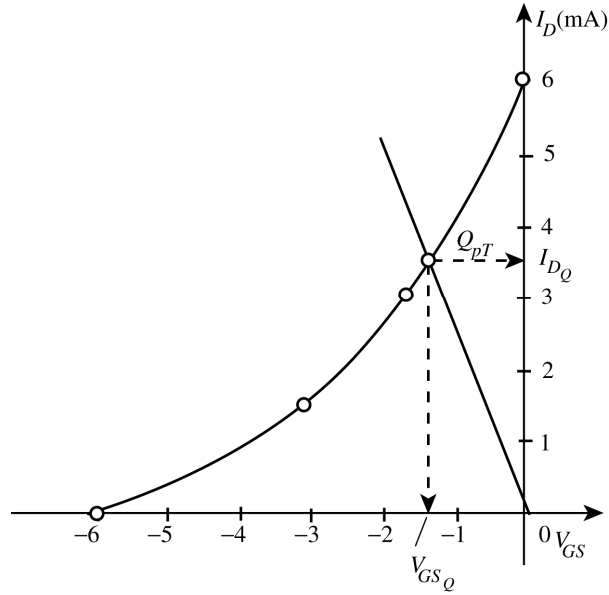


$$\begin{aligned}
 V_{GS} &= 0 \text{ V}, I_D = I_{DSS} = 6 \text{ mA} \\
 V_{GS} &= V_P = -6 \text{ V}, I_D = 0 \text{ mA} \\
 V_{GS} &= \frac{V_P}{2} = -3 \text{ V}, I_D = 1.5 \text{ mA} \\
 V_{GS} &= 0.3 V_P = -1.8 \text{ V}, I_D = 3 \text{ mA} \\
 V_{GS} &= -I_D R_S = -I_D(0.39 \text{ k}\Omega) \\
 \text{For } I_D &= 5 \text{ mA}, V_{GS} = -1.95 \text{ V}
 \end{aligned}$$

From graph $I_{D_Q} \cong 3.55 \text{ mA}$

$$V_{GS_Q} \cong -1.4 \text{ V}$$

$$V_S = -(V_{GS_Q}) = -(-1.4 \text{ V}) \\ = +1.4 \text{ V}$$



12. (a) $V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{110 \text{ k}\Omega (20 \text{ V})}{910 \text{ k}\Omega + 110 \text{ k}\Omega}$
 $= 2.16 \text{ V}$

$$V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 10 \text{ mA}$$

$$V_{GS} = V_P = -3.5 \text{ V}, I_D = 0 \text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = -1.75 \text{ V}, I_D = 2.5 \text{ mA}$$

$$V_{GS} = 0.3 V_P = -1.05 \text{ V}, I_D = 5 \text{ mA}$$

$$V_{GS_Q} = V_G - I_D R_S$$

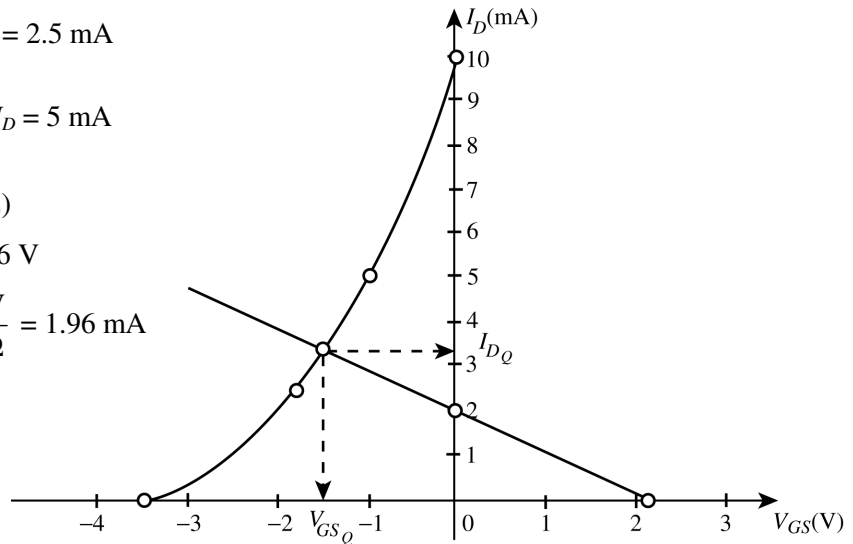
$$V_{GS_Q} = 2.16 - I_D (1.1 \text{ k}\Omega)$$

$$I_D = 0: V_{GS_Q} = V_G = 2.16 \text{ V}$$

$$V_{GS_Q} = 0 \text{ V}, I_D = \frac{2.16 \text{ V}}{1.1 \text{ k}\Omega} = 1.96 \text{ mA}$$

(b) $I_{D_Q} \cong 3.3 \text{ mA}$

$$V_{GS_Q} \cong -1.5 \text{ V}$$



(c) $V_D = V_{DD} - I_{D_Q} R_D$
 $= 20 \text{ V} - (3.3 \text{ mA})(2.2 \text{ k}\Omega)$
 $= 12.74 \text{ V}$

$$V_S = I_S R_S = I_D R_S$$

$$= (3.3 \text{ mA})(1.1 \text{ k}\Omega)$$

$$= 3.63 \text{ V}$$

(d) $V_{DS_Q} = V_{DD} - I_{D_Q} (R_D + R_S)$
 $= 20 \text{ V} - (3.3 \text{ mA})(2.2 \text{ k}\Omega + 1.1 \text{ k}\Omega)$
 $= 20 \text{ V} - 10.89 \text{ V}$
 $= 9.11 \text{ V}$

13. (a) $I_D = I_{DSS} = 10 \text{ mA}$, $V_P = -3.5 \text{ V}$ | $V_G = \frac{110 \text{ k}\Omega(20 \text{ V})}{110 \text{ k}\Omega + 910 \text{ k}\Omega} = \mathbf{2.16 \text{ V}}$
 $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS} = 10 \text{ mA}$
 $V_{GS} = V_P = -3.5 \text{ V}$, $I_D = 0 \text{ mA}$

$$V_{GS} = \frac{V_P}{2} = \frac{-3.5 \text{ V}}{2} = -1.75 \text{ V}, I_D = 2.5 \text{ mA}$$

$$V_{GS} = 0.3 V_P = -1.05 \text{ V}, I_D = 5 \text{ mA}$$

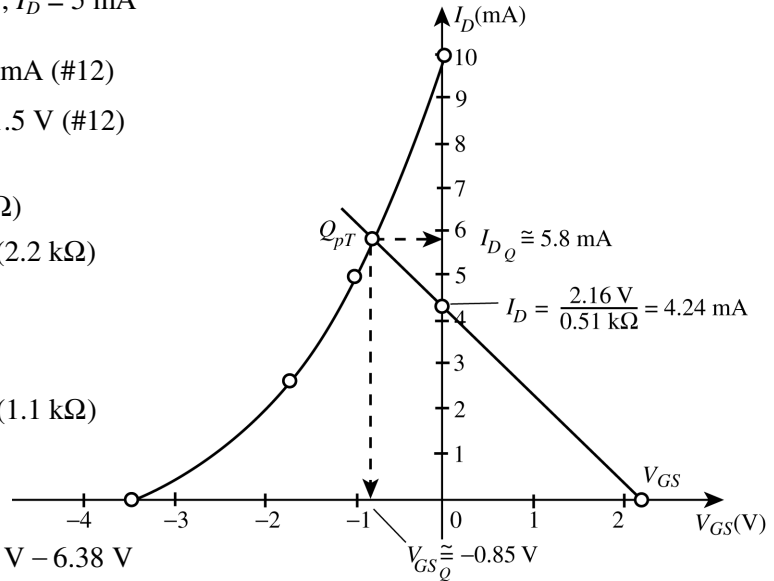
$$I_{D_Q} \cong \mathbf{5.8 \text{ mA}} \text{ vs. } 3.3 \text{ mA} \text{ (\#12)}$$

$$V_{GS_Q} \cong \mathbf{-0.85 \text{ V}} \text{ vs. } -1.5 \text{ V} \text{ (\#12)}$$

$$\begin{aligned} V_D &= 20 \text{ V} - I_D(2.2 \text{ k}\Omega) \\ &= 20 \text{ V} - (5.8 \text{ mA})(2.2 \text{ k}\Omega) \\ &= 20 \text{ V} - 12.76 \text{ V} \\ &= \mathbf{7.24 \text{ V}} \end{aligned}$$

$$\begin{aligned} V_S &= I_D R_S = (5.8 \text{ mA})(1.1 \text{ k}\Omega) \\ &= \mathbf{6.38 \text{ V}} \end{aligned}$$

$$\begin{aligned} V_{DS_Q} &= V_D - V_S = 7.24 \text{ V} - 6.38 \text{ V} \\ &= \mathbf{0.86 \text{ V}} \end{aligned}$$



- (b) As R_S decreases, the intersection on the vertical axis increases. The maximum occurs at $I_D = I_{DSS} = 10 \text{ mA}$.

$$\therefore R_{S_{\min}} = \frac{V_G}{I_{DSS}} = \frac{2.16 \text{ V}}{10 \text{ mA}} = \mathbf{216 \Omega}$$

14. (a) $I_D = \frac{V_{R_D}}{R_D} = \frac{V_{DD} - V_D}{R_D} = \frac{18 \text{ V} - 12 \text{ V}}{2 \text{ k}\Omega} = \frac{6 \text{ V}}{2 \text{ k}\Omega} = \mathbf{3 \text{ mA}}$

(b) $V_S = I_S R_S = I_D R_S = (3 \text{ mA})(0.68 \text{ k}\Omega) = \mathbf{2.04 \text{ V}}$
 $V_{DS} = V_{DD} - I_D(R_D + R_S) = 18 \text{ V} - (3 \text{ mA})(2 \text{ k}\Omega + 0.68 \text{ k}\Omega)$
 $= 18 \text{ V} - 8.04 \text{ V} = \mathbf{9.96 \text{ V}}$

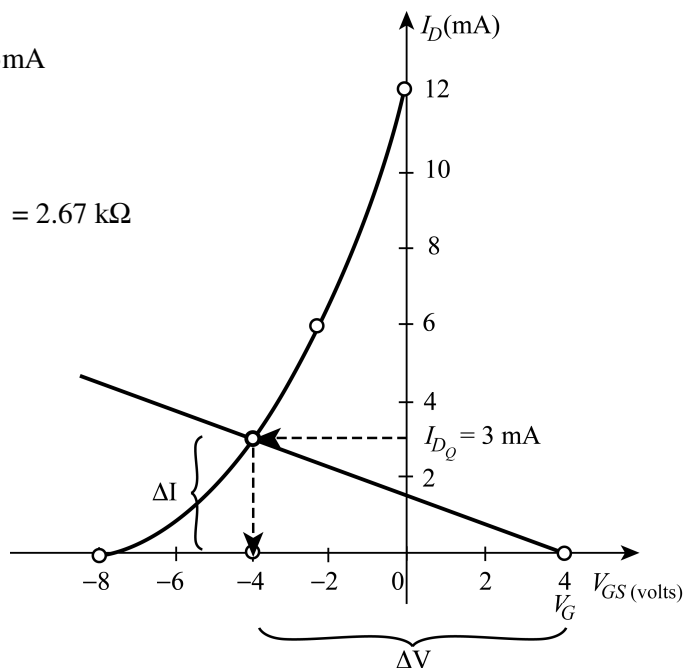
(c) $V_G = \frac{110 \text{ k}\Omega(12 \text{ V})}{110 \text{ k}\Omega + 680 \text{ k}\Omega} = \mathbf{1.67 \text{ V}}$
 $V_{GS} = V_G - I_S R_S = V_G - I_D R_S = V_G - V_S = 1.67 \text{ V} - 2.04 \text{ V}$
 $= \mathbf{-0.37 \text{ V}}$

(d) $V_P = \frac{V_{GS}}{1 - \sqrt{\frac{I_D}{I_{DSS}}}} = \frac{-0.37 \text{ V}}{1 - \sqrt{\frac{3 \text{ mA}}{8 \text{ mA}}}} = \frac{-0.37 \text{ V}}{1 - 0.612} = \frac{-0.37 \text{ V}}{0.388}$
 $= \mathbf{-0.954 \text{ V}}$

15.
$$I_{D_Q} = \frac{16 \text{ V} - 10 \text{ V}}{2 \text{ k}\Omega} = \frac{6 \text{ V}}{2 \text{ k}\Omega} = 3 \text{ mA}$$

$$V_G = \frac{12 \text{ k}\Omega(16 \text{ V})}{12 \text{ k}\Omega + 36 \text{ k}\Omega} = 4 \text{ V}$$

$$R_S = \frac{\Delta V}{\Delta I} = \frac{4 \text{ V} + 4 \text{ V}}{3 \text{ mA}} = \frac{8 \text{ V}}{3 \text{ mA}} = 2.67 \text{ k}\Omega$$



16. (a) $V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 6 \text{ mA}$
 $V_{GS} = V_P = -6 \text{ V}, I_D = 0 \text{ mA}$
 $V_{GS} = V_P / 2 = -3 \text{ V}, I_D = 1.5 \text{ mA}$
 $V_{GS} = 0.3V_P = -1.8 \text{ V}, I_D = 3 \text{ mA}$

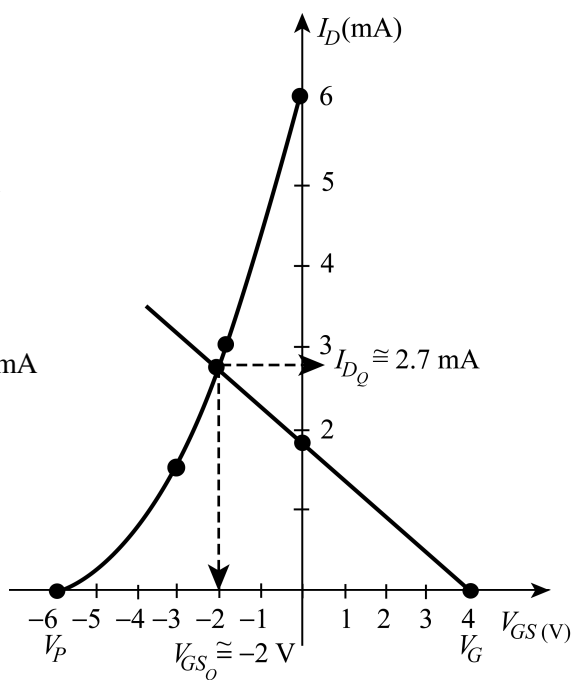
$$V_{GS} = V_{SS} - I_D R_S$$

$$V_{GS} = 4 \text{ V} - I_D (2.2 \text{ k}\Omega)$$

$$V_{GS} = 0 \text{ V}, I_D = \frac{4 \text{ V}}{2.2 \text{ k}\Omega} = 1.818 \text{ mA}$$

$$I_D = 0 \text{ mA}, V_{GS} = 4 \text{ V}$$

$$I_{D_Q} \cong 2.7 \text{ mA}, V_{GS_Q} \cong -2 \text{ V}$$



(b) $V_{DS} = V_{DD} + V_{SS} - I_D (R_D + R_S)$
 $= 16 \text{ V} + 4 \text{ V} - (2.7 \text{ mA})(4.4 \text{ k}\Omega)$
 $= 8.12 \text{ V}$

$$V_S = -V_{SS} + I_D R_S = -4 \text{ V} + (2.7 \text{ mA})(2.2 \text{ k}\Omega)$$

$$= 1.94 \text{ V}$$

or $V_S = -(V_{GS_Q}) = -(2 \text{ V}) = 2 \text{ V}$

$$17. \quad (a) \quad I_D = \frac{V_{DD} + V_{SS} - V_{DS}}{R_D + R_S} = \frac{20 \text{ V} - 2 \text{ V} - 4 \text{ V}}{3 \text{ k}\Omega + 1.2 \text{ k}\Omega} = \frac{14 \text{ V}}{4.2 \text{ k}\Omega} = \mathbf{3.3 \text{ mA}}$$

$$(b) \quad V_D = V_{DD} - I_D R_D = 20 \text{ V} - (3.33 \text{ mA})(3 \text{ k}\Omega) \\ = \mathbf{10 \text{ V}}$$

$$V_S = I_S R_S + V_{SS} = I_D R_S + V_{SS} \\ = (3.33 \text{ mA})(1.2 \text{ k}\Omega) + (2 \text{ V}) \\ = 4 \text{ V} + 2 \text{ V} \\ = \mathbf{6 \text{ V}}$$

$$(c) \quad V_{GS} = V_G - V_S \\ = 0 \text{ V} - 6 \text{ V} \\ = \mathbf{-6 \text{ V}}$$

$$18. \quad (a) \quad I_{DQ} = \mathbf{4 \text{ mA}}$$

$$(b) \quad V_{DQ} = 18 \text{ V} - 4 \text{ mA}(1.8 \text{ k}\Omega) = 18 \text{ V} - 7.2 \text{ V} = \mathbf{10.8 \text{ V}}$$

$$V_{DSQ} = V_D - V_S = 10.8 \text{ V} - I_D R_S = 10.8 \text{ V} - (4 \text{ mA})(1.2 \text{ k}\Omega) = 10.8 \text{ V} - 4.8 \text{ V} = \mathbf{6 \text{ V}}$$

$$(c) \quad P_s = (18 \text{ V})(4 \text{ mA}) = \mathbf{72 \text{ mW}} \\ P_d = (6 \text{ V})(4 \text{ mA}) = \mathbf{24 \text{ mW}}$$

$$19. \quad V_G = \frac{(1.2 \text{ k}\Omega)(16 \text{ V})}{1.2 \text{ k}\Omega + 3.6 \text{ k}\Omega} = 4 \text{ V}$$

$$I_S = I_D = \frac{4 \text{ V}}{1 \text{ k}\Omega} = 4 \text{ mA}$$

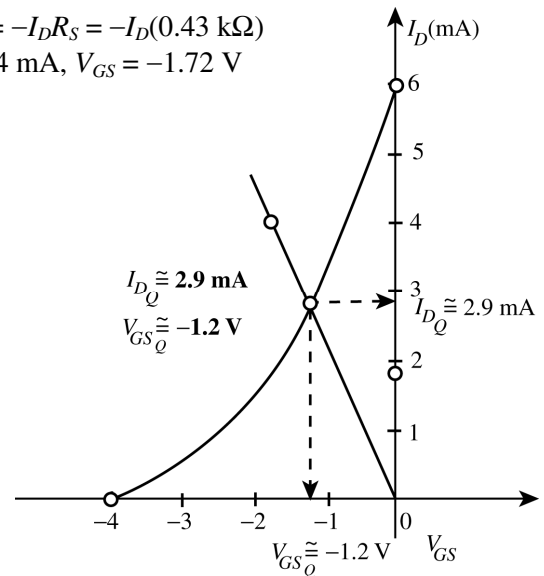
$$V_D = 16 \text{ V} - I_D(1.8 \text{ k}\Omega) = 16 \text{ V} - (4 \text{ mA})(1.8 \text{ k}\Omega) \\ = \mathbf{8.8 \text{ V}}$$

$$V_{GS} = V_G - V_S = 4 \text{ V} - 4 \text{ V} = \mathbf{0 \text{ V}} \quad (\text{requiring } I_D = I_{DSS} = 4 \text{ mA})$$

$$20. \quad \begin{aligned} V_{GS} &= 0 \text{ V}, I_D = I_{DSS} = 6 \text{ mA} \\ V_{GS} &= V_P = -4 \text{ V}, I_D = 0 \text{ mA} \\ V_{GS} &= V_P/2 = -2 \text{ V}, I_D = I_{DSS}/4 = 1.5 \text{ mA} \\ V_{GS} &= 0.3V_P = -1.2 \text{ V}, I_D = I_{DSS}/2 = 3 \text{ mA} \end{aligned}$$

$$\left. \begin{aligned} V_{GS} &= -I_D R_S = -I_D(0.43 \text{ k}\Omega) \\ I_D &= 4 \text{ mA}, V_{GS} = -1.72 \text{ V} \end{aligned} \right\}$$

$$(b) \quad V_{DS} = V_{DD} - I_D(R_D + R_S) \\ = 14 \text{ V} - 2.9 \text{ mA}(1.2 \text{ k}\Omega + 0.43 \text{ k}\Omega) \\ = \mathbf{9.27 \text{ V}} \\ V_D = V_{DD} - I_D R_D \\ = 14 \text{ V} - (2.9 \text{ mA})(1.2 \text{ k}\Omega) \\ = \mathbf{10.52 \text{ V}}$$



21. (a) $V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 8 \text{ mA}$

$V_{GS} = V_P = -8 \text{ V}, I_D = 0 \text{ mA}$

$V_{GS} = \frac{V_P}{2} = -4 \text{ V}, I_D = 2 \text{ mA}$

$V_{GS} = 0.3V_P = -2.4 \text{ V}, I_D = 4 \text{ mA}$

$V_{GS} = +1 \text{ V}, I_D = 10.125 \text{ mA}$

$V_{GS} = +2 \text{ V}, I_D = 12.5 \text{ mA}$

$V_{GS} = -V_{SS} - I_D R_S$

$= -(-4 \text{ V}) - I_D(0.39 \text{ k}\Omega)$

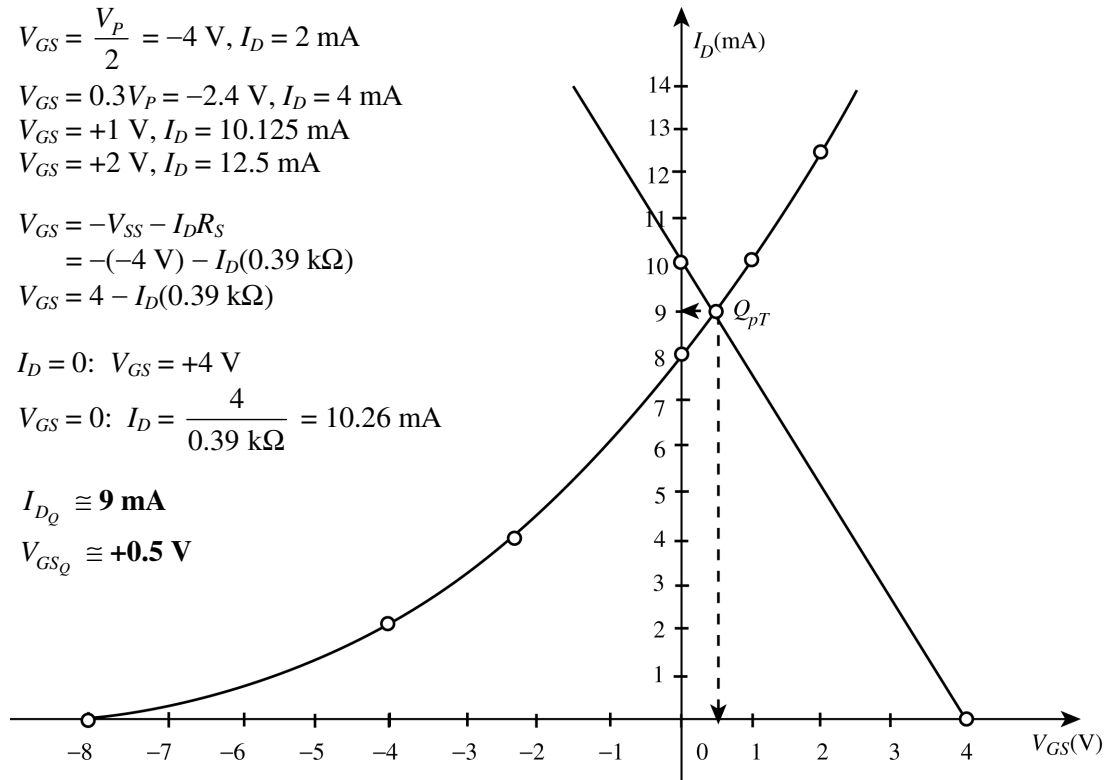
$V_{GS} = 4 - I_D(0.39 \text{ k}\Omega)$

$I_D = 0: V_{GS} = +4 \text{ V}$

$V_{GS} = 0: I_D = \frac{4}{0.39 \text{ k}\Omega} = 10.26 \text{ mA}$

$I_{DQ} \cong \mathbf{9 \text{ mA}}$

$V_{GSQ} \cong \mathbf{+0.5 \text{ V}}$



(b) $V_{DS} = V_{DD} - I_D(R_D + R_S) + V_{SS}$

$= 18 \text{ V} - 9 \text{ mA}(1.2 \text{ k}\Omega + 0.39 \text{ k}\Omega) + 4 \text{ V}$

$= 22 \text{ V} - 14.31 \text{ V}$

$= \mathbf{7.69 \text{ V}}$

$V_S = -(V_{GSQ}) = \mathbf{-0.5 \text{ V}}$

$$22. \quad I_D = k(V_{GS} - V_T)^2$$

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{Th})^2} =$$

$$\frac{5 \text{ mA}}{(7 \text{ V} - 4 \text{ V})^2} = \frac{5 \text{ mA}}{9 \text{ V}^2}$$

$$I_e = 0.556 \times 10^{-3} \text{ A/V}^2$$

and $I_D = \mathbf{0.556 \times 10^{-3} (V_{GS} - 4 \text{ V})^2}$

$$V_{DS} = V_{DD} - I_D(R_D + R_S)$$

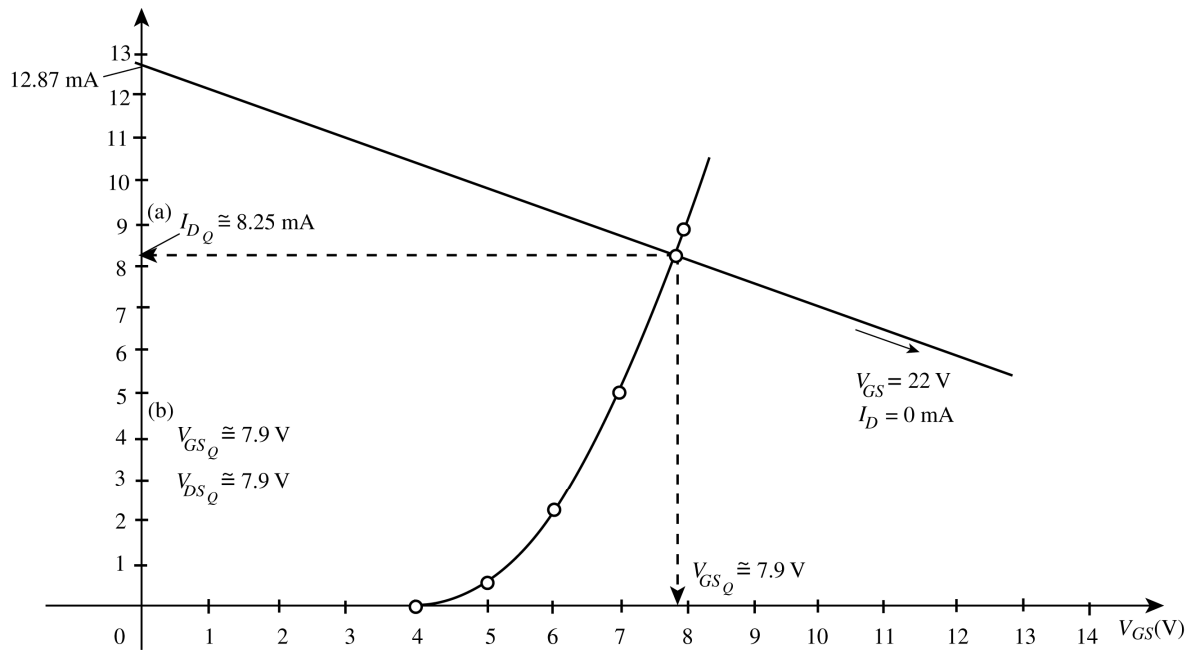
$$V_{DS} = 0 \text{ V}; I_D = \frac{V_{DD}}{R_D + R_S}$$

$$= \frac{22 \text{ V}}{1.2 \text{ k}\Omega + 0.51 \text{ k}\Omega}$$

$$= 12.87 \text{ mA}$$

$$I_D = 0 \text{ mA}, V_{DS} = V_{DD}$$

$$= 22 \text{ V}$$



$$(c) \quad V_D = V_{DD} - I_D R_D$$

$$= 22 \text{ V} - (8.25 \text{ mA})(1.2 \text{ k}\Omega)$$

$$= \mathbf{12.1 \text{ V}}$$

$$V_S = I_S R_S = I_D R_S$$

$$= (8.25 \text{ mA})(0.51 \text{ k}\Omega)$$

$$= \mathbf{4.21 \text{ V}}$$

$$(d) \quad V_{DS} = V_D - V_S$$

$$= 12.1 \text{ V} - 4.21 \text{ V}$$

$$= \mathbf{7.89 \text{ V}}$$

vs. 7.9 V obtained graphically

23. (a) $V_G = \frac{R_2}{R_1 + R_2} V_{DD} = \frac{6.8 \text{ M}\Omega}{10 \text{ M}\Omega + 6.8 \text{ M}\Omega} (24 \text{ V}) = 9.71 \text{ V}$

$$V_{GS} = V_G - I_D R_S$$

$$V_{GS} = 9.71 - I_D (0.75 \text{ k}\Omega)$$

$$\text{At } I_D = 0 \text{ mA, } V_{GS} = 9.71 \text{ V}$$

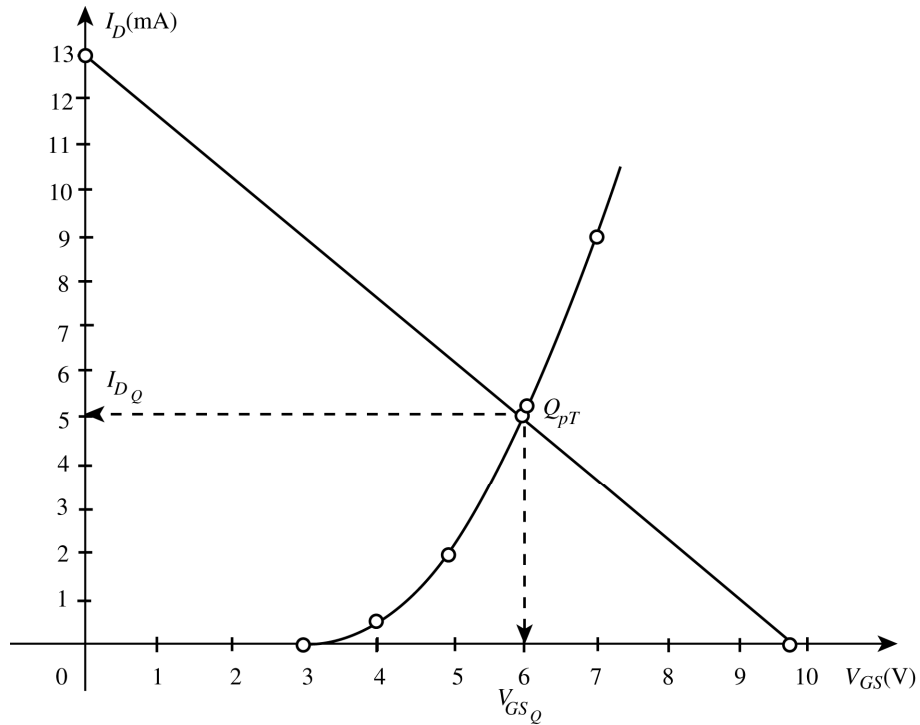
$$\text{At } V_{GS} = 0 \text{ V, } I_D = \frac{9.71 \text{ V}}{0.75 \text{ k}\Omega} = 12.95 \text{ mA}$$

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_{GS(Th)})^2} = \frac{5 \text{ mA}}{(6 \text{ V} - 3 \text{ V})^2} = \frac{5 \text{ mA}}{(3 \text{ V})^2}$$

$$= 0.556 \times 10^{-3} \text{ A/V}^2$$

$$\therefore I_D = \mathbf{0.556 \times 10^{-3} (V_{GS} - 3 \text{ V})^2}$$

V_{GS}	I_D
3 V	0 mA
4 V	0.556 mA
5 V	2.22 mA
6 V	5 mA
7 V	8.9 mA



$$I_{D_Q} \cong \mathbf{5 \text{ mA}}$$

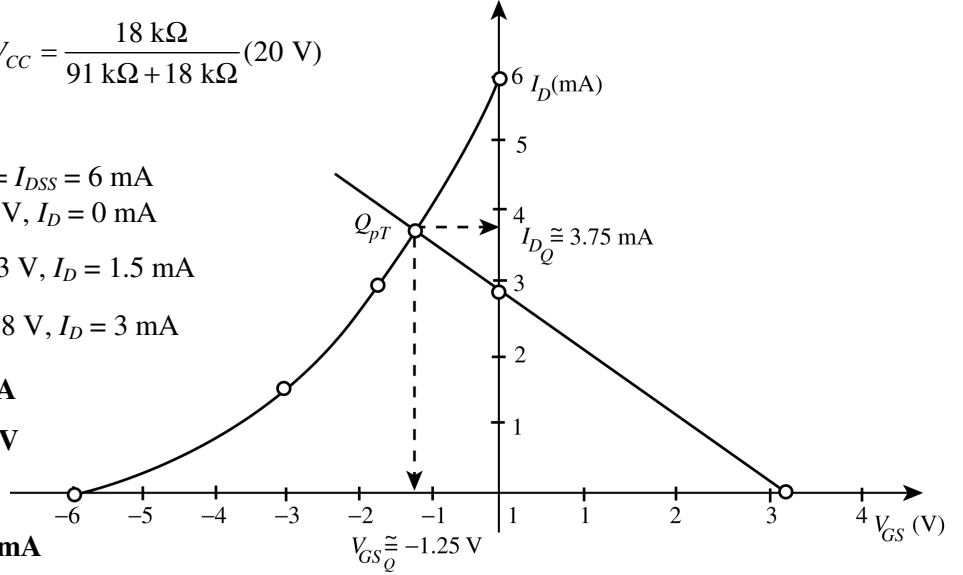
$$V_{GS_Q} \cong \mathbf{6 \text{ V}}$$

$$(b) \quad V_D = V_{DD} - I_D R_D = 24 \text{ V} - (5 \text{ mA})(2.2 \text{ k}\Omega) \\ = \mathbf{13 \text{ V}}$$

$$V_S = I_S R_S = I_D R_S \\ = (5 \text{ mA})(0.75 \text{ k}\Omega) \\ = \mathbf{3.75 \text{ V}}$$

$$24. \quad (a) \quad V_G = \frac{R_2}{R_1 + R_2} V_{CC} = \frac{18 \text{ k}\Omega}{91 \text{ k}\Omega + 18 \text{ k}\Omega} (20 \text{ V}) \\ = \mathbf{3.3 \text{ V}}$$

$$(b) \quad V_{GS} = 0 \text{ V}, I_D = I_{DSS} = 6 \text{ mA} \\ V_{GS} = V_P = -6 \text{ V}, I_D = 0 \text{ mA} \\ V_{GS} = \frac{V_P}{2} = -3 \text{ V}, I_D = 1.5 \text{ mA} \\ V_{GS} = V_P = -1.8 \text{ V}, I_D = 3 \text{ mA} \\ I_{D_Q} \cong \mathbf{3.75 \text{ mA}} \\ V_{GS_Q} \cong \mathbf{-1.25 \text{ V}}$$



$$(c) \quad I_E = I_D = \mathbf{3.75 \text{ mA}}$$

$$(d) \quad I_B = \frac{I_C}{\beta} = \frac{3.75 \text{ mA}}{160} = \mathbf{23.44 \mu A}$$

$$(e) \quad V_D = V_E = V_B - V_{BE} = V_{CC} - I_B R_B - V_{BE} = 20 \text{ V} - (23.44 \mu A)(330 \text{ k}\Omega) - 0.7 \text{ V} \\ = \mathbf{11.56 \text{ V}}$$

$$(f) \quad V_C = V_{CC} - I_C R_C = 20 \text{ V} - (3.75 \text{ mA})(1.1 \text{ k}\Omega) \\ = \mathbf{15.88 \text{ V}}$$

25. Testing:

$$\beta R_E \geq 10 R_2 \\ (100)(1.2 \text{ k}\Omega) \geq 10(10 \text{ k}\Omega) \\ 120 \text{ k}\Omega > 100 \text{ k}\Omega \text{ (satisfied)}$$

$$(a) \quad V_B = V_G = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{10 \text{ k}\Omega (16 \text{ V})}{40 \text{ k}\Omega + 10 \text{ k}\Omega} \\ = \mathbf{3.2 \text{ V}}$$

$$(b) \quad V_E = V_B - V_{BE} = 3.2 \text{ V} - 0.7 \text{ V} = \mathbf{2.5 \text{ V}}$$

$$(c) \quad I_E = \frac{V_E}{R_E} = \frac{2.5 \text{ V}}{1.2 \text{ k}\Omega} = \mathbf{2.08 \text{ mA}} \\ I_C \cong I_E = \mathbf{2.08 \text{ mA}} \\ I_D = I_C = \mathbf{2.08 \text{ mA}}$$

$$(d) \quad I_B = \frac{I_C}{\beta} = \frac{2.08 \text{ mA}}{100} = \mathbf{20.8 \mu A}$$

$$\begin{aligned} (e) \quad V_C &= V_G - V_{GS} \\ V_{GS} &= V_P - 1 - \sqrt{\frac{I_D}{I_{DSS}}} \\ &= (-6 \text{ V}) - 1 - \sqrt{\frac{2.08 \text{ mA}}{6 \text{ mA}}} \\ &= -2.47 \text{ V} \\ V_C &= 3.2 - (-2.47 \text{ V}) \\ &= \mathbf{5.67 \text{ V}} \\ V_S &= V_C = \mathbf{5.67 \text{ V}} \\ V_D &= V_{DD} - I_D R_D \\ &= 16 \text{ V} - (2.08 \text{ mA})(2.2 \text{ k}\Omega) \\ &= \mathbf{11.42 \text{ V}} \end{aligned}$$

$$(f) \quad V_{CE} = V_C - V_E = 5.67 \text{ V} - 2.5 \text{ V} = \mathbf{3.17 \text{ V}}$$

$$(g) \quad V_{DS} = V_D - V_S = 11.42 \text{ V} - 5.67 \text{ V} = \mathbf{5.75 \text{ V}}$$

$$\begin{aligned} 26. \quad V_{GS} &= V_P - 1 - \sqrt{\frac{I_D}{I_{DSS}}} = (-6 \text{ V}) - 1 - \sqrt{\frac{4 \text{ mA}}{8 \text{ mA}}} \\ &= -1.75 \text{ V} \\ V_{GS} &= -I_D R_S: R_S = -\frac{V_{GS}}{I_D} = \frac{-(-1.75 \text{ V})}{4 \text{ mA}} = \mathbf{0.44 \text{ k}\Omega} \\ R_D &= 3R_S = 3(0.44 \text{ k}\Omega) = \mathbf{1.32 \text{ k}\Omega} \\ \text{Standard values: } R_S &= \mathbf{0.43 \text{ k}\Omega} \\ R_D &= \mathbf{1.3 \text{ k}\Omega} \end{aligned}$$

$$\begin{aligned} 27. \quad V_{GS} &= V_P - 1 - \sqrt{\frac{I_D}{I_{DSS}}} = (-4 \text{ V}) - 1 - \sqrt{\frac{2.5 \text{ mA}}{10 \text{ mA}}} \\ &= \mathbf{-2 \text{ V}} \\ V_{GS} &= V_G - V_S \\ \text{and } V_S &= V_G - V_{GS} = 4 \text{ V} - (-2 \text{ V}) \\ &= 6 \text{ V} \\ R_S &= \frac{V_S}{I_D} = \frac{6 \text{ V}}{2.5 \text{ mA}} = \mathbf{2.4 \text{ k}\Omega} \text{ (a standard value)} \\ R_D &= 2.5R_S = 2.5(2.4 \text{ k}\Omega) = 6 \text{ k}\Omega \Rightarrow \text{use } \mathbf{6.2 \text{ k}\Omega} \\ V_G &= \frac{R_2 V_{DD}}{R_1 + R_2} \Rightarrow 4 \text{ V} = \frac{R_2 (24 \text{ V})}{22 \text{ M}\Omega + R_2} \Rightarrow 88 \text{ M}\Omega + 4R_2 = 24R_2 \\ & \quad 20R_2 = 88 \text{ M}\Omega \\ & \quad R_2 = 4.4 \text{ M}\Omega \\ & \quad \text{Use } R_2 = \mathbf{4.3 \text{ M}\Omega} \end{aligned}$$

28. $I_D = k(V_{GS} - V_T)^2$
 $\frac{I_D}{k} = (V_{GS} - V_T)^2$
 $\sqrt{\frac{I_D}{k}} = V_{GS} - V_T$
and $V_{GS} = V_T + \sqrt{\frac{I_D}{k}} = 4 \text{ V} + \sqrt{\frac{6 \text{ mA}}{0.5 \times 10^{-3} \text{ A/V}^2}} = 7.46 \text{ V}$
 $R_D = \frac{V_{R_D}}{I_D} = \frac{V_{DD} - V_{DS}}{I_D} = \frac{V_{DD} - V_{GS}}{I_D} = \frac{16 \text{ V} - 7.46 \text{ V}}{6 \text{ mA}} = \frac{8.54 \text{ V}}{6 \text{ mA}}$
 $= 1.42 \text{ k}\Omega$
Standard value: $R_D = \mathbf{0.75 \text{ k}\Omega}$
 $R_G = \mathbf{10 \text{ M}\Omega}$

29. (a) $I_D = I_S = \frac{V_S}{R_S} = \frac{4 \text{ V}}{1 \text{ k}\Omega} = 4 \text{ mA}$
 $V_{DS} = V_{DD} - I_D(R_D + R_S) = 12 \text{ V} - (4 \text{ mA})(2 \text{ k}\Omega + 1 \text{ k}\Omega)$
 $= 12 \text{ V} - (4 \text{ mA})(3 \text{ k}\Omega)$
 $= 12 \text{ V} - 12 \text{ V}$
 $= 0 \text{ V}$
JFET in saturation!
- (b) $V_S = 0 \text{ V}$ reveals that the JFET is nonconducting and the JFET is either defective or an open-circuit exists in the output circuit. V_S is at the same potential as the grounded side of the $1 \text{ k}\Omega$ resistor.
- (c) Typically, the voltage across the $1 \text{ M}\Omega$ resistor is $\cong 0 \text{ V}$. The fact that the voltage across the $1 \text{ M}\Omega$ resistor is equal to V_{DD} suggests that there is a short-circuit connection from gate to drain with $I_D = 0 \text{ mA}$. Either the JFET is defective or an improper circuit connection was made.

30. $V_G = \frac{75 \text{ k}\Omega(20 \text{ V})}{75 \text{ k}\Omega + 330 \text{ k}\Omega} = 3.7 \text{ V}$ (seems correct!)
 $V_{GS} = 3.7 \text{ V} - 6.25 \text{ V} = -2.55 \text{ V}$ (possibly okay)
 $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$
 $= 10 \text{ mA}(1 - (-2.55 \text{ V})/(-6 \text{ V}))^2$
 $= 3.3 \text{ mA}$ (reasonable)

However, $I_S = \frac{V_S}{R_S} = \frac{6.25 \text{ V}}{1 \text{ k}\Omega} = 6.25 \text{ mA} \neq 3.3 \text{ mA}$
 $V_{R_D} = I_D R_D = I_S R_D = (6.25 \text{ mA})(2.2 \text{ k}\Omega)$
 $= 13.75 \text{ V}$
and $V_{R_S} + V_{R_D} = 6.25 \text{ V} + 13.75 \text{ V}$
 $= \mathbf{20 \text{ V}} = V_{DD}$
 $\therefore V_{DS} = 0 \text{ V}$

1. Possible short-circuit from D-S.
2. Actual I_{DSS} and/or V_P may be larger in magnitude than specified.

$$\begin{aligned}
 31. \quad I_D = I_S &= \frac{V_S}{R_S} = \frac{6.25 \text{ V}}{1 \text{ k}\Omega} = 6.25 \text{ mA} \\
 V_{DS} &= V_{DD} - I_D(R_D + R_S) \\
 &= 20 \text{ V} - (6.25 \text{ mA})(2.2 \text{ k}\Omega + 1 \text{ k}\Omega) \\
 &= 20 \text{ V} - 20 \text{ V} \\
 &= 0 \text{ V (saturation condition)}
 \end{aligned}$$

$$V_G = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{75 \text{ k}\Omega(20 \text{ V})}{330 \text{ k}\Omega + 75 \text{ k}\Omega} = 3.7 \text{ V (as it should be)}$$

$$V_{GS} = V_G - V_S = 3.7 \text{ V} - 6.25 \text{ V} = -2.55 \text{ V}$$

$$\begin{aligned}
 I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2 = 10 \text{ mA} \left(1 - (-2.55 \text{ V})/(6 \text{ V})\right)^2 \\
 &= 3.3 \text{ mA} \neq 6.25 \text{ mA}
 \end{aligned}$$

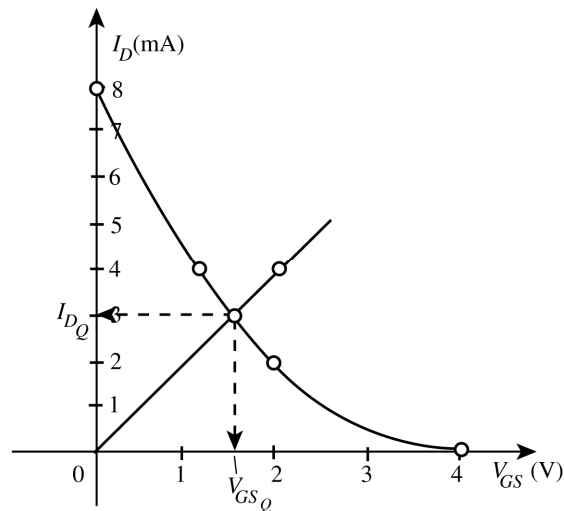
In all probability, an open-circuit exists between the voltage divider network and the gate terminal of the JFET with the transistor exhibiting saturation conditions.

$$\begin{aligned}
 32. \quad (a) \quad V_{GS} &= 0 \text{ V}, I_D = I_{DSS} = 8 \text{ mA} \\
 V_{GS} &= V_P = +4 \text{ V}, I_D = 0 \text{ mA} \\
 V_{GS} &= \frac{V_P}{2} = +2 \text{ V}, I_D = 2 \text{ mA} \\
 V_{GS} &= 0.3 V_P = 1.2 \text{ V}, I_D = 4 \text{ mA}
 \end{aligned}$$

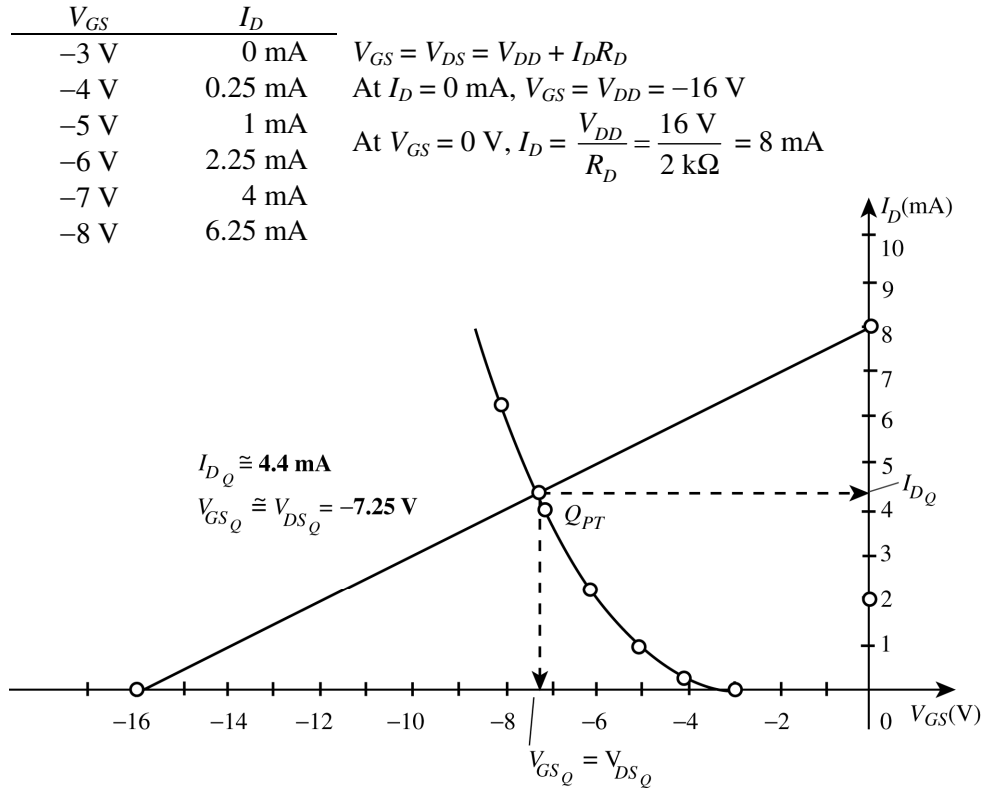
$$\begin{aligned}
 V_{GS} &= I_D R_S \\
 I_D &= 4 \text{ mA}; \\
 V_{GS} &= (4 \text{ mA})(0.51 \text{ k}\Omega) \\
 &= 2.04 \text{ V} \\
 I_{D_Q} &= \mathbf{3 \text{ mA}}, V_{GS_Q} = \mathbf{1.55 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad V_{DS} &= V_{DD} + I_D(R_D + R_S) \\
 &= -18 \text{ V} + (3 \text{ mA})(2.71 \text{ k}\Omega) \\
 &= \mathbf{-9.87 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad V_D &= V_{DD} - I_D R_D \\
 &= -18 \text{ V} - (3 \text{ mA})(2.2 \text{ k}\Omega) \\
 &= \mathbf{-11.4 \text{ V}}
 \end{aligned}$$



$$\begin{aligned}
 33. \quad k &= \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2} = \frac{4 \text{ mA}}{(-7 \text{ V} - (-3 \text{ V}))^2} = \frac{4 \text{ mA}}{(-4 \text{ V})^2} \\
 &= 0.25 \times 10^{-3} \text{ A/V}^2 \\
 I_D &= \mathbf{0.25 \times 10^{-3} (V_{GS} + 3 \text{ V})^2}
 \end{aligned}$$



(b) $V_{DS} = V_{GS} = \mathbf{-7.25 \text{ V}}$

(c) $V_D = V_{DS} = \mathbf{-7.25 \text{ V}}$
 or $V_{DS} = V_{DD} + I_D R_D$
 $= -16 \text{ V} + (4.4 \text{ mA})(2 \text{ k}\Omega)$
 $= -16 \text{ V} + 8.8 \text{ V}$
 $V_{DS} = \mathbf{-7.2 \text{ V} = V_D}$

$$34. \quad \frac{V_{GS}}{|V_P|} = \frac{-1.5 \text{ V}}{4 \text{ V}} = -0.375$$

Find -0.375 on the horizontal axis.

Then move vertically to the $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ curve.

Finally, move horizontally from the intersection with the curve to the left to the I_D/I_{DSS} axis.

$$\frac{I_D}{I_{DSS}} = 0.39$$

and $I_D = 0.39(12 \text{ mA}) = \mathbf{4.68 \text{ mA}}$ vs. 4.7 mA (#1)

$$\begin{aligned}
 V_{DS_Q} &= V_{DD} - I_D R_D = 14 \text{ V} - (4.7 \text{ mA})(1.8 \text{ k}\Omega) \\
 &= \mathbf{5.54 \text{ V}} \text{ vs. } 5.54 \text{ V} (\#1)
 \end{aligned}$$

$$35. \quad m = \frac{|V_P|}{I_{DSS} R_S} = \frac{4 \text{ V}}{(10 \text{ mA})(0.75 \text{ k}\Omega)} = \mathbf{0.533}$$

$$M = m \frac{V_{GG}}{|V_P|} = \frac{0.533(0)}{4 \text{ V}} = \mathbf{0}$$

Draw a straight line from $M = 0$ through $m = 0.533$ until it crosses the normalized curve of I_D

$$= I_{DSS} \left(1 - \frac{V_{GS}}{V_P}\right)^2. \text{ At the intersection with the curve drop a line down to determine}$$

$$\frac{V_{GS}}{|V_P|} = -0.49$$

$$\text{so that } V_{GS_Q} = -0.49V_P = -0.49(4 \text{ V})$$

$$= \mathbf{-1.96 \text{ V}} \text{ (vs. } -1.9 \text{ V \#6)}$$

If a horizontal line is drawn from the intersection to the left vertical axis we find

$$\frac{I_D}{I_{DSS}} = 0.27$$

$$\text{and } I_D = 0.27(I_{DSS}) = 0.27(10 \text{ mA}) = \mathbf{2.7 \text{ mA}}$$

(vs. 2.7 mA from #6)

$$(a) \quad V_{GS_Q} = \mathbf{-1.96 \text{ V}}, \quad I_{D_Q} = \mathbf{2.7 \text{ mA}}$$

$$(b) \quad -$$

$$(c) \quad -$$

$$(d) \quad V_{DS} = V_{DD} - I_D(R_D + R_S) = \mathbf{11.93 \text{ V}} \text{ (like \#6)}$$

$$V_D = V_{DD} - I_D R_D = \mathbf{13.95 \text{ V}} \text{ (like \#6)}$$

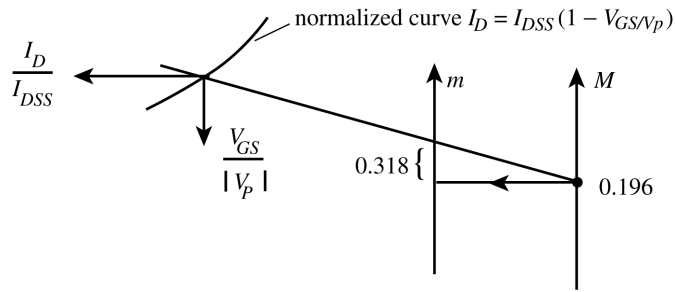
$$V_G = 0 \text{ V}, \quad V_S = I_D R_S = \mathbf{2.03 \text{ V}} \text{ (like \#6)}$$

$$36. \quad V_{GG} = \frac{R_2 V_{DD}}{R_1 + R_2} = \frac{110 \text{ k}\Omega(20 \text{ V})}{110 \text{ k}\Omega + 910 \text{ k}\Omega} = 2.16 \text{ V}$$

$$m = \frac{|V_P|}{I_{DSS} R_S} = \frac{3.5 \text{ V}}{(10 \text{ mA})(1.1 \text{ k}\Omega)} = 0.318$$

$$M = m \times \frac{V_{GG}}{|V_P|} = 0.318 \frac{(2.16 \text{ V})}{3.5} = 0.196$$

Find 0.196 on the vertical axis labeled M and mark the location. Move horizontally to the vertical axis labeled m and then add $m = 0.318$ to the vertical height ($\cong 1.318$ in total)—mark the spot. Draw a straight line through the two points located above, as shown below.



Continue the line until it intersects the $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ curve. At the intersection move horizontally to obtain the I_D/I_{DSS} ratio and move down vertically to obtain the $V_{GS}/|V_P|$ ratio.

$$\frac{I_D}{I_{DSS}} = 0.33 \text{ and } I_{D_Q} = 0.33(10 \text{ mA}) = \mathbf{3.3 \text{ mA}}$$

vs. 3.3 mA (#12)

$$\frac{V_{GS}}{|V_P|} = -0.425 \text{ and } V_{GS_Q} = -0.425(3.5 \text{ V})$$

$$= \mathbf{-1.49 \text{ V}}$$

vs. 1.5 V (#12)

37.
$$m = \frac{|V_P|}{I_{DSS} R_S} = \frac{6 \text{ V}}{(6 \text{ mA})(2.2 \text{ k}\Omega)}$$

$$= \mathbf{0.4545}$$

$$M = m \frac{V_{GG}}{|V_P|} = 0.4545 \frac{(4 \text{ V})}{(6 \text{ V})}$$

$$= \mathbf{0.303}$$

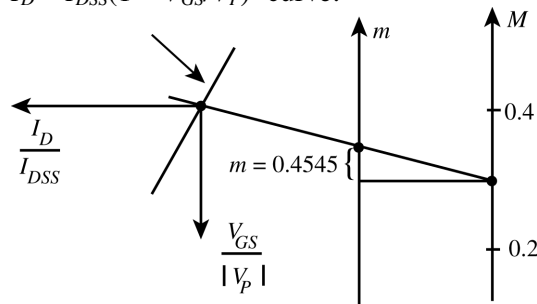
Find 0.303 on the vertical M axis.

Draw a horizontal line from $M = 0.303$ to the vertical m axis.

Add 0.4545 to the vertical location on the m axis defined by the horizontal line.

Draw a straight line between $M = 0.303$ and the point on the m axis resulting from the addition of $m = 0.4545$.

Continue the straight line as shown below until it crosses the normalized $I_D = I_{DSS}(1 - V_{GS}/V_P)^2$ curve:



At the intersection drop a vertical line to determine

$$\frac{V_{GS}}{|V_P|} = -0.34$$

$$\text{and } V_{GS_Q} = -0.34(6 \text{ V})$$

$$= \mathbf{-2.04 \text{ V}} \text{ (vs. } -2 \text{ V from problem 15)}$$

At the intersection draw a horizontal line to the I_D/I_{DSS} axis to determine

$$\frac{I_D}{I_{DSS}} = 0.46$$

$$\text{and } I_{D_Q} = 0.46(6 \text{ mA})$$

$$= \mathbf{2.76 \text{ mA}} \text{ (vs. } 2.7 \text{ mA from problem 15)}$$

$$\text{(a) } I_{D_Q} = \mathbf{2.76 \text{ mA}}, V_{GS_Q} = \mathbf{-2.04 \text{ V}}$$

$$\begin{aligned} \text{(b) } V_{DS} &= V_{DD} + V_{SS} - I_D(R_D + R_S) \\ &= 16 \text{ V} + 4 \text{ V} - (2.76 \text{ mA})(4.4 \text{ k}\Omega) \\ &= \mathbf{7.86 \text{ V}} \text{ (vs. } 8.12 \text{ V from problem 15)} \end{aligned}$$

$$\begin{aligned} V_S &= -V_{SS} + I_D R_S = -4 \text{ V} + (2.76 \text{ mA})(2.2 \text{ k}\Omega) \\ &= -4 \text{ V} + 6.07 \text{ V} \\ &= \mathbf{2.07 \text{ V}} \text{ (vs. } 1.94 \text{ V from problem 15)} \end{aligned}$$

Chapter 8

$$1. \quad g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(12 \text{ mA})}{|-4 \text{ V}|} = \mathbf{6 \text{ mS}}$$

$$2. \quad g_{m0} = \frac{2I_{DSS}}{|V_P|} \quad |V_P| = \frac{2I_{DSS}}{g_{m0}} = \frac{2(12 \text{ mA})}{10 \text{ mS}} = 2.4 \text{ V}$$

$$V_P = \mathbf{-2.4 \text{ V}}$$

$$3. \quad g_{m0} = \frac{2I_{DSS}}{|V_P|} \Rightarrow I_{DSS} = \frac{(g_{m0})(|V_P|)}{2} = \frac{5 \text{ mS}(4 \text{ V})}{2} = \mathbf{10 \text{ mA}}$$

$$4. \quad g_m = g_{m0} \left(1 - \frac{V_{GSQ}}{V_P}\right) = \frac{2(12 \text{ mA})}{|-3 \text{ V}|} \left(1 - \frac{-0.5 \text{ V}}{-3 \text{ V}}\right) = \mathbf{6.67 \text{ mS}}$$

$$5. \quad g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GSQ}}{V_P}\right)$$

$$6 \text{ mS} = \frac{2I_{DSS}}{2.5 \text{ V}} \left(1 - \frac{-1 \text{ V}}{-2.5 \text{ V}}\right)$$

$$I_{DSS} = \mathbf{12.5 \text{ mA}}$$

$$6. \quad g_m = g_{m0} \sqrt{\frac{I_D}{I_{DSS}}} = \frac{2I_{DSS}}{|V_P|} \sqrt{\frac{I_{DSS}/4}{I_{DSS}}} = \frac{2(10 \text{ mA})}{5 \text{ V}} \sqrt{\frac{1}{4}}$$

$$= \frac{20 \text{ mA}}{5 \text{ V}} \cdot \frac{1}{2} = \mathbf{2 \text{ mS}}$$

$$7. \quad g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(8 \text{ mA})}{5 \text{ V}} = 3.2 \text{ mS}$$

$$g_m = g_{m0} \left(1 - \frac{V_{GSQ}}{V_P}\right) = 3.2 \text{ mS} \left(1 - \frac{V_P/4}{V_P}\right) = 3.2 \text{ mS} \left(1 - \frac{1}{4}\right) = 3.2 \text{ mS} \cdot \frac{3}{4}$$

$$= \mathbf{2.4 \text{ mS}}$$

$$8. \quad (a) \quad g_m = y_{fs} = \mathbf{4.5 \text{ mS}}$$

$$(b) \quad r_d = \frac{1}{y_{os}} = \frac{1}{25 \mu\text{S}} = \mathbf{40 \text{ k}\Omega}$$

$$9. \quad g_m = g_{fs} = 4.5 \text{ mS}$$

$$r_d = \frac{1}{g_{os}} = \frac{1}{25 \mu\text{S}} = 40 \text{ k}\Omega$$

$$Z_o = r_d = \mathbf{40 \text{ k}\Omega}$$

$$A_v(\text{FET}) = -g_m r_d = -(4.5 \text{ mS})(40 \text{ k}\Omega) = \mathbf{-180}$$

10. $A_v = -g_m r_d \Rightarrow g_m = \frac{-A_v}{r_d} = -\frac{(-200)}{(100 \text{ k}\Omega)} = \mathbf{2 \text{ mS}}$
11. (a) $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10 \text{ mA})}{5 \text{ V}} = \mathbf{4 \text{ mS}}$
- (b) $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{9 \text{ mA} - 7 \text{ mA}}{0.85 \text{ V} - 0.3 \text{ V}} = \mathbf{3.64 \text{ mS}}$
- (c) Eq. 8.6: $g_m = g_{m0} \left(1 - \frac{V_{GSQ}}{V_P}\right) = 4 \text{ mS} \left(1 - \frac{-0.5 \text{ V}}{-5 \text{ V}}\right) = \mathbf{3.6 \text{ mS}}$
- (d) $g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{8 \text{ mA} - 5 \text{ mA}}{1.5 \text{ V} - 0.5 \text{ V}} = \mathbf{3 \text{ mS}}$
- (e) $g_m = g_{m0} \left(1 - \frac{V_{GSQ}}{V_P}\right) = 4 \text{ mS} \left(1 - \frac{-1 \text{ V}}{-5 \text{ V}}\right) = \mathbf{3.2 \text{ mS}}$
12. (a) $r_d = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS} \text{ constant}} = \frac{(15 \text{ V} - 5 \text{ V})}{(9.1 \text{ mA} - 8.8 \text{ mA})} = \frac{10 \text{ V}}{0.3 \text{ mA}} = \mathbf{33.33 \text{ k}\Omega}$
- (b) At $V_{DS} = 10 \text{ V}$, $I_D = 9 \text{ mA}$ on $V_{GS} = 0 \text{ V}$ curve
 $\therefore g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(9 \text{ mA})}{4 \text{ V}} = \mathbf{4.5 \text{ mS}}$
13. From 2N4220 data:
 (a) $g_m = y_{fs} = 750 \mu\text{S} = \mathbf{0.75 \text{ mS}}$
 (b) $r_d = \frac{1}{y_{os}} = \frac{1}{10 \mu\text{S}} = \mathbf{100 \text{ k}\Omega}$
14. (a) $g_m (@ V_{GS} = -6 \text{ V}) = \mathbf{0}$, $g_m (@ V_{GS} = 0 \text{ V}) = g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(12 \text{ mA})}{6 \text{ V}} = \mathbf{4 \text{ mS}}$
- (b) $g_m (@ I_D = 0 \text{ mA}) = \mathbf{0}$, $g_m (@ I_D = I_{DSS} = 12 \text{ mA}) = g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(12 \text{ mA})}{6 \text{ V}} = \mathbf{4 \text{ mS}}$
15. $g_m = y_{fs} = \mathbf{5.6 \text{ mS}}$, $r_d = \frac{1}{y_{os}} = \frac{1}{15 \mu\text{S}} = \mathbf{66.67 \text{ k}\Omega}$

$$16. \quad g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(10 \text{ mA})}{4 \text{ V}} \left(1 - \frac{-2 \text{ V}}{-4 \text{ V}}\right) = \mathbf{2.5 \text{ mS}}$$

$$r_d = \frac{1}{y_{os}} = \frac{1}{25 \mu\text{S}} = \mathbf{40 \text{ k}\Omega}$$

$$17. \quad \text{Graphically, } V_{GS_Q} = -1.5 \text{ V}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(10 \text{ mA})}{6 \text{ V}} \left(1 - \frac{-1.5 \text{ V}}{-6 \text{ V}}\right) = 2.5 \text{ mS}$$

$$Z_i = R_G = \mathbf{1 \text{ M}\Omega}$$

$$Z_o = R_D \parallel r_d = 1.8 \text{ k}\Omega \parallel 40 \text{ k}\Omega = \mathbf{1.72 \text{ k}\Omega}$$

$$A_v = -g_m(R_D \parallel r_d) = -(2.5 \text{ mS})(1.72 \text{ k}\Omega) \\ = \mathbf{-4.3}$$

$$18. \quad (a) \quad V_{GS_Q} = -1.5 \text{ V, Graphically, } I_{D_Q} = 1.25 \text{ mA}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(5 \text{ mA})}{3 \text{ V}} \left(1 - \frac{-1.5 \text{ V}}{-3 \text{ V}}\right) = 1.67 \text{ mS}$$

$$Z_i = R_G = \mathbf{1 \text{ M}\Omega}$$

$$Z_o = R_D \parallel r_d \\ = 1.8 \text{ k}\Omega \parallel 40 \text{ k}\Omega \\ = \mathbf{1.72 \text{ k}\Omega}$$

$$A_v = -g_m(R_D \parallel r_d) = -(1.67 \text{ mS})(1.72 \text{ k}\Omega) = \mathbf{-2.87}$$

$$(b) \quad Z_i \text{ and } Z_o \text{ the same, 33\% drop in } A_v.$$

$$19. \quad (a) \quad V_{GS_Q} = -2.5 \text{ V}$$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2(10 \text{ mA})}{4 \text{ V}} \left(1 - \frac{-2.5 \text{ V}}{-4 \text{ V}}\right) \\ = 5 \text{ mS}[1 - 0.625] = 5 \text{ mS}[0.375] = 1.875 \text{ mS}$$

$$Z_i = \mathbf{2 \text{ M}\Omega}$$

$$Z_o = 4.7 \text{ k}\Omega \parallel 20 \text{ k}\Omega = \mathbf{3.81 \text{ k}\Omega}$$

$$A_v = -g_m(r_d \parallel R_D) = -1.875 \text{ mS}(3.81 \text{ k}\Omega) \\ = \mathbf{-7.14}$$

$$(b) \quad Z_i = \mathbf{2 \text{ M}\Omega} \text{ (the same), } Z_o = 4.7 \text{ k}\Omega \parallel 40 \text{ k}\Omega = \mathbf{4.21 \text{ k}\Omega} \text{ (increased)}$$

$$A_v = -g_m(R_D \parallel r_d) = -1.875 \text{ mS}[4.21 \text{ k}\Omega] = \mathbf{-7.89} \text{ (} A_v \text{ increased)}$$

20. $g_m = y_{fs} = 3000 \mu S = 3 \text{ mS}$
 $r_d = \frac{1}{y_{os}} = \frac{1}{50 \mu S} = 20 \text{ k}\Omega$
 $Z_i = R_G = \mathbf{10 \text{ M}\Omega}$
 $Z_o = r_d \parallel R_D = 20 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = \mathbf{2.83 \text{ k}\Omega}$
 $A_v = -g_m(r_d \parallel R_D)$
 $= -(3 \text{ mS})(2.83 \text{ k}\Omega)$
 $= \mathbf{-8.49}$
21. $g_m = 3 \text{ mS}, r_d = 20 \text{ k}\Omega$
 $Z_i = \mathbf{10 \text{ M}\Omega}$
 $Z_o = \frac{R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}} = \frac{3.3 \text{ k}\Omega}{1 + (3 \text{ mS})(1.1 \text{ k}\Omega) + \frac{3.3 \text{ k}\Omega + 1.1 \text{ k}\Omega}{20 \text{ k}\Omega}}$
 $= \frac{3.3 \text{ k}\Omega}{1 + 3.3 + 0.22} = \frac{3.3 \text{ k}\Omega}{4.52} = \mathbf{730 \Omega}$
 $A_v = \frac{-g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}} = \frac{-(3 \text{ mS})(3.3 \text{ k}\Omega)}{1 + (3 \text{ mS})(1.1 \text{ k}\Omega) + \frac{3.3 \text{ k}\Omega + 1.1 \text{ k}\Omega}{20 \text{ k}\Omega}}$
 $= \frac{-9.9}{1 + 3.3 + 0.22} = -\frac{9.9}{4.52} = \mathbf{-2.19}$
22. $g_m = y_{fs} = 3000 \mu S = 3 \text{ mS}$
 $r_d = \frac{1}{y_{os}} = \frac{1}{10 \mu S} = 100 \text{ k}\Omega$
 $Z_i = R_G = \mathbf{10 \text{ M}\Omega}$ (the same)
 $Z_o = r_d \parallel R_D = 100 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega = \mathbf{3.195 \text{ k}\Omega}$ (higher)
 $A_v = -g_m(r_d \parallel R_D)$
 $= -(3 \text{ mS})(3.195 \text{ k}\Omega)$
 $= \mathbf{-9.59}$ (higher)
23. (a) $A_v = -g_m R_D$
 $-2 = -g_m (2.7 \text{ k}\Omega)$
 $g_m = \frac{2}{2.7 \text{ k}\Omega} = 0.74 \text{ mS}$
 $g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right)$
 $0.74 \text{ mS} = \frac{2(9 \text{ mA})}{8 \text{ V}} \left(1 - \frac{V_{GS}}{-8 \text{ V}}\right)$
 $0.74 \text{ mS} = 2.25 \text{ mS} \left(1 - \frac{V_{GS}}{-8 \text{ V}}\right) = 2.25 \text{ mS} + 2.25 \text{ mS} \frac{V_{GS}}{8 \text{ V}}$
 $-1.51 \text{ mS} = 2.25 \text{ mS} \frac{V_{GS}}{8 \text{ V}}$

$$-0.671 = \frac{V_{GS}}{8 \text{ V}}, \quad V_{GS_Q} = -(0.671)(8 \text{ V}) = \mathbf{-5.37 \text{ V}}$$

$$V_{GS_Q} = I_D R_D$$

$$R_S = \frac{V_{GS_Q}}{I_D} = \frac{5.37 \text{ V}}{1.4 \text{ mA}} \text{ (from } Q \text{ pt. on graph)} = \mathbf{3.83 \text{ k}\Omega}$$

$$(b) \quad A_v = -g_m R_D$$

$$-2 = -g_m (2.7 \text{ k}\Omega \parallel 30 \text{ k}\Omega) = -g_m (2.48 \text{ k}\Omega)$$

$$g_m = \frac{2}{2.48 \text{ k}\Omega} = 0.81 \text{ mS}$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2(9 \text{ mA})}{8 \text{ V}} \left(1 - \frac{V_{GS}}{-8 \text{ V}}\right) = 2.25 \text{ mS} + 2.25 \text{ mS} \frac{V_{GS}}{8 \text{ V}}$$

$$0.81 \text{ mS} = 2.25 \text{ mS} + 2.25 \text{ mS} \frac{V_{GS}}{8 \text{ V}}$$

$$-1.44 \text{ mS} = 2.25 \text{ mS} \frac{V_{GS}}{8 \text{ V}}$$

$$-0.640 \frac{V_{GS}}{8 \text{ V}}, \quad V_{GS_Q} = -(0.640)(8 \text{ V}) = \mathbf{-5.12 \text{ V}}$$

$$V_{GS_Q} = I_D R_D$$

$$R_S = \frac{V_{GS_Q}}{I_{D_Q}} = \frac{5.12 \text{ V}}{1.5 \text{ mA}} \text{ (from } Q \text{ pt. on graph)} = \mathbf{3.41 \text{ k}\Omega}$$

Repeat with new value of g_m . R_S decreased to 3.41 k Ω from 3.83 k Ω .

$$24. \quad V_{GS_Q} = 0 \text{ V}, \quad g_m = g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(6 \text{ mA})}{6 \text{ V}} = 2 \text{ mS}, \quad r_d = \frac{1}{y_{os}} = \frac{1}{40 \mu\text{S}} = 25 \text{ k}\Omega$$

$$Z_i = \mathbf{1 \text{ M}\Omega}$$

$$Z_o = r_d \parallel R_D = 25 \text{ k}\Omega \parallel 2 \text{ k}\Omega = \mathbf{1.852 \text{ k}\Omega}$$

$$A_v = -g_m (r_d \parallel R_D) = -(2 \text{ mS})(1.852 \text{ k}\Omega) \cong \mathbf{-3.7}$$

$$25. \quad V_{GS_Q} = -0.95 \text{ V}$$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right)$$

$$= \frac{2(12 \text{ mA})}{3 \text{ V}} \left(1 - \frac{-0.95 \text{ V}}{-3 \text{ V}}\right)$$

$$= 5.47 \text{ mS}$$

$$Z_i = 82 \text{ M}\Omega \parallel 11 \text{ M}\Omega = \mathbf{9.7 \text{ M}\Omega}$$

$$Z_o = r_d \parallel R_D = 50 \text{ k}\Omega \parallel 2 \text{ k}\Omega = \mathbf{1.92 \text{ k}\Omega}$$

$$A_v = -g_m (r_d \parallel R_D) = -(5.47 \text{ mS})(1.92 \text{ k}\Omega) = \mathbf{-10.5}$$

$$V_o = A_v V_i = (-10.5)(20 \text{ mV}) = \mathbf{-210 \text{ mV}}$$

26. $V_{GS_Q} = -0.95 \text{ V}$ (as before), $g_m = 5.47 \text{ mS}$ (as before)

$Z_i = 9.7 \text{ M}\Omega$ as before

$$Z_o = \frac{R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

but $r_d \geq 10(R_D + R_S)$

$$\therefore Z_o = \frac{R_D}{1 + g_m R_S} = \frac{2 \text{ k}\Omega}{1 + (5.47 \text{ mS})(0.61 \text{ k}\Omega)} = \frac{2 \text{ k}\Omega}{1 + 3.337} = \frac{2 \text{ k}\Omega}{4.337} = 461.1 \text{ }\Omega$$

$$A_v = \frac{-g_m R_D}{1 + g_m R_S} \text{ since } r_d \geq 10(R_D + R_S)$$

$$= \frac{-(5.47 \text{ mS})(2 \text{ k}\Omega)}{4.337 \text{ (from above)}} = -\frac{10.94}{4.337} = -2.52 \text{ (a big reduction)}$$

$V_o = A_v V_i = (-2.52)(20 \text{ mV}) = -50.40 \text{ mV}$ (compared to -214.4 mV earlier)

27. $V_{GS_Q} = -0.95 \text{ V}$, g_m (problem 23) = 5.47 mS

Z_i (the same) = $9.7 \text{ M}\Omega$

Z_o (reduced) = $r_d \parallel R_D = 20 \text{ k}\Omega \parallel 2 \text{ k}\Omega = 1.82 \text{ k}\Omega$

A_v (reduced) = $-g_m(r_d \parallel R_D) = -(5.47 \text{ mS})(1.82 \text{ k}\Omega) = -9.94$

V_o (reduced) = $A_v V_i = (-9.94)(20 \text{ mV}) = -198.8 \text{ mV}$

28. $V_{GS_Q} = -0.95 \text{ V}$ (as before), $g_m = 5.47 \text{ mS}$ (as before)

$Z_i = 9.7 \text{ M}\Omega$ as before

$$Z_o = \frac{R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}} \text{ since } r_d < 10(R_D + R_S)$$

$$= \frac{2 \text{ k}\Omega}{1 + (5.47 \text{ mS})(0.61 \text{ k}\Omega) + \frac{2 \text{ k}\Omega + 0.61 \text{ k}\Omega}{20 \text{ k}\Omega}}$$

$$= \frac{2 \text{ k}\Omega}{1 + 3.33 + 0.13} = \frac{2 \text{ k}\Omega}{4.46}$$

= $448.4 \text{ }\Omega$ (slightly less than $461.1 \text{ }\Omega$ obtained in problem 24)

$$A_v = \frac{-g_m R_D}{1 + g_m R_S + \frac{R_D + R_S}{r_d}}$$

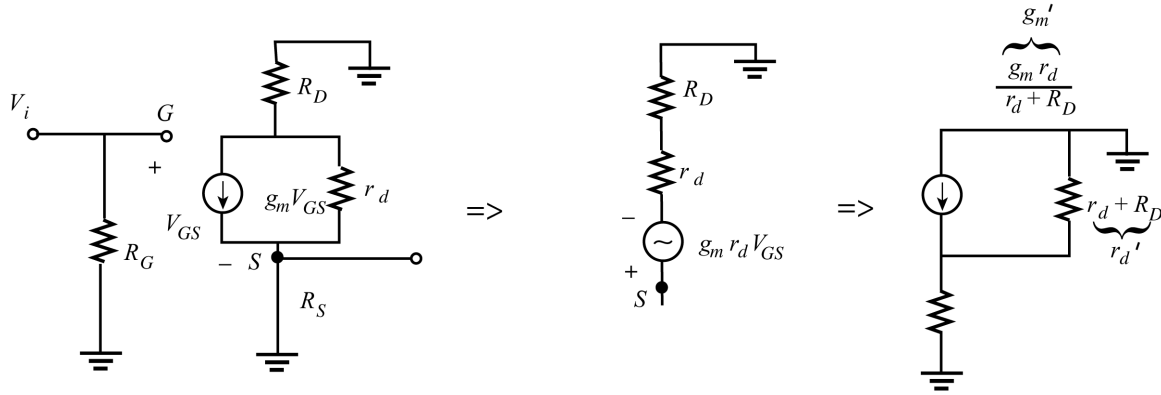
$$= \frac{-(5.47 \text{ mS})(2 \text{ k}\Omega)}{1 + (5.47 \text{ mS})(0.61 \text{ k}\Omega) + \frac{2 \text{ k}\Omega + 0.61 \text{ k}\Omega}{20 \text{ k}\Omega}}$$

$$= \frac{-10.94}{1 + 3.33 + 0.13} = \frac{-10.94}{4.46} = -2.45 \text{ slightly less than } -2.52 \text{ obtained in problem 24)}$$

29. $V_{GS_Q} = -1.75 \text{ V}$, $g_m = 2.14 \text{ mS}$
 $r_d \geq 10R_D$, $\therefore Z_i \cong R_S \parallel 1/g_m = 1.5 \text{ k}\Omega \parallel 1/2.14 \text{ mS}$
 $= 1.5 \text{ k}\Omega \parallel 467.29 \Omega$
 $= \mathbf{356.3 \Omega}$
 $r_d \geq 10R_D$, $\therefore Z_o \cong R_D = \mathbf{3.3 \text{ k}\Omega}$
 $r_d \geq 10R_D$, $\therefore A_v \cong g_m R_D = (2.14 \text{ mS})(3.3 \text{ k}\Omega) = \mathbf{7.06}$
 $V_o = A_v V_i = (7.06)(4 \text{ mV}) = \mathbf{28.24 \text{ mV}}$
30. $V_{GS_Q} = -1.75 \text{ V}$, $g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(8 \text{ mA})}{2.8 \text{ V}} \left(1 - \frac{-1.75 \text{ V}}{-2.8 \text{ V}}\right) = 2.14 \text{ mS}$
 $Z_i = R_S \parallel \frac{r_d + R_D}{1 + g_m r_d} = 1.5 \text{ k}\Omega \parallel \frac{25 \text{ k}\Omega + 3.3 \text{ k}\Omega}{1 + (2.14 \text{ mS})(25 \text{ k}\Omega)} = 1.5 \text{ k}\Omega \parallel \frac{28.3 \text{ k}\Omega}{54.5}$
 $= 1.5 \text{ k}\Omega \parallel 0.52 \text{ k}\Omega = \mathbf{386.1 \Omega}$
 $Z_o = R_D \parallel r_d = 3.3 \text{ k}\Omega \parallel 25 \text{ k}\Omega = \mathbf{2.92 \text{ k}\Omega}$
 $A_v = \frac{g_m R_D + R_D / r_d}{1 + R_D / r_d} = \frac{(2.14 \text{ mS})(3.3 \text{ k}\Omega) + 3.3 \text{ k}\Omega / 25 \text{ k}\Omega}{1 + 3.3 \text{ k}\Omega / 25 \text{ k}\Omega}$
 $= \frac{7.062 + 0.132}{1 + 0.132} = \frac{7.194}{1.132} = 6.36$
 $V_o = A_v V_i = (6.36)(4 \text{ mV}) = \mathbf{25.44 \text{ mV}}$
31. $V_{GS_Q} \cong -1.2 \text{ V}$, $g_m = 2.63 \text{ mS}$
 $r_d \geq 10R_D$, $\therefore Z_i \cong R_S \parallel 1/g_m = 1 \text{ k}\Omega \parallel 1/2.63 \text{ mS} = 1 \text{ k}\Omega \parallel 380.2 \Omega = \mathbf{275.5 \Omega}$
 $Z_o \cong R_D = \mathbf{2.2 \text{ k}\Omega}$
 $A_v \cong g_m R_D = (2.63 \text{ mS})(2.2 \text{ k}\Omega) = \mathbf{5.79}$
32. $V_{GS_Q} = -2.85 \text{ V}$, $g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(9 \text{ mA})}{4.5 \text{ V}} \left(1 - \frac{-2.85 \text{ V}}{-4.5 \text{ V}}\right) = 1.47 \text{ mS}$
 $Z_i = R_G = \mathbf{10 \text{ M}\Omega}$
 $Z_o = r_d \parallel R_S \parallel 1/g_m = 40 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel \underbrace{1/1.47 \text{ mS}}_{680.27 \Omega} = \mathbf{512.9 \Omega}$
 $A_v = \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)} = \frac{(1.47 \text{ mS})(40 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)}{1 + (1.47 \text{ mS})(40 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)} = \frac{3.065}{1 + 3.065}$
 $= \mathbf{0.754}$
33. $V_{GS_Q} = -2.85 \text{ V}$, $g_m = 1.47 \text{ mS}$
 $Z_i = \mathbf{10 \text{ M}\Omega}$ (as in problem 32)
 $Z_o = r_d \parallel R_S \parallel 1/g_m = \underbrace{20 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega}_{1.982 \text{ k}\Omega} \parallel 680.27 \Omega = \mathbf{506.4 \Omega} < 512.9 \Omega \text{ (#27)}$
 $A_v = \frac{g_m (r_d \parallel R_S)}{1 + g_m (r_d \parallel R_S)} = \frac{1.47 \text{ mS}(20 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)}{1 + 1.47 \text{ mS}(20 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)} = \frac{2.914}{1 + 2.914}$
 $= \mathbf{0.745} < 0.754 \text{ (#32)}$

34. $V_{GS_Q} = -3.8 \text{ V}$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(6 \text{ mA})}{6 \text{ V}} \left(1 - \frac{-3.8 \text{ V}}{-6 \text{ V}}\right) = 0.733 \text{ mS}$$



The network now has the format examined in the text and

$$Z_i = R_G = \mathbf{10 \text{ M}\Omega} \quad r'_d = r_d + R_D = 30 \text{ k}\Omega + 3.3 \text{ k}\Omega = 33.3 \text{ k}\Omega$$

$$\begin{aligned} Z_o = r'_d \parallel R_S \parallel 1/g'_m &= g'_m = \frac{g_m r_d}{r_d + R_D} = \frac{(0.733 \text{ mS})(30 \text{ k}\Omega)}{30 \text{ k}\Omega + 3.3 \text{ k}\Omega} = \frac{21.99}{33.3 \text{ k}\Omega} = 0.66 \text{ mS} \\ &= 33.3 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega \parallel 1/0.66 \text{ mS} \\ &= 3 \text{ k}\Omega \parallel 1.52 \text{ k}\Omega \\ &\cong \mathbf{1 \text{ k}\Omega} \end{aligned}$$

$$\begin{aligned} A_v &= \frac{g'_m (r'_d \parallel R_S)}{1 + g'_m (r'_d \parallel R_S)} = \frac{0.66 \text{ mS}(3 \text{ k}\Omega)}{1 + 0.66 \text{ mS}(3 \text{ k}\Omega)} = \frac{1.98}{1 + 1.98} = \frac{1.98}{2.98} \\ &= \mathbf{0.66} \end{aligned}$$

35. $r_d = \frac{1}{y_{os}} = \frac{1}{20 \mu\text{S}} = 50 \text{ k}\Omega, V_{GS_Q} = 0 \text{ V}$

$$g_m = g_{m0} = \frac{2I_{DSS}}{V_P} = \frac{2(8 \text{ mA})}{3} = 5.33 \text{ mS}$$

$$A_v = -g_m R_D = -(5.33 \text{ mS})(1.1 \text{ k}\Omega) = -5.863$$

$$V_o = A_v V_i = (-5.863)(2 \text{ mV}) = \mathbf{11.73 \text{ mV}}$$

36. $V_{GS_Q} = -0.75 \text{ V}, g_m = 5.4 \text{ mS}$

$$Z_i = \mathbf{10 \text{ M}\Omega}$$

$$r_o \geq 10R_D, \therefore Z_o \cong R_D = \mathbf{1.8 \text{ k}\Omega}$$

$$\begin{aligned} r_o \geq 10R_D, \therefore A_v &\cong -g_m R_D = -(5.4 \text{ mS})(1.8 \text{ k}\Omega) \\ &= \mathbf{-9.72} \end{aligned}$$

37. $Z_i = 10 \text{ M}\Omega$
 $Z_o = r_d \parallel R_D = 25 \text{ k}\Omega \parallel 1.8 \text{ k}\Omega = 1.68 \text{ k}\Omega$
 $A_v = -g_m(r_d \parallel R_D)$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(12 \text{ mA})}{3.5 \text{ V}} \left(1 - \frac{-0.75 \text{ V}}{-3.5 \text{ V}}\right) = 5.4 \text{ mS}$$

 $A_v = -(5.4 \text{ mS})(1.68 \text{ k}\Omega)$
 $= -9.07$
38. $g_m = y_{fs} = 6000 \mu\text{S} = 6 \text{ mS}$
 $r_d = \frac{1}{y_{os}} = \frac{1}{35 \mu\text{S}} = 28.57 \text{ k}\Omega$
 $r_d \leq 10R_D, \therefore A_v = -g_m(r_d \parallel R_D)$

$$= -(6 \text{ mS})(\underbrace{28.57 \text{ k}\Omega \parallel 6.8 \text{ k}\Omega}_{5.49 \text{ k}\Omega})$$

 $= -32.94$
 $V_o = A_v V_i = (-32.94)(1.8 \text{ mV})$
 $= -59.29 \text{ mV}$
39. $Z_i = 10 \text{ M}\Omega \parallel 91 \text{ M}\Omega \cong 9 \text{ M}\Omega$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(12 \text{ mA})}{3 \text{ V}} \left(1 - \frac{-1.45 \text{ V}}{-3 \text{ V}}\right) = 4.13 \text{ mS}$$

 $Z_o = r_d \parallel R_S \parallel 1/g_m = 45 \text{ k}\Omega \parallel 1.1 \text{ k}\Omega \parallel 1/4.13 \text{ mS}$
 $= 1.074 \text{ k}\Omega \parallel 242.1 \Omega$
 $= 197.6 \Omega$

$$A_v = \frac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)} = \frac{(4.13 \text{ mS})(45 \text{ k}\Omega \parallel 1.1 \text{ k}\Omega)}{1 + (4.13 \text{ mS})(45 \text{ k}\Omega \parallel 1.1 \text{ k}\Omega)}$$

$$= \frac{(4.13 \text{ mS})(1.074 \text{ k}\Omega)}{1 + (4.13 \text{ mS})(1.074 \text{ k}\Omega)} = \frac{4.436}{1 + 4.436}$$

 $= 0.816$
40. $g_m = 2k(V_{GS_Q} - V_{GS(Th)})$
 $= 2(0.3 \times 10^{-3})(8 \text{ V} - 3 \text{ V})$
 $= 3 \text{ mS}$

41. $V_{GS_Q} = 6.7 \text{ V}$

$$g_m = 2k(V_{GS_Q} - V_T) = 2(0.3 \times 10^{-3})(6.7 \text{ V} - 3 \text{ V}) = 2.22 \text{ mS}$$

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)} = \frac{10 \text{ M}\Omega + 100 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega}{1 + (2.22 \text{ mS})(100 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)}$$

$$= \frac{10 \text{ M}\Omega + 2.15 \text{ k}\Omega}{1 + 2.22 \text{ mS}(2.15 \text{ k}\Omega)} \cong \mathbf{1.73 \text{ M}\Omega}$$

$$Z_o = R_F \parallel r_d \parallel R_D = 10 \text{ M}\Omega \parallel 100 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega = \mathbf{2.15 \text{ k}\Omega}$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -2.22 \text{ mS}(2.15 \text{ k}\Omega) = \mathbf{-4.77}$$

42. $g_m = 2k(V_{GS_Q} - V_T) = 2(0.2 \times 10^{-3})(6.7 \text{ V} - 3 \text{ V})$

$$= 1.48 \text{ mS}$$

$$Z_i = \frac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)} = \frac{10 \text{ M}\Omega + 100 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega}{1 + (1.48 \text{ mS})(100 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega)}$$

$$= \frac{10 \text{ M}\Omega + 2.15 \text{ k}\Omega}{1 + (1.48 \text{ mS})(2.15 \text{ k}\Omega)} = \mathbf{2.39 \text{ M}\Omega} > 1.73 \text{ M}\Omega \text{ (#39)}$$

$$Z_o = R_F \parallel r_d \parallel R_D = \mathbf{2.15 \text{ k}\Omega} = 2.15 \text{ k}\Omega \text{ (#39)}$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -(1.48 \text{ mS})(2.15 \text{ k}\Omega)$$

$$= \mathbf{-3.182} < -4.77 \text{ (#39)}$$

43. $V_{GS_Q} = 5.7 \text{ V}, g_m = 2k(V_{GS_Q} - V_T) = 2(0.3 \times 10^{-3})(5.7 \text{ V} - 3.5 \text{ V})$

$$= 1.32 \text{ mS}$$

$$r_d = \frac{1}{30 \mu\text{S}} = 33.33 \text{ k}\Omega$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -1.32 \text{ mS}(22 \text{ M}\Omega \parallel 33.33 \text{ k}\Omega \parallel 10 \text{ k}\Omega)$$

$$= -10.15$$

$$V_o = A_v V_i = (-10.15)(20 \text{ mV}) = \mathbf{-203 \text{ mV}}$$

44. $I_D = k(V_{GS} - V_T)^2$

$$\therefore k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2} = \frac{4 \text{ mA}}{(7 \text{ V} - 4 \text{ V})^2} = 0.444 \times 10^{-3}$$

$$g_m = 2k(V_{GS_Q} - V_{GS(Th)}) = 2(0.444 \times 10^{-3})(7 \text{ V} - 4 \text{ V})$$

$$= 2.66 \text{ mS}$$

$$A_v = -g_m(R_F \parallel r_d \parallel R_D) = -(2.66 \text{ mS})(22 \text{ M}\Omega \parallel \underbrace{50 \text{ k}\Omega \parallel 10 \text{ k}\Omega}_{8.33 \text{ k}\Omega}) = -22.16$$

$$\underbrace{\hspace{10em}}_{\cong 8.33 \text{ k}\Omega}$$

$$V_o = A_v V_i = (-22.16)(4 \text{ mV}) = \mathbf{-88.64 \text{ mV}}$$

$$45. \quad V_{GS_Q} = 4.8 \text{ V}, g_m = 2k(V_{GS_Q} - V_{GS(th)}) = 2(0.4 \times 10^{-3})(4.8 \text{ V} - 3 \text{ V}) = 1.44 \text{ mS}$$

$$A_v = -g_m(r_d \parallel R_D) = -(1.44 \text{ mS})(40 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega) = -4.39$$

$$V_o = A_v V_i = (-4.39)(0.8 \text{ mV}) = \mathbf{-3.51 \text{ mV}}$$

$$46. \quad r_d = \frac{1}{y_{os}} = \frac{1}{20 \mu\text{S}} = 50 \text{ k}\Omega$$

$$V_{GS_Q} = 0 \text{ V}, \therefore g_m = g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(8 \text{ mA})}{2.5 \text{ V}} = 6.4 \text{ mS}$$

$$|A_v| = g_m(r_d \parallel R_D)$$

$$8 = (6.4 \text{ mS})(50 \text{ k}\Omega \parallel R_D)$$

$$\frac{8}{6.4 \text{ mS}} = 1.25 \text{ k}\Omega = \frac{50 \text{ k}\Omega \cdot R_D}{50 \text{ k}\Omega + R_D}$$

$$\text{and } R_D = \mathbf{1.28 \text{ k}\Omega}$$

$$\text{Use } R_D = \mathbf{1.3 \text{ k}\Omega}$$

$$47. \quad V_{GS_Q} = \frac{1}{3}V_P = \frac{1}{3}(-3 \text{ V}) = -1 \text{ V}$$

$$I_{D_Q} = I_{DSS} \left(1 - \frac{V_{GS_Q}}{V_P}\right)^2 = 12 \text{ mA} \left(1 - \frac{-1 \text{ V}}{-3 \text{ V}}\right)^2 = 5.33 \text{ mA}$$

$$R_S = \frac{V_S}{I_{D_Q}} = \frac{1 \text{ V}}{5.33 \text{ mA}} = 187.62 \Omega \therefore \text{Use } R_S = \mathbf{180 \Omega}$$

$$g_m = \frac{2I_{DSS}}{V_P} \left(1 - \frac{V_{GS_Q}}{V_P}\right) = \frac{2(12 \text{ mA})}{3 \text{ V}} \left(1 - \frac{-1 \text{ V}}{-3 \text{ V}}\right) = 5.33 \text{ mS}$$

$$A_v = -g_m(R_D \parallel r_d) = -10$$

$$\text{or } R_D \parallel 40 \text{ k}\Omega = \frac{-10}{5.33 \text{ mS}} = 1.876 \text{ k}\Omega$$

$$= 1.876 \text{ k}\Omega$$

$$40 \text{ k}\Omega R_D = 1.876 \text{ k}\Omega R_D + 75.04 \text{ k}\Omega^2$$

$$38.124 R_D = 75.04 \text{ k}\Omega$$

$$R_D = 1.97 \text{ k}\Omega \Rightarrow R_D = \mathbf{2 \text{ k}\Omega}$$

$$48. \quad Z_i = R_G = \mathbf{1 \text{ M}\Omega}$$

$$Z_o = r_d \parallel R_D \cong R_D = \mathbf{2.7 \text{ k}\Omega}$$

$$\text{From graph } V_{GS_Q} \cong -3.4 \text{ V}$$

$$g_m = \frac{2(I_{DSS})}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right)$$

$$= \frac{2[10 \text{ mA}]}{6 \text{ V}} \left(1 - \frac{-3.4 \text{ V}}{-6 \text{ V}}\right) = 3.33 \text{ mS}[1 - 0.567]$$

$$= 3.33 \text{ mS}[0.433] = \mathbf{-1.44 \text{ mS}}$$

$$A_{v_{NL}} = -g_m R_D = (-1.44 \text{ mS})(2.7 \text{ k}\Omega) = \mathbf{-3.89}$$

(b) –

$$(c) \quad A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 2.7 \text{ k}\Omega} (-3.89) \\ = \mathbf{-2.47}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{1 \text{ M}\Omega}{1 \text{ M}\Omega + 0.6 \text{ k}\Omega} (-2.47) \\ \cong \mathbf{-2.47}$$

(d) A_{v_L} the same

$$A_{v_s} = \frac{1 \text{ M}\Omega}{1 \text{ M}\Omega + 10 \text{ k}\Omega} (-2.47) \\ \cong \mathbf{-2.45}$$

Slight drop in A_{v_s}

(e) remain the same

49. (a) $Z_i = R_G = \mathbf{2 \text{ M}\Omega}$

From graph $V_{GS_Q} \cong -3.5 \text{ V}$

$$g_m = \frac{2(I_{DSS})}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2[6 \text{ mA}]}{6 \text{ V}} \left(1 - \frac{-3.5 \text{ V}}{-6 \text{ V}}\right) \\ = \mathbf{0.834 \text{ mS}}$$

$$Z_o \cong R_S \parallel 1/g_m = 3.3 \text{ k}\Omega \parallel 1/0.834 \text{ mS} = \mathbf{0.72 \text{ k}\Omega}$$

$$A_{v_{NL}} \cong \frac{g_m R_S}{1 + g_m R_S} = \frac{(0.834)(3.3 \text{ k}\Omega)}{1 + (0.834)(3.3 \text{ k}\Omega)} \\ \cong \mathbf{0.733}$$

$$(c) \quad A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 0.72 \text{ k}\Omega} = \mathbf{0.552}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{2 \text{ M}\Omega}{2 \text{ M}\Omega + 0.5 \text{ k}\Omega} (0.552) \\ \cong \mathbf{0.552}$$

$$(d) \quad A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 0.72 \text{ k}\Omega} (0.733) \\ = \mathbf{0.670} \text{ slight drop in gain}$$

A_{v_s} the same

(e) A_{v_L} the same

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{2 \text{ M}\Omega}{2 \text{ M}\Omega + 20 \text{ k}\Omega} (0.552) \\ \cong \mathbf{0.546} \text{ slight drop in } A_{v_s}$$

(e) Z_i and Z_o the same.

50. (a) From graph $V_{GS_Q} \cong -1.75 \text{ V}$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2[5 \text{ mA}]}{4 \text{ V}} \left(1 - \frac{1.75 \text{ V}}{4 \text{ V}}\right) \\ = 1.41 \text{ mS}$$

$$Z_i \cong R_S \parallel 1/g_m = 1.2 \text{ k}\Omega \parallel 1/1.41 \text{ mS} = 1.2 \text{ k}\Omega \parallel 709.22 \Omega \\ = \mathbf{446 \Omega}$$

$$Z_o \cong R_D = \mathbf{3.3 \text{ k}\Omega}$$

$$A_{v_{NL}} = g_m R_D = (1.41 \text{ mS})(3.3 \text{ k}\Omega) = \mathbf{4.65}$$

(b) –

$$(c) A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} (4.65) = \mathbf{2.73}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{446 \Omega}{446 \Omega + 500 \Omega} (2.73) = \mathbf{1.29}$$

$$(d) A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{2.2 \text{ k}\Omega}{2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} (4.65) = \mathbf{1.86}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{446 \Omega}{446 \Omega + 500 \Omega} (1.86) = \mathbf{0.877}$$

severe drop in $A_{v_L} + A_{v_s}$

$$(e) A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{4.7 \text{ k}\Omega}{4.7 \text{ k}\Omega + 3.3 \text{ k}\Omega} (4.65) = \mathbf{2.73} \text{ (no change)}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{446 \Omega}{446 \Omega + 100 \Omega} (2.73) = \mathbf{2.23} \text{ (improved gain)}$$

$$(f) A_{v_L} = \frac{R_L}{R_L + R_o} A_{v_{NL}} = \frac{2.2 \text{ k}\Omega (4.65)}{2.2 \text{ k}\Omega + 3.3 \text{ k}\Omega} = \mathbf{1.86} \text{ (as in part (d))}$$

$$A_{v_s} = \frac{R_i}{R_i + R_{sig}} A_{v_L} = \frac{446 \Omega}{446 \Omega + 100 \Omega} (1.86) = \mathbf{1.52}$$

(g) Increasing R_L and decreasing R_{sig} will improve the gain.

51. From the graph: $V_{GS_Q} \cong -1.45 \text{ V}$, $I_{D_Q} = 3.7 \text{ mA}$
 $V_D = V_{DD} - I_D R_D = 18 \text{ V} - (3.7 \text{ mA})(2.2 \text{ k}\Omega) = \mathbf{9.86 \text{ V}}$
 $V_S = I_D R_S = (3.7 \text{ mA})(390 \Omega) = \mathbf{1.44 \text{ V}}$
 $V_{DS} = V_D - V_S = 9.86 \text{ V} - 1.44 \text{ V} = \mathbf{8.42 \text{ V}}$
 $V_G = \mathbf{0 \text{ V}}$

52. From Problem 51, $V_{GS_Q} = -1.45 \text{ V}$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2[8 \text{ mA}]}{4.5 \text{ V}} \left(1 - \frac{-1.45 \text{ V}}{4.5 \text{ V}}\right)$$

$$= 2.41 \text{ mS}$$

 $A_{v_1} = A_{v_2} = -g_m R_D = -(2.41 \text{ mS})(2.2 \text{ k}\Omega) = \mathbf{-5.30}$
 $A_{v_T} = A_{v_1} \cdot A_{v_2} = (-5.30)(-5.30) = \mathbf{28.1}$

53. From graph $V_{GS_Q} \cong -1.4 \text{ V}$, $I_{D_Q} \cong 3.6 \text{ mA}$
 $V_D = V_{DD} - I_D R_D = 18 \text{ V} - (3.6 \text{ mA})(2.2 \text{ k}\Omega)$
 $= 18 \text{ V} - 7.92 \text{ V} = \mathbf{10.08 \text{ V}}$
 $V_S = I_D R_S = (3.6 \text{ mA})(390 \Omega) = \mathbf{1.4 \text{ V}}$
 $V_{DS} = V_D - V_S = 10.08 \text{ V} - 1.4 \text{ V} = \mathbf{8.68 \text{ V}}$
 $V_G = \mathbf{0 \text{ V}}$

54. From problem 53: $V_{GS_Q} = -1.4 \text{ V}$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2[12 \text{ mA}]}{3} \left(1 - \frac{-1.4 \text{ V}}{3 \text{ V}}\right)$$

$$= 8 \text{ mS}[1 - 0.467] = 0.533(8 \text{ mS})$$

$$= 4.26 \text{ mS}$$

 $A_{v_1} = A_{v_2} = -g_m (R_D \parallel r_d) = -(4.26 \text{ mS})(2.2 \text{ k}\Omega \parallel 40 \text{ k}\Omega)$

$$= \mathbf{-8.9}$$

 $A_{v_T} = A_{v_1} \cdot A_{v_2} = (-8.9)(-8.9) = \mathbf{79.21}$
 Increase in A_{v_T} compared to problem 52.

55. $g_{os} = 25 \mu\text{S}$, $r_d = 1/25 \mu\text{S} = 40 \text{ k}\Omega$
 $Z_i = \mathbf{10 \text{ M}\Omega}$
 $Z_o = \mathbf{2.7 \text{ k}\Omega}$

56. From graph $V_{GS_Q} = -0.95 \text{ V}$, $I_{D_Q} = \mathbf{2.9 \text{ mA}}$

$$V_D = 10 \text{ V} - I_D(1.8 \text{ k}\Omega) = 10 \text{ V} - (2.9 \text{ mA})(1.8 \text{ k}\Omega) = 10 \text{ V} - 5.22 \text{ V} \\ = \mathbf{4.78 \text{ V}}$$

$$V_S = I_D R_S = (2.9 \text{ mA})(330 \text{ }\Omega) = \mathbf{0.957 \text{ V}}$$

$$V_{DS} = V_D - V_S = 4.78 \text{ V} - 0.957 \text{ V} = \mathbf{3.82 \text{ V}}$$

$$V_G = \mathbf{0 \text{ V}}$$

$$I_D = I_S = \mathbf{2.9 \text{ mA}}$$

$$I_G = \mathbf{0 \text{ A}}$$

$$B_{RE} \geq 10R_2$$

$$(150)(2.2 \text{ k}\Omega) \geq 10(8.2 \text{ k}\Omega)$$

$$330 \text{ k}\Omega \geq 82 \text{ k}\Omega \text{ checks}$$

$$V_B = \frac{8.2 \text{ k}\Omega(10 \text{ V})}{8.2 \text{ k}\Omega + 24 \text{ k}\Omega} = \mathbf{2.55 \text{ V}}$$

$$V_E = V_B - V_{BE} = 2.55 \text{ V} - 0.7 \text{ V} = \mathbf{1.85 \text{ V}}$$

$$I_E \cong I_C = \frac{V_E}{R_E} = \frac{1.85 \text{ V}}{2.2 \text{ k}\Omega} = \mathbf{0.84 \text{ mA}}$$

$$V_C = 10 \text{ V} - I_C 2.7 \text{ k}\Omega = 10 \text{ V} - (0.84 \text{ mA})(2.7 \text{ k}\Omega) \\ = 10 \text{ V} - 2.27 \text{ V} = \mathbf{7.73 \text{ V}}$$

$$V_{CE} = V_C - V_E = 7.73 \text{ V} - 1.85 \text{ V} = \mathbf{5.88 \text{ V}}$$

$$I_B = \frac{I_C}{\beta} = \frac{0.84 \text{ mA}}{150} = \mathbf{5.6 \mu\text{A}}$$

57. $Z_{i_2} = 8.2 \text{ k}\Omega \parallel 24 \text{ k}\Omega \parallel \beta(2.2 \text{ k}\Omega)$

$$= 6.11 \text{ k}\Omega \parallel 330 \text{ k}\Omega \cong 6 \text{ k}\Omega$$

$$A_{v_1} = -g_m(R_D \parallel Z_{i_2})$$

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right) = \frac{2(6 \text{ mA})}{3} \left(1 - \frac{0.95 \text{ V}}{3 \text{ V}}\right) \\ = \mathbf{2.73 \text{ mS}}$$

$$A_{v_1} = -(2.73 \text{ mS})(1.8 \text{ k}\Omega \parallel 6 \text{ k}\Omega)$$

$$= \mathbf{-3.77}$$

$$A_{v_2} = -\frac{R_C}{r_e}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.84 \text{ mA}} = 30.95 \text{ }\Omega$$

$$A_{v_2} = -\frac{2.7 \text{ k}\Omega}{30.95 \text{ }\Omega} = \mathbf{-87.2}$$

$$A_{v_T} = A_{v_1} \cdot A_{v_2} = (-3.77)(-87.2) = \mathbf{328.74}$$

58. $Z_i = \mathbf{10 \text{ M}\Omega}$

$$Z_o \cong R_C = \mathbf{2.7 \text{ k}\Omega}$$

Chapter 9

1. (a) **3, 1.699, -1.151**
 (b) **6.908, 3.912, -0.347**
 (c) results differ by magnitude of 2.3
2. (a) $\log_{10} 0.24 \times 10^6 = \mathbf{5.38}$
 (b) $\log_e (0.24 \times 10^6) = 2.3 \log_{10}(0.24 \times 10^6) = \mathbf{12.37}$
 (c) $\log_e (0.24 \times 10^6) = \mathbf{12.39}$
3. (a) same **22.92**
 (b) same **23.98**
 (c) same **0.903**
4. (a) $\text{dB} = 10 \log_{10} \frac{P_o}{P_i} = 10 \log_{10} \frac{100 \text{ W}}{5 \text{ W}} = 10 \log_{10} 20 = 10(1.301)$
 $= \mathbf{13.01 \text{ dB}}$
 (b) $\text{dB} = 10 \log_{10} \frac{100 \text{ mW}}{5 \text{ mW}} = 10 \log_{10} 20 = 10(1.301)$
 $= \mathbf{13.01 \text{ dB}}$
 (c) $\text{dB} = 10 \log_{10} \frac{100 \mu\text{W}}{20 \mu\text{W}} = 10 \log_{10} 5 = 10(0.6987)$
 $= \mathbf{6.99 \text{ dB}}$
5. $G_{\text{dBm}} = 10 \log_{10} \frac{P_2}{1 \text{ mW}} \Big|_{600 \Omega} = 10 \log_{10} \frac{25 \text{ W}}{1 \text{ mW}} \Big|_{600 \Omega}$
 $= \mathbf{43.98 \text{ dBm}}$
6. $G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} = 20 \log_{10} \frac{220 \text{ V}}{110 \text{ V}} = 20 \log_{10} 2 = 20(0.301)$
 $= \mathbf{6.02 \text{ dB}}$
7. $G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} = 20 \log_{10} \frac{25 \text{ V}}{10 \text{ mV}} = 20 \log_{10} 2500$
 $= 20(3.398) = \mathbf{67.96 \text{ dB}}$
8. (a) Gain of stage 1 = A dB
 Gain of stage 2 = 2 A dB
 Gain of stage 3 = 2.7 A dB
 $A + 2A + 2.7A = 120$
 $A = \mathbf{21.05 \text{ dB}}$

(b) Stage 1: $A_{v_1} = 21.05 \text{ dB} = 20 \log_{10} \frac{V_{o_1}}{V_{i_1}}$

$$\frac{21.05}{20} = 1.0526 = \log_{10} \frac{V_{o_1}}{V_{i_1}}$$

$$10^{1.0526} = \frac{V_{o_1}}{V_{i_1}}$$

and $\frac{V_{o_1}}{V_{i_1}} = \mathbf{11.288}$

Stage 2: $A_{v_2} = 42.1 \text{ dB} = 20 \log_{10} \frac{V_{o_2}}{V_{i_2}}$

$$2.105 = \log_{10} \frac{V_{o_2}}{V_{i_2}}$$

$$10^{2.105} = \frac{V_{o_2}}{V_{i_2}}$$

and $\frac{V_{o_2}}{V_{i_2}} = \mathbf{127.35}$

Stage 3: : $A_{v_3} = 56.835 \text{ dB} = 20 \log_{10} \frac{V_{o_3}}{V_{i_3}}$

$$2.8418 = \log_{10} \frac{V_{o_3}}{V_{i_3}}$$

$$10^{2.8418} = \frac{V_{o_3}}{V_{i_3}}$$

and $\frac{V_{o_3}}{V_{i_3}} = \mathbf{694.624}$

$$A_{v_T} = A_{v_1} \cdot A_{v_2} \cdot A_{v_3} = (11.288)(127.35)(694.624) = \mathbf{99,8541.1}$$

?

$$A_T = 120 \text{ dB} = 20 \log_{10} 99,8541.1$$

120 dB $\cong$ 119.99 dB (difference due to level of accuracy carried through calculations)

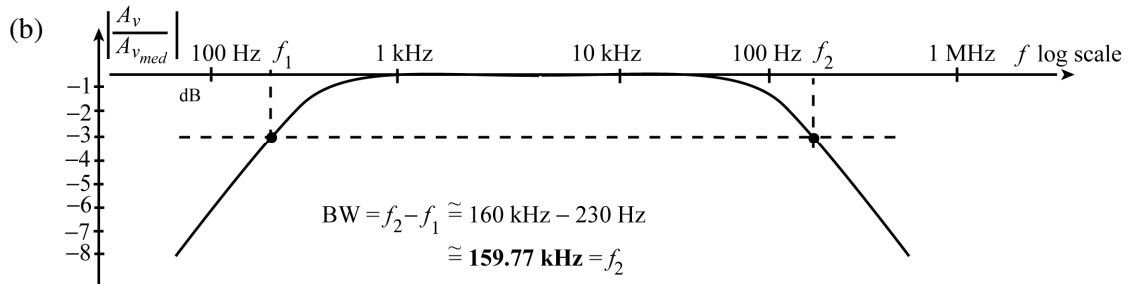
9. (a) $G_{\text{dB}} = 20 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{48 \text{ W}}{5 \mu\text{W}} = \mathbf{69.83 \text{ dB}}$

(b) $G_v = 20 \log_{10} \frac{V_o}{V_i} = 20 \log_{10} \frac{\sqrt{P_o R_o}}{V_i} = \frac{20 \log_{10} \sqrt{(48 \text{ W})(40 \text{ k}\Omega)}}{100 \text{ mV}}$
 $= \mathbf{82.83 \text{ dB}}$

$$(c) \quad R_i = \frac{V_i^2}{P} = \frac{(100 \text{ mV})^2}{5 \mu\text{W}} = \mathbf{2 \text{ k}\Omega}$$

$$(d) \quad P_o = \frac{V_o^2}{R_o} \Rightarrow V_o = \sqrt{P_o R_o} = \sqrt{(48 \text{ W})(40 \text{ k}\Omega)} = \mathbf{1385.64 \text{ V}}$$

10. (a) Same shape except $A_v = 190$ is now level of 1. In fact, all levels of A_v are divided by 190 to obtain normalized plot.
 $0.707(190) = \mathbf{134.33}$ defining cutoff frequencies
 at low end $f_1 \cong \mathbf{230 \text{ Hz}}$ (remember this is a log scale)
 at high end $f_2 \cong \mathbf{160 \text{ kHz}}$



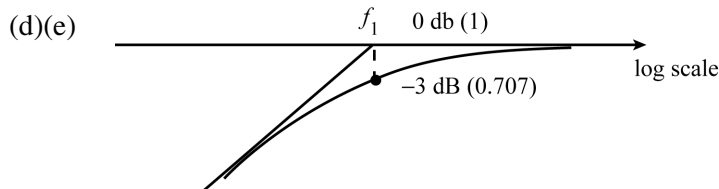
11. (a) $|A_v| = \left| \frac{V_o}{V_i} \right| = \frac{1}{\sqrt{1 + (f_1/f)^2}} \quad f_1 = \frac{1}{2\pi RC} = \frac{1}{2\pi(1.2 \text{ k}\Omega)(0.068 \mu\text{F})}$
 $= 1950.43 \text{ Hz}$

$$|A_v| = \frac{1}{\sqrt{1 + \frac{1950.43 \text{ Hz}^2}{f^2}}}$$

(b)

	$A_{v_{dB}}$
100 Hz: $ A_v = 0.051$	-25.8
1 kHz: $ A_v = 0.456$	-6.81
2 kHz: $ A_v = 0.716$	-2.90
5 kHz: $ A_v = 0.932$	-0.615
10 kHz: $ A_v = 0.982$	-0.162

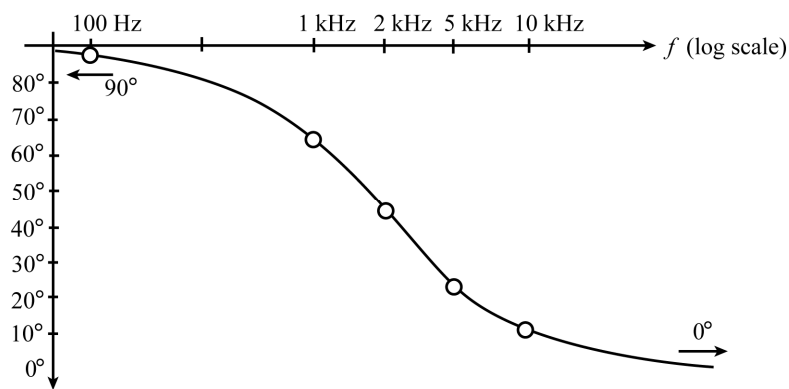
(c) $f_1 \cong \mathbf{1950 \text{ Hz}}$



12. (a) $f_1 = \frac{1}{2\pi RC} = 1.95 \text{ kHz}$
 $\theta = \tan^{-1} \frac{f_1}{f} = \tan^{-1} \frac{1.95 \text{ kHz}}{f}$

(b) $\theta = \tan^{-1} \frac{1.95 \text{ kHz}}{f}$

f	$\theta = \tan^{-1} \frac{1.95 \text{ kHz}}{f}$
100 Hz	87.06°
1 kHz	62.85°
2 kHz	44.27°
5 kHz	21.3°
10 kHz	11.03°



(c) $f_1 = \frac{1}{2\pi RC} = 1.95 \text{ kHz}$

(d) First find $\theta = 45^\circ$ at $f_1 = 1.95 \text{ kHz}$. Then sketch an approach to 90° at low frequencies and 0° at high frequencies. Use an expected shape for the curve noting that the greatest change in θ occurs near f_1 . The resulting curve should be quite close to that plotted above.

13. (a) **10 kHz**

(b) **1 kHz**

(c) **20 kHz $\rightarrow$ 10 kHz $\rightarrow$ 5 kHz**

(d) **1 kHz $\rightarrow$ 10 kHz $\rightarrow$ 100 kHz**

14. From example 9.9, $r_e = 15.76 \Omega$

$$A_v = \frac{-R_C \parallel R_L \parallel r_o}{r_e} = \frac{-4 \text{ k}\Omega \parallel 2.2 \text{ k}\Omega \parallel 40 \text{ k}\Omega}{15.76 \Omega}$$

$$= -86.97 \text{ (vs. } -90 \text{ for Ex. 9.9)}$$

$$f_{L_s} : r_o \text{ does not affect } R_i \therefore f_{L_s} = \frac{1}{2\pi(R_s + R_i)C_s} \text{ the same } \cong \mathbf{6.86 \text{ Hz}}$$

$$f_{L_c} = \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(R_C \parallel r_o + R_L)C_C}$$

$$R_C \parallel r_o = 4 \text{ k}\Omega \parallel 40 \text{ k}\Omega = 5.636 \text{ k}\Omega$$

$$f_{L_c} = \frac{1}{2\pi(5.636 \text{ k}\Omega + 2 \text{ k}\Omega)(1 \mu\text{F})}$$

$$= \mathbf{28.23 \text{ Hz}} \text{ (vs. } 25.68 \text{ Hz for Ex. 9.9)}$$

$$f_{L_E} : R_e \text{ not affected by } r_o, \text{ therefore, } f_{L_E} = \frac{1}{2\pi R_e C_E} \cong \mathbf{327 \text{ Hz}} \text{ is the same.}$$

In total, the effect of r_o on the frequency response was to slightly reduce the mid-band gain.

15. (a) $\beta R_E \geq 10R_2$

$$(120)(1.2 \text{ k}\Omega) \geq 10(10 \text{ k}\Omega)$$

$$144 \text{ k}\Omega \geq 100 \text{ k}\Omega \text{ (checks!)}$$

$$V_B = \frac{10 \text{ k}\Omega(14 \text{ V})}{10 \text{ k}\Omega + 68 \text{ k}\Omega} = 1.795 \text{ V}$$

$$V_E = V_B - V_{BE} = 1.795 \text{ V} - 0.7 \text{ V}$$

$$= 1.095 \text{ V}$$

$$I_E = \frac{V_E}{R_E} = \frac{1.095 \text{ V}}{1.2 \text{ k}\Omega} = 0.913 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{0.913 \text{ mA}} = \mathbf{28.48 \Omega}$$

$$(b) A_{V_{\text{mid}}} = -\frac{(R_L \parallel R_C)}{r_e} = \frac{-(3.3 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega)}{28.48 \Omega}$$

$$= \mathbf{-72.91}$$

$$(c) Z_i = R_1 \parallel R_2 \parallel \beta r_e$$

$$= 68 \text{ k}\Omega \parallel 10 \text{ k}\Omega \parallel \underbrace{(120)(28.48 \Omega)}_{3.418 \text{ k}\Omega}$$

$$= \mathbf{2.455 \text{ k}\Omega}$$

$$\begin{aligned}
(d) \quad f_{L_s} &= \frac{1}{2\pi R_i C_s} = \frac{1}{2\pi(2.455 \text{ k}\Omega)(0.47 \text{ }\mu\text{F})} = \mathbf{137.93 \text{ Hz}} \\
f_{L_c} &= \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(5.6 \text{ k}\Omega + 3.3 \text{ k}\Omega)(0.47 \text{ }\mu\text{F})} = \mathbf{38.05 \text{ Hz}} \\
f_{L_E} &= \frac{1}{2\pi R_e C_E} \text{ with } R_e = R_E \parallel \frac{R_1 \parallel R_2}{\beta} + r_e \\
R_1 \parallel R_2 &= 68 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 8.72 \text{ k}\Omega \\
\frac{R_1 \parallel R_2}{\beta} &= \frac{8.72 \text{ k}\Omega}{120} = 72.67 \Omega \\
R_e &= 1.2 \text{ k}\Omega \parallel 72.67 \Omega + 28.48 \Omega = 1.2 \text{ k}\Omega \parallel 101.15 \Omega = 93.29 \Omega \\
f_{L_E} &= \frac{1}{2\pi R_e C_E} = \frac{1}{2\pi(93.29 \Omega)(20 \text{ }\mu\text{F})} = \mathbf{85.30 \text{ Hz}}
\end{aligned}$$

$$(e) \quad f_1 = f_{L_s} = 137.93 \text{ Hz}$$

$$(f) \quad -$$

$$(g) \quad -$$

$$\begin{aligned}
16. (a) \quad I_B &= \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (111)(0.91 \text{ k}\Omega)} = \frac{19.3 \text{ V}}{470 \text{ k}\Omega + 101.01 \text{ k}\Omega} \\
&= 33.8 \text{ }\mu\text{A} \\
I_E &= (\beta + 1)I_B = (111)(33.8 \text{ }\mu\text{A}) \\
&= 3.752 \text{ mA} \\
r_e &= \frac{26 \text{ mV}}{3.752 \text{ mA}} = \mathbf{6.93 \Omega}
\end{aligned}$$

$$\begin{aligned}
(b) \quad A_{v_{\text{mid}}} &= \frac{V_o}{V_i} = \frac{-(R_C \parallel R_L)}{r_e} = \frac{-(3 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega)}{6.93 \Omega} = \frac{-1.831 \text{ k}\Omega}{6.93 \Omega} \\
&= \mathbf{-264.24}
\end{aligned}$$

$$\begin{aligned}
(c) \quad Z_i &= R_B \parallel \beta r_e = 470 \text{ k}\Omega \parallel (110)(6.93 \Omega) = 470 \text{ k}\Omega \parallel 762.3 \Omega \\
&= \mathbf{761.07 \Omega}
\end{aligned}$$

$$\begin{aligned}
(d) \quad f_{L_s} &= \frac{1}{2\pi Z_i C_s} = \frac{1}{2\pi(761.07 \Omega)(1 \text{ }\mu\text{F})} \\
&= \mathbf{209.12 \text{ Hz}}
\end{aligned}$$

$$\begin{aligned}
f_{L_c} &= \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(3 \text{ k}\Omega + 4.7 \text{ k}\Omega)(1 \text{ }\mu\text{F})} \\
&= \mathbf{20.67 \text{ Hz}}
\end{aligned}$$

$$\begin{aligned}
 f_{L_E} &= \frac{1}{2\pi R_e C_E} \\
 &= \frac{1}{2\pi(762.65 \, \Omega)(6.8 \, \mu\text{F})} \\
 &= \mathbf{30.69 \, \text{Hz}}
 \end{aligned}$$

$$\begin{aligned}
 R_e &= R_E \parallel \frac{R_B}{\beta} = 0.91 \, \text{k}\Omega \parallel \frac{470 \, \text{k}\Omega}{100} + 6.93 \, \Omega \\
 &= 0.91 \, \text{k}\Omega \parallel 4.71 \, \text{k}\Omega \\
 &= 762.65 \, \Omega
 \end{aligned}$$

(e) $f_1 \cong f_{L_s} = \mathbf{209.12 \, \text{Hz}}$

(f) –

(g) –

17. (a) $\beta R_E \geq 10R_2$
 $(100)(2.2 \, \text{k}\Omega) \geq 10(30 \, \text{k}\Omega)$
 $220 \, \text{k}\Omega \not\geq 300 \, \text{k}\Omega \text{ (No!)}$
 $R_{Th} = R_1 \parallel R_2 = 120 \, \text{k}\Omega \parallel 30 \, \text{k}\Omega = 24 \, \text{k}\Omega$
 $E_{Th} = \frac{30 \, \text{k}\Omega(14 \, \text{V})}{30 \, \text{k}\Omega + 120 \, \text{k}\Omega} = 2.8 \, \text{V}$
 $I_B = \frac{E_{Th} - V_{BE}}{R_{Th} + (\beta + 1)R_E} = \frac{2.8 \, \text{V} - 0.7 \, \text{V}}{24 \, \text{k}\Omega + 222.2 \, \text{k}\Omega}$
 $= 8.53 \, \mu\text{A}$
 $I_E = (\beta + 1)I_B = (101)(8.53 \, \mu\text{A})$
 $= 0.86 \, \text{mA}$
 $r_e = \frac{26 \, \text{mV}}{I_E} = \frac{26 \, \text{mV}}{0.86 \, \text{mA}} = \mathbf{30.23 \, \Omega}$

(b) $A_{v_{\text{mid}}} = \frac{R_E \parallel R_L}{r_e + R_E \parallel R_L}$
 $= \frac{2.2 \, \text{k}\Omega \parallel 8.2 \, \text{k}\Omega}{30.23 \, \Omega + 2.2 \, \text{k}\Omega \parallel 8.2 \, \text{k}\Omega}$
 $= \mathbf{0.983}$

(c) $Z_i = R_1 \parallel R_2 \parallel \beta(r_e + R'_E)$ $R'_E = R_E \parallel R_L = 2.2 \, \text{k}\Omega \parallel 8.2 \, \text{k}\Omega = 1.735 \, \text{k}\Omega$
 $= 120 \, \text{k}\Omega \parallel 30 \, \text{k}\Omega \parallel (100)(30.23 \, \Omega + 1.735 \, \text{k}\Omega)$
 $= \mathbf{21.13 \, \text{k}\Omega}$

$$\begin{aligned}
 \text{(d)} \quad f_{L_s} &= \frac{1}{2\pi R_i C_s} \\
 &= \frac{1}{2\pi(21.13 \text{ k}\Omega)(0.1 \text{ }\mu\text{F})} \\
 &= \mathbf{75.32 \text{ Hz}}
 \end{aligned}$$

$$\begin{aligned}
 f_{L_C} &= \frac{1}{2\pi(R_o + R_L)C_C} & R'_s &= R_1 \parallel R_2 \\
 & & &= 120 \text{ k}\Omega \parallel 30 \text{ k}\Omega \\
 & & &= 24 \text{ k}\Omega
 \end{aligned}$$

$$\begin{aligned}
 R_o &= R_E \parallel \frac{R'_s}{\beta} + r_e \\
 &= 2.2 \text{ k}\Omega \parallel \frac{24 \text{ k}\Omega}{100} + 30.23 \text{ }\Omega \\
 &= 2.2 \text{ k}\Omega \parallel 270.23 \text{ }\Omega = 240.69 \text{ }\Omega
 \end{aligned}$$

$$\begin{aligned}
 f_{L_C} &= \frac{1}{2\pi(240.69 \text{ }\Omega + 8.2 \text{ k}\Omega)(0.1 \text{ }\mu\text{F})} \\
 &= \mathbf{188.57 \text{ Hz}}
 \end{aligned}$$

$$\text{(e)} \quad f_{\text{low}} = f_{L_C} \cong \mathbf{188.57 \text{ Hz}}$$

(f) –

(g) –

$$\begin{aligned}
 18. \quad \text{(a)} \quad I_E &= \frac{V_{EE} - V_{EB}}{R_E} = \frac{4 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega} = 2.75 \text{ mA} \\
 r_e &= \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.75 \text{ mA}} = 9.45 \text{ }\Omega
 \end{aligned}$$

$$\begin{aligned}
 \text{(b)} \quad A_{v_{\text{mid}}} &= \frac{R_C \parallel R_L}{r_e} = \frac{3.3 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega}{9.45 \text{ }\Omega} \\
 &= \mathbf{205.1}
 \end{aligned}$$

$$\begin{aligned}
 \text{(c)} \quad Z_i &= R_E \parallel r_e = 1.2 \text{ k}\Omega \parallel 9.45 \text{ }\Omega \\
 &= \mathbf{9.38 \text{ }\Omega}
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad f_{L_s} &= \frac{1}{2\pi Z_i C_s} = \frac{1}{2\pi(9.38 \text{ }\Omega)(10 \text{ }\mu\text{F})} \\
 &\cong \mathbf{1.7 \text{ kHz}} \\
 f_{L_C} &= \frac{1}{2\pi(R_o + R_L)C_E} = \frac{1}{2\pi(3.3 \text{ k}\Omega + 4.7 \text{ k}\Omega)(10 \text{ }\mu\text{F})} \\
 &= \mathbf{1.989 \text{ Hz}}
 \end{aligned}$$

$$\text{(e)} \quad f_1 = f_{L_s} \cong \mathbf{1.7 \text{ kHz}}$$

(f) –

(g) –

19. From problem 15

(a) $r_e = \mathbf{28.48 \, \Omega}$

(b) $A_{V_{\text{mid}}} = \mathbf{-72.91}$

(c) $Z_i = \mathbf{2.455 \, k\Omega}$

(d)
$$f_{L_s} = \frac{1}{2\pi(R_i + R_s)C_s} = \frac{1}{2\pi(2.455 \, \text{k}\Omega + 0.82 \, \text{k}\Omega)(0.47 \, \mu\text{F})}$$

$f_{L_s} = \mathbf{103.4 \, \text{Hz}}$

$f_{L_c} = \mathbf{38.05 \, \text{Hz}}$ from problem 15

$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

with $R_e = R_E \parallel \left(\frac{R'_S}{\beta} + r_e \right)$ and $R'_S = R_S \parallel R_1 \parallel R_2$

$$R'_S = 0.82 \, \text{k}\Omega \parallel 68 \, \text{k}\Omega \parallel 10 \, \text{k}\Omega = 810.2 \, \text{k}\Omega \parallel 10 \, \text{k}\Omega = 749.48 \, \Omega$$

$$\frac{R'_S}{\beta} + r_e = \frac{749.48 \, \Omega}{120} + 28.48 \, \Omega = 6.25 \, \Omega + 28.48 \, \Omega = 34.73 \, \Omega$$

$$f_{L_E} = \frac{1}{2\pi(33.75 \, \Omega)(20 \, \mu\text{F})} = \mathbf{235.79 \, \text{Hz}}$$

(e) $f_1 = f_{L_E} = \mathbf{235.79 \, \text{Hz}}$

(f) –

(g) –

20. From problem 16

(a) $r_e = \mathbf{6.93 \, \Omega}$

(b) $A_{v_{\text{mid}}} = \mathbf{-264.24}$

(c) $Z_i = \mathbf{761.07 \, \Omega}$

$$(d) \quad f_{L_s} = \frac{1}{2\pi(R_s + Z_i)C_s} = \frac{1}{2\pi(600 \, \Omega + 761.07 \, \Omega)(1 \, \mu\text{F})}$$

$$= \mathbf{116.93 \, \text{Hz}}$$

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(3 \, \text{k}\Omega + 4.7 \, \text{k}\Omega)(1 \, \mu\text{F})}$$

$$= \mathbf{20.67 \, \text{Hz}}$$

$$f_{L_E} = \frac{1}{2\pi R_e C_E}$$

$$= \frac{1}{2\pi(12.21 \, \Omega)(6.8 \, \mu\text{F})}$$

$$= \mathbf{1.917 \, \text{Hz}}$$

$$R_e = R_E \parallel \frac{R'_s}{\beta} + r_e = 0.91 \, \text{k}\Omega \parallel \frac{R_s \parallel R_B}{\beta} + r_e$$

$$= 0.91 \, \text{k}\Omega \parallel \frac{0.6 \, \text{k}\Omega \parallel 470 \, \text{k}\Omega}{100} + 6.93 \, \Omega$$

$$= 0.91 \, \text{k}\Omega \parallel 12.38 \, \Omega$$

$$= 12.21 \, \Omega$$

$$(e) \quad f_1 = f_{L_E} \cong \mathbf{1.917 \, \text{kHz}}$$

(f) –

(g) –

21. From problem 17

$$(a) \quad r_e = \mathbf{30.23 \, \Omega}$$

$$(b) \quad A_{v_{\text{mid}}} = \mathbf{0.983}$$

$$(c) \quad Z_i = \mathbf{21.13 \, \text{k}\Omega}$$

$$(d) \quad f_{L_s} = \frac{1}{2\pi(R_s + R_i)C_s} = \frac{1}{2\pi(1 \, \text{k}\Omega + 21.13 \, \text{k}\Omega)(0.1 \, \mu\text{F})} = \mathbf{71.92 \, \text{Hz}}$$

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C}$$

$$= \frac{1}{2\pi(39.12 \, \Omega + 8.2 \, \text{k}\Omega)(0.1 \, \mu\text{F})}$$

$$= \mathbf{193.16 \, \text{Hz}}$$

$$R_o = R_E \parallel \frac{R'_s}{\beta} + r_e$$

$$R'_s = R_s \parallel R_1 \parallel R_2$$

$$= 1 \, \text{k}\Omega \parallel 120 \, \text{k}\Omega \parallel 30 \, \text{k}\Omega$$

$$= 0.96 \, \text{k}\Omega$$

$$R_o = (2.2 \, \text{k}\Omega) \parallel \frac{(0.96 \, \text{k}\Omega)}{100} + 30.23 \, \Omega$$

$$= 39.12 \, \Omega$$

$$(e) \quad f_{1_{\text{low}}} = f_{L_C} \cong \mathbf{193.16 \, \text{Hz}}$$

(f) –

(g) –

22. From problem 18

(a) $r_e = \mathbf{9.45 \, \Omega}$

(b) $A_{v_{\text{mid}}} = \mathbf{205.1}$

(c) $Z_i = \mathbf{9.38 \, \Omega}$

(d) $f_{L_s} = \frac{1}{2\pi(R_s + Z_i)C_s} = \frac{1}{2\pi(100 \, \Omega + 9.38 \, \Omega)(10 \, \mu\text{F})} = \mathbf{145.5 \, \text{kHz}}$
 $f_{L_c} = \frac{1}{2\pi(R_o + R_L)C_E} = \frac{1}{2\pi(3.3 \, \text{k}\Omega + 4.7 \, \text{k}\Omega)(10 \, \mu\text{F})} = \mathbf{1.989 \, \text{Hz}}$

(e) $f_1 = f_{L_s} = \mathbf{145.5 \, \text{Hz}}$

(f) –

(g) –

23. (a) $V_{GS} = -I_D R_S$

$$\left. \begin{aligned} I_D &= I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \\ V_{GS} &\cong \mathbf{-2.45 \, \text{V}} \\ I_{D_Q} &\cong \mathbf{2.1 \, \text{mA}} \end{aligned} \right\}$$

(b) $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(6 \, \text{mA})}{6 \, \text{V}} = \mathbf{2 \, \text{mS}}$

$$g_m = g_{m0} \left(1 - \frac{V_{GS_Q}}{V_P} \right) = 2 \, \text{mS} \left(1 - \frac{(-2.45 \, \text{V})}{(-6 \, \text{V})} \right) = \mathbf{1.18 \, \text{mS}}$$

(c) $A_{v_{\text{mid}}} = -g_m(R_D \parallel R_L)$
 $= -1.18 \, \text{mS}(3 \, \text{k}\Omega \parallel 3.9 \, \text{k}\Omega) = -1.18 \, \text{mS}(1.6956 \, \text{k}\Omega)$
 $= \mathbf{-2}$

(d) $Z_i = R_G = \mathbf{1 \, \text{M}\Omega}$

(e) $A_{v_s} = A_v = \mathbf{-2}$

$$(f) \quad f_{LG} = \frac{1}{2\pi(R_{sig} + R_i)C_G} = \frac{1}{2\pi(1 \text{ k}\Omega + 1 \text{ M}\Omega)(0.1 \text{ }\mu\text{F})}$$

$$= \mathbf{1.59 \text{ Hz}}$$

$$f_{LC} = \frac{1}{2\pi(R_o + R_L)C_C}$$

$$= \frac{1}{2\pi(3 \text{ k}\Omega + 3.9 \text{ k}\Omega)(4.7 \text{ }\mu\text{F})}$$

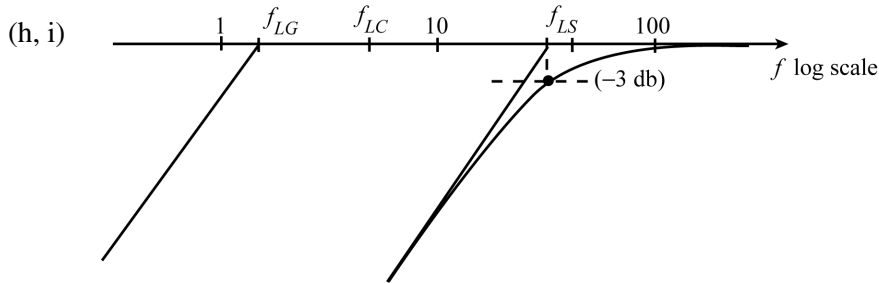
$$= \mathbf{4.91 \text{ Hz}}$$

$$f_{LS} = \frac{1}{2\pi R_{eq} C_S} \quad R_{eq} = R_S \parallel \frac{1}{g_m} = 1.2 \text{ k}\Omega \parallel \frac{1}{1.18 \text{ mS}} = 1.2 \text{ k}\Omega \parallel 847.46 \text{ }\Omega$$

$$= \frac{1}{2\pi(496.69 \text{ }\Omega)(10 \text{ }\mu\text{F})} = 496.69 \text{ }\Omega$$

$$= \mathbf{32.04 \text{ Hz}}$$

$$(g) \quad f_1 \cong f_{LS} \cong \mathbf{32 \text{ Hz}}$$



24. (a) same as problem 19
 $V_{GS_Q} \cong \mathbf{-2.45 \text{ V}}, I_{D_Q} \cong \mathbf{2.1 \text{ mA}}$

(b) $g_{m0} = \mathbf{2 \text{ mS}}, g_m = \mathbf{1.18 \text{ mS}}$ (r_d has no effect!)

(c) $A_{v_{mid}} = -g_m(R_D \parallel R_L \parallel r_d)$
 $= -1.18 \text{ mS}(3 \text{ k}\Omega \parallel 3.9 \text{ k}\Omega \parallel 100 \text{ k}\Omega)$
 $= -1.18 \text{ mS}(1.67 \text{ k}\Omega)$
 $= \mathbf{-1.971}$ (vs. -2 for problem 19)

(d) $Z_i = R_G = \mathbf{1 \text{ M}\Omega}$ (the same)

(e) $A_{v_{s(mid)}} = \frac{Z_i}{Z_i + R_{sig}}(A_{v_{mid}}) = \frac{1 \text{ M}\Omega}{1 \text{ M}\Omega + 1 \text{ k}\Omega}(-1.971)$
 $= \mathbf{-1.969}$ vs. -2 for problem 19

(f) $f_{L_G} = \mathbf{1.59 \text{ Hz}}$ (no effect)

$f_{L_C} : R_o = R_D \parallel r_d = 3 \text{ k}\Omega \parallel 100 \text{ k}\Omega = 2.91 \text{ k}\Omega$

$$f_{L_C} = \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(2.91 \text{ k}\Omega + 3.9 \text{ k}\Omega)(4.7 \text{ }\mu\text{F})}$$

$= \mathbf{4.97 \text{ Hz}}$ vs. 4.91 Hz for problem 19

$$f_{L_S} : R_{eq} = \frac{R_S}{1 + R_S(1 + g_m r_d)/(r_d + (R_D \parallel R_L))}$$

$$= \frac{1.2 \text{ k}\Omega}{1 + (1.2 \text{ k}\Omega)(1 + (1.18 \text{ mS})(100 \text{ k}\Omega))/(100 \text{ k}\Omega + 3 \text{ k}\Omega \parallel 3.9 \text{ k}\Omega)}$$

$$= \frac{1.2 \text{ k}\Omega}{1 + 1.404}$$

$$\cong 499.2 \text{ }\Omega$$

$$f_{L_S} : = \frac{1}{2\pi R_{eq} C_S} = \frac{1}{2\pi(499.2 \text{ }\Omega)(10 \text{ }\mu\text{F})}$$

$= \mathbf{31.88 \text{ Hz}}$ vs. 32.04 for problem 19.

Effect of $r_d = 100 \text{ k}\Omega$ insignificant!

25. (a) $V_G = \frac{68 \text{ k}\Omega(20 \text{ V})}{68 \text{ k}\Omega + 220 \text{ k}\Omega} = 4.72 \text{ V}$

$$\left. \begin{array}{l} V_{GS} = V_G - I_D R_S \\ V_{GS} = 4.72 \text{ V} - I_D(2.2 \text{ k}\Omega) \\ I_D = I_{DSS}(1 - V_{GS}/V_P)^2 \end{array} \right\} \begin{array}{l} V_{GS_Q} \cong \mathbf{-2.55 \text{ V}} \\ I_{D_Q} \cong \mathbf{3.3 \text{ mA}} \end{array}$$

(b) $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10 \text{ mA})}{6 \text{ V}} = 3.33 \text{ mS}$

$$g_m = g_{m0} \left(1 - \frac{V_{GS}}{V_P}\right) = 3.33 \text{ mS} \left(1 - \frac{-2.55 \text{ V}}{-6 \text{ V}}\right)$$

$= \mathbf{1.91 \text{ mS}}$

(c) $A_{v_{mid}} = -g_m(R_D \parallel R_L)$

$$= -(1.91 \text{ mS})(3.9 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega)$$

$= \mathbf{-4.39}$

(d) $Z_i = 68 \text{ k}\Omega \parallel 220 \text{ k}\Omega = \mathbf{51.94 \text{ k}\Omega}$

(e) $A_{v_{s(mid)}} = \frac{V_o}{V_i} \cdot \frac{V_i}{V_s}$

$$\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_s} = \frac{51.94 \text{ k}\Omega}{51.94 \text{ k}\Omega + 1.5 \text{ k}\Omega} = 0.972$$

$$A_{v_{s(mid)}} = (-4.39)(0.972) = \mathbf{-4.27}$$

$$(f) \quad f_{LG} = \frac{1}{2\pi(R_{sig} + R_i)C_G} = \frac{1}{2\pi(1.5 \text{ k}\Omega + 51.94 \text{ k}\Omega)(1 \text{ }\mu\text{F})}$$

$$= \mathbf{2.98 \text{ Hz}}$$

$$f_{LC} = \frac{1}{2\pi(R_o + R_L)C_C} = \frac{1}{2\pi(3.9 \text{ k}\Omega + 5.6 \text{ k}\Omega)(6.8 \text{ }\mu\text{F})}$$

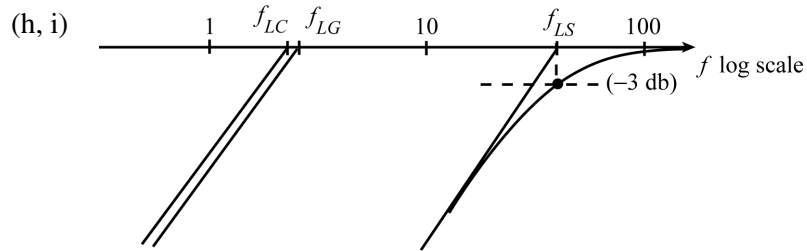
$$= \mathbf{2.46 \text{ Hz}}$$

$$f_{LS} = \frac{1}{2\pi R_{eq} C_S} \qquad R_{eq} = R_S \parallel \frac{1}{g_m} = 1.5 \text{ k}\Omega \parallel \frac{1}{1.91 \text{ mS}}$$

$$= \frac{1}{2\pi(388.1 \text{ }\Omega)(10 \text{ }\mu\text{F})} \qquad = 1.5 \text{ k}\Omega \parallel 523.56 \text{ }\Omega$$

$$= \mathbf{41 \text{ Hz}} \qquad = 388.1 \text{ }\Omega$$

$$(g) \quad f_1 = f_{LS} \cong \mathbf{41 \text{ Hz}}$$



26. (a) $C_{M_i} = (1 - A_v)C_f$

$$= (1 - (-120))(10 \text{ pF})$$

$$= \mathbf{1210 \text{ pF}}$$

(b) $C_{M_o} = (1 - \frac{1}{A_v})C_f = 1 - \frac{1}{(-120)} 10 \text{ pF}$

$$= \mathbf{10.08 \text{ pF}}$$

(c) absolutely

27. (a)
$$f_{H_i} = \frac{1}{2\pi R_{Th_1} C_i}$$

$$= \frac{1}{2\pi(614.56 \Omega)(931.92 \text{ pF})}$$

$$= \mathbf{277.89 \text{ kHz}}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_2} C_o}$$

$$= \frac{1}{2\pi(2.08 \text{ k}\Omega)(28 \text{ pF})}$$

$$= \mathbf{2.73 \text{ MHz}}$$

$$R_{Th_1} = R_s \parallel R_1 \parallel R_2 \parallel R_i$$

$$= \underbrace{0.82 \text{ k}\Omega \parallel 68 \text{ k}\Omega}_{0.81 \text{ k}\Omega} \parallel \underbrace{10 \text{ k}\Omega \parallel 3.418 \text{ k}\Omega}_{2.547 \text{ k}\Omega}$$

$$= 614.56 \Omega$$

$$C_i = C_{W_1} + C_{be} + C_{bc}(1 - A_v)$$

$$= 5 \text{ pF} + 40 \text{ pF} + 12 \text{ pF}(1 - (-72.91))$$

$$= 931.92 \text{ pF} \quad \uparrow \text{Prob. 15}$$

$$R_{Th_2} = R_C \parallel R_L = 5.6 \text{ k}\Omega \parallel 3.3 \text{ k}\Omega$$

$$= 2.08 \text{ k}\Omega$$

$$C_o = C_{W_o} + C_{ce} + C_{M_o}$$

$$= 8 \text{ pF} + 8 \text{ pF} + 12 \text{ pF}$$

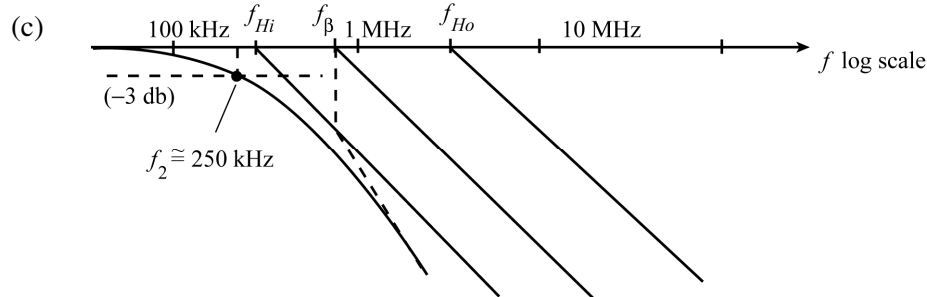
$$= 28 \text{ pF}$$

(b)
$$f_\beta \cong \frac{1}{2\pi\beta_{\text{mid}} r_e (C_{be} + C_{bc})} = \frac{1}{2\pi(120)(28.48 \Omega)(40 \text{ pF} + 12 \text{ pF})}$$

$$= \mathbf{895.56 \text{ kHz}} \quad \uparrow \text{Prob. 15}$$

$$f_T = \beta f_\beta = (120)(895.56 \text{ kHz})$$

$$= \mathbf{107.47 \text{ MHz}}$$



(d)
$$\text{GBP} = (A_{v_{\text{mid}}})(\text{BW})$$

$$= (72.91)(250 \text{ kHz})$$

$$= \mathbf{18.23 \text{ MHz}}$$

28. (a)
$$f_{H_i} = \frac{1}{2\pi R_{Th_1} C_i}$$

$$R_{Th_1} = R_s \parallel R_B \parallel R_i$$

$$R_i: I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{20 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (111)(0.91 \text{ k}\Omega)}$$

$$= 33.8 \mu\text{A}$$

$$I_E = (\beta + 1)I_B = (110 + 1)(33.8 \mu\text{A})$$

$$= 3.75 \text{ mA}$$

$$r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{3.75 \text{ mA}} = 6.93 \Omega$$

$$R_i = \beta r_e = (110)(6.93 \Omega) \\ = 762.3 \Omega$$

$$R_{Th_1} = R_s \parallel R_B \parallel R_i = 0.6 \text{ k}\Omega \parallel 470 \text{ k}\Omega \parallel 762.3 \Omega \\ = 335.50 \Omega$$

$$f_{H_i} = \frac{1}{2\pi(335.50 \Omega)(C_i)}$$

$$C_i: C_i = C_{W_i} + C_{be} + (1 - A_v)C_{bc}$$

$$A_v: A_{v_{mid}} = \frac{-(R_L \parallel R_C)}{r_e} = \frac{-(4.7 \text{ k}\Omega \parallel 3 \text{ k}\Omega)}{6.93 \Omega} \\ = -264.2$$

$$C_i = 7 \text{ pF} + 20 \text{ pF} + (1 - (-264.2))6 \text{ pF} \\ = 1.62 \text{ nF}$$

$$f_{H_i} = \frac{1}{2\pi(335.50 \Omega)(1.62 \text{ nF})} \\ \cong \mathbf{293 \text{ kHz}}$$

$$f_{H_o} = \frac{1}{2\pi R_{Th_2} C_o}$$

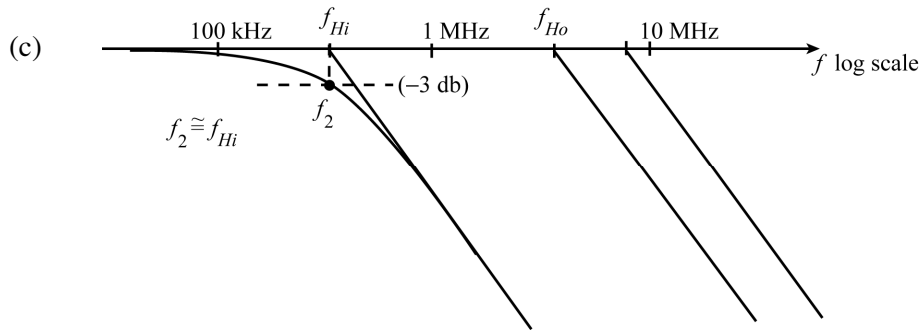
$$R_{Th_2} = R_C \parallel R_L = 3 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 1.831 \text{ k}\Omega$$

$$C_o = C_{W_o} + C_{ce} + \underbrace{C_{M_o}}_{\cong C_f = C_{bc}} \\ = 11 \text{ pF} + 10 \text{ pF} + 6 \text{ pF} \\ = 27 \text{ pF}$$

$$f_{H_o} = \frac{1}{2\pi(1.831 \text{ k}\Omega)(27 \text{ pF})} \\ = \mathbf{3.22 \text{ MHz}}$$

$$(b) f_\beta = \frac{1}{2\pi\beta_{mid}r_e(C_{be} + C_{bc})} \\ = \frac{1}{2\pi(110)(6.93 \Omega)(20 \text{ pF} + 6 \text{ pF})} \\ = \mathbf{8.03 \text{ MHz}}$$

$$f_T = \beta_{mid}f_\beta = (110)(8.03 \text{ MHz}) \\ = \mathbf{883.3 \text{ MHz}}$$



(d) $GBP = (A_{v_{mid}})(BW)$
 $= (264.24)(293 \text{ kHz})$
 $= \mathbf{77.42 \text{ MHz}}$

29.

(a) $f_{Hi} = \frac{1}{2\pi R_{Th_1} C_i}$
 $= \frac{1}{2\pi(955 \Omega)(58 \text{ pF})}$
 $= \mathbf{2.87 \text{ MHz}}$

$$R_{Th_1} = R_s \parallel R_1 \parallel R_2 \parallel Z_b$$

$$Z_b = \beta r_e + (\beta + 1)(R_E \parallel R_L)$$

$$= (100)(30.23 \Omega) + (101)(2.2 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega)$$

$$= 3.023 \text{ k}\Omega + 175.2 \text{ k}\Omega$$

$$= 178.2 \text{ k}\Omega$$

$$R_{Th_1} = 1 \text{ k}\Omega \parallel 120 \text{ k}\Omega \parallel 30 \text{ k}\Omega \parallel 178.2 \text{ k}\Omega$$

$$= 955 \Omega$$

$$C_i = C_{W_i} + C_{be} + C_{bc} \text{ (No Miller effect)}$$

$$= 8 \text{ pF} + 30 \text{ pF} + 20 \text{ pF}$$

$$= 58 \text{ pF}$$

$$f_{Ho} = \frac{1}{2\pi R_{Th_2} C_o}$$

$$= \frac{1}{2\pi(38.94 \Omega)(32 \text{ pF})}$$

$$= \mathbf{127.72 \text{ MHz}}$$

$$R_{Th_2} = R_E \parallel R_L \parallel r_e + \frac{\overbrace{R_1 \parallel R_2 \parallel R_3}^{24 \text{ k}\Omega}}{\beta}$$

$$= 2.2 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega \parallel 30.23 \Omega + \frac{24 \text{ k}\Omega \parallel 1 \text{ k}\Omega}{100}$$

$$= 1.735 \text{ k}\Omega \parallel (30.23 \Omega + 9.6 \Omega)$$

$$= 1.735 \text{ k}\Omega \parallel 39.83 \Omega$$

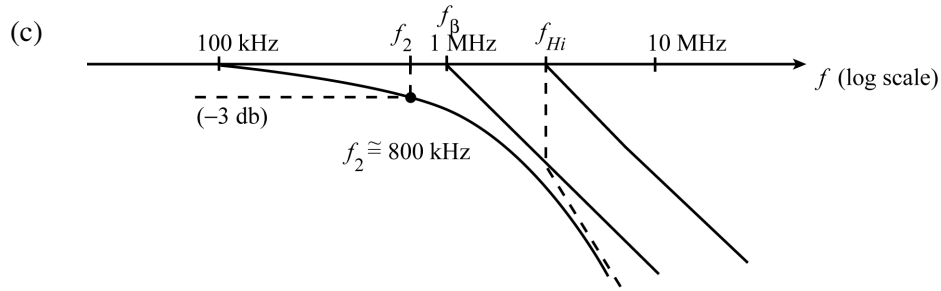
$$= 38.94 \Omega$$

$$C_o = C_{W_o} + C_{ce}$$

$$= 10 \text{ pF} + 12 \text{ pF}$$

$$= 32 \text{ pF}$$

(b) $f_\beta = \frac{1}{2\pi\beta_{mid}r_e(C_{be} + C_{bc})}$
 $= \frac{1}{2\pi(100)(30.23 \Omega)(30 \text{ pF} + 20 \text{ pF})}$
 $= \mathbf{1.05 \text{ MHz}}$
 $f_T = \beta_{mid}f_\beta = 100(1.05 \text{ MHz}) = \mathbf{105 \text{ MHz}}$



(d) $GBP = (A_{v_{mid}})(BW)$
 $= (0.983)(800 \text{ kHz})$
 $= \mathbf{786.4 \text{ MHz}}$

30. (a) $f_{H_i} = \frac{1}{2\pi R_{Th_1} C_i}$
 $R_{Th_1} = R_s \parallel R_E \parallel R_i$
 $R_i: I_E = \frac{V_{EE} - V_{BE}}{R_E} = \frac{4 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega} = 2.75 \text{ mA}$
 $r_e = \frac{26 \text{ mV}}{I_E} = \frac{26 \text{ mV}}{2.75 \text{ mA}} = 9.45 \text{ }\Omega$
 $R_i = R_E \parallel r_e = 1.2 \text{ k}\Omega \parallel 9.45 \text{ }\Omega$
 $= 9.38 \text{ }\Omega$

$C_i: C_i = C_{W_i} + C_{be}$ (no Miller cap-noninverting!)
 $= 8 \text{ pF} + 24 \text{ pF}$
 $= 32 \text{ pF}$
 $R_i = 0.1 \text{ k}\Omega \parallel 1.2 \text{ k}\Omega \parallel 9.38 \text{ }\Omega = 8.52 \text{ }\Omega$

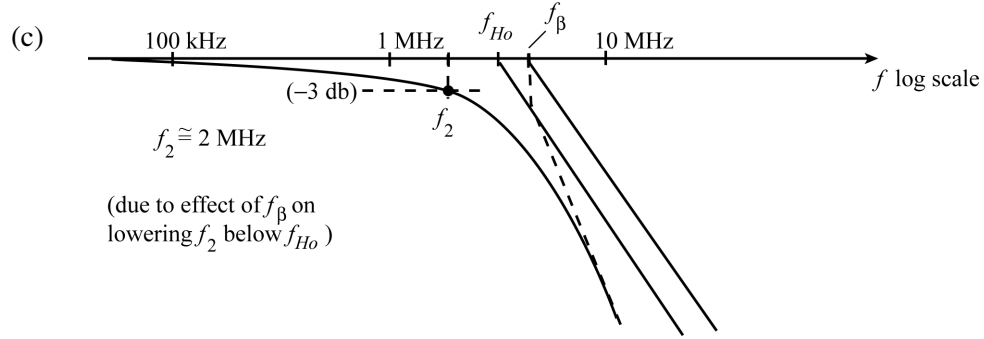
$f_{H_i} = \frac{1}{2\pi(8.52 \text{ }\Omega)(32 \text{ pF})} \cong \mathbf{584 \text{ MHz}}$

$f_{H_o} = \frac{1}{2\pi R_{Th_2} C_o}$ $R_{Th_2} = R_C \parallel R_L = 3.3 \text{ k}\Omega \parallel 4.7 \text{ k}\Omega = 1.94 \text{ k}\Omega$

$C_o = C_{W_o} + C_{bc}$ (no Miller)
 $= 10 \text{ pF} + 18 \text{ pF}$
 $= 28 \text{ pF}$

$f_{H_o} = \frac{1}{2\pi(1.94 \text{ k}\Omega)(28 \text{ pF})}$
 $= \mathbf{2.93 \text{ MHz}}$

$$\begin{aligned}
 (b) \quad f_{\beta} &= \frac{1}{2\pi\beta_{\text{mid}}r_e(C_{be} + C_{bc})} \\
 &= \frac{1}{2\pi(80)(9.45 \Omega)(24 \text{ pF} + 18 \text{ pF})} \\
 &= \mathbf{5.01 \text{ MHz}} \\
 f_T &= \beta_{\text{mid}}f_{\beta} = (80)(5.01 \text{ MHz}) \\
 &= \mathbf{400.8 \text{ MHz}}
 \end{aligned}$$



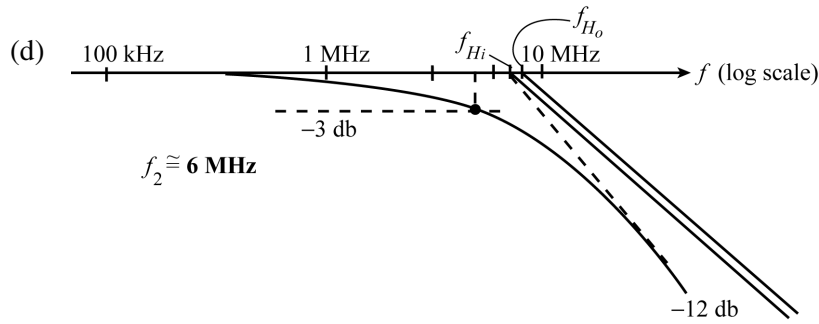
$$\begin{aligned}
 (d) \quad \text{GBP} &= (A_{v_{\text{mid}}})(\text{BW}) \\
 &= (205.1)(2 \text{ MHz}) \\
 &= \mathbf{410.2 \text{ MHz}}
 \end{aligned}$$

31. (a) From problem 23 $g_{m0} = \mathbf{2 \text{ mS}}$, $g_m = \mathbf{1.18 \text{ mS}}$

(b) From problem 23 $A_{v_{\text{mid}}} \cong A_{v_s(\text{mid})} = \mathbf{-2}$

$$\begin{aligned}
 (c) \quad f_{H_i} &= \frac{1}{2\pi R_{Th} C_i} \\
 f_{H_i} &= \frac{1}{2\pi(999 \Omega)(21 \text{ pF})} \\
 &= \mathbf{7.59 \text{ MHz}} \\
 f_{H_o} &= \frac{1}{2\pi R_{Th_2} C_o} \\
 &= \frac{1}{2\pi(1.696 \text{ k}\Omega)(12 \text{ pF})} \\
 &= \mathbf{7.82 \text{ MHz}}
 \end{aligned}$$

$$\begin{aligned}
 R_{Th_1} &= R_{\text{sig}} \parallel R_G \\
 &= 1 \text{ k}\Omega \parallel 1 \text{ M}\Omega \\
 &= 999 \Omega \\
 C_i &= C_{W_i} + C_{gs} + C_{M_i} \\
 C_{M_i} &= (1 - A_v)C_{gd} \\
 &= (1 - (-2))4 \text{ pF} \\
 &= 12 \text{ pF} \\
 C_i &= 3 \text{ pF} + 6 \text{ pF} + 12 \text{ pF} \\
 &= 21 \text{ pF} \\
 R_{Th_2} &= R_D \parallel R_L \\
 &= 3 \text{ k}\Omega \parallel 3.9 \text{ k}\Omega \\
 &= 1.696 \text{ k}\Omega \\
 C_o &= C_{W_o} + C_{ds} + C_{M_o} \\
 C_{M_o} &= 1 - \frac{1}{-2} 4 \text{ pF} \\
 &= (1.5)(4 \text{ pF}) \\
 &= 6 \text{ pF} \\
 C_o &= 5 \text{ pF} + 1 \text{ pF} + 6 \text{ pF} \\
 &= 12 \text{ pF}
 \end{aligned}$$



(e) $GBP = (A_{v_{mid}})(BW)$
 $= (2)(6 \text{ MHz})$
 $= \mathbf{12 \text{ MHz}}$

32. (a) $g_{m0} = \frac{2I_{DSS}}{|V_P|} = \frac{2(10 \text{ mA})}{6 \text{ V}} = \mathbf{3.33 \text{ mS}}$

From problem #21 $V_{GS_Q} \approx -2.55 \text{ V}$, $I_{D_Q} \approx 3.3 \text{ mA}$

$$g_m = g_{m0} \left(1 - \frac{V_{GS_Q}}{V_P} \right) = 3.33 \text{ mS} \left(1 - \frac{-2.55 \text{ V}}{-6 \text{ V}} \right) = \mathbf{1.91 \text{ mS}}$$

(b) $A_{v_{mid}} = -g_m(R_D \parallel R_L)$
 $= -(1.91 \text{ mS})(3.9 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega)$
 $= \mathbf{-4.39}$

$$Z_i = 68 \text{ k}\Omega \parallel 220 \text{ k}\Omega = 51.94 \text{ k}\Omega$$

$$\frac{V_i}{V_s} = \frac{Z_i}{Z_i + R_{sig}} = \frac{51.94 \text{ k}\Omega}{51.94 \text{ k}\Omega + 1.5 \text{ k}\Omega} = 0.972$$

$$A_{v_s(\text{mid})} = (-4.39)(0.972)$$

$$= \mathbf{-4.27}$$

(c) $f_{Hi} = \frac{1}{2\pi R_{Th_1} C_i}$ $R_{Th_1} = R_{sig} \parallel R_1 \parallel R_2$
 $= 1.5 \text{ k}\Omega \parallel 51.94 \text{ k}\Omega$
 $= 1.46 \text{ k}\Omega$

$$C_i = C_{W_i} + C_{gs} + (1 - A_v)C_{gd}$$

$$= 4 \text{ pF} + 12 \text{ pF} + (1 - (-4.39))8 \text{ pF}$$

$$= 59.12 \text{ pF}$$

$$f_{Hi} = \frac{1}{2\pi(1.46 \text{ k}\Omega)(59.12 \text{ pF})}$$

$$= \mathbf{1.84 \text{ MHz}}$$

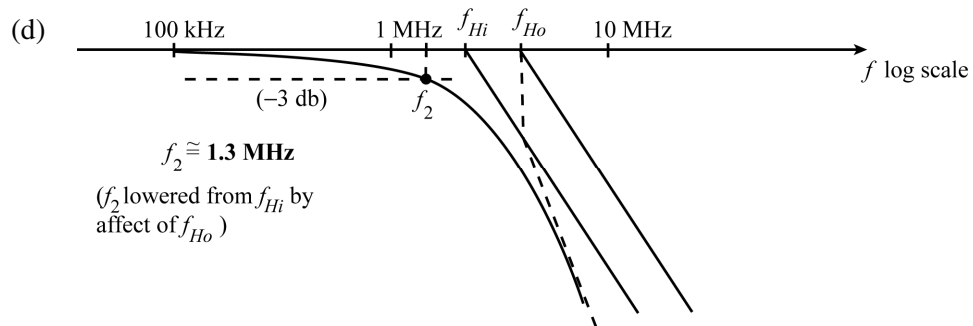
$$f_{Ho} = \frac{1}{2\pi R_{Th_2} C_o}$$

$$R_{Th_2} = R_D \parallel R_L = 3.9 \text{ k}\Omega \parallel 5.6 \text{ k}\Omega$$

$$= 2.3 \text{ k}\Omega$$

$$\begin{aligned}
 C_o &= C_{w_o} + C_{ds} + \left(1 - \frac{1}{A_v}\right) C_{gd} \\
 &= 6 \text{ pF} + 3 \text{ pF} + \left(1 - \frac{1}{(-4.39)}\right) 8 \text{ pF} \\
 &= 18.82 \text{ pF}
 \end{aligned}$$

$$\begin{aligned}
 f_{H_o} &= \frac{1}{2\pi(2.3 \text{ k}\Omega)(18.82 \text{ pF})} \\
 &= \mathbf{3.68 \text{ MHz}}
 \end{aligned}$$



(e) $\text{GBP} = (A_{v_{\text{mid}}})(\text{BW})$

$$\begin{aligned}
 &= (4.27)(1.3 \text{ MHz}) \\
 &= \mathbf{5.55 \text{ MHz}}
 \end{aligned}$$

33. $A_{v_T} = A_{v_1} \cdot A_{v_2} \cdot A_{v_3} \cdot A_{v_4}$

$$\begin{aligned}
 &= A_v^4 \\
 &= (20)^4 \\
 &= \mathbf{16 \times 10^4}
 \end{aligned}$$

34. $f_2' = \left(\sqrt{2^{1/n}} - 1\right) f_2$

$$\begin{aligned}
 &= \left(\underbrace{\sqrt{2^{1/4}} - 1}_{1.18}\right) (2.5 \text{ MHz}) \\
 &= 0.435(2.5 \text{ MHz}) \\
 &= \mathbf{1.09 \text{ MHz}}
 \end{aligned}$$

35. $f_1' = \frac{f_1}{\sqrt{2^{1/n}} - 1} = \frac{40 \text{ Hz}}{\sqrt{2^{1/4}} - 1}$

$$\begin{aligned}
 &= \frac{40 \text{ Hz}}{0.435} \\
 &= \mathbf{91.96 \text{ Hz}}
 \end{aligned}$$

$$\begin{aligned}
36. \quad (a) \quad v &= \frac{4}{\pi} V_m \sin 2\pi f_s t + \frac{1}{3} \sin 2\pi(3f_s)t + \frac{1}{5} \sin 2\pi(5f_s)t \\
&\quad + \frac{1}{7} \sin 2\pi(7f_s)t + \frac{1}{9} \sin 2\pi(9f_s)t + \dots \\
&= 12.73 \times 10^{-3} (\sin 2\pi(100 \times 10^3)t + \frac{1}{3} \sin 2\pi(300 \times 10^3)t \\
&\quad + \frac{1}{5} \sin 2\pi(500 \times 10^3)t + \frac{1}{7} \sin 2\pi(700 \times 10^3)t + \frac{1}{9} \sin 2\pi(900 \times 10^3)t)
\end{aligned}$$

$$\begin{aligned}
(b) \quad BW &\cong \frac{0.35}{t_r} && \text{At 90\% or 81 mV, } t \cong 0.75 \mu\text{s} \\
&\cong \frac{0.35}{0.7 \mu\text{s}} && \text{At 10\% or 9 mV, } t \cong 0.05 \mu\text{s} \\
&\cong 500 \text{ kHz} && t_r \cong 0.75 \mu\text{s} - 0.05 \mu\text{s} = 0.7 \mu\text{s}
\end{aligned}$$

$$\begin{aligned}
(c) \quad P &= \frac{V - V'}{V} = \frac{90 \text{ mV} - 80 \text{ mV}}{90 \text{ mV}} = 0.111 \\
f_{L_o} &= \frac{P}{\pi} f_s = \frac{(0.111)(100 \text{ kHz})}{\pi} \cong \mathbf{3.53 \text{ kHz}}
\end{aligned}$$

Chapter 10

$$1. \quad V_o = -\frac{R_F}{R_1} V_1 = -\frac{250 \text{ k}\Omega}{20 \text{ k}\Omega} (1.5 \text{ V}) = \mathbf{-18.75 \text{ V}}$$

$$2. \quad A_v = \frac{V_o}{V_i} = -\frac{R_F}{R_1}$$

For $R_1 = 10 \text{ k}\Omega$:

$$A_v = -\frac{500 \text{ k}\Omega}{10 \text{ k}\Omega} = \mathbf{-50}$$

For $R_1 = 20 \text{ k}\Omega$:

$$A_v = -\frac{500 \text{ k}\Omega}{20 \text{ k}\Omega} = \mathbf{-25}$$

$$3. \quad V_o = -\frac{R_f}{R_1} V_1 = -\frac{1 \text{ M}\Omega}{20 \text{ k}\Omega} \quad V_1 = 2 \text{ V}$$

$$V_1 = \frac{2 \text{ V}}{-50} = \mathbf{-40 \text{ mV}}$$

$$4. \quad V_o = -\frac{R_F}{R_1} V_1 = -\frac{200 \text{ k}\Omega}{20 \text{ k}\Omega} V_1 = -10 V_1$$

For $V_1 = 0.1 \text{ V}$:

$$V_o = -10(0.1 \text{ V}) = \mathbf{-1 \text{ V}}$$

For $V_1 = 0.5 \text{ V}$:

$$V_o = -10(0.5 \text{ V}) = \mathbf{-5 \text{ V}}$$

} V_o ranges
from
 $\mathbf{-1 \text{ V to } -5 \text{ V}}$

$$5. \quad V_o = 1 + \frac{R_F}{R_1} \quad V_1 = 1 + \frac{360 \text{ k}\Omega}{12 \text{ k}\Omega} (-0.3 \text{ V})$$

$$= 31(-0.3 \text{ V}) = \mathbf{-9.3 \text{ V}}$$

$$6. \quad V_o = 1 + \frac{R_F}{R_1} \quad V_1 = 1 + \frac{360 \text{ k}\Omega}{12 \text{ k}\Omega} \quad V_1 = 2.4 \text{ V}$$

$$V_1 = \frac{2.4 \text{ V}}{31} = \mathbf{77.42 \text{ mV}}$$

$$7. \quad V_o = 1 + \frac{R_F}{R_1} \quad V_1$$

For $R_1 = 10 \text{ k}\Omega$:

$$V_o = 1 + \frac{200 \text{ k}\Omega}{10 \text{ k}\Omega} (0.5 \text{ V}) = 21(0.5 \text{ V}) = \mathbf{10.5 \text{ V}}$$

For $R_1 = 20 \text{ k}\Omega$:

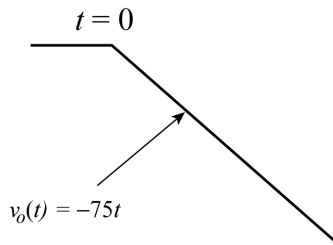
$$V_o = 1 + \frac{200 \text{ k}\Omega}{20 \text{ k}\Omega} (0.5 \text{ V}) = 11(0.5 \text{ V}) = \mathbf{5.5 \text{ V}}$$

V_o ranges from 5.5 V to 10.5 V.

$$\begin{aligned} 8. \quad V_o &= -\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \\ &= -\frac{330 \text{ k}\Omega}{33 \text{ k}\Omega} (0.2 \text{ V}) + \frac{330 \text{ k}\Omega}{22 \text{ k}\Omega} (-0.5 \text{ V}) + \frac{330 \text{ k}\Omega}{12 \text{ k}\Omega} (0.8 \text{ V}) \\ &= -[10(0.2 \text{ V}) + 15(-0.5 \text{ V}) + 27.5(0.8 \text{ V})] \\ &= -[2 \text{ V} + (-7.5 \text{ V}) + 2.2 \text{ V}] \\ &= -[24 \text{ V} - 7.5 \text{ V}] = \mathbf{-16.5 \text{ V}} \end{aligned}$$

$$\begin{aligned} 9. \quad V_o &= -\frac{R_F}{R_1} V_1 + \frac{R_F}{R_2} V_2 + \frac{R_F}{R_3} V_3 \\ &= -\frac{68 \text{ k}\Omega}{33 \text{ k}\Omega} (0.2 \text{ V}) + \frac{68 \text{ k}\Omega}{22 \text{ k}\Omega} (-0.5 \text{ V}) + \frac{68 \text{ k}\Omega}{12 \text{ k}\Omega} (+0.8 \text{ V}) \\ &= -[0.41 \text{ V} - 1.55 \text{ V} + 4.53 \text{ V}] \\ &= \mathbf{-3.39 \text{ V}} \end{aligned}$$

$$\begin{aligned} 10. \quad v_o(t) &= -\frac{1}{RC} \int v_1(t) dt \\ &= -\frac{1}{(200 \text{ k}\Omega)(0.1 \mu\text{F})} \int 1.5 dt \\ &= -50(1.5t) = \mathbf{-75t} \end{aligned}$$



$$11. \quad V_o = V_1 = \mathbf{+0.5 \text{ V}}$$

$$\begin{aligned} 12. \quad V_o &= -\frac{R_f}{R_1} V_1 = -\frac{100 \text{ k}\Omega}{20 \text{ k}\Omega} (1.5 \text{ V}) \\ &= -5(1.5 \text{ V}) = \mathbf{-7.5 \text{ V}} \end{aligned}$$

$$\begin{aligned} 13. \quad V_2 &= -\frac{200 \text{ k}\Omega}{20 \text{ k}\Omega} (0.2 \text{ V}) = \mathbf{-2 \text{ V}} \\ V_3 &= 1 + \frac{200 \text{ k}\Omega}{10 \text{ k}\Omega} (0.2 \text{ V}) = \mathbf{+4.2 \text{ V}} \end{aligned}$$

$$\begin{aligned}
14. \quad V_o &= 1 + \frac{400 \text{ k}\Omega}{20 \text{ k}\Omega} (0.1 \text{ V}) \cdot \frac{-100 \text{ k}\Omega}{20 \text{ k}\Omega} + \frac{-100 \text{ k}\Omega}{10 \text{ k}\Omega} (0.1 \text{ V}) \\
&= (2.1 \text{ V})(-5) + (-10)(0.1 \text{ V}) \\
&= -10.5 \text{ V} - 1 \text{ V} = \mathbf{-11.5 \text{ V}}
\end{aligned}$$

$$\begin{aligned}
15. \quad V_o &= -\frac{600 \text{ k}\Omega}{15 \text{ k}\Omega}(25 \text{ mV}) + \frac{600 \text{ k}\Omega}{30 \text{ k}\Omega}(-20 \text{ mV}) - \frac{300 \text{ k}\Omega}{30 \text{ k}\Omega} \\
&\quad + -\frac{300 \text{ k}\Omega}{15 \text{ k}\Omega}(-20 \text{ mV}) \\
&= -[40(25 \text{ mV}) + (20)(-20 \text{ mV})](-10) + (-20)(-20 \text{ mV}) \\
&= -[1 \text{ V} - 0.4 \text{ V}](-10) + 0.4 \text{ V} \\
&= 6 \text{ V} + 0.4 \text{ V} = \mathbf{6.4 \text{ V}}
\end{aligned}$$

$$\begin{aligned}
16. \quad V_o &= 1 + \frac{R_f}{R_1} V_{Io} + I_{Io} R_f \\
&= 1 + \frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} (6 \text{ mV}) + (120 \text{ nA})(200 \text{ k}\Omega) \\
&= 101(6 \text{ mV}) + 24 \text{ mV} \\
&= 606 \text{ mV} + 24 \text{ mV} = \mathbf{630 \text{ mV}}
\end{aligned}$$

$$\begin{aligned}
17. \quad I_{IB}^+ &= I_{IB} + \frac{I_{Io}}{2} = 20 \text{ nA} + \frac{4 \text{ nA}}{2} = \mathbf{22 \text{ nA}} \\
I_{IB}^- &= I_{IB} - \frac{I_{Io}}{2} = 20 \text{ nA} - \frac{4 \text{ nA}}{2} = \mathbf{18 \text{ nA}}
\end{aligned}$$

$$\begin{aligned}
18. \quad f_1 &= 800 \text{ kHz} \\
f_c &= \frac{f_1}{A_{v_2}} = \frac{800 \text{ kHz}}{150 \times 10^3} = \mathbf{5.3 \text{ Hz}}
\end{aligned}$$

$$19. \quad A_{CL} = \frac{SR}{\Delta V_i / \Delta t} = \frac{2.4 \text{ V}/\mu\text{s}}{0.3 \text{ V}/10 \mu\text{s}} = \mathbf{80}$$

$$\begin{aligned}
20. \quad A_{CL} &= \frac{R_f}{R_1} = \frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} = 100 \\
K &= A_{CL} V_i = 100(50 \text{ mV}) = 5 \text{ V} \\
w_s &\leq \frac{SR}{K} = \frac{0.4 \text{ V}/\mu\text{s}}{5 \text{ V}} = \mathbf{80 \times 10^3 \text{ rad/s}} \\
f_s &= \frac{w_s}{2\pi} = \frac{80 \times 10^3}{2\pi} = \mathbf{12.73 \text{ kHz}}
\end{aligned}$$

21. $V_{Io} = 1 \text{ mV}$, typical
 $I_{Io} = 20 \text{ nA}$, typical

$$\begin{aligned} V_o(\text{offset}) &= 1 + \frac{R_f}{R_1} V_{Io} + I_{Io} R_f \\ &= 1 + \frac{200 \text{ k}\Omega}{20 \text{ k}\Omega} (1 \text{ mV}) + (200 \text{ k}\Omega)(20 \text{ nA}) \\ &= 101(1 \text{ mV}) + 4000 \times 10^{-6} \\ &= 101 \text{ mV} + 4 \text{ mV} = \mathbf{105 \text{ mV}} \end{aligned}$$

22. Typical characteristics for 741
 $R_o = 25 \text{ }\Omega$, $A = 200 \text{ K}$

$$(a) \quad A_{CL} = -\frac{R_f}{R_1} = -\frac{200 \text{ k}\Omega}{2 \text{ k}\Omega} = \mathbf{-100}$$

$$(b) \quad Z_i = R_1 = \mathbf{2 \text{ k}\Omega}$$

$$\begin{aligned} (c) \quad Z_o &= \frac{R_o}{1 + \beta A} = \frac{25 \text{ }\Omega}{1 + \frac{1}{100}(200,000)} \\ &= \frac{25 \text{ }\Omega}{2001} = \mathbf{0.0125 \text{ }\Omega} \end{aligned}$$

$$23. \quad A_d = \frac{V_o}{V_d} = \frac{120 \text{ mV}}{1 \text{ mV}} = 120$$

$$A_c = \frac{V_o}{V_c} = \frac{20 \text{ }\mu\text{V}}{1 \text{ mV}} = 20 \times 10^{-3}$$

$$\begin{aligned} \text{Gain (dB)} &= 20 \log \frac{A_d}{A_c} = 20 \log \frac{120}{20 \times 10^{-3}} \\ &= 20 \log(6 \times 10^3) = \mathbf{75.56 \text{ dB}} \end{aligned}$$

$$24. \quad V_d = V_{i1} - V_{i2} = 200 \mu\text{V} - 140 \mu\text{V} = 60 \mu\text{V}$$

$$V_c = \frac{V_{i1} + V_{i2}}{2} = \frac{(200 \mu\text{V} + 140 \mu\text{V})}{2} = 170 \mu\text{V}$$

$$(a) \quad \text{CMRR} = \frac{A_d}{A_c} = 200$$

$$A_c = \frac{A_d}{200} = \frac{6000}{200} = \mathbf{30}$$

$$(b) \quad \text{CMRR} = \frac{A_d}{A_c} = 10^5$$

$$A_c = \frac{A_d}{10^5} = \frac{6000}{10^5} = 0.06 = \mathbf{60 \times 10^{-3}}$$

$$\text{Using } V_o = A_d V_d \left(1 + \frac{1}{\text{CMRR}} \frac{V_c}{V_d} \right)$$

$$(a) \quad V_o = 6000(60 \mu\text{V}) \left(1 + \frac{1}{200} \frac{170 \mu\text{V}}{60 \mu\text{V}} \right) = \mathbf{365.1 \text{ mV}}$$

$$(b) \quad V_o = 6000(60 \mu\text{V}) \left(1 + \frac{1}{10^5} \frac{170 \mu\text{V}}{60 \mu\text{V}} \right) = \mathbf{360.01 \text{ mV}}$$

Chapter 11

$$1. \quad V_o = -\frac{R_F}{R_1} V_1 = -\frac{180 \text{ k}\Omega}{3.6 \text{ k}\Omega} (3.5 \text{ mV}) = \mathbf{-175 \text{ mV}}$$

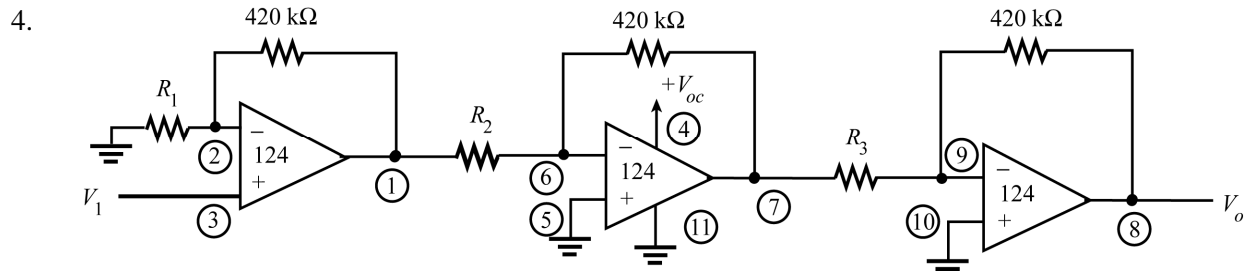
$$2. \quad V_o = 1 + \frac{R_F}{R_1} V_1 = 1 + \frac{750 \text{ k}\Omega}{36 \text{ k}\Omega} (150 \text{ mV, rms})$$

$$= \mathbf{3.275 \text{ V, rms } \angle 0^\circ}$$

$$3. \quad V_o = 1 + \frac{510 \text{ k}\Omega}{18 \text{ k}\Omega} (20 \mu\text{V}) - \frac{680 \text{ k}\Omega}{22 \text{ k}\Omega} - \frac{750 \text{ k}\Omega}{33 \text{ k}\Omega}$$

$$= (29.33)(-30.91)(-22.73)(20 \mu\text{V})$$

$$= \mathbf{412 \text{ mV}}$$



$$1 + \frac{420 \text{ k}\Omega}{R_1} = +15 \quad -\frac{420 \text{ k}\Omega}{R_2} = -22 \quad \frac{420 \text{ k}\Omega}{R_2} = -30$$

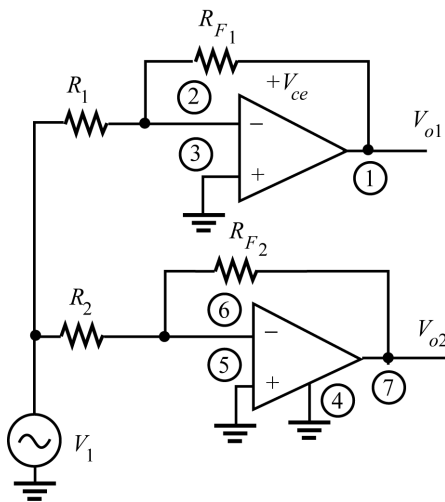
$$R_1 = \frac{420 \text{ k}\Omega}{14} \quad R_2 = \frac{420 \text{ k}\Omega}{22} \quad R_3 = \frac{420 \text{ k}\Omega}{30}$$

$$\mathbf{R_1 = 71.4 \text{ k}\Omega} \quad \mathbf{R_2 = 19.1 \text{ k}\Omega} \quad \mathbf{R_3 = 14 \text{ k}\Omega}$$

$$V_o = (+15)(-22)(-30)V_1 = 9000(80 \mu\text{V}) = 792 \text{ mV}$$

$$= \mathbf{0.792 \text{ V}}$$

5.



$$V_{o1} = -\frac{R_{F1}}{R_1} V_1 = -\frac{150 \text{ k}\Omega}{R_1} V_1$$

$$\frac{V_{o1}}{V_1} = A_{v1} = -15 = -\frac{150 \text{ k}\Omega}{R_1}$$

$$R_1 = \frac{150 \text{ k}\Omega}{15} = \mathbf{10 \text{ k}\Omega}$$

$$V_{o2} = -\frac{R_{F2}}{R_2} V_1 = -\frac{150 \text{ k}\Omega}{R_2} V_1$$

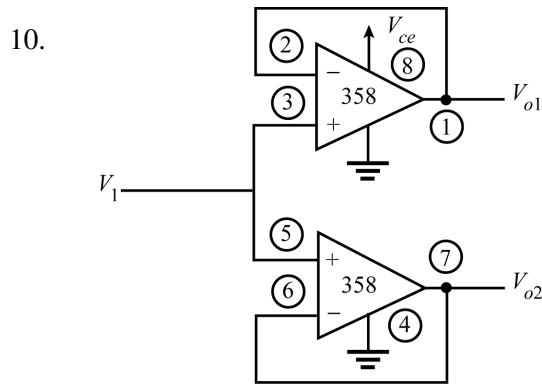
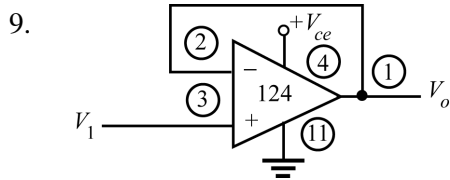
$$\frac{V_{o2}}{V_1} = A_{v2} = -30 = -\frac{150 \text{ k}\Omega}{R_2}$$

$$R_2 = \frac{150 \text{ k}\Omega}{30} = \mathbf{5 \text{ k}\Omega}$$

$$\begin{aligned}
 6. \quad V_o &= -\frac{R_F}{R_1}V_1 + \frac{R_F}{R_2}V_2 = -\frac{470\text{ k}\Omega}{47\text{ k}\Omega}(40\text{ mV}) + \frac{470\text{ k}\Omega}{12\text{ k}\Omega}(20\text{ mV}) \\
 &= -[400\text{ mV} + 783.3\text{ mV}] = \mathbf{-1.18\text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 7. \quad V_o &= \frac{10\text{ k}\Omega}{10\text{ k}\Omega + 10\text{ k}\Omega} \frac{150\text{ k}\Omega + 300\text{ k}\Omega}{150\text{ k}\Omega} V_1 - \frac{300\text{ k}\Omega}{150\text{ k}\Omega} V_2 \\
 &= 0.5(3)(1\text{ V}) - 2(2\text{ V}) = 1.5\text{ V} - 4\text{ V} = \mathbf{-2.5\text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 8. \quad V_o &= -\frac{330\text{ k}\Omega}{33\text{ k}\Omega}(12\text{ mV}) - \frac{470\text{ k}\Omega}{47\text{ k}\Omega} + \frac{470\text{ k}\Omega}{47\text{ k}\Omega}(18\text{ mV}) \\
 &= -[(-120\text{ mV})(10) + 180\text{ mV}] = -[-1.2\text{ V} + 0.18\text{ V}] \\
 &= \mathbf{+1.02\text{ V}}
 \end{aligned}$$



$$11. \quad I_L = \frac{V_1}{R_1} = \frac{12\text{ V}}{2\text{ k}\Omega} = \mathbf{6\text{ mA}}$$

$$12. \quad V_o = -I_L R_1 = -(2.5\text{ mA})(10\text{ k}\Omega) = \mathbf{-25\text{ V}}$$

$$\begin{aligned}
 13. \quad \frac{I_o}{V_1} &= \frac{R_F}{R_1} \frac{1}{R_s} \\
 I_o &= \frac{100\text{ k}\Omega}{200\text{ k}\Omega} \frac{1}{10\text{ }\Omega} (10\text{ mV}) = \mathbf{0.5\text{ mA}}
 \end{aligned}$$

$$\begin{aligned}
 14. \quad V_o &= 1 + \frac{2R}{R_p} [V_2 - V_1] \\
 &= 1 + \frac{2(5000)}{1000} [1\text{ V} - 3\text{ V}] = \mathbf{-22\text{ V}}
 \end{aligned}$$

$$15. \quad f_{OH} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(2.2 \text{ k}\Omega)(0.05 \text{ }\mu\text{F})}$$

$$= \mathbf{1.45 \text{ kHz}}$$

$$16. \quad f_{OL} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(20 \text{ k}\Omega)(0.02 \text{ }\mu\text{F})}$$

$$= \mathbf{397.9 \text{ Hz}}$$

$$17. \quad f_{OL} = \frac{1}{2\pi R_1 C_1} = \frac{1}{2\pi(10 \text{ k}\Omega)(0.05 \text{ }\mu\text{F})} = \mathbf{318.3 \text{ Hz}}$$

$$f_{OH} = \frac{1}{2\pi R_2 C_2} = \frac{1}{2\pi(20 \text{ k}\Omega)(0.02 \text{ }\mu\text{F})}$$

$$= \mathbf{397.9 \text{ Hz}}$$

Chapter 12

2.5 mA

$$1. \quad I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{18 \text{ V} - 0.7 \text{ V}}{1.2 \text{ k}\Omega} = 14.42 \text{ mA}$$

$$I_{C_Q} = \beta I_{B_Q} = 40(14.42 \text{ mA}) = 576.67 \text{ mA}$$

$$P_i = V_{CC} I_{dc} \cong V_{CC} I_{C_Q} = (18 \text{ V})(576.67 \text{ mA}) \\ \cong \mathbf{10.4 \text{ W}}$$

$$I_C(\text{rms}) = \beta I_B(\text{rms}) \\ = 40(5 \text{ mA}) = 200 \text{ mA}$$

$$P_o = I_C^2(\text{rms}) R_C = (200 \text{ mA})^2 (16 \Omega) = \mathbf{640 \text{ mW}}$$

$$2. \quad I_{B_Q} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{18 \text{ V} - 0.7 \text{ V}}{1.5 \text{ k}\Omega} = 11.5 \text{ mA}$$

$$I_{C_Q} = \beta I_{B_Q} = 40(11.5 \text{ mA}) = 460 \text{ mA}$$

$$P_i(\text{dc}) = V_{CC} I_{dc} = V_{CC} (I_{C_Q} + I_{B_Q}) \\ = 18 \text{ V}(460 \text{ mA} + 11.5 \text{ mA}) \\ = \mathbf{8.5 \text{ W}}$$

$$P_i \approx V_{CC} I_{C_Q} = 18 \text{ V}(460 \text{ mA}) = 8.3 \text{ W}$$

$$3. \quad \text{From problem 2: } I_{C_Q} = 460 \text{ mA}, P_i = 8.3 \text{ W.}$$

For maximum efficiency of 25%:

$$\% \eta = 100\% \times \frac{P_o}{P_i} = \frac{P_o}{8.3 \text{ W}} \times 100\% = 25\%$$

$$P_o = 0.25(8.3 \text{ W}) = \mathbf{2.1 \text{ W}}$$

[If dc bias condition also is considered:

$$V_C = V_{CC} - I_{C_Q} R_C = 18 \text{ V} - (460 \text{ mA})(16 \Omega) = 10.64 \text{ V}$$

collector may vary $\pm 7.36 \text{ V}$ about Q-point, resulting in maximum output power:

$$P_o = \frac{V_{CE}^2(P)}{2R_C} = \frac{(7.36 \text{ V})^2}{2(16)} = \mathbf{1.69 \text{ W}}$$

4. Assuming maximum efficiency of 25%
with $P_o(\text{max}) = 1.5 \text{ W}$

$$\% \eta = \frac{P_o}{P_i} \times 100\%$$

$$P_i = \frac{1.5 \text{ W}}{0.25} = 6 \text{ W}$$

Assuming dc bias at mid-point, $V_C = 9 \text{ V}$

$$I_{C_Q} = \frac{V_{CC} - V_C}{R_C} = \frac{18 \text{ V} - 9 \text{ V}}{16 \Omega} = 0.5625 \text{ A}$$

$$P_i(\text{dc}) = V_{CC} I_{C_Q} = (18 \text{ V})(0.5625 \text{ A}) \\ = \mathbf{10.38 \text{ W}}$$

at this input:

$$\% \eta = \frac{P_o}{P_i} \times 100\% = \frac{1.5 \text{ W}}{10.38 \text{ W}} \times 100\% = \mathbf{14.45\%}$$

5. $R_p = \frac{N_1}{N_2}^2 R_s = \frac{25}{1}^2 (4 \Omega) = \mathbf{2.5 \text{ k}\Omega}$

6. $R_2 = a^2 R_1$
 $a^2 = \frac{R_2}{R_1} = \frac{8 \text{ k}\Omega}{8 \Omega} = 1000$
 $a = \sqrt{1000} = \mathbf{31.6}$

7. $R_2 = a^2 R_1$
 $8 \text{ k}\Omega = a^2 (4 \Omega)$
 $a^2 = \frac{8 \text{ k}\Omega}{4 \Omega} = 2000$
 $a = \sqrt{2000} = \mathbf{44.7}$

8. (a) $P_{pri} = P_L = \mathbf{2 \text{ W}}$

(b) $P_L = \frac{V_L^2}{R_L}$

$$V_L = \sqrt{P_L R_L} = \sqrt{(2 \text{ W})(16 \Omega)} \\ = \sqrt{32} = \mathbf{5.66 \text{ V}}$$

(c) $R_2 = a^2 R_1 = (3.87)^2 (16 \Omega) = \mathbf{239.6 \Omega}$

$$P_{pri} = \frac{V_{pri}^2}{R_{pri}} = 2 \text{ W}$$

$$V_{pri}^2 = (2 \text{ W})(239.6 \Omega)$$

$$V_{pri} = \sqrt{479.2} = \mathbf{21.89 \text{ V}}$$

$$[\text{or, } V_{pri} = a V_L = (3.87)(5.66 \text{ V}) = 21.9 \text{ V}]$$

$$(d) \quad P_L = I_L^2 R_L$$

$$I_L = \sqrt{\frac{P_L}{R_L}} = \sqrt{\frac{2 \text{ W}}{16 \, \Omega}} = \mathbf{353.55 \text{ mA}}$$

$$P_{pri} = 2 \text{ W} = I_{pri}^2 R_{pri} = (239.6 \, \Omega) I_{pri}^2$$

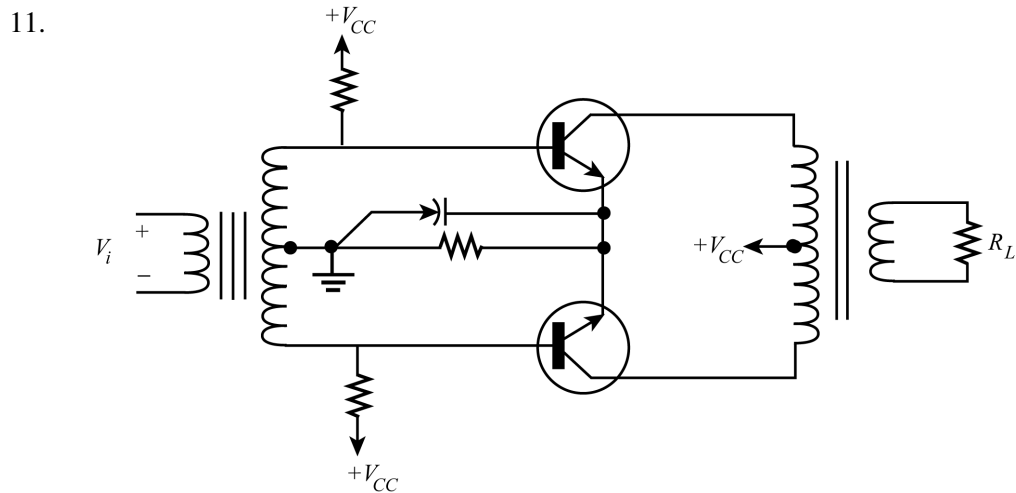
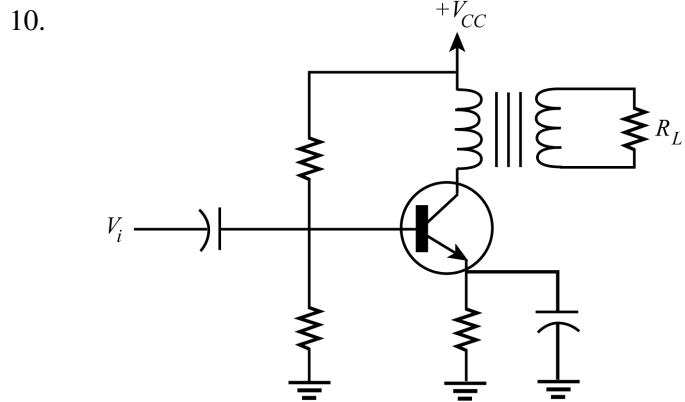
$$I_{pri} = \sqrt{\frac{2 \text{ W}}{239.6 \, \Omega}} = \mathbf{91.36 \text{ mA}}$$

$$\text{or, } I_{pri} = \frac{I_L}{a} = \frac{353.55 \text{ mA}}{3.87} = \mathbf{91.36 \text{ mA}}$$

9. $I_{dc} = I_{C_Q} = 150 \text{ mA}$

$$P_i = V_{CC} I_{C_Q} = (36 \text{ V})(150 \text{ mA}) = 5.4 \text{ W}$$

$$\% \eta = \frac{P_o}{P_i} \times 100\% = \frac{2 \text{ W}}{5.4 \text{ W}} \times 100\% = \mathbf{37\%}$$



12. (a) $P_i = V_{CC}I_{dc} = (25 \text{ V})(1.75 \text{ A}) = \mathbf{43.77 \text{ W}}$
Where, $I_{dc} = \frac{2}{\pi} I_p = \frac{2}{\pi} \frac{V_p}{R_L} = \frac{2}{\pi} \cdot \frac{22 \text{ V}}{8 \Omega} = 1.75 \text{ A}$

(b) $P_o = \frac{V_p^2}{2R_L} = \frac{(22 \text{ V})^2}{2(8 \Omega)} = \mathbf{30.25 \text{ W}}$

(c) $\% \eta = \frac{P_o}{P_i} \times 100\% = \frac{30.25 \text{ W}}{43.77 \text{ W}} \times 100\% = \mathbf{69\%}$

13. (a) $\max P_i = V_{CC}I_{dc}$
 $= V_{CC} \cdot \frac{2}{\pi} \cdot \frac{V_{CC}}{R_L} = (25 \text{ V}) \frac{2}{\pi} \cdot \frac{25 \text{ V}}{8 \Omega}$
 $= \mathbf{49.74 \text{ W}}$

(b) $\max P_o = \frac{V_{CC}^2}{2R_L} = \frac{(25 \text{ V})^2}{2(8 \Omega)} = \mathbf{39.06 \text{ W}}$

(c) $\max \% \eta = \frac{\max P_o}{\max P_i} \times 100\% = \frac{39.06 \text{ W}}{49.74 \text{ W}} \times 100\%$
 $= \mathbf{78.5\%}$

14. (a) $V_{L(\text{peak})} = 20 \text{ V}$

$$P_i = V_{CC}I_{dc} = V_{CC} \frac{2}{\pi} \cdot \frac{V_L}{R_L}$$

$$= (22 \text{ V}) \frac{2}{\pi} \cdot \frac{20 \text{ V}}{4 \Omega} = 70 \text{ W}$$

$$P_o = \frac{V_L^2}{2R_L} = \frac{(20 \text{ V})^2}{2(4 \Omega)} = 50 \text{ W}$$

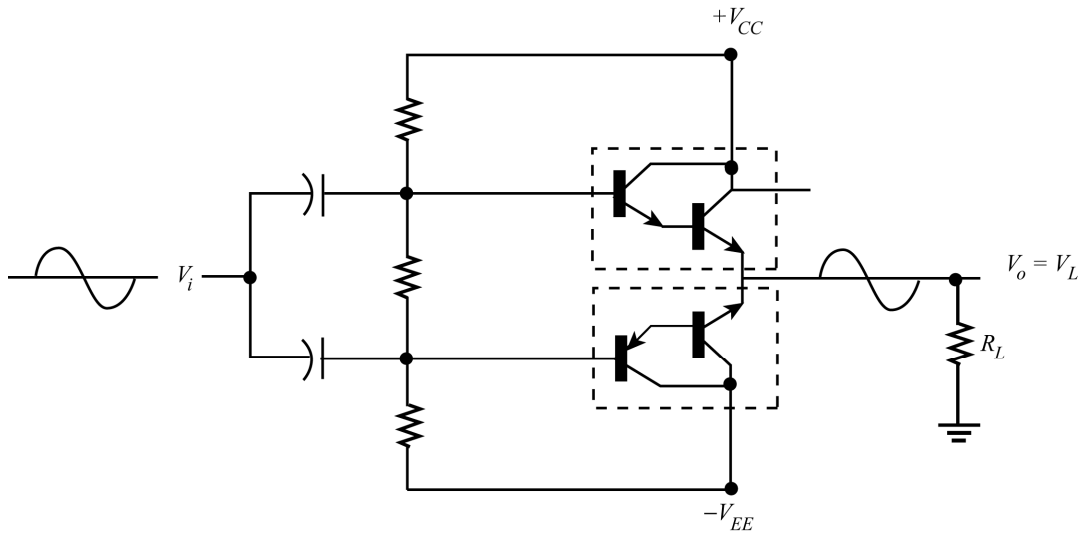
$$\% \eta = \frac{P_o}{P_i} \times 100\% = \frac{50 \text{ W}}{70 \text{ W}} \times 100\% = \mathbf{71.4\%}$$

(b) $P_i = (22 \text{ V}) \frac{2}{\pi} \cdot \frac{4 \text{ V}}{4 \Omega} = 14 \text{ W}$

$$P_o = \frac{(4)^2}{2(4)} = 2 \text{ W}$$

$$\% \eta = \frac{P_o}{P_i} \times 100\% = \frac{2 \text{ W}}{14 \text{ W}} \times 100\% = \mathbf{14.3\%}$$

15.



16. (a) $\max P_o(\text{ac})$ for $V_{L_{\text{peak}}} = 30 \text{ V}$:

$$\max P_o(\text{ac}) = \frac{V_L^2}{2R_L} = \frac{(30 \text{ V})^2}{2(8 \Omega)} = \mathbf{56.25 \text{ W}}$$

$$(b) \quad \max P_i(\text{dc}) = V_{CC} I_{\text{dc}} = V_{CC} \frac{2}{\pi} \cdot \frac{V_o}{R_L} = V_{CC} \frac{2}{\pi} \cdot \frac{30 \text{ V}}{8 \Omega} = \mathbf{71.62 \text{ W}}$$

$$(c) \quad \max \% \eta = \frac{\max P_o}{\max P_i} \times 100\% = \frac{56.25 \text{ W}}{71.62 \text{ W}} \times 100\% = \mathbf{78.54\%}$$

$$(d) \quad \max P_{Z_Q} = \frac{2}{\pi^2} \cdot \frac{V_{CC}^2}{R_L} = \frac{2}{\pi^2} \cdot \frac{(30)^2}{8} = \mathbf{22.8 \text{ W}}$$

$$\begin{aligned} 17. \quad (a) \quad P_i(\text{dc}) &= V_{CC} I_{\text{dc}} = V_{CC} \cdot \frac{2}{\pi} \cdot \frac{V_o}{R_L} \\ &= 30 \text{ V} \cdot \frac{2}{\pi} \cdot \frac{\sqrt{2} \cdot 8}{8} = \mathbf{27 \text{ W}} \end{aligned}$$

$$(b) \quad P_o(\text{ac}) = \frac{V_L^2(\text{rms})}{R_L} = \frac{(8 \text{ V})^2}{8 \Omega} = \mathbf{8 \text{ W}}$$

$$(c) \quad \% \eta = \frac{P_o}{P_i} \times 100\% = \frac{8 \text{ W}}{27 \text{ W}} \times 100\% = \mathbf{29.6\%}$$

$$(d) \quad P_{2Q} = P_i - P_o = 27 \text{ W} - 8 \text{ W} = \mathbf{19 \text{ W}}$$

$$\begin{aligned}
18. \quad (a) \quad P_o(\text{ac}) &= \frac{V_L^2(\text{rms})}{R_L} = \frac{(18 \text{ V})^2}{8 \, \Omega} = \mathbf{40.5 \text{ W}} \\
(b) \quad P_i(\text{dc}) &= V_{CC} I_{\text{dc}} = V_{CC} \frac{2}{\pi} \cdot \frac{V_{L_{\text{peak}}}}{R_L} \\
&= (40 \text{ V}) \frac{2}{\pi} \cdot \frac{18\sqrt{2} \text{ V}}{8 \, \Omega} = \mathbf{81 \text{ W}} \\
(c) \quad \% \eta &= \frac{P_o}{P_i} \times 100\% = \frac{40.5 \text{ W}}{81 \text{ W}} \times 100\% = \mathbf{50\%} \\
(d) \quad P_{2Q} &= P_i - P_o = 81 \text{ W} - 40.5 \text{ W} = \mathbf{40.5 \text{ W}}
\end{aligned}$$

$$\begin{aligned}
19. \quad \%D_2 &= \left| \frac{A_2}{A_1} \right| \times 100\% = \left| \frac{0.3 \text{ V}}{2.1 \text{ V}} \right| \times 100\% \cong \mathbf{14.3\%} \\
\%D_3 &= \left| \frac{A_3}{A_1} \right| \times 100\% = \frac{0.1 \text{ V}}{2.1 \text{ V}} \times 100\% \cong \mathbf{4.8\%} \\
\%D_4 &= \left| \frac{A_4}{A_1} \right| \times 100\% = \frac{0.05 \text{ V}}{2.1 \text{ V}} \times 100\% \cong \mathbf{2.4\%}
\end{aligned}$$

$$\begin{aligned}
20. \quad \%THD &= \sqrt{D_2^2 + D_3^2 + D_4^2} \times 100\% \\
&= \sqrt{(0.143)^2 + (0.048)^2 + (0.024)^2} \times 100\% \\
&= \mathbf{15.3\%}
\end{aligned}$$

$$\begin{aligned}
21. \quad D_2 &= \left| \frac{\frac{1}{2}(V_{CE_{\text{max}}} + V_{CE_{\text{min}}})}{V_{CE_{\text{max}}} - V_{CE_{\text{min}}}} \right| \times 100\% \\
&= \left| \frac{\frac{1}{2}(20 \text{ V} + 2.4 \text{ V}) - 10 \text{ V}}{20 \text{ V} - 2.4 \text{ V}} \right| \times 100\% \\
&= \frac{1.2 \text{ V}}{17.6 \text{ V}} \times 100\% = \mathbf{6.8\%}
\end{aligned}$$

$$\begin{aligned}
22. \quad THD &= \sqrt{D_2^2 + D_3^2 + D_4^2} = \sqrt{(0.15)^2 + (0.01)^2 + (0.05)^2} \\
&\cong 0.16 \\
P_I &= \frac{I_1^2 R_C}{2} = \frac{(3.3 \text{ A})^2 (4 \, \Omega)}{2} = \mathbf{21.8 \text{ W}} \\
P &= (1 + THD^2) P_I = [1 + (0.16)^2] 21.8 \text{ W} \\
&= \mathbf{22.36 \text{ W}}
\end{aligned}$$

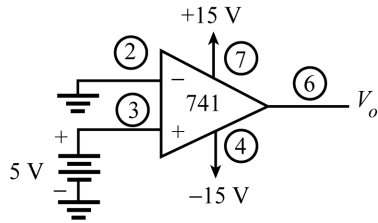
$$\begin{aligned}
 23. \quad P_D(150^\circ\text{C}) &= P_D(25^\circ\text{C}) - (T_{150} - T_{25}) (\text{Derating Factor}) \\
 &= 100 \text{ W} - (150^\circ\text{C} - 25^\circ\text{C})(0.6 \text{ W}/^\circ\text{C}) \\
 &= 100 \text{ W} - 125(0.6) = 100 - 75 \\
 &= \mathbf{25 \text{ W}}
 \end{aligned}$$

$$\begin{aligned}
 24. \quad P_D &= \frac{T_J - T_A}{\theta_{JC} + \theta_{CS} + \theta_{SA}} = \frac{200^\circ\text{C} - 80^\circ\text{C}}{0.5 \text{ }^\circ\text{C}/\text{W} + 0.8 \text{ }^\circ\text{C}/\text{W} + 1.5 \text{ }^\circ\text{C}/\text{W}} \\
 &= \frac{120^\circ\text{C}}{2.8 \text{ }^\circ\text{C}/\text{W}} = \mathbf{42.9 \text{ W}}
 \end{aligned}$$

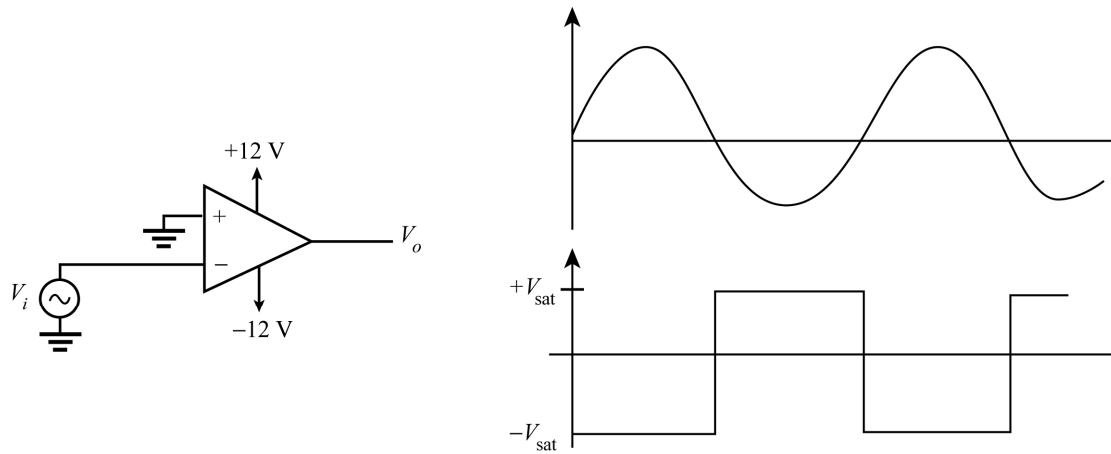
$$\begin{aligned}
 25. \quad P_D &= \frac{T_J - T_A}{\theta_{JA}} \\
 &= \frac{200^\circ\text{C} - 80^\circ\text{C}}{(40^\circ\text{C}/\text{W})} = \frac{120^\circ\text{C}}{40^\circ\text{C}/\text{W}} \\
 &= \mathbf{3 \text{ W}}
 \end{aligned}$$

Chapter 13

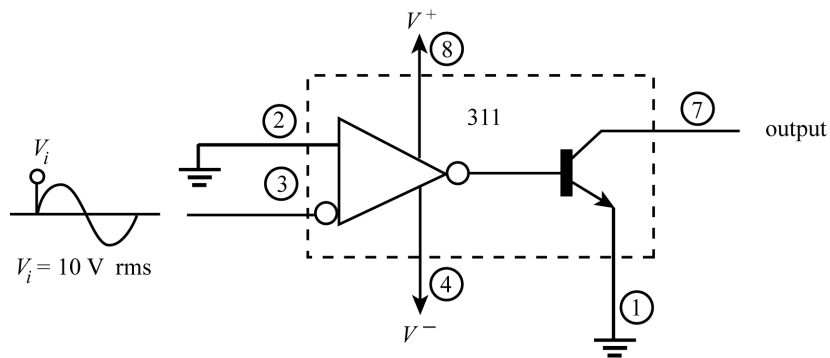
1.



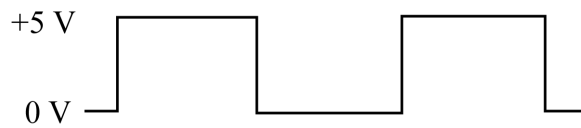
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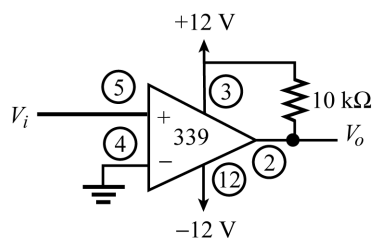
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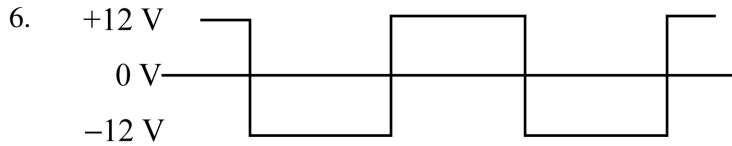


4.



5.



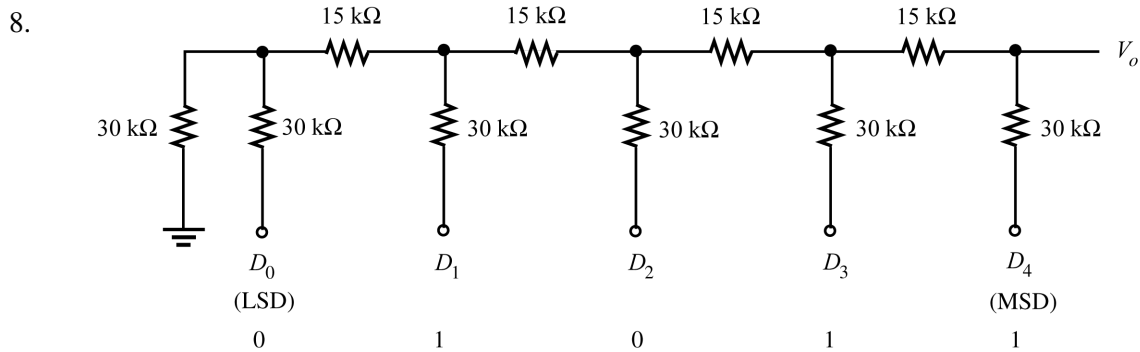


7. Circuit operates as a window detector.

Output goes *low* for input *above* $\frac{9.1 \text{ k}\Omega}{9.1 \text{ k}\Omega + 6.2 \text{ k}\Omega} (+12 \text{ V}) = \mathbf{7.1 \text{ V}}$

Output goes *low* for input *below* $\frac{1 \text{ k}\Omega}{1 \text{ k}\Omega + 6.2 \text{ k}\Omega} (+12 \text{ V}) = \mathbf{1.7 \text{ V}}$

Output is *high* for input between +1.7 V and +7.1 V.



9. $\frac{11010}{2^5} (16 \text{ V}) = \frac{26}{32} (16 \text{ V}) = \mathbf{13 \text{ V}}$

10. Resolution = $\frac{V_{REF}}{2^n} = \frac{10 \text{ V}}{2^{12}} = \frac{10 \text{ V}}{4096} = \mathbf{2.4 \text{ mV/count}}$

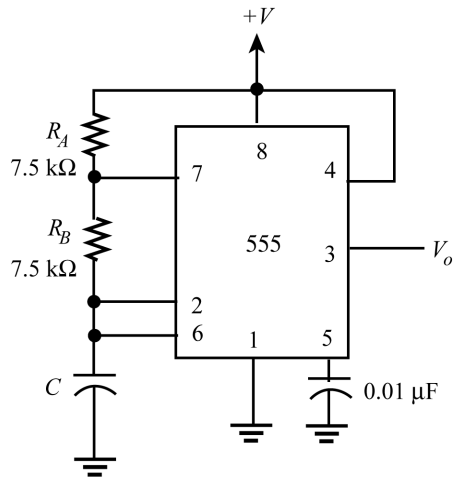
11. See section 13.3.

12. Maximum number of count steps = $2^{12} = \mathbf{4096}$

13. $2^{12} = 4096$ steps at $T = \frac{1}{f} = \frac{1}{20 \text{ MHz}} = 50 \text{ ns/count}$

Period = $4096 \text{ counts} \times 50 \frac{\text{ns}}{\text{count}} = \mathbf{204.8 \mu\text{s}}$

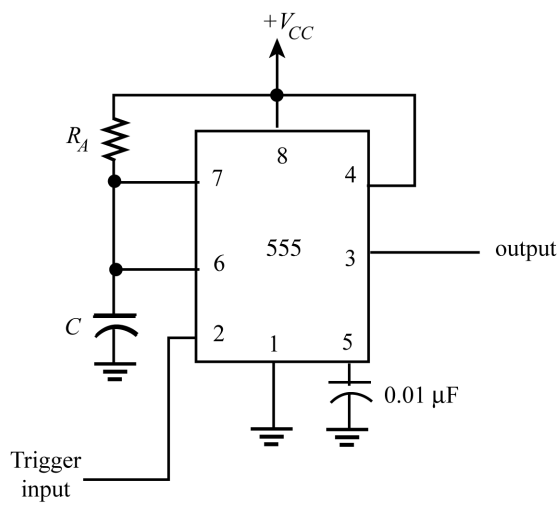
14.



$$f = \frac{1.44}{(R_A + 2R_B)C} = 350 \text{ kHz}$$

$$C = \frac{1.44}{7.5 \text{ k}\Omega + 2(7.5 \text{ k}\Omega)(350 \text{ kHz})} \cong \mathbf{183 \text{ pF}}$$

15.



$$T = 1.1 R_A C$$

$$20 \mu\text{s} = 1.1(7.5 \text{ k}\Omega)C$$

$$C = \frac{20 \times 10^{-6}}{1.1(7.5 \times 10^3)}$$

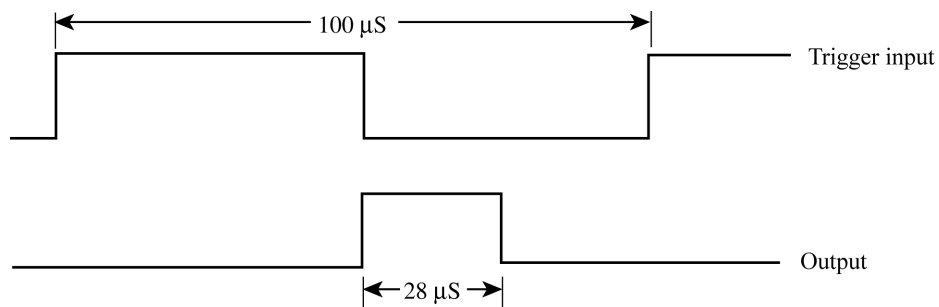
$$= 2.4 \times 10^{-9}$$

$$= 2400 \times 10^{-12}$$

$$= \mathbf{2400 \text{ pF}}$$

16. $T = \frac{1}{f} = \frac{1}{10 \text{ kHz}} = 100 \mu\text{s}$

$$T = 1.1 R_A C = 1.1(5.1 \text{ k}\Omega)(5 \text{ nF}) = \mathbf{28 \mu\text{s}}$$



$$17. \quad f_o = \frac{2}{R_1 C_1} \frac{V^+ - V_C}{V^+}$$

$$V^+ = 12 \text{ V}$$

$$V_C = \frac{R_3}{R_2 + R_3} (V^+) = \frac{11 \text{ k}\Omega}{1.8 \text{ k}\Omega + 11 \text{ k}\Omega} (+12 \text{ V}) = 10.3 \text{ V}$$

$$f_o = \frac{2}{(4.7 \text{ k}\Omega)(0.001 \text{ }\mu\text{F})} \frac{12 \text{ V} - 10.3 \text{ V}}{12 \text{ V}}$$

$$= 60.3 \times 10^3 \cong \mathbf{60 \text{ kHz}}$$

18. With potentiometer set at top:

$$V_C = \frac{R_3 + R_4}{R_2 + R_3 + R_4} V^+ = \frac{5 \text{ k}\Omega + 18 \text{ k}\Omega}{510 \text{ }\Omega + 5 \text{ k}\Omega + 18 \text{ k}\Omega} (12 \text{ V}) = 11.74 \text{ V}$$

resulting in a lower cutoff frequency of

$$f_o = \frac{2}{R_1 C_1} \frac{V^+ - V_C}{V^+} = \frac{2}{(10 \times 10^3)(0.001 \text{ }\mu\text{F})} \frac{12 \text{ V} - 11.74 \text{ V}}{12 \text{ V}}$$

$$= \mathbf{4.3 \text{ kHz}}$$

With potentiometer set at bottom:

$$V_C = \frac{R_4}{R_2 + R_3 + R_4} V^+ = \frac{18 \text{ k}\Omega}{510 \text{ }\Omega + 5 \text{ k}\Omega + 18 \text{ k}\Omega} (12 \text{ V})$$

$$= 9.19 \text{ V}$$

resulting in a higher cutoff frequency of

$$f_o = \frac{2}{R_1 C_1} \frac{V^+ - V_C}{V^+} = \frac{2}{(10 \text{ k}\Omega)(0.001 \text{ }\mu\text{F})} \frac{12 \text{ V} - 9.19 \text{ V}}{12 \text{ V}}$$

$$= \mathbf{61.2 \text{ kHz}}$$

19. $V^+ = 12 \text{ V}$

$$V_C = \frac{R_3}{R_2 + R_3} V^+ = \frac{10 \text{ k}\Omega}{1.5 \text{ k}\Omega + 10 \text{ k}\Omega} (12 \text{ V}) = 10.4 \text{ V}$$

$$f_o = \frac{2}{R_1 C_1} \frac{V^+ - V_C}{V^+} = \frac{2}{10 \text{ k}\Omega(C_1)} \frac{12 \text{ V} - 10.4 \text{ V}}{12 \text{ V}}$$

$$= 200 \text{ kHz}$$

$$C_1 = \frac{2}{10 \text{ k}\Omega(200 \text{ kHz})} (0.133)$$

$$= 133 \times 10^{-12} = \mathbf{133 \text{ pF}}$$

$$20. \quad f_o = \frac{0.3}{R_1 C_1} = \frac{0.3}{(4.7 \text{ k}\Omega)(0.001 \text{ }\mu\text{F})}$$

$$= \mathbf{63.8 \text{ kHz}}$$

$$21. \quad C_1 = \frac{0.3}{R_1 f} = \frac{0.3}{(10 \text{ k}\Omega)(100 \text{ kHz})} = \mathbf{300 \text{ pF}}$$

$$\begin{aligned}
 22. \quad f_L &= \pm \frac{8f_o}{V} \\
 &= \pm \frac{8(63.8 \times 10^3)}{6 \text{ V}} \quad f_o = \frac{0.3}{R_1 C_1} = \frac{0.3}{4.7 \text{ k}\Omega(0.001 \text{ }\mu\text{F})} \\
 &= \mathbf{85.1 \text{ kHz}} \quad \quad \quad = 63.8 \text{ kHz}
 \end{aligned}$$

23. For current loop: mark = 20 mA
space = 0 mA

For RS – 232 C: mark = –12 V
space = +12 V

24. A line (or lines) onto which data bits are connected.

25. Open-collector is active-LOW only.
Tri-state is active-HIGH or active-LOW.

Chapter 14

$$1. \quad A_f = \frac{A}{1 + \beta A} = \frac{-2000}{1 + -\frac{1}{10}(-2000)} = \frac{-2000}{201} = \mathbf{-9.95}$$

$$2. \quad \frac{dA_f}{A_f} = \frac{1}{\beta A} \frac{dA}{A} = \frac{1}{-\frac{1}{20}(-1000)}(10\%) = \mathbf{0.2\%}$$

$$3. \quad A_f = \frac{A}{1 + \beta A} = \frac{-300}{1 + -\frac{1}{15}(-300)} = \frac{-300}{21} = \mathbf{-14.3}$$

$$R_{if} = (1 + \beta A)R_i = 21(1.5 \text{ k}\Omega) = \mathbf{31.5 \text{ k}\Omega}$$

$$R_{of} = \frac{R_o}{1 + \beta A} = \frac{50 \text{ k}\Omega}{21} = \mathbf{2.4 \text{ k}\Omega}$$

$$4. \quad R_L = \frac{R_o R_D}{R_o + R_D} = 40 \text{ k}\Omega \parallel 8 \text{ k}\Omega = 6.7 \text{ k}\Omega$$

$$A = -g_m R_L = -(5000 \times 10^{-6})(6.7 \times 10^3) = \mathbf{-33.5}$$

$$\beta = \frac{-R_2}{R_1 + R_2} = \frac{-200 \text{ k}\Omega}{200 \text{ k}\Omega + 800 \text{ k}\Omega} = \mathbf{-0.2}$$

$$A_f = \frac{A}{1 + \beta A} = \frac{-33.5}{1 + (-0.2)(-33.5)} = \frac{-33.5}{7.7}$$

$$\quad \quad \quad = \mathbf{-4.4}$$

5. DC bias:

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1)R_E} = \frac{16 \text{ V} - 0.7 \text{ V}}{600 \text{ k}\Omega + 76(1.2 \text{ k}\Omega)}$$

$$= \frac{15.3 \text{ V}}{691.2 \text{ k}\Omega} = 22.1 \mu\text{A}$$

$$I_E = (1 + \beta)I_B$$

$$= 76(22.1 \mu\text{A}) = 1.68 \text{ mA}$$

$$[V_{CE} = V_{CC} - I_C(R_C + R_E) = 16 \text{ V} - 1.68 \text{ mA}(4.7 \text{ k}\Omega + 1.2 \text{ k}\Omega) \cong 6.1 \text{ V}]$$

$$r_e = \frac{26 \text{ mV}}{I_E(\text{mA})} = \frac{26 \text{ mV}}{1.68 \text{ mA}} \cong 15.5 \Omega$$

$$h_{ie} = (1 + \beta)r_e = 76(15.5 \Omega) = \mathbf{1.18 \text{ k}\Omega} = Z_i$$

$$Z_o = R_C = \mathbf{4.7 \text{ k}\Omega}$$

$$A_v = \frac{-h_{fe}}{h_{ie} + R_E} = \frac{-75}{1.18 \text{ k}\Omega + 1.2 \text{ k}\Omega} = -31.5 \times 10^{-3}$$

$$\beta = R_E = -1.2 \times 10^3$$

$$(1 + \beta A) = 1 + (-1.2 \times 10^3)(-31.5 \times 10^{-3}) \\ = 38.8$$

$$A_f = \frac{A_v}{1 + \beta A_v} = \frac{-31.5 \times 10^{-3}}{38.8} = 811.86 \times 10^{-6}$$

$$A_{v_f} = -A_f R_C = -(811.86 \times 10^{-6})(4.7 \times 10^3) = \mathbf{-3.82}$$

$$Z_{i_f} = (1 + \beta A_v) Z_i = (38.8)(1.18 \text{ k}\Omega) = \mathbf{45.8 \text{ k}\Omega}$$

$$Z_{o_f} = (1 + \beta A_v) Z_o = (38.8)(4.7 \text{ k}\Omega) = \mathbf{182.4 \text{ k}\Omega}$$

without feedback (R_E bypassed):

$$A_v = \frac{-R_C}{r_e} = -\frac{4.7 \text{ k}\Omega}{15.5 \Omega} = \mathbf{-303.2}$$

$$6. \quad C = \frac{1}{2\pi R f \sqrt{6}} = \frac{1}{2\pi(10 \times 10^3)(2.5 \times 10^3) \sqrt{6}} \\ = 2.6 \times 10^{-9} = \mathbf{2600 \text{ pF}} = 0.0026 \mu\text{F}$$

$$7. \quad f_o = \frac{1}{2\pi RC} \cdot \frac{1}{\sqrt{6 + 4 \frac{R_c}{R}}} \\ = \frac{1}{2\pi(6 \times 10^3)(1500 \times 10^{-12})} \cdot \frac{1}{\sqrt{6 + 4(18 \times 10^3 / 6 \times 10^3)}} \\ = 4.17 \text{ kHz} \cong \mathbf{4.2 \text{ kHz}}$$

$$8. \quad f_o = \frac{1}{2\pi RC} = \frac{1}{2\pi(10 \times 10^3)(2400 \times 10^{-12})} \\ = \mathbf{6.6 \text{ kHz}}$$

$$9. \quad C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{(750 \text{ pF})(2000 \text{ pF})}{750 \text{ pF} + 2000 \text{ pF}} = 577 \text{ pF} \\ f_o = \frac{1}{2\pi \sqrt{LC_{eq}}} = \frac{1}{2\pi \sqrt{40 \times 10^{-6} (577 \times 10^{-12})}} \\ = \mathbf{1.05 \text{ MHz}}$$

$$10. \quad f_o = \frac{1}{2\pi \sqrt{LC_{eq}}}, \quad \text{where } C_{eq} = \frac{C_1 C_2}{C_1 + C_2} \\ = \frac{1}{2\pi \sqrt{(100 \mu\text{H})(3300 \text{ pF})}} \\ = \mathbf{277 \text{ kHz}} \\ = \frac{(0.005 \mu\text{F})(0.01 \mu\text{F})}{0.005 \mu\text{F} + 0.01 \mu\text{F}} \\ = 3300 \text{ pF}$$

$$\begin{aligned}
 11. \quad f_o &= \frac{1}{2\pi\sqrt{L_{\text{eq}}C}}, \\
 &= \frac{1}{2\pi\sqrt{(4 \times 10^{-3})(250 \times 10^{-12})}} \\
 &= \mathbf{159.2 \text{ kHz}}
 \end{aligned}$$

$$\begin{aligned}
 L_{\text{eq}} &= L_1 + L_2 + 2 M \\
 &= 1.5 \text{ mH} + 1.5 \text{ mH} + 2(0.5 \text{ mH}) \\
 &= 4 \text{ mH}
 \end{aligned}$$

$$\begin{aligned}
 12. \quad f_o &= \frac{1}{2\pi\sqrt{LC_{\text{eq}}}}, \\
 &= \frac{1}{2\pi\sqrt{(1800 \mu\text{H})(150 \text{ pF})}} \\
 &= 306.3 \text{ kHz}
 \end{aligned}$$

$$\begin{aligned}
 \text{where } L_{\text{eq}} &= L_1 + L_2 + 2 M \\
 &= 750 \mu\text{H} + 750 \mu\text{H} + 2(150 \mu\text{H}) \\
 &= 1800 \mu\text{H}
 \end{aligned}$$

13. See Fig. 14.33a and Fig. 14.34.

$$14. \quad f_o = \frac{1}{R_T C_T \ln(1/(1 - \eta))}$$

for $\eta = 0.5$:

$$f_o \cong \frac{1.5}{R_T C_T}$$

(a) Using $R_T = 1 \text{ k}\Omega$

$$C_T = \frac{1.5}{R_T f_o} = \frac{1.5}{(1 \text{ k}\Omega)(1 \text{ kHz})} = \mathbf{1.5 \mu\text{F}}$$

(b) Using $R_T = 10 \text{ k}\Omega$

$$C_T = \frac{1.5}{R_T f_o} = \frac{1.5}{(10 \text{ k}\Omega)(150 \text{ kHz})} = \mathbf{1000 \text{ pF}}$$

Chapter 15

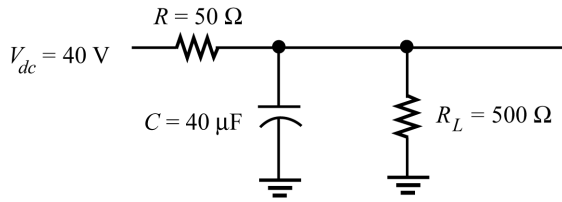
1. $\text{ripple factor} = \frac{V_r(\text{rms})}{V_{\text{dc}}} = \frac{2 \text{ V} / \sqrt{2}}{50 \text{ V}} = \mathbf{0.028}$
2. $\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{28 \text{ V} - 25 \text{ V}}{25 \text{ V}} \times 100\% = \mathbf{12\%}$
3. $V_{\text{dc}} = 0.318V_m$
 $V_m = \frac{V_{\text{dc}}}{0.318} = \frac{20 \text{ V}}{0.318} = 62.89 \text{ V}$
 $V_r = 0.385V_m = 0.385(62.89 \text{ V}) = \mathbf{24.2 \text{ V}}$
4. $V_{\text{dc}} = 0.636V_m$
 $V_m = \frac{V_{\text{dc}}}{0.636} = \frac{8 \text{ V}}{0.636} = 12.6 \text{ V}$
 $V_r = 0.308V_m = 0.308(12.6 \text{ V}) = \mathbf{3.88 \text{ V}}$
5. $\%r = \frac{V_r(\text{rms})}{V_{\text{dc}}} \times 100\%$
 $V_r(\text{rms}) = rV_{\text{dc}} = \frac{8.5}{100} \times 14.5 \text{ V} = \mathbf{1.2 \text{ V}}$
6. $V_{NL} = V_m = 18 \text{ V}$
 $V_{FL} = 17 \text{ V}$
 $\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{18 \text{ V} - 17 \text{ V}}{17 \text{ V}} \times 100\%$
 $\quad \quad \quad = \mathbf{5.88\%}$
7. $V_m = 18 \text{ V}$
 $C = 400 \mu\text{F}$
 $I_L = 100 \text{ mA}$
 $V_r = \frac{2.4I_{\text{dc}}}{C} = \frac{2.4(100)}{400} = 0.6 \text{ V, rms}$
 $V_{\text{dc}} = V_m - \frac{4.17I_{\text{dc}}}{C}$
 $\quad \quad \quad = 18 \text{ V} - \frac{4.17(100)}{400} = \mathbf{16.96 \text{ V}}$
 $\quad \quad \quad \cong \mathbf{17 \text{ V}}$
8. $V_r = \frac{2.4I_{\text{dc}}}{C} = \frac{2.4(120)}{200} = \mathbf{1.44 \text{ V}}$
9. $C = 100 \mu\text{F}$
 $\left. \begin{array}{l} V_{\text{dc}} = 12 \text{ V} \\ R_L = 2.4 \text{ k}\Omega \end{array} \right\} I_{\text{dc}} = \frac{V_{\text{dc}}}{R_L} = \frac{12 \text{ V}}{2.4 \text{ k}\Omega} = 5 \text{ mA}$

- $$V_r(\text{rms}) = \frac{2.4I_{\text{dc}}}{C} = \frac{2.4(5)}{100} = \mathbf{0.12 \text{ V}}$$
10. $C = \frac{2.4I_{\text{dc}}}{rV_{\text{dc}}} = \frac{2.4(150)}{(0.15)(24)} = \mathbf{100 \mu\text{F}}$
11. $C = 500 \mu\text{F}$
 $I_{\text{dc}} = 200 \text{ mA}$
 $R = 8\% = 0.08$
- Using $r = \frac{2.4I_{\text{dc}}}{CV_{\text{dc}}}$
 $V_{\text{dc}} = \frac{2.4I_{\text{dc}}}{rC} = \frac{2.4(200)}{0.08(500)} = 12 \text{ V}$
- $$V_m = V_{\text{dc}} + \frac{4.17I_{\text{dc}}}{C} = 12 \text{ V} + \frac{(200)(4.17)}{500}$$
- $$= 12 \text{ V} + 1.7 \text{ V} = \mathbf{13.7 \text{ V}}$$
12. $C = \frac{2.4I_{\text{dc}}}{V_r} = \frac{2.4(200)}{(0.07)} = \mathbf{6857 \mu\text{F}}$
13. $C = 120 \mu\text{F}$
 $I_{\text{dc}} = 80 \text{ mA}$
 $V_m = 25 \text{ V}$
 $V_{\text{dc}} = V_m - \frac{4.17I_{\text{dc}}}{C} = 25 \text{ V} - \frac{4.17(80)}{120}$

$$= 22.2 \text{ V}$$
- $$\%r = \frac{2.4I_{\text{dc}}}{CV_{\text{dc}}} \times 100\% = \frac{2.4(80)}{(120)(22.2)} \times 100\%$$
- $$= \mathbf{7.2\%}$$
14. $V_r' = \frac{r \cdot V_{\text{dc}}'}{100} = \frac{2(80)}{100} = \mathbf{1.6 \text{ V, rms}}$
15. $V_r = 2 \text{ V}$
 $V_{\text{dc}} = 24 \text{ V}$
 $R = 33 \Omega, C = 120 \mu\text{F}$
 $X_C = \frac{1.3}{C} = \frac{1.3}{120} = 10.8 \Omega$
- $$\%r = \frac{V_r}{V_{\text{dc}}} \times 100\% = \frac{2 \text{ V}}{24 \text{ V}} \times 100\%$$
- $$= \mathbf{8.3\%}$$
- $$V_r' = \frac{X_C}{R} V_r = \frac{10.8}{33} (2 \text{ V}) = 0.65 \text{ V}$$
- $$V_{\text{dc}}' = V_{\text{dc}} - I_{\text{dc}}R = 24 \text{ V} - 33 \Omega (100 \text{ mA})$$
- $$= 20.7 \text{ V}$$

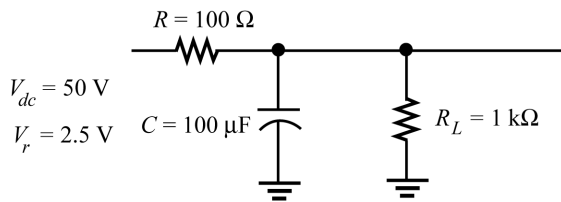
$$\%r' = \frac{V_r'}{V_{dc}'} \times 100\% = \frac{0.65 \text{ V}}{20.7 \text{ V}} \times 100\% = \mathbf{3.1\%}$$

16.



$$\begin{aligned} V_{dc}' &= \frac{R_L}{R + R_L} V_{dc} \\ &= \frac{500}{50 + 500} (40 \text{ V}) \\ &= 36.4 \text{ V} \\ I_{dc} &= \frac{V_{dc}'}{R_L} = \frac{36.4 \text{ V}}{500 \Omega} = \mathbf{72.8 \text{ mA}} \end{aligned}$$

17.



$$\begin{aligned} X_C &= \frac{1.3}{C} = \frac{1.3}{100} = 13 \Omega \\ V_r' &= \frac{X_C}{R} V_r = \frac{13}{100} (2.5 \text{ V}) \\ &= \mathbf{0.325 \text{ V, rms}} \end{aligned}$$

18.

$$\begin{aligned} V_{NL} &= 60 \text{ V} \\ V_{FL} &= \frac{R_L}{R + R_L} V_{dc} = \frac{1 \text{ k}\Omega}{100 \Omega + 1 \text{ k}\Omega} (50 \text{ V}) = 45.46 \text{ V} \\ \%VR &= \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100\% = \frac{50 \text{ V} - 45.46 \text{ V}}{45.46 \text{ V}} \times 100\% \\ &= \mathbf{10\%} \end{aligned}$$

19.

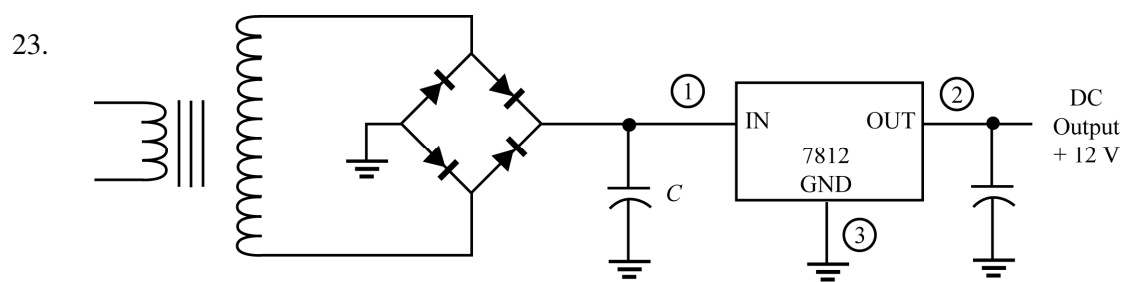
$$\begin{aligned} V_o &= V_Z - V_{BE} = 8.3 \text{ V} - 0.7 \text{ V} = \mathbf{7.6 \text{ V}} \\ V_{CE} &= V_i - V_o = 15 \text{ V} - 7.6 \text{ V} = 7.4 \text{ V} \\ I_R &= \frac{V_i - V_Z}{R} = \frac{15 \text{ V} - 8.3 \text{ V}}{1.8 \text{ k}\Omega} = 3.7 \text{ mA} \\ I_L &= \frac{V_o}{R_L} = \frac{7.6 \text{ V}}{2 \text{ k}\Omega} = 3.8 \text{ mA} \\ I_B &= \frac{I_C}{\beta} = \frac{3.8 \text{ mA}}{100} = 38 \mu\text{A} \end{aligned}$$

$$I_Z = I_R - I_B = 3.7 \text{ mA} - 38 \text{ } \mu\text{A} = \mathbf{3.66 \text{ mA}}$$

$$\begin{aligned}
 20. \quad V_o &= \frac{R_1 + R_2}{R_2} (V_z + V_{BE_2}) \\
 &= \frac{33 \text{ k}\Omega + 22 \text{ k}\Omega}{22 \text{ k}\Omega} (10 \text{ V} + 0.7 \text{ V}) \\
 &= \mathbf{26.75 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 21. \quad V_o &= 1 + \frac{R_1}{R_2} V_z = 1 + \frac{12 \text{ k}\Omega}{8.2 \text{ k}\Omega} 10 \text{ V} \\
 &= \mathbf{24.6 \text{ V}}
 \end{aligned}$$

$$22. \quad V_o = V_L = 10 \text{ V} + 0.7 \text{ V} = \mathbf{10.7 \text{ V}}$$



$$\begin{aligned}
 24. \quad I_L &= 250 \text{ mA} \\
 V_m &= V_r(\text{rms}) \cdot \sqrt{2} = \sqrt{2} (20 \text{ V}) = 28.3 \text{ V} \\
 V_{r_{\text{peak}}} &= \sqrt{3} V_r(\text{rms}) = \sqrt{3} \frac{2.4 I_{\text{dc}}}{C} \\
 &= \sqrt{3} \frac{2.4(250)}{500} = 2.1 \text{ V} \\
 V_{\text{dc}} &= V_m - V_{r_{\text{peak}}} = 28.3 \text{ V} - 2.1 \text{ V} = 26.2 \text{ V} \\
 V_i(\text{low}) &= V_{\text{dc}} - V_{r_{\text{peak}}} = 26.2 \text{ V} - 2.1 \text{ V} = \mathbf{24.1 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 25. \quad &\text{To maintain } V_i(\text{min}) \geq 7.3 \text{ V (see Table 15.1)} \\
 &V_{r_{\text{peak}}} \leq V_m - V_i(\text{min}) = 12 \text{ V} - 7.3 \text{ V} = 4.7 \text{ V} \\
 &\text{so that}
 \end{aligned}$$

$$V_r(\text{rms}) = \frac{V_{r_{\text{peak}}}}{\sqrt{3}} = \frac{4.7 \text{ V}}{1.73} = 2.7 \text{ V}$$

The maximum value of load current is then

$$I_{\text{dc}} = \frac{V_r(\text{rms})C}{2.4} = \frac{(2.7 \text{ V})(200)}{2.4} = \mathbf{225 \text{ mA}}$$

$$\begin{aligned}
 26. \quad V_o &= V_{\text{ref}} \left(1 + \frac{R_2}{R_1} \right) + I_{\text{adj}} R_L \\
 &= 1.25 \text{ V} \left(1 + \frac{1.8 \text{ k}\Omega}{240 \text{ }\Omega} \right) + 100 \text{ }\mu\text{A}(2.4 \text{ k}\Omega) \\
 &= 1.25 \text{ V}(8.5) + 0.24 \text{ V} \\
 &= \mathbf{10.87 \text{ V}}
 \end{aligned}$$

$$\begin{aligned}
 27. \quad V_o &= V_{\text{ref}} \left(1 + \frac{R_2}{R_1} \right) + I_{\text{adj}} R_2 \\
 &= 1.25 \text{ V} \left(1 + \frac{1.5 \text{ k}\Omega}{220 \text{ }\Omega} \right) + 100 \text{ }\mu\text{A}(1.5 \text{ k}\Omega) \\
 &= \mathbf{9.9 \text{ V}}
 \end{aligned}$$

Chapter 16

1. (a) The Schottky Barrier diode is constructed using an n -type semiconductor material and a metal contact to form the diode junction, while the conventional p - n junction diode uses both p - and n -type semiconductor materials to form the junction.

(b) –

2. (a) In the forward-biased region the dynamic resistance is about the same as that for a p - n junction diode. Note that the slope of the curves in the forward-biased region is about the same at different levels of diode current.

(b) In the reverse-biased region the reverse saturation current is larger in magnitude than for a p - n junction diode, and the Zener breakdown voltage is lower for the Schottky diode than for the conventional p - n junction diode.

$$3. \quad \frac{\Delta I_R}{\Delta^\circ C} = \frac{100 \mu\text{A} - 0.5 \mu\text{A}}{75^\circ \text{C}} = 1.33 \mu\text{A}/^\circ\text{C}$$

$$\Delta I_R = (1.33 \mu\text{A}/^\circ\text{C})\Delta C = (1.33 \mu\text{A}/^\circ\text{C})(25^\circ\text{C}) = 33.25 \mu\text{A}$$

$$I_R = 0.5 \mu\text{A} + 33.25 \mu\text{A} = \mathbf{33.75 \mu\text{A}}$$

$$4. \quad X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1 \text{ MHz})(7 \text{ pF})} = \mathbf{22.7 \text{ k}\Omega}$$

$$R_{DC} = \frac{V_F}{I_F} = \frac{400 \text{ mV}}{10 \text{ mA}} = \mathbf{40 \Omega}$$

5. –

$$6. \quad (a) \quad V_F \cong \mathbf{0.7 \text{ V}}$$

$$(b) \quad V_F \cong \mathbf{0.5 \text{ V}}$$

(c) As $T \uparrow$, $V_F \downarrow$

$$7. \quad C_D \cong 6.2 \text{ pF}, \quad X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1 \text{ MHz})(6.2 \text{ pF})} = \mathbf{25.67 \text{ k}\Omega}$$

$$8. \quad (a) \quad C_T(V_R) = \frac{C(0)}{\left(1 + |V_R/V_T|\right)^n} = \frac{80 \text{ pF}}{1 + \frac{4.2 \text{ V}}{0.7 \text{ V}}^{1/3}}$$

$$= \frac{80 \text{ pF}}{1.912} = \mathbf{41.85 \text{ pF}}$$

$$(b) \quad k = C_T(V_T + V_R)^n$$

$$= 41.85 \text{ pF} \underbrace{(0.7 \text{ V} + 4.2 \text{ V})^{1/3}}_{1.698}$$

$$\cong \mathbf{71 \times 10^{-12}}$$

$$9. \quad (a) \quad \text{At } -3 \text{ V, } C = \mathbf{40 \text{ pF}}$$

$$\text{At } -12 \text{ V, } C = \mathbf{20 \text{ pF}}$$

$$\Delta C = 40 \text{ pF} - 20 \text{ pF} = \mathbf{20 \text{ pF}}$$

$$(b) \quad \text{At } -8 \text{ V, } \frac{\Delta C}{\Delta V_R} = \frac{40 \text{ pF}}{20 \text{ V}} = \mathbf{2 \text{ pF/V}}$$

$$\text{At } -2 \text{ V, } \frac{\Delta C}{\Delta V_R} = \frac{60 \text{ pF}}{9 \text{ V}} = \mathbf{6.67 \text{ pF/V}}$$

$$\frac{\Delta C}{\Delta V_R} \text{ increases at less negative values of } V_R.$$

$$10. \quad \text{Ratio} = \frac{C_t(-1 \text{ V})}{C_t(-8 \text{ V})} = \frac{92 \text{ pF}}{5.5 \text{ pF}} = \mathbf{16.73}$$

$$\frac{C_t(-1.25 \text{ V})}{C_t(-7 \text{ V})} = \mathbf{13}$$

$$11. \quad C_t \cong 15 \text{ pF}$$

$$Q = \frac{1}{2\pi f R_s C_t} = \frac{1}{2\pi(10 \text{ MHz})(3 \Omega)(15 \text{ pF})}$$

$$= \mathbf{354.61} \text{ vs } \mathbf{350} \text{ on chart}$$

$$12. \quad TC_C = \frac{\Delta C}{C_o(T_1 - T_0)} \times 100\% \Rightarrow T_1 = \frac{\Delta C \times 100\%}{TC_C(C_o)} + T_o$$

$$= \frac{(0.11 \text{ pF})(100)}{(0.02)(22 \text{ pF})} + 25$$

$$= \mathbf{50^\circ\text{C}}$$

$$13. \quad V_R \text{ from } -2 \text{ V to } -8 \text{ V}$$

$$C_t(-2 \text{ V}) = 60 \text{ pF, } C_t(-8 \text{ V}) = 6 \text{ pF}$$

$$\text{Ratio} = \frac{C_t(-2 \text{ V})}{C_t(-8 \text{ V})} = \frac{60 \text{ pF}}{6 \text{ pF}} = \mathbf{10}$$

14. $Q(-1 \text{ V}) = 82, Q(-10 \text{ V}) = 5000$

$$\text{Ratio} = \frac{Q(-10 \text{ V})}{Q(-1 \text{ V})} = \frac{5000}{82} = 60.98$$

$$BW = \frac{f_o}{Q} = \frac{10 \times 10^6 \text{ Hz}}{82} = \mathbf{121.95 \text{ kHz}}$$

$$BW = \frac{f_o}{Q} = \frac{10 \times 10^6 \text{ Hz}}{5000} = \mathbf{2 \text{ kHz}}$$

15. $C_i \cong 55 \text{ pF}, C_T = \frac{C_i C_C}{C_i + C_C} = \frac{(55 \text{ pF})(40 \text{ pF})}{55 \text{ pF} + 40 \text{ pF}} = 23.16 \text{ pF}$

$$f_r = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(2\text{mH})(23.16 \text{ pF})}} \cong \mathbf{739.5 \text{ kHz}}$$

16. $\eta\% = \frac{P_{\max}}{(A_{\text{cm}^2})(100 \text{ mW/cm}^2)} \times 100\%$

$$9\% = \frac{P_{\max}}{(2 \text{ cm}^2)(100 \text{ mW/cm}^2)} \times 100\%$$

$$P_{\max} = \mathbf{18 \text{ mW}}$$

17. The greatest rate of increase in power will occur at low illumination levels. At higher illumination levels, the change in V_{OC} drops to nearly zero, while the current continues to rise linearly. At low illumination levels the voltage increases logarithmically with the linear increase in current.

18. (a) $\frac{\Delta I_{SC}}{\Delta f_c} = \frac{100 \text{ mA}}{2(20f_c) - 20f_c} = \frac{100 \text{ mA}}{20f_c} = \mathbf{5 \text{ mA}/f_c}$

(b) At $f_{c1} = 20f_c$, $x I_{SC1} = 100 \text{ mA}$

$$\Delta I_{SC} = (5 \text{ mA}/f_c)(\Delta f_c) = (5 \text{ mA}/f_c)(8f_c) = 40 \text{ mA}$$

$$I_{SC} = I_{SC1} + 40 \text{ mA} = 100 \text{ mA} + 40 \text{ mA} = \mathbf{140 \text{ mA}}$$

19. (a) $\Delta V_{OC} = 0.57 \text{ V} - 0.54 \text{ V} = 0.03 \text{ V}$

$$\frac{\Delta V_{OC}}{\Delta f_c} = \frac{0.03 \text{ V}}{80f_c} = \mathbf{0.375 \text{ mV}/f_c}$$

(b) $\Delta V_{OC} = (0.375 \text{ mV}/f_c)(\Delta f_c) = (0.375 \text{ mV}/f_c)(20f_c) = 7.5 \text{ mV}$

$$V_{OC} = 0.54 \text{ V} + 7.5 \text{ mV} = \mathbf{547.5 \text{ mV}}$$

20. (a) At knee, $I_{SC} \cong \mathbf{90 \text{ mA}}, V_{OC} \cong \mathbf{0.5 \text{ V}}$

At $V_{OC} = 0 \text{ V}, I_{SC} = \mathbf{100 \text{ mA}}$

At $I_{SC} = 0 \text{ A}, V_{OC} = \mathbf{0.56 \text{ V}}$

- (b) Plotting $P = VI$ we find $P_{\max} = VI = (0.5 \text{ V})(90 \text{ mA}) = \mathbf{45 \text{ mW}}$
- (c) $P_{\max} = 45 \text{ mW}$ vs. $P = VI = (0.5 \text{ V})(180 \text{ mA}) = \mathbf{90 \text{ mW}}$ for f_{c_2}
21. (a) $W = \frac{h\nu}{\lambda} = \frac{(6.624 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ m/s})}{470 \text{ nm(blue)}} = \mathbf{422.81 \times 10^{-21} \text{ J}}$
- (b) $W = \frac{h\nu}{\lambda} = \frac{(6.624 \times 10^{-34} \text{ Js})(3 \times 10^8 \text{ m/s})}{650 \text{ nm(red)}} = \mathbf{305.72 \times 10^{-21} \text{ J}}$
- (c) yes
- (d) Yes, higher energy level.
- (e) Low wavelength of emitted light has high energy content.
22. $\frac{4 \times 10^{-9} \text{ W/m}^2}{1.609 \times 10^{-12}} = 2,486 f_c$
 From the intersection of $V_A = 30 \text{ V}$ and $2,486 f_c$ we find
 $I_\lambda \cong \mathbf{440 \mu A}$
23. Note that V_λ is given and not V .
 At the intersection of $V_\lambda = 25 \text{ V}$ and $3000 f_c$ we find $I_\lambda \cong 500 \mu A$ and
 $V_R = I_\lambda R = (500 \times 10^{-6} \text{ A})(100 \times 10^3 \Omega) = \mathbf{50 \text{ V}}$
24. Drawing a straight line approximation to the curve through the intersection of the axis will result in
- $$m = \frac{\Delta y}{\Delta x} = \frac{200 \mu A}{1000 f_c} = 0.2 \mu A/f_c$$
- $$y = mx + b \quad I_\lambda \quad (0.2 \mu A/f_c)(f_c) \quad 0$$
- $$I_\lambda = \mathbf{(0.2 \mu A/f_c)(f_c)}$$
25. (a) Extending the curve:
 $0.1 \text{ k}\Omega \rightarrow 1000 f_c, 1 \text{ k}\Omega \rightarrow 25 f_c$
 $\frac{\Delta R}{\Delta f_c} = \frac{(1 - 0.1) \times 10^3 \Omega}{(1000 - 25) f_c} = \mathbf{0.92 \Omega/f_c} \cong \mathbf{0.9 \Omega/f_c}$
- (b) $1 \text{ k}\Omega \rightarrow 25 f_c, 10 \text{ k}\Omega \rightarrow 1.3 f_c$
 $\frac{\Delta R}{\Delta f_c} = \frac{(10 - 1) \times 10^3 \Omega}{(25 - 1.3) f_c} = \mathbf{379.75 \Omega/f_c} \cong \mathbf{380 \Omega/f_c}$

(c) $10 \text{ k}\Omega \rightarrow 1.3f_c$, $100 \text{ k}\Omega \rightarrow 0.15f_c$

$$\frac{\Delta R}{\Delta f_c} = \frac{(100 - 10) \times 10^3}{(1.3 - 0.15)f_c} = 78,260.87 \Omega/f_c \cong 78 \times 10^3 \Omega/f_c$$

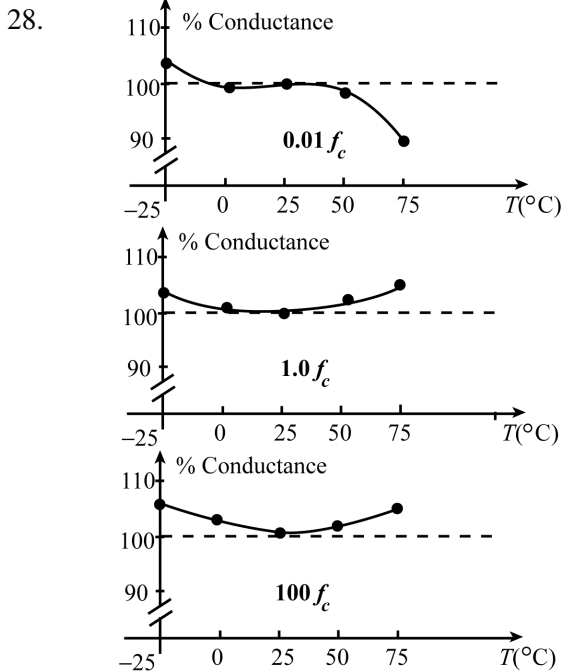
The greatest rate of change in resistance occurs in the low illumination region.

26. The “dark current” of a photodiode is the diode current level when no light is striking the diode. It is essentially the reverse saturation leakage current of the diode, comprised mainly of minority carriers.

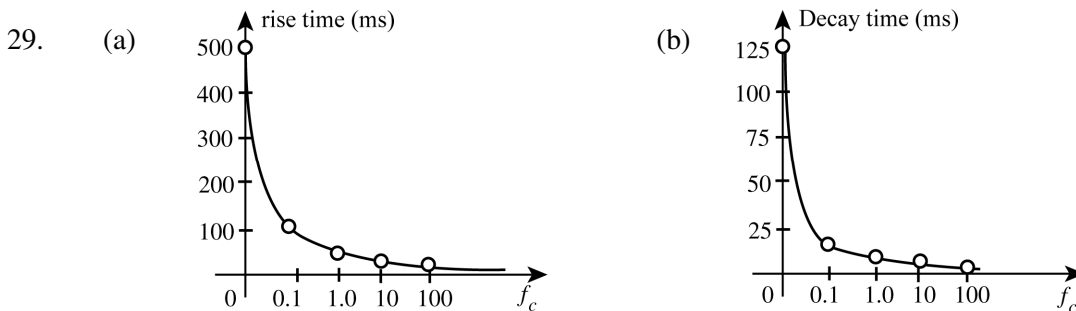
27. $10f_c \rightarrow R \cong 2 \text{ k}\Omega$

$$V_o = 6 \text{ V} = \frac{(2 \times 10^3 \Omega)V_i}{2 \times 10^3 \Omega + 5 \times 10^3 \Omega}$$

$$V_i = 21 \text{ V}$$



Except for low illumination levels ($0.01f_c$) the % conductance curves appear above the 100% level for the range of temperature. In addition, it is interesting to note that for other than the low illumination levels the % conductance is higher above and below room temperature (25°C). In general, the % conductance level is not adversely affected by temperature for the illumination levels examined.



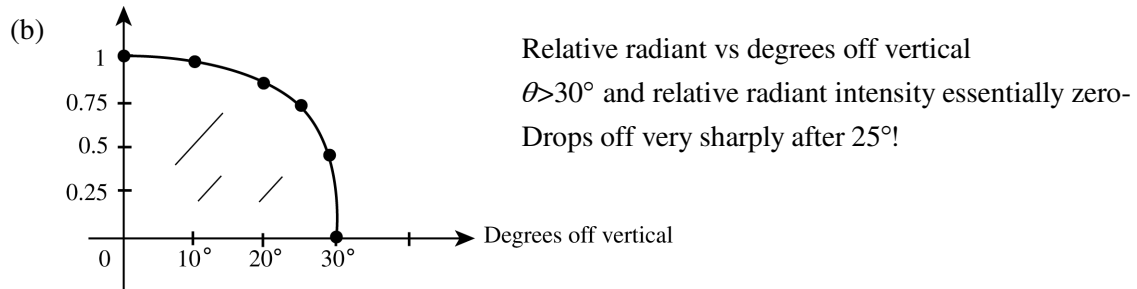
- (c) Increased levels of illumination result in reduced rise and decay times.

30. The highest % sensitivity occurs between 5250Å and 5750Å. Fig 16.20 reveals that the CdS unit would be most sensitive to *yellow*. The % sensitivity of the CdS unit of Fig. 16.30 is at the 30% level for the range 4800Å → 7000Å. This range includes green, yellow, and orange in Fig. 16.20.

31. (a) $\cong 5 \text{ mW}$ radiant flux

(b) $\cong 3.5 \text{ mW}$ $\frac{3.5 \text{ mW}}{1.496 \times 10^{-13} \text{ W/lm}} = 2.34 \times 10^{10} \text{ lms}$

32. (a) Relative radiant intensity $\cong 0.8$.



33. At $I_F = 60 \text{ mA}$, $\Phi \cong 4.4 \text{ mW}$
 At 5° , relative radiant intensity = 0.8
 $(0.8)(4.4 \text{ mW}) = 3.52 \text{ mW}$

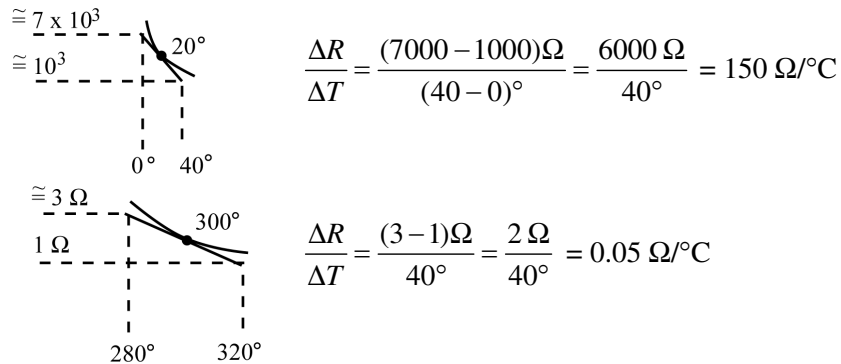
34. 6, 7, 8

35. —

36. The LED generates a light source in response to the application of an electric voltage. The LCD depends on ambient light to utilize the change in either reflectivity or transmissivity caused by the application of an electric voltage.

37. The LCD display has the advantage of using approximately 1000 times less power than the LED for the same display, since much of the power in the LED is used to produce the light, while the LCD utilizes ambient light to see the display. The LCD is usually more visible in daylight than the LED since the sun's brightness makes the LCD easier to see. The LCD, however, requires a light source, either internal or external, and the temperature range of the LCD is limited to temperatures above freezing.

38. Since log scales are present, the differentials must be as small as possible.



From the above $150 \Omega/^\circ\text{C}$: $0.05 \Omega/^\circ\text{C} = 3000:1$

Therefore, the highest rate of change occurs at lower temperatures such as 20°C .

39. No. 1 Fenwall Electronics Thermistor material.

Specific resistance $\approx 10^4 = 10,000 \Omega \text{ cm}$

$$R = \frac{\rho \ell}{A} \quad \therefore R = 2 \times (10,000 \Omega) = \mathbf{20 \text{ k}\Omega}$$

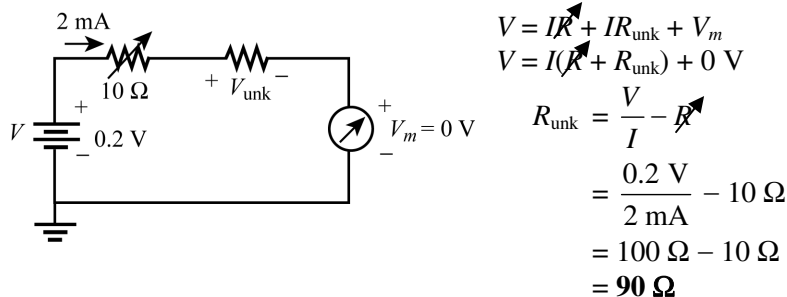
twice

40. (a) $\approx 10^{-5} \text{ A} = \mathbf{10 \mu\text{A}}$

(b) Power $\approx \mathbf{0.1 \text{ mW}}$, $R \approx 10^7 \Omega = \mathbf{10 \text{ M}\Omega}$

(c) Log scale $\approx \mathbf{0.3 \text{ mW}}$

- 41.



42. The primary difference between the standard p - n junction diode and the tunnel diode is that the tunnel diode is doped at a level from 100 to several thousand times the doping level of a p - n junction diode, thus producing a diode with a “negative resistance” region in its characteristic curve.

43. At 1 MHz: $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(1 \times 10^6 \text{ Hz})(5 \times 10^{-12} \text{ F})}$
 $= \mathbf{31.83 \text{ k}\Omega}$

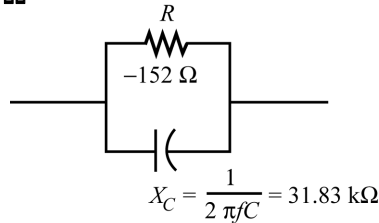
At 100 MHz: $X_C = \frac{1}{2\pi(100 \times 10^6 \text{ Hz})(5 \times 10^{-12} \text{ F})}$
 $= \mathbf{318.3 \Omega}$

At 1 MHz: $X_{L_S} = 2\pi fL = 2\pi(1 \times 10^6 \text{ Hz})(6 \times 10^{-9} \text{ H})$
 $= \mathbf{0.0337 \Omega}$

At 100 MHz: $X_{L_S} = 2\pi(100 \times 10^6 \text{ Hz})(6 \times 10^{-9} \text{ H})$
 $= \mathbf{3.769 \Omega}$

L_S effect is negligible!

R and C in parallel:
 $f = 1 \text{ MHz}$



$$Z_T = \frac{(152 \Omega \angle 180^\circ)(31.83 \text{ k}\Omega \angle -90^\circ)}{-152 \Omega - j31.83 \text{ k}\Omega}$$

$$= -152.05 \Omega \angle 0.27^\circ \cong -152 \Omega \angle 0^\circ$$

$f = 100 \text{ MHz}$

$$Z_T = \frac{(152 \Omega \angle 180^\circ)(318.3 \Omega \angle -90^\circ)}{-152 \Omega - j318.3}$$

$$= -137.16 \Omega \angle 25.52^\circ \neq -152 \Omega \angle 0^\circ$$

At very high frequencies X_C has some impact!

44. The heavy doping greatly reduces the width of the depletion region resulting in lower levels of Zener voltage. Consequently, small levels of reverse voltage can result in a significant current levels.

45. At $V_T = 0.1 \text{ V}$,
 $I_F \cong 5.5 \text{ mA}$
 At $V_T = 0.3 \text{ V}$
 $I_F \cong 2.3 \text{ mA}$

$$R = \frac{\Delta V}{\Delta I} = \frac{0.3 \text{ V} - 0.1 \text{ V}}{2.3 \text{ mA} - 5.5 \text{ mA}}$$

$$= \frac{0.2 \text{ V}}{-3.2 \text{ mA}} = \mathbf{-62.5 \Omega}$$

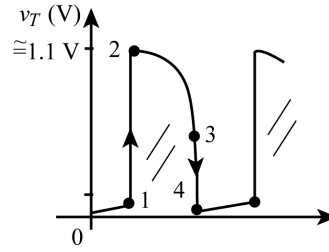
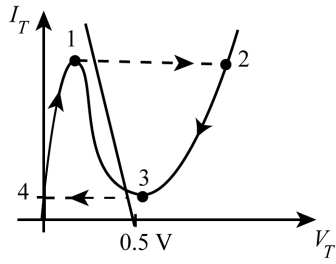
46. $I_{\text{sat}} = \frac{E}{R} = \frac{2 \text{ V}}{0.39 \text{ k}\Omega} \cong 5.13 \text{ mA}$

From graph: Stable operating points: $I_T \cong \mathbf{5 \text{ mA}}$, $V_T \cong \mathbf{60 \text{ mV}}$

$I_T \cong \mathbf{2.8 \text{ mA}}$, $V_T = \mathbf{900 \text{ mV}}$

47.
$$I_{\text{sat}} = \frac{E}{R} = \frac{0.5 \text{ V}}{51 \Omega} = 9.8 \text{ mA}$$

Draw load line on characteristics.



48.
$$f_s = \frac{1}{2\pi\sqrt{LC}} \sqrt{1 - \frac{R_l^2 C}{L}}$$

$$= \frac{1}{2\pi\sqrt{(5 \times 10^{-3} \text{ H})(1 \times 10^{-6} \text{ F})}} \sqrt{1 - \frac{(10 \Omega)^2 (1 \times 10^{-6} \text{ F})}{5 \times 10^{-3} \text{ H}}}$$

$$= (2250.79 \text{ Hz})(0.9899)$$

$$\cong \mathbf{2228 \text{ Hz}}$$

Chapter 17

1. –
2. –
3. –
4.
 - (a) p - n junction diode
 - (b) The SCR will not fire once the gate current is reduced to a level that will cause the forward blocking region to extend beyond the chosen anode-to-cathode voltage. In general, as I_G decreases, the blocking voltage required for conduction increases.
 - (c) The SCR will fire once the anode-to-cathode voltage is less than the forward blocking region determined by the gate current chosen.
 - (d) The holding current increases with decreasing levels of gate current.
5.
 - (a) Yes
 - (b) No
 - (c) No. As noted in Fig. 17.8b the minimum gate voltage required to trigger all units is 3 V.
 - (d) $V_G = 6$ V, $I_G = 800$ mA is a good choice (center of preferred firing area).
 $V_G = 4$ V, $I_G = 1.6$ A is less preferable due to higher power dissipation in the gate. Not in preferred firing area.
6. In the conduction state, the SCR has characteristics very similar to those of a p - n junction diode (where $V_T = 0.7$ V).
7. The smaller the level of R_1 , the higher the peak value of the gate current. The higher the peak value of the gate current the sooner the triggering level will be reached and conduction initiated.
8.
 - (a)

$$V_P = \frac{V_{sec}(rms)}{2} \sqrt{2}$$

$$= \frac{117 \text{ V}}{2} (\sqrt{2}) = 82.78 \text{ V}$$

$$V_{DC} = 0.636(82.78 \text{ V})$$

$$= \mathbf{52.65 \text{ V}}$$
 - (b) $V_{AK} = V_{DC} - V_{Batt} = 52.65 \text{ V} - 11 \text{ V} = \mathbf{41.65 \text{ V}}$

- (c) $V_R = V_Z + V_{GK}$
 $= 11 \text{ V} + 3 \text{ V}$
 $= 14 \text{ V}$
 At 14 V, SCR₂ conducts and stops the charging process.
- (d) At least 3 V to turn on SCR₂.
- (e) $V_2 \cong \frac{1}{2}V_P = \frac{1}{2}(82.78 \text{ V}) = \mathbf{41.39 \text{ V}}$
9. (a) Full-wave rectified waveform with peak value of 168.28 V
- (b) $P = VI = 100 \text{ W} = (0.707)(168.28 \text{ V})(I_{\text{rms}})$
 $I_{\text{rms}} = 840.5 \text{ mA}, I_{\text{peak}} = \mathbf{1.19 \text{ A}}$
- (c) $\mathbf{1.19 \text{ A}}$
- (d) $t_r = \frac{1}{4}T = \frac{1}{4} \frac{1}{60 \text{ Hz}} = \mathbf{4.17 \text{ ms}}$
- (e) $\tau = RC(510 \text{ k}\Omega)(0.1 \mu\text{F}) = \mathbf{51 \text{ ms}}$
- (f) open
- (g) dc level established $= 0.636(V_{\text{peak}}) = 0.636(168.28 \text{ V}) \cong 107 \text{ V}$
 $v_C = 107 \text{ V}(1 - e^{-t/51 \text{ ms}}) = 40 \text{ V}$
 $t = \mathbf{23.86 \text{ ms}}$
- (h) turn on
- (g) forced commutation
10. (a) $V_{\text{peak}} = (1.414)(6.3 \text{ V}) = 8.91 \text{ V}$
 $V_{\text{peak across 6 V lamp}} = 8.91 \text{ V} - 0.7 \text{ V} = \mathbf{8.21 \text{ V}}$
- (b) $V_{R_2} = \frac{150 \Omega(6 \text{ V})}{150 \Omega + 1 \text{ k}\Omega} = 0.783 \text{ V}$
 $V_{C_1(\text{peak})} = 8.21 \text{ V} - 0.783 \text{ V} = \mathbf{7.43 \text{ V}}$
- (c) $V_{\text{peak}} = 8.21 \text{ V} - 0.7 \text{ V} - 5 \text{ V} = \mathbf{2.51 \text{ V}}$
- (d) $V_{\text{lamp}} = 6 \text{ V}$
- (e) $P = VI = 2 \text{ W} = (6 \text{ V})(I)$
 $I = \frac{2 \text{ W}}{6 \text{ V}} = \mathbf{0.333 \text{ A}}$

11. –
12. Negative pulse at the cathode terminal.
13. (a) $V_{GK} = -12 \text{ V} + V'_R, V'_R = \frac{R'(24 \text{ V})}{R' + R_S}$
- $$\therefore V_{GK} = -12 \text{ V} + \frac{R'(24 \text{ V})}{R' + R_S}$$
- (b) $R_S = R', V_{GK} = 0 \text{ V}$, scs off
- (c) $V_{GK} = 2 \text{ V} = -12 \text{ V} + \frac{10 \text{ k}\Omega(24 \text{ V})}{10 \text{ k}\Omega + R_S}$
- $$R_S = \mathbf{14 \text{ k}\Omega}$$
- (d) $I = \frac{12 \text{ V}}{200 \Omega} = \mathbf{60 \text{ mA}}$
- (e) $I = \frac{12 \text{ V}}{100 \text{ k}\Omega} = \mathbf{0.12 \text{ mA}}$
- (f) Inductive element in alarm could establish spikes. Use protective capacitive element.
14. (a) Charge toward 200 V but will be limited by the development of a negative voltage $V_{GK}(=V_Z - V_{C_1})$ that will eventually turn the GTO off.
- (b) $\tau = R_3 C_1 = (20 \text{ k}\Omega)(0.1 \mu\text{F})$
 $= 2 \text{ ms}$
 $5\tau = \mathbf{10 \text{ ms}}$
- (c) $5\tau' = \frac{1}{2}(5\tau) = 5 \text{ ms} = 5R_{GTO} C_1$
- $$R_{GTO} = \frac{5 \text{ ms}}{5C_1} = \frac{5 \text{ ms}}{5(0.1 \times 10^{-6} \text{ F})} = \mathbf{10 \text{ k}\Omega} = \frac{1}{2}(20 \text{ k}\Omega - \text{above})$$
15. (a) $\cong \mathbf{0.7 \text{ mW/cm}^2}$
- (b) $0^\circ\text{C} \rightarrow 0.82 \text{ mW/cm}^2$
 $100^\circ\text{C} \rightarrow 0.16 \text{ mW/cm}^2$
 $\frac{0.82 - 0.16}{0.82} \times 100\% \cong \mathbf{80.5\%}$

$$\begin{aligned}
16. \quad V_C &= V_{BR} + V_{GK} = 6 \text{ V} + 3 \text{ V} = 9 \text{ V} \\
V_C &= 40(1 - e^{-t/RC}) = 9 \\
40 - 40e^{-t/RC} &= 9 \\
40e^{-t/RC} &= 31 \\
e^{-t/RC} &= 31/40 = 0.775 \\
RC &= (10 \times 10^3 \Omega)(0.2 \times 10^{-6} \text{ F}) = 2 \times 10^{-3} \text{ s} \\
\log_e(e^{-t/RC}) &= \log_e 0.775 \\
-t/RC &= -t/2 \times 10^{-3} = -0.255 \\
\text{and } t &= 0.255(2 \times 10^{-3}) = \mathbf{0.51 \text{ ms}}
\end{aligned}$$

$$17. \quad -$$

$$\begin{aligned}
18. \quad V_{BR_1} &= V_{BR_2} \pm 10\% V_{BR_2} \\
&= 6.4 \text{ V} \pm 0.64 \text{ V} \Rightarrow \mathbf{5.76 \text{ V} \rightarrow 7.04 \text{ V}}
\end{aligned}$$

$$19. \quad \mathbf{V_G} = \frac{(1 \text{ M}\Omega - jX_{C_b})V_i \angle 0^\circ}{10 \text{ M}\Omega + 1 \text{ M}\Omega - jX_{C_b}} = \frac{(1 \text{ M}\Omega - jX_{C_b})V_i \angle 0^\circ}{11 \text{ M}\Omega - jX_{C_b}}$$

For 45° shift in denominator $X_{C_b} = 11 \text{ M}\Omega$. In the numerator the result is $1 \text{ M}\Omega - j11 \text{ M}\Omega \cong -j11 \text{ M}\Omega = 11 \text{ M}\Omega \angle -90^\circ$.

$$\therefore \mathbf{V_G} = \frac{(11 \text{ M}\Omega \angle -90^\circ)(V_i \angle 0^\circ)}{15.56 \text{ M}\Omega \angle -45^\circ} = 0.707V_i \angle -45^\circ$$

$\therefore$ A 45° shift is established with $X_{C_b} = 11 \text{ M}\Omega$

$$X_{C_b} = \frac{1}{2\pi f C_b} \quad C_b = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(60 \text{ Hz})(11 \text{ M}\Omega)} \quad \mathbf{241 \text{ pF}}$$

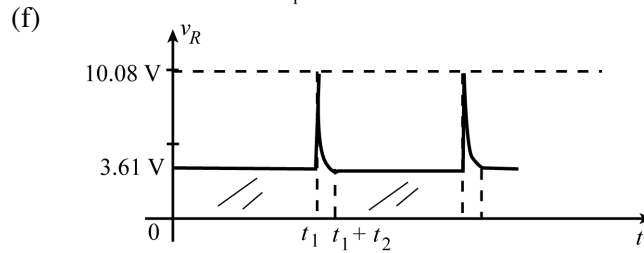
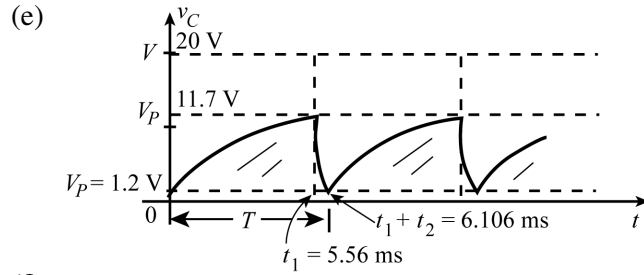
$$\begin{aligned}
20. \quad \mathbf{V_{x_c}} &= \frac{(X_C \angle -90^\circ)(170 \text{ V} \angle 0^\circ)}{R - jX_C} = 12 \text{ V} \angle \theta \\
&= \frac{(X_C)((170 \text{ V})(\angle -90^\circ))}{\sqrt{R^2 + X_C^2} \angle -\theta'} = 12 \text{ V} \angle \theta \\
&\therefore \frac{(X_C)(170 \text{ V})}{\sqrt{R^2 + X_C^2}} = 12 \text{ V}
\end{aligned}$$

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(60 \text{ Hz})(1 \mu\text{F})} = 2.65 \text{ k}\Omega$$

$$\frac{(2.65 \text{ k}\Omega)(170 \text{ V})}{\sqrt{R^2 + (2.65 \text{ k}\Omega)^2}} = 12 \text{ V}$$

$$\mathbf{R = 25.25 \text{ k}\Omega}$$

21. $\frac{V - V_P}{I_P} > R_1$
 $\frac{40 \text{ V} - [0.6(40 \text{ V}) + 0.7 \text{ V}]}{10 \times 10^{-6}} = \mathbf{1.53 \text{ M}\Omega} > R_1$
 $\frac{V - V_V}{I_V} < R_1 \Rightarrow \frac{40 \text{ V} - 1 \text{ V}}{8 \text{ mA}} = 4.875 \text{ k}\Omega < R_1$
 $\therefore \mathbf{1.53 \text{ M}\Omega > R_1 > 4.875 \text{ k}\Omega}$
22. (a) $\eta = \frac{R_{B_1}}{R_{B_1} + R_{B_2}} \Big|_{I_E=0} \Rightarrow 0.65 = \frac{2 \text{ k}\Omega}{2 \text{ k}\Omega + R_{B_2}} \quad R_{B_2} = \mathbf{1.08 \text{ k}\Omega}$
- (b) $R_{BB} = (R_{B_1} + R_{B_2}) \Big|_{I_E=0} = 2 \text{ k}\Omega + 1.08 \text{ k}\Omega = \mathbf{3.08 \text{ k}\Omega}$
- (c) $V_{R_{B_1}} = \eta V_{BB} = 0.65(20 \text{ V}) = \mathbf{13 \text{ V}}$
- (d) $V_P = \eta V_{BB} + V_D = 13 \text{ V} + 0.7 \text{ V} = \mathbf{13.7 \text{ V}}$
23. (a) $\eta = \frac{R_{B_1}}{R_{BB}} \Big|_{I_E=0}$
 $0.55 = \frac{R_{B_1}}{10 \text{ k}\Omega}$
 $R_{B_1} = \mathbf{5.5 \text{ k}\Omega}$
 $R_{BB} = R_{B_1} + R_{B_2}$
 $10 \text{ k}\Omega = 5.5 \text{ k}\Omega + R_{B_2}$
 $R_{B_2} = \mathbf{4.5 \text{ k}\Omega}$
- (b) $V_P = \eta V_{BB} + V_D = (0.55)(20 \text{ V}) + 0.7 \text{ V} = \mathbf{11.7 \text{ V}}$
- (c) $R_1 < \frac{V - V_P}{I_P} = \frac{20 \text{ V} - 11.7 \text{ V}}{50 \mu\text{A}} = 166 \text{ k}\Omega$
ok: $68 \text{ k}\Omega < 166 \text{ k}\Omega$
- (d) $t_1 = R_1 C \log_e \frac{V - V_V}{V - V_P} = (68 \times 10^3)(0.1 \times 10^{-6}) \log_e \frac{18.8}{8.3} = 5.56 \text{ ms}$
 $t_2 = (R_{B_1} + R_2) C \log_e \frac{V_P}{V_V} = (0.2 \text{ k}\Omega + 2.2 \text{ k}\Omega)(0.1 \times 10^{-6}) \log_e \frac{11.7}{1.2}$
 $= 0.546 \text{ ms}$
 $T = t_1 + t_2 = 6.106 \text{ ms}$
 $f = \frac{1}{T} = \frac{1}{6.106 \text{ ms}} = \mathbf{163.77 \text{ Hz}}$



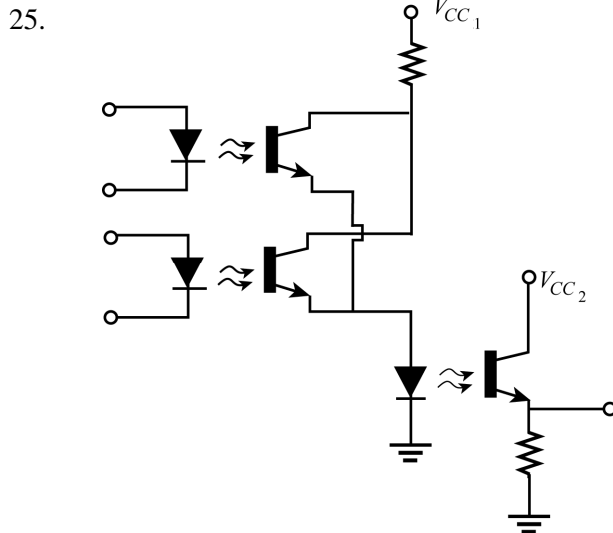
$$V_{R_2} = \frac{R_2 V}{R_2 + R_{BB}} = \frac{2.2 \text{ k}\Omega (20 \text{ V})}{2.2 \text{ k}\Omega + 10 \text{ k}\Omega} = 3.61 \text{ V}$$

$$V_{R_2} \cong \frac{R_2 (V_P - 0.7 \text{ V})}{R_2 + R_{B_1}} = \frac{2.2 \text{ k}\Omega (11.7 \text{ V} - 0.7 \text{ V})}{2.2 \text{ k}\Omega + 0.2 \text{ k}\Omega} = 10.08 \text{ V}$$

(g) $f \cong \frac{1}{R_1 C \log_e (1/(1 - \eta))} = \frac{1}{(6.8 \text{ k}\Omega)(0.1 \text{ }\mu\text{F}) \log_e 2.22} = 184.16 \text{ Hz}$

difference in frequency levels is partly due to the fact that $t_2 \cong 10\%$ of t_1 .

24. $I_B = 25 \text{ }\mu\text{A}$
 $I_C = h_{fe} I_B = (40)(25 \text{ }\mu\text{A}) = 1 \text{ mA}$



$$26. \quad (a) \quad D_F = \frac{\Delta I}{\Delta T} \\ = \frac{0.95 - 0}{25 - (-50)} = \frac{0.95}{75} = \mathbf{1.26\%/^{\circ}\text{C}}$$

(b) Yes, curve flattens after 25°C.

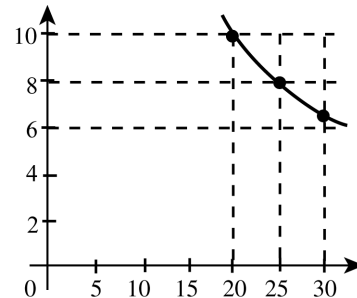
$$27. \quad (a) \quad \text{At } 25^{\circ}\text{C}, I_{CEO} \cong 2 \text{ nA} \\ \text{At } 50^{\circ}\text{C}, I_{CEO} \cong 30 \text{ nA} \\ \frac{\Delta I_{CEO}}{\Delta T} = \frac{(30 - 2) \times 10^{-9} \text{ A}}{(50 - 25)^{\circ}\text{C}} = \frac{28 \text{ nA}}{25^{\circ}\text{C}} = \mathbf{1.12 \text{ nA}/^{\circ}\text{C}} \\ I_{CEO}(35^{\circ}\text{C}) = I_{CEO}(25^{\circ}\text{C}) + (1.12 \text{ nA}/^{\circ}\text{C})(35^{\circ}\text{C} - 25^{\circ}\text{C}) \\ = 2 \text{ nA} + 11.2 \text{ nA} \\ = \mathbf{13.2 \text{ nA}} \\ \text{From Fig. 17.55 } I_{CEO}(35^{\circ}\text{C}) \cong \mathbf{4 \text{ nA}}$$

Derating factors, therefore, cannot be defined for large regions of non-linear curves. Although the curve of Fig. 17.55 appears to be linear, the fact that the vertical axis is a log scale reveals that I_{CEO} and $T(^{\circ}\text{C})$ have a non-linear relationship.

$$28. \quad \frac{I_o}{I_i} = \frac{I_C}{I_F} = \frac{20 \text{ mA}}{\cong 45 \text{ mA}} = \mathbf{0.44}$$

Yes, relatively efficient.

$$29. \quad (a) \quad P_D = V_{CE} I_C = 200 \text{ mW} \\ I_C = \frac{P_D}{V_{CE_{\max}}} = \frac{200 \text{ mW}}{30 \text{ V}} = 6.67 \text{ mA @ } V_{CE} = 30 \text{ V} \\ V_{CE} = \frac{P_D}{I_C} = \frac{200 \text{ mW}}{10 \text{ mA}} = 20 \text{ V @ } I_C = 10 \text{ mA} \\ I_C = \frac{P_D}{V_{CE}} = \frac{200 \text{ mW}}{25 \text{ V}} = 8.0 \text{ mA @ } V_{CE} = 25 \text{ V}$$



Almost the entire area of Fig. 17.57 falls within the power limits.

$$(b) \quad \beta_{dc} = \frac{I_C}{I_F} = \frac{4 \text{ mA}}{10 \text{ mA}} = \mathbf{0.4}, \text{ Fig. 17.56 } \frac{I_C}{I_F} \cong \frac{4 \text{ mA}}{10 \text{ mA}} = \mathbf{0.4}$$

The fact that the I_F characteristics of Fig. 17.57 are fairly horizontal reveals that the level of I_C is somewhat unaffected by the level of V_{CE} except for very low or high values. Therefore, a plot of I_C vs. I_F as shown in Fig. 17.56 can be provided without any reference to the value of V_{CE} . As noted above, the results are essentially the same.

30. (a) $I_C \geq 3 \text{ mA}$

(b) At $I_C = 6 \text{ mA}$; $R_L = 1 \text{ k}\Omega$, $t = 8.6 \mu\text{s}$
 $R_L = 100 \Omega$; $t = 2 \mu\text{s}$
 $1 \text{ k}\Omega : 100 \Omega = 10:1$
 $8.6 \mu\text{s} : 2 \mu\text{s} = 4.3:1$
 $\Delta R : \Delta t \cong \mathbf{2.3:1}$

31. $\eta = \frac{3R_{B_2}}{3R_{B_2} + R_{B_1}} = \frac{3}{4} = \mathbf{0.75}$, $V_G = \eta V_{BB} = 0.75(20 \text{ V}) = \mathbf{15 \text{ V}}$

32. $V_P = 8.7 \text{ V}$, $I_P = 100 \mu\text{A}$ $Z_P = \frac{V_P}{I_P} = \frac{8.7 \text{ V}}{100 \mu\text{A}} = \mathbf{87 \text{ k}\Omega}$ ($\cong$ open)

$V_V = 1 \text{ V}$, $I_V = 5.5 \text{ mA}$ $Z_V = \frac{V_V}{I_V} = \frac{1 \text{ V}}{5.5 \text{ mA}} = \mathbf{181.8 \Omega}$ (relatively low)

$87 \text{ k}\Omega : 181.8 \Omega = 478.55:1 \cong \mathbf{500:1}$

33. Eq. 17.23: $T = RC \log_e \frac{V_{BB}}{V_{BB} - V_P} = RC \log_e \frac{V_{BB}}{V_{BB} - (\eta V_{BB} + V_D)}$

Assuming $\eta V_{BB} \gg V_D$, $T = RC \log_e \frac{V_{BB}}{V_{BB}(1 - \eta)} = RC \log_e (1/(1 - \eta)) = RC \log_e \frac{1}{1 - \frac{R_{B_1}}{R_{B_1} + R_{B_2}}}$

$= RC \log_e \frac{R_{B_1} + R_{B_2}}{R_{B_2}} = RC \log_e 1 + \frac{R_{B_1}}{R_{B_2}}$ Eq. 17.24

34. (a) Minimum V_{BB} :

$$R_{\max} = \frac{V_{BB} - V_P}{I_P} \geq 20 \text{ k}\Omega$$

$$\frac{V_{BB} - (\eta V_{BB} + V_D)}{I_P} = 20 \text{ k}\Omega$$

$$V_{BB} - \eta V_{BB} - V_D = I_P 20 \text{ k}\Omega$$

$$V_{BB}(1 - \eta) = I_P 20 \text{ k}\Omega + V_D$$

$$V_{BB} = \frac{I_P 20 \text{ k}\Omega + V_D}{1 - \eta}$$

$$= \frac{(100 \mu\text{A})(20 \text{ k}\Omega) + 0.7 \text{ V}}{1 - 0.67}$$

$$= \mathbf{8.18 \text{ V}}$$

10 V OK

$$(b) \quad R < \frac{V_{BB} - V_V}{I_V} = \frac{12 \text{ V} - 1 \text{ V}}{5.5 \text{ mA}} = 2 \text{ k}\Omega$$

$$R < \mathbf{2 \text{ k}\Omega}$$

$$(c) \quad T \cong RC \log_e 1 + \frac{R_{B_1}}{R_{B_2}}$$

$$2 \times 10^{-3} = R(1 \times 10^{-6}) \log_e 1 + \underbrace{\frac{10 \text{ k}\Omega}{5 \text{ k}\Omega}}_{\log_e 3 = 1.0986}$$

$$R = \frac{2 \times 10^{-3}}{(1 \times 10^{-6})(1.0986)}$$

$$R = \mathbf{1.82 \text{ k}\Omega}$$